Engineering Mathematics

Previous Year Questions

GATE — ALL BRANCHES

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Engineering Mathematics

Previous Year Questions

Engineering Mathematics (EC/EE/IN/CS/ME/CE/CH/PI)

Authors: Himanshu Agarwal & Anish Saha

Published by
PrepFusion LLP
www.PrepFusion.in

First Edition: August 2026

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Engineering Mathematics (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7^{th} question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○ ○ ○   ● ○ ○ Easy   ● ● ○ Medium   ● ● ● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions the real total for that branch.

MCQ, MSQ, NAT that same total, split by question type.

1M, 2M how many of those questions carry 1 or 2 marks.

Descriptive questions worth more than 2 marks, or whose marks value isn't recorded.

Avg Weightage the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.

Range the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

Visit our Website

prepfusion.in

Solutions, practice & report

pyq.prepfusion.in

Subscribe: English Channel

youtube.com/@PrepFusion_GATE

Subscribe: Hindi Channel

youtube.com/@prepfusion-hindi

Join our Telegram group

t.me/All_About_Learning

Follow us on LinkedIn

linkedin.com/company/prepfusionee

Subscribe: Himanshu Agarwal

youtube.com/@HimanshuAgarwal_

Subscribe: Anish Saha

youtube.com/@AnishSaha_

Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

UNIT

I

Linear Algebra

Topics

1. Systems of linear equations
2. Matrix algebra
3. Rank
4. Determinants
5. Eigenvalues and eigenvectors
6. Cayley–Hamilton theorem
7. Linear independence
8. Vector spaces
9. Basis and dimension
10. Linear transformations

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 51	**MCQ:** 35	**MSQ:** 2	**NAT:** 14	
1M: 31	**2M:** 20	**Desc:** 0	**Avg Weightage** ~3M	**Range:** 1M(2026)–3M(2024)

EE — Electrical Engineering

Questions: 55	**MCQ:** 42	**MSQ:** 3	**NAT:** 10	
1M: 30	**2M:** 24	**Desc:** 1	**Avg Weightage** ~3M	**Range:** 1M(2023)–8M(2026)

IN — Instrumentation Engineering

Questions: 36	**MCQ:** 27	**MSQ:** 2	**NAT:** 7	
1M: 24	**2M:** 12	**Desc:** 0	**Avg Weightage** ~3M	**Range:** 1M(2019)–3M(2023)

CS — Computer Science & IT

Questions: 57	**MCQ:** 34	**MSQ:** 6	**NAT:** 17	
1M: 33	**2M:** 23	**Desc:** 1	**Avg Weightage** ~3M	**Range:** 1M(2023)–4M(2025)

ME — Mechanical Engineering

Questions: 45	**MCQ:** 37	**MSQ:** 2	**NAT:** 6	
1M: 30	**2M:** 15	**Desc:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–4M(2022)

CE — Civil Engineering

Questions: 47	**MCQ:** 35	**MSQ:** 6	**NAT:** 6	
1M: 31	**2M:** 16	**Desc:** 0	**Avg Weightage** ~2M	**Range:** 1M(2019)–4M(2023)

CH — Chemical Engineering

Questions: 22	**MCQ:** 14	**MSQ:** 1	**NAT:** 7	
1M: 13	**2M:** 8	**Desc:** 1	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2021)

PI — Production & Industrial Engineering

Questions: 25	**MCQ:** 22	**MSQ:** 0	**NAT:** 3	
1M: 16	**2M:** 9	**Desc:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–4M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q1.26.1 MSQ 2026 1M

Consider the matrix $M = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{bmatrix}$. Which of the following options is/ are TRUE if $\det(M) \neq 0$?

- **A.** $a = -\frac{1}{2}$ and $b = -\frac{1}{2}$
- **B.** $a = \frac{1}{2}$ and $b = \frac{1}{2}$
- **C.** $a = -3$ and $b = 0$
- **D.** $a = \frac{1}{2}$ and $b = -3$

Q1.25.1 MCQ 2025 1M

Consider the matrix A below:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \gamma \end{bmatrix}$$

For which of the following combinations of α, β, and γ, is the rank of A at least three?

(i) $\alpha = 0$ and $\beta = \gamma \neq 0$.

(ii) $\alpha = \beta = \gamma = 0$.

(iii) $\beta = \gamma = 0$ and $\alpha \neq 0$.

(iv) $\alpha = \beta = \gamma \neq 0$.

A. Only (i), (iii), and (iv)

B. Only (iv)

C. Only (ii)

D. Only (i) and (iii)

Q1.24.1 NAT 2024 1M

Let $\mathbb{R}$ and $\mathbb{R}^3$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors

$$\{[2 \quad -3 \quad \alpha], \quad [3 \quad -1 \quad 3], \quad [1 \quad -5 \quad 7]\}$$

does not form a basis of $\mathbb{R}^3$ is ________.

Q1.24.2 MSQ 2024 2M

Consider the matrix $\begin{bmatrix} 1 & k \\ 2 & 1 \end{bmatrix}$, where k is a positive real number. Which of the following vectors is/are eigenvector(s) of this matrix?

- **A.** $\begin{bmatrix} 1 \\ -\sqrt{2/k} \end{bmatrix}$
- **B.** $\begin{bmatrix} 1 \\ \sqrt{2/k} \end{bmatrix}$
- **C.** $\begin{bmatrix} \sqrt{2k} \\ 1 \end{bmatrix}$
- **D.** $\begin{bmatrix} \sqrt{2k} \\ -1 \end{bmatrix}$

Q1.23.1 MCQ 2023 1M

Let the sets of eigenvalues and eigenvectors of a matrix B be $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{v_k \mid 1 \leq k \leq n\}$, respectively. For any invertible matrix P, the sets of eigenvalues and eigenvectors of the matrix A, where
$B = P^{-1}AP$, respectively, are

A. $\{\lambda_k \det(A) \mid 1 \leq k \leq n\}$ and $\{Pv_k \mid 1 \leq k \leq n\}$

B. $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{v_k \mid 1 \leq k \leq n\}$

C. $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{Pv_k \mid 1 \leq k \leq n\}$

D. $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{P^{-1}v_k \mid 1 \leq k \leq n\}$

Q1.23.2 MCQ 2023 2M

Let x be an $n \times 1$ real column vector with length $l = \sqrt{x^T x}$. The trace of the matrix $P = xx^T$ is

A. l^2 **B.** $\frac{l^2}{4}$ **C.** l **D.** $\frac{l^2}{2}$

Q1.22.1 MCQ 2022 1M

Consider a system of linear equations $Ax = b$, where

$$A = \begin{bmatrix} 1 & -\sqrt{2} & 3 \\ -1 & \sqrt{2} & -3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

This system of equations admits ________.

A. a unique solution for x

B. infinitely many solutions for x

C. no solutions for x

D. exactly two solutions for x

Q1.22.2 MCQ 2022 2M

Let α, β be two non-zero real numbers and v_1, v_2 be two non-zero real vectors of size 3×1. Suppose that v_1 and v_2 satisfy $v_1^T v_2 = 0$, $v_1^T v_1 = 1$, and $v_2^T v_2 = 1$.
Let A be the 3×3 matrix given by:

$$A = \alpha v_1 v_1^T + \beta v_2 v_2^T$$

The eigenvalues of A are ________.

A. $0, \alpha, \beta$

B. $0, \alpha + \beta, \alpha - \beta$

C. $0, \dfrac{\alpha + \beta}{2}, \sqrt{\alpha\beta}$

D. $0, 0, \sqrt{\alpha^2 + \beta^2}$

Q1.21.1 NAT 2021 1M

If the vectors $(1.0, -1.0, 2.0)$, $(7.0, 3.0, x)$ and $(2.0, 3.0, 1.0)$ in $\mathbb{R}^3$ are linearly dependent, the value of x is ________

Q1.21.2 NAT 2021 2M

A real 2×2 non-singular matrix A with repeated eigenvalue is given as

$$\mathbf{A} = \begin{bmatrix} x & -3.0 \\ 3.0 & 4.0 \end{bmatrix}$$

where x is a real positive number. The value of x (rounded off to one decimal place) is ________

Q1.20.1 MCQ 2020 1M

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_6$ are six vectors in $\mathbb{R}^4$, which one of the following statements is FALSE?

A. It is not necessary that these vectors span $\mathbb{R}^4$.

B. These vectors are not linearly independent.

C. Any four of these vectors form a basis for $\mathbb{R}^4$.

D. If $\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5, \mathbf{v}_6\}$ spans $\mathbb{R}^4$, then it forms a basis for $\mathbb{R}^4$.

Q1.20.2 MCQ 2020 2M

Consider the following system of linear equations.

$$x_1 + 2x_2 = b_1; \quad 2x_1 + 4x_2 = b_2;$$
$$3x_1 + 7x_2 = b_3; \quad 3x_1 + 9x_2 = b_4$$

Which one of the following conditions ensures that a solution exists for the above system?

A. $b_2 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

B. $b_3 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

C. $b_2 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

D. $b_3 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

Q1.19.1 NAT 2019 1M

The number of distinct eigenvalues of the matrix

$$A = \begin{bmatrix} 2 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

is equal to ________.

Q1.18.1 MCQ 2018 1M

Let M be a real 4×4 matrix. Consider the following statements:

S1: M has 4 linearly independent eigenvectors.

S2: M has 4 distinct eigenvalues.

S3: M is non-singular (invertible).

Which one among the following is TRUE?

A. S1 implies S2 **B.** S1 implies S3

C. S2 implies S1 **D.** S3 implies S2

Q1.18.2 NAT 2018 1M

Consider matrix $A = \begin{bmatrix} k & 2k \\ k^2 - k & k^2 \end{bmatrix}$ and vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
The number of distinct real values of k for which the equation $Ax = 0$ has infinitely many solutions is ______.

Q1.17.1 NAT 2017 1M

The rank of the matrix

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

is ________.

Q1.17.2 MCQ 2017 1M

Consider the 5×5 matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix}.$$

It is given that A has only one real eigenvalue. Then the real eigenvalue of A is

A. -2.5 **B.** 0 **C.** 15 **D.** 25

Q1.17.3 MCQ 2017 1M

The rank of the matrix $M = \begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.16.1 MCQ 2016 1M

Let $M^4 = I$ (where I denotes the identity matrix) and $M \neq I$, $M^2 \neq I$ and $M^3 \neq I$. Then, for any natural number k, M^{-1} equals:

A. M^{4k+1} **B.** M^{4k+2} **C.** M^{4k+3} **D.** M^{4k}

Q1.16.2 NAT 2016 1M

The value of x for which the matrix

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 9 & 7 & 13 \\ -6 & -4 & -9+x \end{bmatrix}$$

has zero as an eigenvalue is ________.

Q1.16.3 NAT 2016 2M

The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has $\det(A) = 100$ and $\text{trace}(A) = 14$. The value of $|a - b|$ is ________.

Q1.16.4 MCQ 2016 1M

Consider a 2×2 square matrix

$$\mathbf{A} = \begin{bmatrix} \sigma & x \\ \omega & \sigma \end{bmatrix},$$

where x is unknown. If the eigenvalues of the matrix $\mathbf{A}$ are $(\sigma + j\omega)$ and $(\sigma - j\omega)$, then x is equal to

A. $+j\omega$ **B.** $-j\omega$ **C.** $+\omega$ **D.** $-\omega$

Q1.16.5 MCQ 2016 2M

If the vectors $e_1 = (1, 0, 2)$, $e_2 = (0, 1, 0)$ and $e_3 = (-2, 0, 1)$ form an orthogonal basis of the three-dimensional real space $\mathbb{R}^3$, then the vector $u = (4, 3, -3) \in \mathbb{R}^3$ can be expressed as

A. $u = -\frac{2}{5}e_1 - 3e_2 - \frac{11}{5}e_3$

B. $u = -\frac{2}{5}e_1 - 3e_2 + \frac{11}{5}e_3$

C. $u = -\frac{2}{5}e_1 + 3e_2 + \frac{11}{5}e_3$

D. $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

Q1.15.1 NAT 2015 1M

Consider a system of linear equations:

$$x - 2y + 3z = -1,$$

$$x - 3y + 4z = 1, \text{ and}$$

$$-2x + 4y - 6z = k.$$

The value of k for which the system has infinitely many solutions is ______.

Q1.15.2 NAT 2015 1M

The value of p such that the vector $\begin{bmatrix}1\\2\\3\end{bmatrix}$ is an eigenvector of the matrix $\begin{bmatrix}4 & 1 & 2\\ p & 2 & 1\\ 14 & -4 & 10\end{bmatrix}$ is ________.

Q1.15.3 MCQ 2015 1M

The value of x for which all the eigen-values of the matrix given below are real is

$$\begin{bmatrix}10 & 5+j & 4\\ x & 20 & 2\\ 4 & 2 & -10\end{bmatrix}$$

A. $5+j$ **B.** $5-j$ **C.** $1-5j$ **D.** $1+5j$

Q1.15.4 MCQ 2015 1M

For $A = \begin{bmatrix}1 & \tan x\\ -\tan x & 1\end{bmatrix}$, the determinant of A^TA^{-1} is

A. $\sec^2 x$ **B.** $\cos 4x$ **C.** 1 **D.** 0

Q1.14.1 NAT 2014 1M

A real (4×4) matrix A satisfies the equation $A^2 = I$, where I is the (4×4) identity matrix. The positive eigen value of A is ________.

Q1.14.2 MCQ 2014 1M

For matrices of same dimension M, N and scalar c, which one of these properties DOES NOT ALWAYS hold?

A. $(M^T)^T = M$ **B.** $(cM)^T = c(M)^T$
C. $(M+N)^T = M^T + N^T$ **D.** $MN = NM$

Q1.14.3 NAT 2014 2M

Consider the matrix

$$J_6 = \begin{bmatrix}0&0&0&0&0&1\\0&0&0&0&1&0\\0&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&0&0&0\\1&0&0&0&0&0\end{bmatrix}$$

which is obtained by reversing the order of the columns of the identity matrix I_6. Let $P = I_6 + \alpha J_6$, where α is a non-negative real number. The value of α for which $\det(P) = 0$ is ________.

Q1.14.4 NAT 2014 1M

The determinant of matrix A is 5 and the determinant of matrix B is 40. The determinant of matrix AB is ________.

Q1.14.5 MCQ 2014 2M

The system of linear equations

$$\begin{pmatrix}2&1&3\\3&0&1\\1&2&5\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix} = \begin{pmatrix}5\\-4\\14\end{pmatrix} \text{ has}$$

A. a unique solution

B. infinitely many solutions

C. no solution

D. exactly two solutions

Q1.14.6 NAT 2014 2M

The maximum value of the determinant among all 2×2 real symmetric matrices with trace 14 is ________.

Q1.14.7 MCQ 2014 2M

Which one of the following statements is NOT true for a square matrix A?

A. If A is upper triangular, the eigenvalues of A are the diagonal elements of it

B. If A is real symmetric, the eigenvalues of A are always real and positive

C. If A is real, the eigenvalues of A and A^T are always the same

D. If all the principal minors of A are positive, all the eigenvalues of A are also positive

Q1.13.1 MCQ 2013 1M

The minimum eigenvalue of the following matrix is

$$\begin{bmatrix}3&5&2\\5&12&7\\2&7&5\end{bmatrix}$$

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.13.2 MCQ 2013 2M

Let A be an $m \times n$ matrix and B an $n \times m$ matrix. It is given that determinant $(I_m + AB)$ = determinant $(I_n + BA)$, where I_k is the $k \times k$ identity matrix. Using the above property, the determinant of the matrix given below is

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

A. 2 **B.** 5 **C.** 8 **D.** 16

Q1.12.1 MCQ 2012 2M

Given that

$A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the value of A^3 is

A. $15A + 12I$
B. $19A + 30I$
C. $17A + 15I$
D. $17A + 21I$

Q1.11.1 MCQ 2011 2M

The system of equations

$$\begin{aligned} x + y + z &= 6 \\ x + 4y + 6z &= 20 \\ x + 4y + \lambda z &= \mu \end{aligned}$$

has **NO** solution for values of λ and μ given by

A. $\lambda = 6, \mu = 20$
B. $\lambda = 6, \mu \neq 20$
C. $\lambda \neq 6, \mu = 20$
D. $\lambda \neq 6, \mu \neq 20$

Q1.10.1 MCQ 2010 1M

The eigenvalues of a skew-symmetric matrix are

A. always zero

B. always pure imaginary

C. either zero or pure imaginary

D. always real

Q1.09.1 MCQ 2009 2M

The eigen values of the following matrix are

$$\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

A. $3, 3 + 5j, 6 - j$
B. $-6 + 5j, 3 + j, 3 - j$
C. $3 + j, 3 - j, 5 + j$
D. $3, -1 + 3j, -1 - 3j$

Q1.08.1 MCQ 2008 1M

All the four entries of the 2×2 matrix $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$ are nonzero, and one of its eigenvalues is zero. Which of the following statements is true?

A. $p_{11}p_{22} - p_{12}p_{21} = 1$
B. $p_{11}p_{22} - p_{12}p_{21} = -1$
C. $p_{11}p_{22} - p_{12}p_{21} = 0$
D. $p_{11}p_{22} + p_{12}p_{21} = 0$

Q1.08.2 MCQ 2008 1M

The system of linear equations

$$4x + 2y = 7$$

$$2x + y = 6$$

has

A. a unique solution

B. no solution

C. an infinite number of solutions

D. exactly two distinct solutions

Q1.06.1 MCQ 2006 1M

The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is:

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.06.2 MCQ 2006 2M

The eigenvalues and the corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	**Eigenvector**
$\lambda_1 = 8$	$v_1 = \begin{bmatrix}1\\1\end{bmatrix}$
$\lambda_2 = 4$	$v_2 = \begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is:

A. $\begin{bmatrix}6 & 2\\2 & 6\end{bmatrix}$ **B.** $\begin{bmatrix}4 & 6\\6 & 4\end{bmatrix}$ **C.** $\begin{bmatrix}2 & 4\\4 & 2\end{bmatrix}$ **D.** $\begin{bmatrix}4 & 8\\8 & 4\end{bmatrix}$

Q1.06.3 MCQ 2006 2M

For the matrix $\begin{bmatrix}4 & 2\\2 & 4\end{bmatrix}$ the eigenvalue corresponding to the eigenvector $\begin{bmatrix}101\\101\end{bmatrix}$ is:

A. 2 **B.** 4 **C.** 6 **D.** 8

Q1.05.1 MCQ 2005 2M

Given an orthogonal matrix $\begin{bmatrix}1 & 1 & 1 & 1\\1 & 1 & -1 & -1\\1 & -1 & 0 & 0\\0 & 0 & 1 & -1\end{bmatrix}$, $[AA^T]^{-1}$ is:

A. $\begin{bmatrix}\frac{1}{4} & 0 & 0 & 0\\0 & \frac{1}{4} & 0 & 0\\0 & 0 & \frac{1}{2} & 0\\0 & 0 & 0 & \frac{1}{2}\end{bmatrix}$ **B.** $\begin{bmatrix}\frac{1}{2} & 0 & 0 & 0\\0 & \frac{1}{2} & 0 & 0\\0 & 0 & \frac{1}{2} & 0\\0 & 0 & 0 & \frac{1}{2}\end{bmatrix}$

C. $\begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & 1\end{bmatrix}$ **D.** $\begin{bmatrix}\frac{1}{4} & 0 & 0 & 0\\0 & \frac{1}{4} & 0 & 0\\0 & 0 & \frac{1}{4} & 0\\0 & 0 & 0 & \frac{1}{4}\end{bmatrix}$

Q1.05.2 MCQ 2005 2M

Given the matrix $\begin{bmatrix}-4 & 2\\4 & 3\end{bmatrix}$, the eigen vector is

A. $\begin{bmatrix}3\\2\end{bmatrix}$ **B.** $\begin{bmatrix}4\\3\end{bmatrix}$ **C.** $\begin{bmatrix}2\\-1\end{bmatrix}$ **D.** $\begin{bmatrix}-1\\2\end{bmatrix}$

Q1.00.1 MCQ 2000 1M

The eigen values of the matrix $\begin{bmatrix}2 & -1 & 0 & 0\\0 & 3 & 0 & 0\\0 & 0 & -2 & 0\\0 & 0 & -1 & 4\end{bmatrix}$ are

A. 2, -2, 1, -1 **B.** 2, 3, -2, 4
C. 2, 3, 1, 4 **D.** None of the above

Q1.98.1 MCQ 1998 1M

The eigen values of the matrix $\mathbf{A} = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ are

A. 1, 1 **B.** -1, -1 **C.** $j, -j$ **D.** 1, -1

Q1.94.1 MCQ 1994 1M

The rank of $(m \times n)$ matrix $(m < n)$ cannot be more than

A. m **B.** n **C.** mn **D.** None

Q1.94.2 MCQ 1994 2M

The following system of equations

$$x_1 + x_2 + x_3 = 3$$
$$x_1 - x_3 = 0$$
$$x_1 - x_2 + x_3 = 1$$

has

A. Unique solution

B. No solution

C. Infinite number of solutions

D. Only one solution

Electrical Engineering

Q1.26.1 MSQ 2026 1M

Two $n \times n$ matrices A and B have a common eigenvalue 2, and the same corresponding nonzero eigenvector. Which of the following options is/are correct?
(Note: I is the $n \times n$ identity matrix.)

☐ **A.** Determinant $(A - 2I) = 0$

- [ ] **B.** Determinant $(B - 2I) = 0$
- [ ] **C.** Determinant $(A + B - 2I) = 0$
- [ ] **D.** Determinant $(A + B - 4I) = 0$

Q1.26.2 NAT 2026 1M

A is an $m \times m$ skew-symmetric matrix with real-valued entries, and x is an m-dimensional column vector with real-valued entries such that $x^{T}x = 1$. The quantity $x^{T}Ax$ evaluates to ________. (Answer in integer)

Q1.26.3 MCQ 2026 2M

Which one of the following statements is ALWAYS correct about a collection of p column vectors, each having n real-valued entries?

A. If $p > n$, then the column vectors must be linearly dependent

B. If $p > n$, then the column vectors must be linearly independent

C. If $p = n$, then the column vectors must be orthogonal

D. If $p < n$, then the column vectors must be linearly independent

Q1.26.4 MSQ 2026 2M

Consider an $n \times n$ orthogonal matrix A with real entries and each column having unit Euclidean norm. Which of the following statements is/are correct?

- [ ] **A.** The value of the determinant of A is either $+1$ or -1
- [ ] **B.** The eigenvalues of A have modulus 1
- [ ] **C.** $||Ax|| = ||x||$, for all $x \in R^n$, where $||x||$ denotes the Euclidean norm of x, and $(Ax)^T(Ay) \neq x^T y$, for all distinct $x, y \in R^n$
- [ ] **D.** $||Ax|| = ||x||$, for all $x \in R^n$, where $||x||$ denotes the Euclidean norm of x, and $(Ax)^T(Ay) = x^T y$, for all $x, y \in R^n$

Q1.26.5 MSQ 2026 2M

Consider the system of linear equations: $Ax = b$, where A is an $n \times n$ matrix, and x and b are n-dimensional column vectors. Suppose this system of equations has a unique solution. Which of the following statements is/are correct?

- [ ] **A.** A^{-1} exists
- [ ] **B.** The system of equations $A^m x = b$ also has a unique solution for $m = 1, 2, 3, \ldots$
- [ ] **C.** $\text{rank}(A) = \text{rank}(A^m)$, for $m = 1, 2, 3, \ldots$
- [ ] **D.** $\text{rank}(A) < \text{rank}([A \mid b])$, where $[A \mid b]$ denotes the augmented matrix.

Q1.25.1 MCQ 2025 1M

Let v_1 and v_2 be the two eigenvectors corresponding to distinct eigenvalues of a 3×3 real symmetric matrix. Which one of the following statements is true?

A. $v_1^T v_2 \neq 0$ **B.** $v_1^T v_2 = 0$
C. $v_1 + v_2 = 0$ **D.** $v_1 - v_2 = 0$

Q1.25.2 MCQ 2025 1M

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$, and $b = \begin{bmatrix} 1/3 \\ -1/3 \\ 0 \end{bmatrix}$. Then, the system of linear equations $Ax = b$ has

A. a unique solution.

B. infinitely many solutions.

C. a finite number of solutions.

D. no solution.

Q1.25.3 MCQ 2025 1M

Let $P = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and let I be the identity matrix.

Then P^2 is equal to

A. $2P - I$ **B.** P **C.** I **D.** $P + I$

Q1.24.1 MCQ 2024 1M

Which one of the following matrices has an inverse?

A. $\begin{bmatrix} 1 & 4 & 8 \\ 0 & 4 & 2 \\ 0.5 & 2 & 4 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 2 & 9 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 4 & 8 \\ 0 & 4 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 4 & 8 \\ 0 & 4 & 2 \\ 3 & 12 & 24 \end{bmatrix}$

Q1.24.2 NAT 2024 1M

The sum of the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^2$ is _______ .

(rounded off to the nearest integer)

Q1.23.1 MCQ 2023 1M

In the figure, the vectors u and v are related as: $Au = v$ by a transformation matrix A. The correct choice of A is

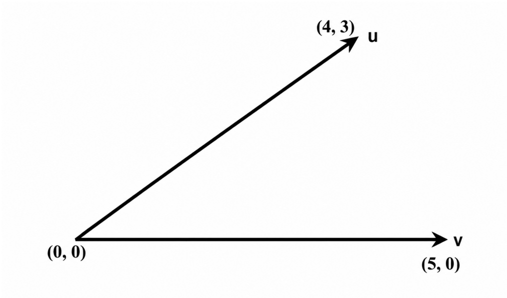


Fig. Q1.23.1

A. $\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{bmatrix}$

B. $\begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$

C. $\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$

D. $\begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & -\frac{4}{5} \end{bmatrix}$

Q1.22.1 MCQ 2022 1M

Consider a 3×3 matrix A whose (i, j)-th element, $a_{i,j} = (i - j)^3$. Then the matrix A will be

A. symmetric **B.** skew-symmetric
C. unitary **D.** null

Q1.22.2 MCQ 2022 2M

e^A denotes the exponential of a square matrix A. Suppose λ is an eigenvalue and v is the corresponding eigen-vector of matrix A. Consider the following two statements:

Statement 1: e^λ is an eigenvalue of e^A.

Statement 2: v is an eigen-vector of e^A.

Which one of the following options is correct?

A. Statement 1 is true and statement 2 is false.

B. Statement 1 is false and statement 2 is true.

C. Both the statements are correct.

D. Both the statements are false.

Q1.22.3 MCQ 2022 2M

Consider a matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -2 \\ 0 & 1 & 1 \end{bmatrix}$. The matrix A satisfies the equation $6A^{-1} = A^2 + cA + dI$, where c and d are scalars and I is the identity matrix. Then $(c + d)$ is equal to

A. 5 **B.** 17 **C.** -6 **D.** 11

Q1.21.1 MCQ 2021 1M

Let p and q be real numbers such that $p^2 + q^2 = 1$. The eigenvalues of the matrix

$$\begin{bmatrix} p & q \\ q & -p \end{bmatrix}$$

are

A. 1 and 1 **B.** 1 and -1
C. j and $-j$ **D.** pq and $-pq$

Q1.21.2 NAT 2021 2M

Let A be a 10×10 matrix such that A^5 is a null matrix, and let I be the 10×10 identity matrix. The determinant of $A + I$ is __________.

Q1.20.1 MCQ 2020 2M

The number of purely real elements in a lower triangular representation of the given 3×3 matrix, obtained through the given decomposition is __________.

$$\begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 1 \\ 3 & 1 & 7 \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{12} & a_{22} & 0 \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} & 0 & 0 \\ a_{12} & a_{22} & 0 \\ a_{13} & a_{23} & a_{33} \end{bmatrix}^T$$

A. 5 **B.** 6 **C.** 8 **D.** 9

Q1.19.1 MCQ 2019 1M

M is a 2×2 matrix with eigenvalues 4 and 9. The eigenvalues of M^2 are

A. 4 and 9 **B.** 2 and 3 **C.** -2 and -3 **D.** 16 and 81

Q1.19.2 MCQ 2019 2M

Consider a 2×2 matrix $M = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$, where, v_1 and v_2 are the column vectors. Suppose $M^{-1} = \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix}$, where u_1^T and u_2^T are the row vectors. Consider the following statements:

Statement 1: $u_1^T v_1 = 1$ and $u_2^T v_2 = 1$

Statement 2: $u_1^T v_2 = 0$ and $u_2^T v_1 = 0$

Which of the following options is correct?

A. Statement 1 is true and statement 2 is false

B. Statement 2 is true and statement 1 is false

C. Both the statements are true

D. Both the statements are false

Q1.18.1 NAT 2018 1M

Consider a non-singular 2×2 square matrix A. If trace$(A) = 4$ and trace$(A^2) = 5$, the determinant of the matrix A is ___________ (up to 1 decimal place).

Q1.18.2 NAT 2018 2M

Let $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ and $B = A^3 - A^2 - 4A + 5I$, where I is the 3×3 identity matrix. The determinant of B is ___________ (up to 1 decimal place).

Q1.17.1 MCQ 2017 1M

The matrix $A = \begin{bmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{bmatrix}$ has three distinct eigenvalues and one of its eigenvectors is $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Which one of the following can be another eigenvector of A?

A. $\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$ **B.** $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ **C.** $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Q1.17.2 MCQ 2017 2M

The eigenvalues of the matrix given below are

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix}$$

A. (0,-1,-3) **B.** (0, -2, -3) **C.** (0, 2, 3) **D.** (0, 1, 3)

Q1.16.1 NAT 2016 1M

Consider a 3×3 matrix with every element being equal to 1. Its only non-zero eigenvalue is _______.

Q1.16.2 MCQ 2016 2M

Let the eigenvalues of a 2×2 matrix A be $1, -2$ with eigenvectors x_1 and x_2 respectively. Then the eigenvalues and eigenvectors of the matrix $A^2 - 3A + 4I$ would, respectively, be

A. 2, 14; x_1, x_2 **B.** 2, 14; $x_1 + x_2, x_1 - x_2$

C. 2, 0; x_1, x_2 **D.** 2, 0; $x_1 + x_2, x_1 - x_2$

Q1.16.3 MCQ 2016 2M

Let A be a 4×3 real matrix with rank 2. Which one of the following statement is TRUE?

A. Rank of $A^T A$ is less than 2.

B. Rank of $A^T A$ is equal to 2.

C. Rank of $A^T A$ is greater than 2.

D. Rank of $A^T A$ can be any number between 1 and 3.

Q1.16.4 MCQ 2016 1M

A 3×3 matrix P is such that, $P^3 = P$. Then the eigenvalues of P are

A. $1, 1, -1$

B. $1, 0.5 + j0.866, 0.5 - j0.866$

C. $1, -0.5 + j0.866, -0.5 - j0.866$

D. $0, 1, -1$

Q1.16.5 MCQ 2016 2M

Let $P = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Consider the set S of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ such that $a^2 + b^2 = 1$ where $\begin{pmatrix} a \\ b \end{pmatrix} = P \begin{pmatrix} x \\ y \end{pmatrix}$. Then S is

A. a circle of radius $\sqrt{10}$

B. a circle of radius $\frac{1}{\sqrt{10}}$

C. an ellipse with major axis along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

D. an ellipse with minor axis along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q1.15.1 NAT 2015 1M

If the sum of the diagonal elements of a 2×2 matrix is -6, then the maximum possible value of determinant of the matrix is ________.

Q1.15.2 MCQ 2015 2M

The maximum value of "a" such that the matrix

$$\begin{pmatrix} -3 & 0 & -2 \\ 1 & -1 & 0 \\ 0 & a & -2 \end{pmatrix}$$

has three linearly independent real eigenvectors is

A. $\frac{2}{3\sqrt{3}}$ **B.** $\frac{1}{3\sqrt{3}}$ **C.** $\frac{1+2\sqrt{3}}{3\sqrt{3}}$ **D.** $\frac{1+\sqrt{3}}{3\sqrt{3}}$

Q1.15.3 MCQ 2015 1M

We have a set of 3 linear equations in 3 unknowns. '$X \equiv Y$' means X and Y are equivalent statements and '$X \not\equiv Y$' means X and Y are not equivalent statements.

P: There is a unique solution.

Q: The equations are linearly independent.

R: All eigenvalues of the coefficient matrix are nonzero.

S: The determinant of the coefficient matrix is nonzero.

Which one of the following is TRUE?

A. $P \equiv Q \equiv R \equiv S$ **B.** $P \equiv R \not\equiv Q \equiv S$

C. $P \equiv Q \not\equiv R \equiv S$ **D.** $P \not\equiv Q \not\equiv R \not\equiv S$

Q1.14.1 NAT 2014 2M

A system matrix is given as follows

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -6 & -11 & 6 \\ -6 & -11 & 5 \end{bmatrix}.$$

The absolute value of the ratio of the maximum eigenvalue to the minimum eigenvalue is ________

Q1.14.2 MCQ 2014 1M

Two matrices A and B are given below:

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}; \qquad B = \begin{bmatrix} p^2+q^2 & pr+qs \\ pr+qs & r^2+s^2 \end{bmatrix}$$

If the rank of matrix A is N, then the rank of matrix B is

A. $N/2$ **B.** $N-1$ **C.** N **D.** $2N$

Q1.14.3 MCQ 2014 1M

Which one of the following statements is true for all real symmetric matrices?

A. All the eigenvalues are real.

B. All the eigenvalues are positive.

C. All the eigenvalues are distinct.

D. Sum of all the eigenvalues is zero.

Q1.13.1 MCQ 2013 2M

A matrix has eigenvalues -1 and -2. The corresponding eigenvectors are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ respectively. The matrix is

A. $\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

Q1.11.1 MCQ 2011 1M

The two vectors $[1, 1, 1]$ and $[1, a, a^2]$, where $a = \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$, are

A. Orthonormal **B.** Orthogonal

C. Parallel **D.** Collinear

Q1.11.2 MCQ 2011 2M

The matrix $[A] = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}$ is decomposed into a product of a lower triangular matrix $[L]$ and an upper triangular matrix $[U]$. The properly decomposed $[L]$ and $[U]$ matrices respectively are

A. $\begin{bmatrix} 1 & 0 \\ 4 & -1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$
B. $\begin{bmatrix} 2 & 0 \\ 4 & -1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
C. $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$
D. $\begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$

Q1.10.1 MCQ 2010 2M

An eigenvector of $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ is

A. $[-1 \;\; 1 \;\; 1]^\top$
B. $[1 \;\; 2 \;\; 1]^\top$
C. $[1 \;\; -1 \;\; 2]^\top$
D. $[2 \;\; 1 \;\; -1]^\top$

Q1.09.1 MCQ 2009 1M

The trace and determinant of a 2×2 matrix are known to be -2 and -35 respectively. Its eigenvalues are

A. -30 and -5
B. -37 and -1
C. -7 and 5
D. 17.5 and -2

Q1.08.1 MCQ 2008 1M

The characteristic equation of a (3×3) matrix P is defined as

$$\alpha(\lambda) = |\lambda I - P| = \lambda^3 + \lambda^2 + 2\lambda + 1 = 0.$$

If I denotes identity matrix, then the inverse of matrix P will be

A. $(P^2 + P + 2I)$
B. $(P^2 + P + I)$
C. $-(P^2 + P + I)$
D. $-(P^2 + P + 2I)$

Q1.08.2 MCQ 2008 1M

If the rank of a (5×6) matrix Q is 4, then which one of the following statements is correct?

A. Q will have four linearly independent rows and four linearly independent columns

B. Q will have four linearly independent rows and five linearly independent columns

C. $QQ^\top$ will be invertible

D. $Q^\top Q$ will be invertible

Q1.08.3 MCQ 2008 2M

Let P be a 2×2 real orthogonal matrix and $\vec{x}$ is a real vector $[x_1, x_2]^\top$ with length $\|\vec{x}\| = (x_1^2 + x_2^2)^{\frac{1}{2}}$. Then, which one of the following statements is correct?

A. $\|P\vec{x}\| \leq \|\vec{x}\|$ where at least one vector satisfies $\|P\vec{x}\| < \|\vec{x}\|$

B. $\|P\vec{x}\| = \|\vec{x}\|$ for all vectors $\vec{x}$

C. $\|P\vec{x}\| \geq \|\vec{x}\|$ where at least one vector satisfies $\|P\vec{x}\| > \|\vec{x}\|$

D. No relationship can be established between $\|\vec{x}\|$ and $\|P\vec{x}\|$

Q1.08.4 MCQ 2008 2M

A is a $m \times n$ full rank matrix with $m > n$ and I is an identity matrix. Let matrix $A^+ = (A^\top A)^{-1} A^\top$. Then, which one of the following statements is FALSE?

A. $AA^+A = A$
B. $(AA^+)^2 = AA^+$
C. $A^+A = I$
D. $AA^+A = A^+$

Q1.07.1 MCQ 2007 1M

$\mathbf{x} = [x_1 \;\; x_2 \;\; \ldots \;\; x_n]^\top$ is an n-tuple nonzero vector. The $n \times n$ matrix $\mathbf{V} = \mathbf{x}\mathbf{x}^\top$

A. has rank zero
B. has rank 1
C. is orthogonal
D. has rank n

Q1.07.2 MCQ 2007 2M

Let x and y be two vectors in a 3 dimensional space and $\langle x, y \rangle$ denote their dot product. Then the determinant

$$\det \begin{bmatrix} \langle x, x \rangle & \langle x, y \rangle \\ \langle y, x \rangle & \langle y, y \rangle \end{bmatrix}$$

A. is zero when x and y are linearly independent

B. is positive when x and y are linearly independent

C. is non-zero for all non-zero x and y

D. is zero only when either x or y is zero

Q1.07.3 MCQ 2007 2M

The linear operation $L(\mathbf{x})$ is defined by the cross product $L(\mathbf{x}) = \mathbf{b} \times \mathbf{x}$, where $\mathbf{b} = [0 \;\; 1 \;\; 0]^\top$ and $\mathbf{x} = [x_1 \;\; x_2 \;\; x_3]^\top$

are three dimensional vectors. The 3×3 matrix M of this operation satisfies

$$L(\mathbf{x}) = M \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Then the eigenvalues of M are

A. $0, +1, -1$ **B.** $1, -1, 1$ **C.** $i, -i, 1$ **D.** $i, -i, 0$

Common Data Questions

Answer Questions Q1.07.4 and Q1.07.5 using the following data. Cayley-Hamilton Theorem states that a square matrix satisfies its own characteristic equation.

Consider a matrix: $A = \begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}$

Q1.07.4 MCQ 2007 1M

A satisfies the relation

A. $A + 3I + 2A^{-1} = 0$ **B.** $A^2 + 2A + 2I = 0$

C. $(A + I)(A + 2I) = 0$ **D.** $\exp(A) = 0$

Q1.07.5 MCQ 2007 1M

A^9 equals

A. $511\,A + 510\,I$ **B.** $309\,A + 104\,I$

C. $154\,A + 155\,I$ **D.** $\exp(9\,A)$

Q1.05.1 MCQ 2005 1M

In the matrix equation $Px = q$, which of the following is a necessary condition for the existence of at least one solution for the unknown vector x:

A. Augmented matrix $[P \quad q]$ must have the same rank as matrix P

B. Vector q must have only non-zero elements

C. Matrix P must be singular

D. Matrix P must be square

Q1.05.2 MCQ 2005 2M

If $R = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{bmatrix}$, the top row of R^{-1} is:

A. $\begin{bmatrix} 5 & 6 & 4 \end{bmatrix}$ **B.** $\begin{bmatrix} 5 & -3 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 2 & 0 & -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 & -1 & \frac{1}{2} \end{bmatrix}$

Q1.02.1 MCQ 2002 1M

The determinant of the matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 100 & 1 & 0 & 0 \\ 100 & 200 & 1 & 0 \\ 100 & 200 & 300 & 1 \end{bmatrix}$ is:

A. 100 **B.** 200 **C.** 1 **D.** 300

Q1.98.1 MCQ 1998 1M

The vector $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ is an eigen vector of $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.

One of the given values of A is:

A. 1 **B.** 2 **C.** 5 **D.** -1

Q1.97.1 NAT 1997 5M

Express the given matrix A as a product of two triangular matrices, L and U, where the diagonal elements of the lower triangular matrix L are unity and U is an upper triangular matrix.

$$a = \begin{bmatrix} 2 & 1 & 5 \\ 4 & 8 & 13 \\ 6 & 27 & 31 \end{bmatrix}$$

Q1.95.1 NAT 1995 2M

A set three linear equations with unknowns X, Y and Z is shown below.

$$\begin{bmatrix} 0 & 21.432 & 34.432 \\ 17.587 & -1.348 & -1.865 \\ 0 & -2.663 & -1.803 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \begin{bmatrix} 56.042 \\ 02.631 \\ 05.952 \end{bmatrix}$$

Decompose the coefficient matrix into an upper triangular matrix and then solve for the unknowns, X, Y and Z.

Q1.94.1 MCQ 1994 1M

A 5×7 matrix has all its entries equal to -1. The rank of the matrix is

A. 7 **B.** 5 **C.** 1 **D.** Zero

Instrumentation Engineering

Q1.26.1 MCQ 2026 1M

A matrix $P \in \mathbb{R}^{3\times3}$ satisfies $P^3-4P^2+5P-2I=0$ where I is a 3×3 identity matrix. The set of all possible eigenvalues of matrix P is ____.

A. $\{1,2,3\}$ **B.** $\{1\}$ **C.** $\{1,2\}$ **D.** $\{2\}$

Q1.26.2 MCQ 2026 1M

Consider a matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Let $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. The number of solutions to the linear system of equations $A\mathbf{x} = \mathbf{b}$ is ____.

A. 0 **B.** 1 **C.** 6 **D.** infinitely many

Q1.25.1 MCQ 2025 1M

A $2n \times 2n$ matrix $A = [a_{ij}]$ has its elements as

$$a_{ij} = \begin{cases} \beta & \text{if } (i+j) \text{ is odd,} \\ -\beta & \text{if } (i+j) \text{ is even,} \end{cases}$$

where n is any integer greater than 2 and β is any non-zero real number. The rank of A is

A. 1 **B.** 2 **C.** n **D.** 2n

Q1.25.2 NAT 2025 1M

If one of the eigenvectors of the matrix $A = \begin{bmatrix} -1 & -1 \\ x & -4 \end{bmatrix}$ is along the direction of $\begin{bmatrix} \alpha \\ 2\alpha \end{bmatrix}$, where α is any non-zero real number, then the value of x is ____ (in integer).

Q1.24.1 MCQ 2024 1M

A matrix M is constructed by stacking three column vectors v_1, v_2, v_3 as

$$M = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}.$$

Choose the set of vectors from the following options such that $\text{rank}(M) = 3$.

A. $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

B. $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

C. $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

D. $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$

Q1.24.2 NAT 2024 1M

A 3×3 matrix P with all real elements has eigenvalues $\frac{1}{4}$, 1, and -2. The value of $|P^{-1}|$ is ____ (rounded off to nearest integer).

Q1.23.1 MCQ 2023 1M

Choose solution set S corresponding to the systems of two equations

$$x - 2y + z = 0$$
$$x - z = 0$$

Note: $\mathscr{R}$ denotes the set of real numbers

A. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \middle| \alpha \in \mathscr{R} \right\}$

B. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha, \beta \in \mathscr{R} \right\}$

C. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \middle| \alpha, \beta \in \mathscr{R} \right\}$

D. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha \in \mathscr{R} \right\}$

Q1.23.2 NAT 2023 2M ◯ ◯ ◯

The rank of the matrix A given below is one. The ratio $\frac{\alpha}{\beta}$ is ______ (rounded off to the nearest integer).

$$A = \begin{bmatrix} 1 & 4 \\ -3 & \alpha \\ \beta & 6 \end{bmatrix}$$

Q1.22.1 MSQ 2022 1M ◯ ◯ ◯

Given $M = \begin{bmatrix} 2 & 3 & 7 \\ 6 & 4 & 7 \\ 4 & 6 & 14 \end{bmatrix}$, which of the following statement(s) is/are correct?

- ☐ **A.** The rank of M is 2
- ☐ **B.** The rank of M is 3
- ☐ **C.** The rows of M are linearly independent
- ☐ **D.** The determinant of M is 0

Q1.22.2 MSQ 2022 2M ◯ ◯ ◯

The matrix $A = \begin{bmatrix} 4 & 3 \\ 9 & -2 \end{bmatrix}$ has eigenvalues -5 and 7. The eigenvector(s) is/are ______

A. $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ **C.** $\begin{bmatrix} 2 \\ -6 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 \\ 8 \end{bmatrix}$

Q1.21.1 MCQ 2021 1M ◯ ◯ ◯

Consider the row vectors $v = (1, 0)$ and $w = (2, 0)$. The rank of the matrix $M = 2v^T v + 3w^T w$, where the superscript T denotes the transpose, is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q1.21.2 NAT 2021 2M ◯ ◯ ◯

Given $A = \begin{pmatrix} 2 & 5 \\ 0 & 3 \end{pmatrix}$. The value of the determinant $|A^4 - 5A^3 + 6A^2 + 2I| =$ ______.

Q1.20.1 MCQ 2020 1M ◯ ◯ ◯

A set of linear equations is given in the form $Ax = b$, where A is a 2×4 matrix with real number entries and $b \neq 0$. Will it be possible to solve for x and obtain a unique solution by multiplying both left and right sides of the equation by A^T (the super script T denotes the transpose) and inverting the matrix $A^T A$? Answer is ______

A. Yes, it is always possible to get a unique solution for any 2×4 matrix A.

B. No, it is not possible to get a unique solution for any 2×4 matrix A.

C. Yes, can obtain a unique solution provided the matrix $A^T A$ is well conditioned

D. Yes, can obtain a unique solution provided the matrix A is well conditioned

Q1.20.2 MCQ 2020 2M ◯ ◯ ◯

Consider the matrix $M = \begin{bmatrix} 1 & -1 & 0 \\ 1 & -2 & 1 \\ 0 & -1 & 1 \end{bmatrix}$. One of the eigenvectors of M is

A. $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ **C.** $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Q1.19.1 NAT 2019 1M ◯ ◯ ◯

A 3×3 matrix has eigenvalues 1, 2 and 5. The determinant of the matrix is ______.

Q1.18.1 MCQ 2018 1M ◯ ◯ ◯

Let N be a 3 by 3 matrix with real number entries. The matrix N is such that $N^2 = 0$. The eigen values of N are

A. 0, 0, 0 **B.** 0,0,1 **C.** 0,1,1 **D.** 1,1,1

Q1.18.2 MCQ 2018 2M ◯ ◯ ◯

Consider the following system of linear equations:

$$3x + 2ky = -2$$
$$kx + 6y = 2$$

Here x and y are the unknowns and k is a real constant. The value of k for which there are infinite number of solutions is

A. 3 **B.** 1 **C.** -3 **D.** -6

Q1.17.1 NAT 2017 1M

If **v** is a non-zero vector of dimension 3×1, then the matrix $A = \mathbf{v}\mathbf{v}^{\mathrm{T}}$ has a rank = ______

Q1.17.2 MCQ 2017 1M

The figure shows a shape ABC and its mirror image $A_1B_1C_1$ across the horizontal axis (X-axis). The coordinate transformation matrix that maps ABC to $A_1B_1C_1$ is

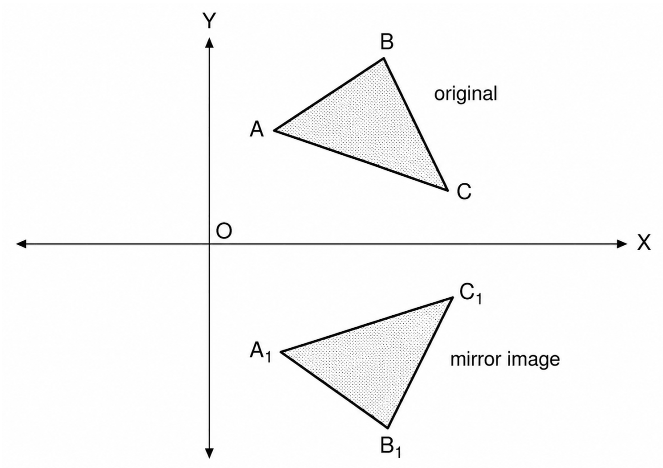


Fig. Q1.17.2

A. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **B.** $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Q1.17.3 MCQ 2017 1M

The eigenvalues of the matrix $\mathbf{A} = \begin{bmatrix} 1 & -1 & 5 \\ 0 & 5 & 6 \\ 0 & -6 & 5 \end{bmatrix}$ are

A. $-1,\ 5,\ 6$ **B.** $1,\ -5 \pm j6$

C. $1,\ 5 \pm j6$ **D.** $1,\ 5,\ 5$

Q1.16.1 NAT 2016 2M

Consider the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{pmatrix}$ whose eigenvalues are $1, -1$ and 3. Then Trace of $(\mathbf{A}^3 - 3\mathbf{A}^2)$ is ______.

Q1.15.1 MCQ 2015 1M

Let **A** be an $n \times n$ matrix with rank r $(0 < r < n)$. Then $\mathbf{Ax} = 0$ has p independent solutions, where p is

A. r **B.** n **C.** $n - r$ **D.** $n + r$

Q1.14.1 MCQ 2014 2M

A scalar valued function is defined as $f(x) = x^T Ax + b^T x + c$, where A is a symmetric positive definite matrix with dimension $n \times n$; b and x are vectors of dimension $n \times 1$. The minimum value of $f(x)$ will occur when x equals

A. $(A^T A)^{-1} b$ **B.** $-(A^T A)^{-1} b$

C. $-\left(\dfrac{A^{-1}b}{2}\right)$ **D.** $\dfrac{A^{-1}b}{2}$

Q1.13.1 MCQ 2013 1M

The dimension of the null space of the matrix $\begin{bmatrix} 0 & 1 & 1 \\ 1 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.13.2 MCQ 2013 2M

One pair of eigenvectors corresponding to the two eigenvalues of the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is

A. $\begin{bmatrix} 1 \\ -j \end{bmatrix}, \begin{bmatrix} j \\ -1 \end{bmatrix}$ **B.** $\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 \\ j \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 \\ j \end{bmatrix}, \begin{bmatrix} j \\ 1 \end{bmatrix}$

Q1.11.1 MCQ 2011 1M

The matrix $M = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ has eigenvalues $-3, -3, 5$.

An eigenvector corresponding to the eigenvalue 5 is $\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}^T$. One of the eigenvectors of the matrix M^3 is

A. $\begin{bmatrix} 1 & 8 & -1 \end{bmatrix}^T$ **B.** $\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}^T$

C. $\begin{bmatrix} 1 & \sqrt[3]{2} & -1 \end{bmatrix}^T$ **D.** $\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}^T$

Q1.10.1 MCQ 2010 1M

A real $n \times n$ matrix $A = [a_{ij}]$ is defined as follows:

$$a_{ij} = i, \quad \text{if } i = j$$
$$= 0, \quad \text{otherwise}$$

The summation of all n eigenvalues of A is

A. $n(n+1)/2$ **B.** $n(n-1)/2$

C. $\frac{n(n+1)(2n+1)}{6}$ **D.** n^2

Q1.10.2 MCQ 2010 2M

X and Y are non-zero square matrices of size $n \times n$. If $XY = 0_{n \times n}$, then

A. $|X| = 0$ and $|Y| \neq 0$ **B.** $|X| \neq 0$ and $|Y| = 0$

C. $|X| = 0$ and $|Y| = 0$ **D.** $|X| \neq 0$ and $|Y| \neq 0$

Q1.09.1 MCQ 2009 1M

Let $P \neq 0$ be a 3×3 real matrix. There exist linearly independent vectors x and y such that $Px = 0$ and $Py = 0$. The dimension of the range space of P is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.09.2 MCQ 2009 2M

The eigenvalues of a (2×2) matrix X are -2 and -3. The eigenvalues of the matrix $(X+I)^{-1}(X+5I)$ are

A. $-3, -4$ **B.** $-1, -2$ **C.** $-1, -3$ **D.** $-2, -4$

Q1.09.3 MCQ 2009 2M

The matrix $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ rotates a vector about the axis $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ by an angle of

A. $30°$ **B.** $60°$ **C.** $90°$ **D.** $120°$

Q1.07.1 MCQ 2007 1M

Let A be an $n \times n$ real matrix such that $A^2 = I$ and y be an n-dimensional vector. Then the linear system of equations $Ax = y$ has

A. no solution

B. a unique solution

C. more than one but finitely many independent solutions

D. infinitely many independent solutions

Q1.07.2 MCQ 2007 2M

Let $A = [a_{ij}]$, $1 \leq i, j \leq n$, with $n \geq 3$ and $a_{ij} = i.j$. Then the rank of A is

A. 0 **B.** 1 **C.** $n-1$ **D.** n

Q1.06.1 MCQ 2006 1M

For a given 2×2 matrix A, it is observed that $A\begin{bmatrix} 1 \\ -1 \end{bmatrix} = -1\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $A\begin{bmatrix} 1 \\ -2 \end{bmatrix} = -2\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ then the matrix A is

A. $A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$

B. $A = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

C. $A = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

D. $A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$

Q1.05.1 MCQ 2005 1M

Let A be a 3×3 matrix with rank 2. Then $AX = 0$ has

A. only the trivial solution $X = 0$

B. one independent solution

C. two independent solutions

D. three independent solutions

Q1.01.1 MCQ 2001 1M

The necessary condition to diagonalize a matrix is that

A. all its eigen values should be distinct

B. its eigen vectors should be independent

C. its eigen values should be real

D. the matrix is non-singular

Computer Science & IT

Q1.26.1 MCQ 2026 1M

Consider 4×4 matrices with their elements from $\{0, 1\}$. The number of such matrices with even number of 1s in every row and every column is

A. 512 **B.** 1025 **C.** 1023 **D.** 255

Q1.26.2 MCQ 2026 1M

For $n > 1$, the maximum multiplicity of any eigenvalue of an $n \times n$ matrix with elements from $\mathbb{R}$ is

A. n **B.** n - 1 **C.** 1 **D.** n + 1

Q1.26.3 MSQ 2026 1M

Let $n > 1$. Consider an $n \times n$ matrix M with its elements from $\mathbb{R}$. Let the vector $(0, 1, 0, 0, \ldots, 0) \in \mathbb{R}^n$ be in the null space of M. Which of the following options is/are always correct?

- ☐ **A.** Determinant of M is 1
- ☐ **B.** Determinant of M is 0
- ☐ **C.** Rank of M is 1
- ☐ **D.** There are at least two non-zero vectors in the null space of M

Q1.26.4 NAT 2026 1M

Consider the system of linear equations given below.

$$ax + y = b$$

$$16x + ay = 24$$

Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is ________. *(answer in integer)*

Q1.26.5 NAT 2026 2M

The determinant of a 4×4 matrix A is 3. The value of the determinant of $2A$ is ________. *(answer in integer)*

Q1.25.1 MSQ 2025 1M

Consider the given system of linear equations for variables x and y, where k is a real-valued constant. Which of the following option(s) is/are CORRECT?

$$x + ky = 1$$

$$kx + y = -1$$

A. There is exactly one value of k for which the above system of equations has no solution.

B. There exist an infinite number of values of k for which the system of equations has no solution.

C. There exists exactly one value of k for which the system of equations has exactly one solution.

D. There exists exactly one value of k for which the system of equations has an infinite number of solutions.

Q1.25.2 MCQ 2025 2M

Let A be a 2×2 matrix as given.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13} ?

A. $1, -1$ **B.** $2\sqrt{2}, -2\sqrt{2}$

C. $4\sqrt{2}, -4\sqrt{2}$ **D.** $64\sqrt{2}, -64\sqrt{2}$

Q1.25.3 MCQ 2025 1M

If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?

A. $\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$ **B.** $\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$

C. $\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$ **D.** $\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$

Q1.25.4 MCQ 2025 1M

Let L, M, and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.25.5 MSQ 2025 2M

Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3\times3}$ and $Q \in \mathbb{R}^{3\times1}$. Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Which of the following statement(s) is/are TRUE?

- [ ] **A.** The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
- [ ] **B.** If P is invertible, then both L and U are invertible.
- [ ] **C.** If P is singular, then at least one of the diagonal elements of U is zero.
- [ ] **D.** If P is symmetric, then both L and U are symmetric.

Q1.24.1 MCQ 2024 1M

The product of all eigenvalues of the matrix $\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$ is

A. -1 **B.** 0 **C.** 1 **D.** 2

Q1.24.2 MSQ 2024 2M

Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

- [ ] **A.** There exist at least $m - n$ linearly independent solutions to this system
- [ ] **B.** There exist $m-n$ linearly independent vectors such that every solution is a linear combination of these vectors
- [ ] **C.** There exists a non-zero solution in which at least $m - n$ variables are 0
- [ ] **D.** There exists a solution in which at least n variables are non-zero

Q1.24.3 MSQ 2024 2M

Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- [ ] **A.** The determinant of B is the negative of the determinant of A
- [ ] **B.** If A is invertible, then B is also invertible
- [ ] **C.** If A is symmetric, then B is also symmetric
- [ ] **D.** If the trace of A is zero, then the trace of B is also zero

Q1.23.1 MCQ 2023 1M

Let $A = \begin{bmatrix}1 & 2 & 3 & 4\\ 4 & 1 & 2 & 3\\ 3 & 4 & 1 & 2\\ 2 & 3 & 4 & 1\end{bmatrix}$ and $B = \begin{bmatrix}3 & 4 & 1 & 2\\ 4 & 1 & 2 & 3\\ 1 & 2 & 3 & 4\\ 2 & 3 & 4 & 1\end{bmatrix}$.

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?

A. $\det(A) = \det(B)$

B. $\det(B) = -\det(A)$

C. $\det(A) = 0$

D. $\det(AB) = \det(A) + \det(B)$

Q1.22.1 MCQ 2022 2M

Consider solving the following system of simultaneous equations using LU decomposition.

$$x_1 + x_2 - 2x_3 = 4$$
$$x_1 + 3x_2 - x_3 = 7$$
$$2x_1 + x_2 - 5x_3 = 7$$

where L and U are denoted as

$$L = \begin{pmatrix}L_{11} & 0 & 0\\ L_{21} & L_{22} & 0\\ L_{31} & L_{32} & L_{33}\end{pmatrix}, \quad U = \begin{pmatrix}U_{11} & U_{12} & U_{13}\\ 0 & U_{22} & U_{23}\\ 0 & 0 & U_{33}\end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32}, U_{33}, and x_1 ?

A. $L_{32} = 2, U_{33} = -\frac{1}{2}, x_1 = -1$

B. $L_{32} = 2, U_{33} = 2, x_1 = -1$

C. $L_{32} = -\frac{1}{2}, U_{33} = 2, x_1 = 0$

D. $L_{32} = -\frac{1}{2}, U_{33} = -\frac{1}{2}, x_1 = 0$

Q1.22.2 MSQ 2022 2M

Which of the following is/are the eigenvector(s) for the matrix given below?

$$\begin{pmatrix}-9 & -6 & -2 & -4\\ -8 & -6 & -3 & -1\\ 20 & 15 & 8 & 5\\ 32 & 21 & 7 & 12\end{pmatrix}$$

A. $\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$ **B.** $\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$ **C.** $\begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$ **D.** $\begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$

Q1.21.1 NAT 2021 2M

Consider the following matrix.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

The largest eigenvalue of the above matrix is ____________.

Q1.21.2 NAT 2021 1M

Suppose that P is a 4×5 matrix such that every solution of the equation $P\mathbf{x} = \mathbf{0}$ is a scalar multiple of $\begin{bmatrix} 2 & 5 & 4 & 3 & 1 \end{bmatrix}^T$. The rank of P is ____________.

Q1.20.1 MCQ 2020 1M

Let A and B be two $n \times n$ matrices over real numbers. Let rank(M) and det(M) denote the rank and determinant of a matrix M, respectively. Consider the following statements.

I. $\text{rank}(AB) = \text{rank}(A)\text{rank}(B)$

II. $\det(AB) = \det(A)\det(B)$

III. $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

IV. $\det(A + B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

A. I and II only **B.** I and IV only

C. II and III only **D.** III and IV only

Q1.19.1 NAT 2019 2M

Consider the following matrix:

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen values of R is ____________.

Q1.18.1 NAT 2018 1M

Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v. The largest eigenvalue of A is ________.

Q1.18.2 MCQ 2018 2M

Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$. Consider the following statements:

(I) P does not have an inverse

(II) P has a repeated eigenvalue

(III) P cannot be diagonalized

Which one of the following options is correct?

A. Only I and III are necessarily true

B. Only II is necessarily true

C. Only I and II are necessarily true

D. Only II and III are necessarily true

Q1.17.1 MCQ 2017 1M

Let $c_1 \ldots c_n$ be scalars, not all zero, such that $\sum_{i=1}^{n} c_i a_i = 0$ where a_i are column vectors in R^n. Consider the set of linear equations $Ax = b$ where $A = a_1 \ldots a_n$ and $b = \sum_{i=1}^{n} a_i$. The set of equations has

A. a unique solution at $x = J_n$ where J_n denotes a n-dimensional vector of all 1

B. no solution

C. infinitely many solutions

D. finitely many solutions

Q1.17.2 MCQ 2017 2M

Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij}^2 = 50$. Consider the following statements:

(I) One eigen value must be in $[-5, 5]$

(II) The eigen value with the largest magnitude must be strictly greater than 5.

Which of the above statements about eigen values of A is/are necessarily CORRECT?

A. Both (I) and (II) **B.** (I) only

C. (II) only **D.** Neither (I) nor (II)

Q1.17.3 NAT 2017 1M

Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices. Then the rank of $P + Q$ is ________.

Q1.17.4 NAT 2017 2M

If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is ________.

Q1.16.1 NAT 2016 1M

Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is ________.

Q1.16.2 MCQ 2016 1M

Consider the systems, each consisting of m linear equations in n variables.

I. If $m < n$, then all such systems have a solution

II. If $m > n$, then none of these systems has a solution

III. If $m = n$, then there exists a system which has a solution

Which one of the following is CORRECT?

A. I, II and III are true **B.** Only II and III are true
C. Only III is true **D.** None of them is true

Q1.16.3 NAT 2016 1M

Suppose that the eigenvalues of matrix A are $1, 2, 4$. The determinant of $(A^{-1})^T$ is ________.

Q1.16.4 NAT 2016 2M

Let A_1, A_2, A_3, and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5, respectively. The minimum number of scalar multiplications required to find the product $A_1A_2A_3A_4$ using the basic matrix multiplication method is ________.

Q1.15.1 MCQ 2015 2M

Consider the following 2×2 matrix A where two elements are unknown and are marked by a and b. The eigenvalues of this matrix are -1 and 7. What are the values of a and b?

$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}$$

A. $a = 6, b = 4$ **B.** $a = 4, b = 6$
C. $a = 3, b = 5$ **D.** $a = 5, b = 3$

Q1.15.2 MCQ 2015 1M

In the given matrix $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, one of the eigenvalues is 1. The eigenvectors corresponding to the eigenvalue 1 are

A. $\{\alpha(4, 2, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

B. $\{\alpha(-4, 2, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

C. $\{\alpha(\sqrt{2}, 0, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

D. $\{\alpha(-\sqrt{2}, 0, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

Q1.15.3 NAT 2015 1M

The larger of the two eigenvalues of the matrix $\begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ is ________.

Q1.15.4 NAT 2015 2M

Perform the following operations on the matrix

$$\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$$

(i)Add the third row to the second row.
(ii)Subtract the third column from the first column.
The determinant of the resultant matrix is ________.

Q1.15.5 MCQ 2015 2M

If the following system has non-trivial solution,

$$px + qy + rz = 0$$
$$qx + ry + pz = 0$$
$$rx + py + qz = 0,$$

then which one of the following options is TRUE?

A. $p - q + r = 0$ or $p = q = -r$

B. $p + q - r = 0$ or $p = -q = r$

C. $p + q + r = 0$ or $p = q = r$

D. $p - q + r = 0$ or $p = -q = -r$

Q1.14.1 NAT 2014 1M

The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a 4x4 symmetric positive definite matrix is ________.

Q1.14.2 NAT 2014 2M

The product of the non-zero eigenvalues of the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

is ________.

Q1.14.3 NAT 2014 1M

If V_1 and V_2 are 4-dimensional subspaces of a 6-dimensional vector space V, then the smallest possible dimension of $V_1 \cap V_2$ is ________.

Q1.14.4 MCQ 2014 1M

Which one of the following statements is TRUE about every $n \times n$ matrix with only real eigenvalues?

A. If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.

B. If the trace of the matrix is positive, all its eigenvalues are positive.

C. If the determinant of the matrix is positive, all its eigenvalues are positive.

D. If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

Q1.13.1 MCQ 2013 1M

Which one of the following does NOT equal $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$?

A. $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$ **B.** $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

C. $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$ **D.** $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

Q1.12.1 MCQ 2012 1M

Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

A. 1024 and −1024 **B.** $1024\sqrt{2}$ and $-1024\sqrt{2}$

C. $4\sqrt{2}$ and $-4\sqrt{2}$ **D.** $512\sqrt{2}$ and $-512\sqrt{2}$

Q1.11.1 MCQ 2011 2M

Four matrices M_1, M_2, M_3 and M_4 of dimensions $p\times q$, $q\times r$, $r\times s$ and $s\times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example when multiplied as $((M_1\times M_2)\times(M_3\times M_4))$, the total number of scalar multiplications is $pqr+rst+prt$. When multiplied as $(((M_1\times M_2)\times M_3)\times M_4)$, the total number of scalar multiplications is $pqr+prs+pst$.

If $p = 10$, $q = 100$, $r = 20$, $s = 5$, and $t = 80$, then the minimum number of scalar multiplications needed is

A. 248000 **B.** 44000 **C.** 19000 **D.** 25000

Q1.10.1 MCQ 2010 2M

Consider the following matrix

$$A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8, then

A. $x = 4, y = 10$ **B.** $x = 5, y = 8$

C. $x = -3, y = 9$ **D.** $x = -4, y = 10$

Q1.08.1 MCQ 2008 1M

The following system of equations

$$x_1 + x_2 + 2x_3 = 1$$

$$x_1 + 2x_2 + 3x_3 = 2$$

$$x_1 + 4x_2 + \alpha x_3 = 4$$

has a unique solution. The only possible value(s) for α is/are

A. 0 **B.** either 0 or 1

C. one of 0, 1 or -1 **D.** any real number

Q1.08.2 MCQ 2008 2M

How many of the following matrices have an eigenvalue 1?

$$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}1 & -1\\1 & 1\end{bmatrix} \text{ and } \begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}$$

A. one **B.** two **C.** three **D.** four

Q1.07.1 MCQ 2007 2M

Let A be a 4×4 matrix with eigenvalues -5, -2, 1, 4.

Which of the following is an eigenvalue of $\begin{bmatrix}A & I\\I & A\end{bmatrix}$, where I is the 4×4 identity matrix?

A. -5 **B.** -7 **C.** 2 **D.** 1

Q1.04.1 MCQ 2004 1M

The number of different $n \times n$ symmetric matrices with each elements being either 0 or 1 is

A. 2^n **B.** 2^{n^2}

C. $2^{\frac{n^2+n}{2}}$ **D.** $2^{\frac{n^2-n}{2}}$

Q1.04.2 MCQ 2004 1M

Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. $ABCD = I$ then $B^{-1} =$

A. $D^{-1}C^{-1}A^{-1}$ **B.** CDA

C. ABC **D.** does not exist

Q1.01.1 MCQ 2001 1M

Consider the following statements:

S1: The sum of two singular $n \times n$ matrices may be non-singular

S2: The sum of two $n \times n$ non-singular matrices may be singular.

Which of the following statements is correct?

A. S1 and S2 are both true **B.** S1 is true, S2 is false

C. S1 is false, S2 is true **D.** S1 and S2 are both false

Q1.00.1 MCQ 2000 1M

An $n \times n$ array v is defined as follows:

$$v[i, j] = i - j \text{ for all } i, j, 1 \le i \le n, 1 \le j \le n$$

The sum of the elements of the array v is

A. 0 **B.** n -1 **C.** $n^2 - 3n + 2$ **D.** $n^2\dfrac{(n+1)}{2}$

Q1.98.1 MCQ 1998 2M

Consider the following determinant $\Delta = \begin{vmatrix}1 & a & bc\\1 & b & ca\\1 & C & ab\end{vmatrix}$. Which of the following is a factor of Δ?

A. a+b **B.** a-b **C.** a+b+c **D.** abc

Q1.97.1 MCQ 1997 1M

Let $A = (a_{ij})$ be an n-rowed square matrix and I_{12} be the matrix obtained by interchanging the first and second rows of the n-rowed Identify matrix. Then AI_{12} is such that its first

A. row is the same as its second row

B. row is the same as the second row of A

C. column is the same as the second column of A

D. row is all zero

Q1.96.1 NAT 1996 5M

Let $A = \begin{bmatrix}a_{11} & a_{12}\\a_{21} & a_{22}\end{bmatrix}$ and $B = \begin{bmatrix}b_{11} & b_{12}\\b_{21} & b_{22}\end{bmatrix}$ be two matrices such that $AB = I$. Let $C = A\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}$ and $CD = I$. Express the elements of D in terms of the elements of B.

Q1.96.2 MCQ 1996 2M

The matrices $\begin{bmatrix}\cos\theta & -\sin\theta\\\sin\theta & \cos\theta\end{bmatrix}$ and $\begin{bmatrix}a & 0\\0 & b\end{bmatrix}$ commute under multiplication

A. if $a = b$ or $\theta = n\pi$, n is an integer

B. always

C. never

D. if $a\cos\theta \ne b\sin\theta$

Q1.96.3 MCQ 1996 1M

Let $Ax = b$ be a system of linear equations where A is an $m \times n$ matrix and b is a $m \times 1$ column vector and X is a $n \times 1$ column vector of unknowns. Which of the following is false?

A. The system has a solution if and only if, both A and the augmented matrix $[A \quad b]$ have the same rank.

B. If $m < n$ and b is the zero vector, then the system has infinitely many solutions.

C. If $m = n$ and b is non-zero vector, then the system has a unique solution.

D. The system will have only a trivial solution when $m = n$, b is the zero vector and $\text{rank}(A) = n$.

Q1.95.1 MCQ 1995 1M

The rank of the following $(n+1) \times (n+1)$ matrix, where a is a real number is

$$\begin{bmatrix} 1 & a & a^2 & \dots & a^n \\ 1 & a & a^2 & \dots & a^n \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a & a^2 & \dots & a^n \end{bmatrix}$$

A. 1 **B.** 2 **C.** n **D.** Depends on the value of a

Q1.94.1 MCQ 1994 1M

Let A and B be real symmetric matrices of size $n \times n$. Then which one of the following is true?

A. $AA' = I$ **B.** $A = A^{-1}$

C. $AB = BA$ **D.** $(AB)' = BA$

Mechanical Engineering

Q1.25.1 MCQ 2025 1M

Let A and B be real symmetric matrices of same size. Which one of the following options is correct?

A. $A^T = A^{-1}$ **B.** $AB = BA$

C. $(AB)^T = B^T A^T$ **D.** $A = A^{-1}$

Q1.24.1 MCQ 2024 1M

Consider the system of linear equations

$$x + 2y + z = 5$$

$$2x + ay + 4z = 12$$

$$2x + 4y + 6z = b$$

The values of a and b such that there exists a non-trivial null space and the system admits infinite solutions are

A. a = 8, b = 14 **B.** a = 4, b = 12

C. a = 8, b = 12 **D.** a = 4, b = 14

Q1.24.2 MCQ 2024 2M

The matrix $\begin{bmatrix} 1 & a \\ 8 & 3 \end{bmatrix}$ (where $a > 0$) has a negative eigenvalue if a is greater than

A. $\frac{3}{8}$ **B.** $\frac{1}{8}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{5}$

Q1.22.1 MCQ 2022 1M

If $A = \begin{bmatrix} 10 & 2k+5 \\ 3k-3 & k+5 \end{bmatrix}$ is a symmetric matrix, the value of k is ________.

A. 8 **B.** 5 **C.** -0.4 **D.** $\frac{1+\sqrt{1561}}{12}$

Q1.22.2 MSQ 2022 2M

The system of linear equations in real (x, y) given by

$$\begin{pmatrix} x & y \end{pmatrix} \begin{bmatrix} 2 & 5-2\alpha \\ \alpha & 1 \end{bmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

involves a real parameter α and has infinitely many non-trivial solutions for special value(s) of α. Which one or more among the following options is/are non-trivial solution(s) of (x, y) for such special value(s) of α?

- [ ] **A.** $x = 2, \quad y = -2$
- [ ] **B.** $x = -1, \quad y = 4$
- [ ] **C.** $x = 1, \quad y = 1$
- [ ] **D.** $x = 4, \quad y = -2$

Q1.22.3 NAT 2022 2M

If the sum and product of eigenvalues of a 2×2 real matrix $\begin{bmatrix} 3 & p \\ p & q \end{bmatrix}$ are 4 and -1 respectively, then $|p|$ is ________ (in integer).

Q1.22.4 MSQ 2022 2M

A is a 3×5 real matrix of rank 2. For the set of homogeneous equations $A\mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is a zero vector and $\mathbf{x}$ is a vector of unknown variables, which of the following is/are true?

- [ ] **A.** The given set of equations will have a unique solution.
- [ ] **B.** The given set of equations will be satisfied by a zero vector of appropriate size.

☐ **C.** The given set of equations will have infinitely many solutions.

☐ **D.** The given set of equations will have many but a finite number of solutions.

Q1.21.1 MCQ 2021 2M

Consider a vector p in 2-dimensional space. Let its direction (counter-clockwise angle with the positive x-axis) be θ. Let p be an eigenvector of a 2×2 matrix A with corresponding eigenvalue λ, $\lambda > 0$. If we denote the magnitude of a vector v by $\|v\|$, identify the VALID statement regarding p', where $p' = Ap$.

A. Direction of $p' = \lambda\theta$, $\|p'\| = \|p\|$

B. Direction of $p' = \theta$, $\|p'\| = \lambda\|p\|$

C. Direction of $p' = \lambda\theta$, $\|p'\| = \lambda\|p\|$

D. Direction of $p' = \theta$, $\|p'\| = \|p\|/\lambda$

Q1.21.2 MCQ 2021 1M

Consider an $n \times n$ matrix A and a non-zero $n \times 1$ vector p. Their product $Ap = \alpha^2 p$, where $\alpha \in \Re$ and $\alpha \notin \{-1, 0, 1\}$. Based on the given information, the eigen value of A^2 is:

A. α **B.** α^2 **C.** $\sqrt{\alpha}$ **D.** α^4

Q1.21.3 MCQ 2021 2M

Let the superscript T represent the transpose operation. Consider the function $f(x) = \frac{1}{2}x^T Qx - r^T x$, where x and r are $n \times 1$ vectors and Q is a symmetric $n \times n$ matrix. The stationary point of $f(x)$ is

A. $Q^T r$ **B.** $Q^{-1} r$ **C.** $\frac{r}{r^T r}$ **D.** r

Q1.20.1 MCQ 2020 1M

Multiplication of real valued square matrices of same dimension is

A. associative

B. commutative

C. always positive definite

D. not always possible to compute

Q1.20.2 MCQ 2020 1M

A matrix P is decomposed into its symmetric part S and skew symmetric part V. If

$$S = \begin{pmatrix} -4 & 4 & 2 \\ 4 & 3 & 7/2 \\ 2 & 7/2 & 2 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & 7/2 \\ -3 & -7/2 & 0 \end{pmatrix},$$

then matrix P is

A. $\begin{pmatrix} -4 & 6 & -1 \\ 2 & 3 & 0 \\ 5 & 7 & 2 \end{pmatrix}$ **B.** $\begin{pmatrix} -4 & 2 & 5 \\ 6 & 3 & 7 \\ -1 & 0 & 2 \end{pmatrix}$

C. $\begin{pmatrix} 4 & -6 & 1 \\ -2 & -3 & 0 \\ -5 & -7 & -2 \end{pmatrix}$ **D.** $\begin{pmatrix} -2 & 9/2 & -1 \\ -1 & 81/4 & 11 \\ -2 & 45/2 & 73/4 \end{pmatrix}$

Q1.20.3 NAT 2020 1M

Let I be a 100 dimensional identity matrix and E be the set of its distinct (no value appears more than once in E) real eigenvalues. The number of elements in E is _______.

Q1.19.1 MCQ 2019 2M

The set of equations

$$x + y + z = 1$$

$$ax - ay + 3z = 5$$

$$5x - 3y + az = 6$$

has infinite solutions, if $a =$

A. -3 **B.** 3 **C.** 4 **D.** -4

Q1.19.2 MCQ 2019 1M

In matrix equation $[A]\{X\} = \{R\}$,

$$[A] = \begin{bmatrix} 4 & 8 & 4 \\ 8 & 16 & -4 \\ 4 & -4 & 15 \end{bmatrix}, \quad \{X\} = \begin{Bmatrix} 2 \\ 1 \\ 4 \end{Bmatrix}, \quad \{R\} = \begin{Bmatrix} 32 \\ 16 \\ 64 \end{Bmatrix}.$$

One of the eigenvalues of matrix $[A]$ is

A. 4 **B.** 8 **C.** 15 **D.** 16

Q1.18.1 NAT 2018 1M

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 1 \end{bmatrix}$ then $\det(A^{-1})$ is ________. (correct to two decimal places).

Q1.17.1 MCQ 2017 2M

Consider the matrix $P = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$.

Which one of the following statements about P is INCORRECT?

A. Determinant of P is equal to 1.

B. P is orthogonal.

C. Inverse of P is equal to its transpose.

D. All Eigenvalues of P are real numbers

Q1.17.2 NAT 2017 1M

The determinant of a 2×2 matrix is 50. If one eigenvalue of the matrix is 10, the other eigenvalue is ________

Q1.17.3 NAT 2017 2M

Consider the matrix $A = \begin{bmatrix} 50 & 70 \\ 70 & 80 \end{bmatrix}$ whose eigenvectors corresponding to eigenvalues λ_1 and λ_2 are $x_1 = \begin{bmatrix} 70 \\ \lambda_1 - 50 \end{bmatrix}$ and $x_2 = \begin{bmatrix} \lambda_2 - 80 \\ 70 \end{bmatrix}$, respectively. The value of $x_1^T x_2$ is ______.

Q1.16.1 MCQ 2016 1M

The solution to the system of equations

$$\begin{bmatrix} 2 & 5 \\ -4 & 3 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} 2 \\ -30 \end{Bmatrix}$$

is

A. 6, 2 **B.** −6, 2 **C.** −6, −2 **D.** 6, −2

Q1.16.2 MCQ 2016 1M

The condition for which the eigenvalues of the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & k \end{bmatrix}$$

are positive, is

A. $k > 1/2$ **B.** $k > -2$ **C.** $k > 0$ **D.** $k < -1/2$

Q1.16.3 MCQ 2016 1M

A real square matrix A is called skew-symmetric if

A. $A^T = A$ **B.** $A^T = A^{-1}$

C. $A^T = -A$ **D.** $A^T = A + A^{-1}$

Q1.16.4 NAT 2016 2M

The number of linearly independent eigenvectors of matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is ________

Q1.15.1 MCQ 2015 1M

At least one eigen value of a singular matrix is

A. Positive **B.** Zero **C.** Negative **D.** Imaginary

Q1.15.2 MCQ 2015 1M

If any two columns of a determinant $P = \begin{vmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{vmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

A. Absolute value remains unchanged but sign will change.

B. Both absolute value and sign will change.

C. Absolute value will change but sign will not change.

D. Both absolute value and sign will remain unchanged.

Q1.15.3 MCQ 2015 2M

For a given matrix $P = \begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$, where $i = \sqrt{-1}$, the inverse of matrix P is

A. $\frac{1}{24}\begin{bmatrix} 4-3i & i \\ -i & 4+3i \end{bmatrix}$ **B.** $\frac{1}{25}\begin{bmatrix} i & 4-3i \\ 4+3i & -i \end{bmatrix}$

C. $\frac{1}{24}\begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$ **D.** $\frac{1}{25}\begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$

Q1.14.1 MCQ 2014 1M

Given that the determinant of the matrix $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{bmatrix}$ is -12, the determinant of the matrix $\begin{bmatrix} 2 & 6 & 0 \\ 4 & 12 & 8 \\ -2 & 0 & 4 \end{bmatrix}$ is

A. -96 **B.** -24 **C.** 24 **D.** 96

Q1.14.2 MCQ 2014 1M

Which one of the following describes the relationship among the three vectors, $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $5\hat{i} + 6\hat{j} + 4\hat{k}$?

A. The vectors are mutually perpendicular

B. The vectors are linearly dependent

C. The vectors are linearly independent

D. The vectors are unit vectors

Q1.14.3 MCQ 2014 1M

One of the eigenvectors of the matrix $\begin{bmatrix} -5 & 2 \\ -9 & 6 \end{bmatrix}$ is

A. $\begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$ **B.** $\begin{Bmatrix} -2 \\ 9 \end{Bmatrix}$ **C.** $\begin{Bmatrix} 2 \\ -1 \end{Bmatrix}$ **D.** $\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

Q1.14.4 MCQ 2014 1M

Consider a 3×3 real symmetric matrix S such that two of its eigen values are $a \neq 0,\ b \neq 0$ with respective eigenvectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$. If $a \neq b$ then $x_1y_1 + x_2y_2 + x_3y_3$ equals

A. a **B.** b **C.** ab **D.** 0

Q1.14.5 MCQ 2014 1M

Which one of the following equations is a correct identity for arbitrary 3×3 real matrices P, Q and R?

A. $P(Q + R) = PQ + RP$

B. $(P - Q)^2 = P^2 - 2PQ + Q^2$

C. $\det(P + Q) = \det P + \det Q$

D. $(P + Q)^2 = P^2 + PQ + QP + Q^2$

Q1.13.1 MCQ 2013 1M

The eigen values of symmetric matrix are all

A. Complex with non-zero positive imaginary part

B. Complex with non-zero negative imaginary part

C. Real

D. Pure imaginary

Q1.13.2 MCQ 2013 1M

Choose the CORRECT set of functions, which are linearly dependent.

A. $\sin x$, $\sin^2 x$ and $\cos^2 x$ **B.** $\cos x$, $\sin x$ and $\tan x$

C. $\cos 2x$, $\sin^2 x$ and $\cos^2 x$ **D.** $\cos 2x$, $\sin x$ and $\cos x$

Q1.12.1 MCQ 2012 2M

For the matrix $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, ONE of the normalized eigen vectors is given as

A. $\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$ **B.** $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$ **C.** $\begin{pmatrix} \frac{3}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} \end{pmatrix}$ **D.** $\begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$

Q1.10.1 MCQ 2010 1M

One of the eigen vectors of the matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$ is

A. $\begin{Bmatrix} 2 \\ -1 \end{Bmatrix}$ **B.** $\begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$ **C.** $\begin{Bmatrix} 4 \\ 1 \end{Bmatrix}$ **D.** $\begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$

Q1.09.1 MCQ 2009 1M

For a matrix $[M] = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$, the transpose of the matrix is equal to the inverse of the matrix $[M]^{\mathrm{T}} = [M]^{-1}$. The value of x is given by

A. $-\frac{4}{5}$ **B.** $-\frac{3}{5}$ **C.** $\frac{3}{5}$ **D.** $\frac{4}{5}$

Q1.08.1 MCQ 2008 1M

The matrix $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 0 & 6 \\ 1 & 1 & p \end{bmatrix}$ has one eigenvalue equal to 3. The sum of the other two eigenvalues is

A. p **B.** $p - 1$ **C.** $p - 2$ **D.** $p - 3$

Q1.08.2 MCQ 2008 2M

The eigenvectors of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ are written in the form $\begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$. What is $a + b$?

A. 0 **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Q1.07.1 MCQ 2007 1M

If a square matrix A is real and symmetric, then the eigen values

A. are always real

B. are always real and positive

C. are always real and non-negative

D. occur in complex conjugate pairs

Q1.07.2 MCQ 2007 2M

The number of linearly independent eigen vectors of $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is

A. 0 **B.** 1 **C.** 2 **D.** infinite

Q1.06.1 MCQ 2006 1M

Eigen values of a matrix $S = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ are 5 and 1. What are the eigen values of the matrix $S^2 = SS$?

A. 1 and 25 **B.** 6 and 4

C. 5 and 1 **D.** 2 and 10

Q1.06.2 MCQ 2006 1M

Match the items in columns I and II.

columns I	columns II
(P) Singular matrix	(1) Determinant is not defined
(Q) Non-square matrix	(2) Determinant is always one
(R) Real symmetric matrix	(3) Determinant is zero
(S) Orthogonal matrix	(4) Eigen values are always real
	(5) Eigen values are nto defined

Fig. Q1.06.2

A. P - 3 Q - 1 R - 4 S - 2 **B.** P - 2 Q - 3 R - 4 S - 1

C. P - 3 Q - 2 R - 5 S - 4 **D.** P - 3 Q - 4 R - 2 S - 1

Q1.06.3 MCQ 2006 2M

Multiplication of matrices E and F is G. Matrices E and G are $E = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

What is the matrix F?

A. $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **B.** $\begin{bmatrix} \sin\theta & \cos\theta & 0 \\ -\cos\theta & \sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} \sin\theta & -\cos\theta & 0 \\ \cos\theta & \sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q1.05.1 MCQ 2005 1M

A is a 3×4 real matrix and $Ax = b$ is an inconsistent system of equations. The highest possible rank of A is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q1.95.1 MCQ 1995 1M

Among the following, the pair of vectors orthogonal to each other is

A. $[3, 4, 7], [3, 4, 7]$ **B.** $[1, 0, 0], [1, 1, 0]$

C. $[1, 0, 2], [0, 5, 0]$ **D.** $[1, 1, 1], [-1, -1, -1]$

Civil Engineering

Q1.26.1 MCQ 2026 1M

Matrix P is given as

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

The TRUE option is

A. Trace of P is equal to the sum of the Eigen values of P.

B. $P^T P$ is an identity matrix.

C. P is a skew-symmetric matrix.

D. Absolute magnitude of each Eigen value is 1.

Q1.26.2 MCQ 2026 1M

Given:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

The above system of equations represents a

A. plane **B.** line **C.** volume **D.** point

Q1.26.3 MCQ 2026 1M

Matrix A has the eigenvalues 1, 2, and 3. The Trace of A^2 is

A. 6 **B.** 14 **C.** 20 **D.** 8

Q1.26.4 NAT 2026 1M

The eigenvalues of $[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$ are $\lambda_1 = -1.547$, $\lambda_2 = 12.330$, and $\lambda_3 = 4.711$. The absolute value of the determinant of matrix A is ________ *(rounded off to two decimal places)*.

Q1.25.1 MSQ 2025 2M

Pick the CORRECT eigen value(s) of the matrix $[A]$ from the following choices.

$$[A] = \begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix}$$

A. 10 **B.** 4 **C.** -2 **D.** -10

Q1.25.2 MCQ 2025 1M

Suppose λ is an eigenvalue of matrix A and x is the corresponding eigenvector. Let x also be an eigenvector of the matrix $B = A - 2I$, where I is the identity matrix. Then, the eigenvalue of B corresponding to the eigenvector x is equal to

A. λ **B.** $\lambda + 2$ **C.** 2λ **D.** $\lambda - 2$

Q1.25.3 MCQ 2025 1M

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ -2 & -3 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. For $Ax = b$ to be solvable, which one of the following options is the correct condition on b_1, b_2, and b_3:

A. $b_1 + b_2 + b_3 = 1$

B. $3b_1 + b_2 + 2b_3 = 0$

C. $b_1 + 3b_2 + b_3 = 2$

D. $b_1 + b_2 + b_3 = 2$

Q1.24.1 MCQ 2024 1M

The statements P and Q are related to matrices A and B, which are conformable for both addition and multiplication.

$$\text{P:}\quad (A + B)^{\mathrm{T}} = A^{\mathrm{T}} + B^{\mathrm{T}}$$
$$\text{Q:}\quad (AB)^{\mathrm{T}} = A^{\mathrm{T}} B^{\mathrm{T}}$$

Which one of the following options is CORRECT?

A. P is TRUE and Q is FALSE

B. Both P and Q are TRUE

C. P is FALSE and Q is TRUE

D. Both P and Q are FALSE

Q1.24.2 NAT 2024 2M

Consider two matrices $A = \begin{bmatrix} 2 & 1 & 4 \\ 1 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 2 & 3 \\ 1 & 4 \end{bmatrix}$.

The determinant of the matrix AB is ________ (in integer).

Q1.23.1 MSQ 2023 1M

If M is an arbitrary real $n \times n$ matrix, then which of the following matrices will have non-negative eigenvalues?

A. M^2 **B.** MM^{T} **C.** $M^{\mathrm{T}}M$ **D.** $(M^{\mathrm{T}})^2$

Q1.23.2 MSQ 2023 2M

For the matrix

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

which of the following statements is/are TRUE?

- **A.** The eigenvalues of $[A]^{\mathrm{T}}$ are same as the eigenvalues of $[A]$
- **B.** The eigenvalues of $[A]^{-1}$ are the reciprocals of the eigenvalues of $[A]$
- **C.** The eigenvectors of $[A]^{\mathrm{T}}$ are same as the eigenvectors of $[A]$

- [ ] **D.** The eigenvectors of $[A]^{-1}$ are same as the eigenvectors of $[A]$

Q1.23.3 MCQ 2023 2M

Two vectors $\begin{bmatrix}2 & 1 & 0 & 3\end{bmatrix}^T$ and $\begin{bmatrix}1 & 0 & 1 & 2\end{bmatrix}^T$ belong to the null space of a 4×4 matrix of rank 2. Which one of the following vectors also belongs to the null space?

A. $\begin{bmatrix}1 & 1 & -1 & 1\end{bmatrix}^T$ **B.** $\begin{bmatrix}2 & 0 & 1 & 2\end{bmatrix}^T$

C. $\begin{bmatrix}0 & -2 & 1 & -1\end{bmatrix}^T$ **D.** $\begin{bmatrix}3 & 1 & 1 & 2\end{bmatrix}^T$

Q1.23.4 MCQ 2023 2M

Cholesky decomposition is carried out on the following square matrix $[A]$.

$$[A] = \begin{bmatrix}8 & -5\\ -5 & a_{22}\end{bmatrix}$$

Let l_{ij} and a_{ij} be the $(i, j)^{th}$ elements of matrices $[L]$ and $[A]$, respectively. If the element l_{22} of the decomposed lower triangular matrix $[L]$ is 1.968, what is the value (rounded off to the nearest integer) of the element a_{22}?

A. 5 **B.** 7 **C.** 9 **D.** 11

Q1.22.1 MSQ 2022 1M

The matrix M is defined as

$$M = \begin{bmatrix}1 & 3\\ 4 & 2\end{bmatrix}$$

and has eigenvalues 5 and -2. The matrix Q is formed as

$$Q = M^3 - 4M^2 - 2M$$

Which of the following is/are the eigenvalue(s) of matrix Q?

A. 15 **B.** 25 **C.** -20 **D.** -30

Q1.22.2 MSQ 2022 1M

P and Q are two square matrices of the same order. Which of the following statement(s) is/are correct?

- [ ] **A.** If P and Q are invertible, then $[PQ]^{-1} = Q^{-1}P^{-1}$.
- [ ] **B.** If P and Q are invertible, then $[QP]^{-1} = P^{-1}Q^{-1}$.
- [ ] **C.** If P and Q are invertible, then $[PQ]^{-1} = P^{-1}Q^{-1}$.
- [ ] **D.** If P and Q are not invertible, then $[PQ]^{-1} = Q^{-1}P^{-1}$.

Q1.22.3 NAT 2022 1M

The components of pure shear strain in a sheared material are given in the matrix form:

$$\varepsilon = \begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$$

Here, Trace$(\varepsilon) = 0$. Given, $P = $ Trace(ε^8) and $Q = $ Trace(ε^{11}). The numerical value of $(P + Q)$ is ________. (*in integer*)

Q1.22.4 MSQ 2022 2M

Let y be a non-zero vector of size 2022×1. Which of the following statement(s) is/are TRUE?

- [ ] **A.** yy^T is a symmetric matrix.
- [ ] **B.** y^Ty is an eigenvalue of yy^T.
- [ ] **C.** yy^T has a rank of 2022.
- [ ] **D.** yy^T is invertible.

Q1.20.1 NAT 2020 2M

Consider the system of equations

$$\begin{bmatrix}1 & 3 & 2\\ 2 & 2 & -3\\ 4 & 4 & -6\\ 2 & 5 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}1\\ 1\\ 2\\ 1\end{bmatrix}$$

The value of x_3 (*round off to the nearest integer*), is ________.

Q1.20.2 MCQ 2020 1M

A 4×4 matrix $[P]$ is given below

$$[P] = \begin{bmatrix}0 & 1 & 3 & 0\\ -2 & 3 & 0 & 4\\ 0 & 0 & 6 & 1\\ 0 & 0 & 1 & 6\end{bmatrix}$$

The eigenvalues of $[P]$ are

A. 0, 3, 6, 6 **B.** 1, 2, 3, 4

C. 3, 4, 5, 7 **D.** 1, 2, 5, 7

Q1.19.1 MCQ 2019 1M

Euclidean norm (length) of the vector $\begin{bmatrix}4 & -2 & -6\end{bmatrix}^T$ is

A. $\sqrt{12}$ **B.** $\sqrt{24}$ **C.** $\sqrt{48}$ **D.** $\sqrt{56}$

Q1.18.1 MCQ 2018 1M

For the given orthogonal matrix Q,

$$Q = \begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$$

The inverse is

A. $\begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$

B. $\begin{bmatrix} -3/7 & -2/7 & -6/7 \\ 6/7 & -3/7 & -2/7 \\ -2/7 & -6/7 & 3/7 \end{bmatrix}$

C. $\begin{bmatrix} 3/7 & -6/7 & 2/7 \\ 2/7 & 3/7 & 6/7 \\ 6/7 & 2/7 & -3/7 \end{bmatrix}$

D. $\begin{bmatrix} -3/7 & 6/7 & -2/7 \\ -2/7 & -3/7 & -6/7 \\ -6/7 & -2/7 & 3/7 \end{bmatrix}$

Q1.18.2 MCQ 2018 1M

The matrix $\begin{pmatrix} 2 & -4 \\ 4 & -2 \end{pmatrix}$ has

A. real eigenvalues and eigenvectors

B. real eigenvalues but complex eigenvectors

C. complex eigenvalues but real eigenvectors

D. complex eigenvalues and eigenvectors

Q1.17.1 MCQ 2017 1M

Consider the following simultaneous equations (with c_1 and c_2 being constants):

$$3x_1 + 2x_2 = c_1$$
$$4x_1 + x_2 = c_2$$

The characteristic equation for these simultaneous equations is

A. $\lambda^2 - 4\lambda - 5 = 0$ **B.** $\lambda^2 - 4\lambda + 5 = 0$

C. $\lambda^2 + 4\lambda - 5 = 0$ **D.** $\lambda^2 + 4\lambda + 5 = 0$

Q1.17.2 MCQ 2017 2M

If $A = \begin{bmatrix} 1 & 5 \\ 6 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 7 \\ 8 & 4 \end{bmatrix}$, AB^{T} is equal to

A. $\begin{bmatrix} 38 & 28 \\ 32 & 56 \end{bmatrix}$ **B.** $\begin{bmatrix} 3 & 40 \\ 42 & 8 \end{bmatrix}$

C. $\begin{bmatrix} 43 & 27 \\ 34 & 50 \end{bmatrix}$ **D.** $\begin{bmatrix} 38 & 32 \\ 28 & 56 \end{bmatrix}$

Q1.17.3 MCQ 2017 1M

The matrix P is the inverse of a matrix Q. If I denotes the identity matrix, which one of the following options is correct?

A. $PQ = I$ but $QP \neq I$

B. $QP = I$ but $PQ \neq I$

C. $PQ = I$ and $QP = I$

D. $PQ - QP = I$

Q1.17.4 MCQ 2017 2M

Consider the matrix $\begin{bmatrix} 5 & -1 \\ 4 & 1 \end{bmatrix}$. Which one of the following statements is TRUE for the eigenvalues and eigenvectors of this matrix?

A. Eigenvalue 3 has a multiplicity of 2, and only one independent eigenvector exists.

B. Eigenvalue 3 has a multiplicity of 2, and two independent eigenvectors exist.

C. Eigenvalue 3 has a multiplicity of 2, and no independent eigenvector exists.

D. Eigenvalues are 3 and −3, and two independent eigenvectors exist.

Q1.16.1 MCQ 2016 1M

If the entries in each column of a square matrix M add up to 1, then an eigenvalue of M is

A. 4 **B.** 3 **C.** 2 **D.** 1

Q1.15.1 MCQ 2015 1M

Let $A = [a_{ij}]$, $1 \leq i, j \leq n$ with $n \geq 3$ and $a_{ij} = i.j$. The rank of A is:

A. 0 **B.** 1 **C.** n-1 **D.** n

Q1.15.2 MCQ 2015 2M

The two Eigen values of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & p \end{bmatrix}$ have a ratio of 3 : 1 for $p = 2$. What is another value of p for which the Eigen values have the same ratio of 3 : 1?

A. -2 **B.** 1 **C.** $\frac{7}{3}$ **D.** $\frac{14}{3}$

Q1.14.1 NAT 2014 1M

Given the matrices $J = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix}$ and $K = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, the product $K^T JK$ is ____________

Q1.13.1 NAT 2013 1M

What is the minimum number of multiplications involved in computing the matrix product PQR? Matrix P has 4 rows and 2 columns, matrix Q has 2 rows and 4 columns, and matrix R has 4 rows and 1 column.________

Q1.11.1 MCQ 2011 1M

$[A]$ is a square matrix which is neither symmetric nor skew-symmetric and $[A]^T$ is it's transpose. The sum and difference of these matrices are defined as $[S] = [A]+[A]^T$ and $[D] = [A]-[A]^T$ respectively. Which of the following statements is true?

A. Both $[S]$ and $[D]$ are symmetric

B. Both $[S]$ and $[D]$ are skew-symmetric

C. $[S]$ is skew-symmetric and $[D]$ is symmetric

D. $[S]$ is symmetric and $[D]$ is skew-symmetric

Q1.10.1 MCQ 2010 2M

The inverse of the matrix $\begin{bmatrix} 3+2i & i \\ -i & 3-2i \end{bmatrix}$ is

A. $\frac{1}{12}\begin{bmatrix} 3+2i & -i \\ i & 3-2i \end{bmatrix}$ **B.** $\frac{1}{12}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

C. $\frac{1}{14}\begin{bmatrix} 3+2i & -i \\ i & 3-2i \end{bmatrix}$ **D.** $\frac{1}{14}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

Q1.09.1 MCQ 2009 2M

In the solution of the following set of linear equations by Gauss elimination using partial pivoting $5x + y + 2z = 34$; $4y - 3z = 12$; and $10x - 2y + z = -4$. The pivots for elimination of x and y are

A. 10 and 4 **B.** 10 and 2 **C.** 5 and 4 **D.** 5 and -4

Q1.08.1 MCQ 2008 1M

The product of matrices $(PQ)^{-1}P$ is

A. P^{-1} **B.** Q^{-1} **C.** $P^{-1}Q^{-1}$ **D.** PQP^{-1}

Q1.05.1 MCQ 2005 1M

Consider the matrices $X_{(4,3)}$, $Y_{(4,3)}$ and $P_{(2,3)}$. The order of $\left(P(X^TY)^{-1}P^T\right)^T$ will be

A. (2×2) **B.** (3×3) **C.** (4×3) **D.** (3×4)

Q1.05.2 MCQ 2005 1M

Consider a non-homogeneous system of linear equations representing mathematically an over-determined system. Such a system will be

A. Consistent having a unique solution

B. Consistent having many solutions

C. Inconsistent having unique solution

D. Inconsistent having no solution

Q1.05.3 MCQ 2005 2M

Consider the system of equations $A_{(n\times n)}X_{(n\times 1)} = l_{(n\times 1)}$ where, 1 is a scalar. Let (l_i, x_i) be an eigen-pair of an eigen value and its corresponding eigen vector for real matrix A. Let I be a $(n \times n)$ unit matrix. Which one of the following statement is NOT correct?

A. For a homogeneous $n \times n$ system of linear equations $(A - l_i I)x = 0$ having a nontrivial solution, the rank of $(A - l_i I)$ is less than n.

B. For matrix A^m, m being a positive integer, (λ_i^x, x_i^m) will be the eigen-pair for all i.

C. If $A^T = A^{-1}$, then $|l_i| = 1$ for all i.

D. If $A^T = A$, then l_i is real for all i.

Q1.04.1 MCQ 2004 1M

Real matrices $\{A\}_{3\times1}$, $[B]_{3\times3}$, $[C]_{3\times5}$, $[D]_{5\times3}$, $[E]_{5\times5}$, and $[F]_{5\times1}$ are given. Matrices $[B]$ and $[E]$ are symmetric. Following statements are made with respect to these matrices.

(I) Matrix product $\{F\}^{\mathrm{T}}[C]^{\mathrm{T}}[B][C]\{F\}$ is a scalar

(II) Matrix product $[D]^{\mathrm{T}}[F][D]$ is always symmetric

With reference to above statements, which of the following applies?

A. Statement I is true but II is false

B. Statement I is false but II is true

C. Both the statements are true

D. Both the statements are false

Q1.01.1 MCQ 2001 2M

The product $[P][Q]^{\mathrm{T}}$ of the following two matrices $[P]$ and $[Q]$ is

$$[P] = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}; [Q] = \begin{bmatrix} 4 & 8 \\ 9 & 2 \end{bmatrix}$$

A. $\begin{bmatrix} 32 & 24 \\ 56 & 46 \end{bmatrix}$ **B.** $\begin{bmatrix} 46 & 56 \\ 24 & 32 \end{bmatrix}$

C. $\begin{bmatrix} 35 & 22 \\ 61 & 42 \end{bmatrix}$ **D.** $\begin{bmatrix} 32 & 56 \\ 24 & 46 \end{bmatrix}$

Q1.00.1 MCQ 2000 1M

Consider the following two statements :

(I) The maximum number of linearly independent column vectors of a matrix A is called the rank of A.

(II) If A is an $n\times n$ square matrix, it will be nonsingular is rank $A = n$.

With reference to the above statements, which of the following applies ?

A. Both the statements are false

B. Both the statements are true.

C. I is true but II is false.

D. I is false but II is true.

Q1.99.1 MCQ 1999 1M

If A is any $n \times n$ matrix and k is a scalar, $|kA| = \alpha|A|$, where α is

A. kn **B.** n^k **C.** k^n **D.** k/n

Q1.99.2 MCQ 1999 1M

Number of terms in the expansion of general determinant of order n is

A. n^2 **B.** $n!$ **C.** n **D.** $(n+1)^2$

Q1.99.3 MCQ 1999 2M

The equation $\begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & -1 \\ y & x^2 & x \end{vmatrix} = 0$, represents a parabola passing through the points

A. $(0,1), (0,2), (0,-1)$ **B.** $(0,0), (-1,1), (1,2)$

C. $(1,1), (0,0), (2,2)$ **D.** $(1,2), (2,1), (0,0)$

Q1.98.1 MCQ 1998 1M

In matrix algebra $AS = AT$ (A, S, T, are matrices of appropriate order) implies $S = T$ only if

A. A is symmetric **B.** A is singular

C. A is non singular **D.** A is skew symmetric

Q1.98.2 MCQ 1998 2M

The real symmetric matrix C corresponding to the Quadratic form

$$Q = 4x_1x_2 - 5x_2^2, \text{ is}$$

A. $\begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix}$ **B.** $\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & 2 \\ 2 & -5 \end{bmatrix}$

Q1.97.1 MCQ 1997 1M

If A and B are two matrices and if AB exists, then BA exists

A. only if A has as many rows as B has columns

B. only if both A and B are square matrices

C. only if A and B are skew matrices

D. only if both A and B are symmetric

Chemical Engineering

Q1.26.1 MCQ 2026 1M

Which one of the following is the pair of eigenvalues of the matrix $\begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$?

A. $-7, 7$ **B.** $-5, 5$ **C.** $-4, 3$ **D.** $-3, 4$

Q1.24.1 MSQ 2024 1M

Consider a linear homogeneous system of equations $\mathbf{Ax} = \mathbf{0}$, where $\mathbf{A}$ is an $n \times n$ matrix, $\mathbf{x}$ is an $n \times 1$ vector and $\mathbf{0}$ is an $n \times 1$ null vector. Let r be the rank of $\mathbf{A}$. For a non-trivial solution to exist, which of the following conditions is/are satisfied?

A. Determinant of $\mathbf{A} = 0$

B. $r = n$

C. $r < n$

D. Determinant of $\mathbf{A} \neq 0$

Q1.24.2 NAT 2024 1M

Consider a matrix $\mathbf{A} = \begin{bmatrix} -5 & a \\ -2 & -2 \end{bmatrix}$, where a is a constant. If the eigenvalues of $\mathbf{A}$ are -1 and -6, then the value of a, rounded off to the nearest integer, is ________

Q1.23.1 NAT 2023 2M

If a matrix M is defined as $M = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$, the sum of all the eigenvalues of M^3 is equal to ________ (in integer).

Q1.22.1 MCQ 2022 1M

Given matrix $A = \begin{bmatrix} x & 1 & 3 \\ y & 2 & 6 \\ 3 & 5 & 7 \end{bmatrix}$, the ordered pair (x, y) for which $\det(A) = 0$ is

A. $(1, 1)$ **B.** $(1, 2)$

C. $(2, 2)$ **D.** $(2, 1)$

Q1.21.1 NAT 2021 1M

$\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are vectors of length 4.

$$\begin{aligned} \mathbf{A} &= [a_1 \quad a_2 \quad a_3 \quad a_4] \\ \mathbf{B} &= [b_1 \quad b_2 \quad b_3 \quad b_4] \\ \mathbf{C} &= [c_1 \quad c_2 \quad c_3 \quad c_4] \\ \mathbf{D} &= [d_1 \quad d_2 \quad d_3 \quad d_4] \end{aligned}$$

It is known that $\mathbf{B}$ is not a scalar multiple of $\mathbf{A}$. Also, $\mathbf{C}$ is linearly independent of $\mathbf{A}$ and $\mathbf{B}$. Further, $\mathbf{D} = 3\mathbf{A} + 2\mathbf{B} + \mathbf{C}$. The rank of the matrix $\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{bmatrix}$ is ________.

Q1.21.2 MCQ 2021 2M

Let A be a square matrix of size $n \times n$ $(n > 1)$. The elements of $A = \{a_{ij}\}$ are given by

$$a_{ij} = \begin{cases} i \times j, & \text{if } i \geq j \\ 0, & \text{if } i < j \end{cases}$$

The determinant of A is

A. 0 **B.** 1 **C.** $n!$ **D.** $(n!)^2$

Q1.19.1 MCQ 2019 1M

A system of n homogeneous linear equations containing n unknowns will have non-trivial solutions if and only if the determinant of the coefficient matrix is

A. 1 **B.** -1 **C.** 0 **D.** ∞

Q1.18.1 NAT 2018 2M

For the matrix $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ if det stands for the determinant and A^T is the transpose of A then the value of $\det(A^T A)$ is ________

Q1.16.1 NAT 2016 2M

A set of simultaneous linear algebraic equations is represented in a matrix form as shown below.

$$\begin{bmatrix} 0 & 0 & 0 & 4 & 13 \\ 2 & 5 & 5 & 2 & 10 \\ 0 & 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 4 & 5 \\ 2 & 3 & 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 46 \\ 161 \\ 61 \\ 30 \\ 81 \end{bmatrix}$$

The value (rounded off to the nearest integer) of x_3 is ________

Q1.15.1 MCQ 2015 1M

The following set of three vectors

$$\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}x\\6\\x\end{pmatrix} \text{ and } \begin{pmatrix}3\\4\\2\end{pmatrix},$$

is linearly dependent when x is equal to

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.15.2 NAT 2015 1M

For the matrix $\begin{pmatrix}4 & 3\\3 & 4\end{pmatrix}$, if $\begin{pmatrix}1\\1\end{pmatrix}$ is an eigenvector, the corresponding eigenvalue is __________.

Q1.13.1 MCQ 2013 1M

Which of the following statements are TRUE?

P. The eigen values of a symmetric matrix are real

Q. The value of the determinant of an orthogonal matrix can only be +1

R. The transpose of a square matrix **A** has the same eigen values as those of **A**

S. The inverse of an '$n \times n$' matrix exists if and only if the rank is less than 'n'

A. P and Q only **B.** P and R only

C. Q and R only **D.** P and S only

Q1.12.1 MCQ 2012 1M

Consider the following (2×2) matrix

$$\begin{pmatrix}4 & 0\\0 & 4\end{pmatrix}$$

Which one of the following vectors is NOT a valid eigen vector of the above matrix?

A. $\begin{pmatrix}1\\0\end{pmatrix}$ **B.** $\begin{pmatrix}-2\\1\end{pmatrix}$ **C.** $\begin{pmatrix}4\\-3\end{pmatrix}$ **D.** $\begin{pmatrix}0\\0\end{pmatrix}$

Q1.11.1 MCQ 2011 1M

Let $\lambda_1 = -1$ and $\lambda_2 = 3$ be the Eigen values and $\underline{V_1} = \begin{pmatrix}1\\0\end{pmatrix}$ and $\underline{V_2} = \begin{pmatrix}1\\1\end{pmatrix}$ be the corresponding Eigen vectors of a real 2×2 matrix R. Given that $P = (\underline{V_1}\ \underline{V_2})$, which ONE of the following matrices represents $P^{-1}RP$?

A. $\begin{pmatrix}0 & -1\\3 & 0\end{pmatrix}$ **B.** $\begin{pmatrix}0 & 3\\-1 & 0\end{pmatrix}$ **C.** $\begin{pmatrix}3 & 0\\0 & -1\end{pmatrix}$ **D.** $\begin{pmatrix}-1 & 0\\0 & 3\end{pmatrix}$

Q1.10.1 MCQ 2010 1M

A real $n \times n$ matrix $A = [a_{ij}]$ is defined as follows:

$$\begin{aligned} a_{ij} &= i, \quad \text{if } i = j \\ &= 0, \quad \text{otherwise} \end{aligned}$$

The summation of all n eigenvalues of A is

A. $n(n+1)/2$ **B.** $n(n-1)/2$

C. $\dfrac{n(n+1)(2n+1)}{6}$ **D.** n^2

Q1.10.2 MCQ 2010 2M

X and **Y** are non-zero square matrices of size $n \times n$. If $\mathbf{XY} = \mathbf{0}_{n \times n}$ then

A. $|\mathbf{X}| = 0$ and $|\mathbf{Y}| \neq 0$ **B.** $|\mathbf{X}| \neq 0$ and $|\mathbf{Y}| = 0$

C. $|\mathbf{X}| = 0$ and $|\mathbf{Y}| = 0$ **D.** $|\mathbf{X}| \neq 0$ and $|\mathbf{Y}| \neq 0$

Q1.09.1 MCQ 2009 1M

The system of linear equations $Ax = 0$, where A is an $n \times n$ matrix, has a non-trivial solution ONLY if

A. rank of $A > n$ **B.** rank of $A = n$

C. rank of $A < n$ **D.** A is an identity matrix

Q1.09.2 MCQ 2009 2M

The eigenvalues of matrix $A = \begin{bmatrix}1 & 2\\4 & 3\end{bmatrix}$ are 5 and -1.

Then the eigenvalues of $-2A + 3I$ (I is a 2×2 identity matrix) are

A. -7 and 5 **B.** 7 and -5

C. $-\dfrac{1}{7}$ and $\dfrac{1}{5}$ **D.** $\dfrac{1}{7}$ and $-\dfrac{1}{5}$

Q1.07.1 MCQ 2007 2M

A and B are two 3×3 matrix such that

$$A = \begin{bmatrix}-2 & 4 & 6\\1 & 2 & 1\\0 & 4 & 4\end{bmatrix}, B \neq 0$$

and $AB = 0$. Then the rank of matrix B is

A. $r = 2$ **B.** $r < 3$ **C.** $r \leq 3$ **D.** $r = 3$

Q1.05.1 MCQ 2005 2M

The matrix A is given by,

$$A = \begin{pmatrix} 1 & 4 \\ a & 2 \end{pmatrix}$$

The eigenvalues of the matrix A are real and non-negative for the condition

A. $(-1/16) \le a \le (1/16)$

B. $(-1/2) \le a \le (1/2)$

C. $(-1/2) \le a \le (1/16)$

D. $(-1/16) \le a \le (1/2)$

Q1.02.1 NAT 2002 5M

Matrix $A = \begin{bmatrix} 0.1 & 0.5 & 0 \\ 0.8 & 0 & 0.4 \\ 0.1 & 0.5 & 0.6 \end{bmatrix}$ has the property that it satisfies $Ax = x$, for some vector x.

a. Write the characteristic equation to be solved for eigenvalues of A.

b. Based on visual observation, find one of the eigenvalues of A.

c. Find the other two eigenvalues of A.

Production & Industrial Engineering

Q1.26.1 MCQ 2026 2M

Which ONE of the following options CORRECTLY matches the matrices to their properties?

Matrix		Property	
P	$\begin{bmatrix} -30 & 17 & 5 \\ 17 & 0 & -12 \\ 5 & -12 & 4 \end{bmatrix}$	1	Singular
Q	$\begin{bmatrix} 0 & 1 & 9 \\ 7 & 0 & 2 \\ 12 & 3 & 0 \end{bmatrix}$	2	Triangular
R	$\begin{bmatrix} 2/3 & 1/3 & 2/3 \\ 0 & 2/3 & -1/3 \\ 0 & 4/3 & -2/3 \end{bmatrix}$	3	Symmetric
S	$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$	4	Trace free

A. P-3; Q-1; R-4; S-2 **B.** P-4; Q-1; R-3; S-2
C. P-3; Q-4; R-1; S-2 **D.** P-4; Q-1; R-2; S-3

Q1.25.1 MCQ 2025 1M

The eigenvalues of the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are

A. $-\sqrt{-1}$ and $\sqrt{-1}$ **B.** -1 and 1
C. $1+\sqrt{-1}$ and $1-\sqrt{-1}$ **D.** $-1+\sqrt{-1}$ and $-1-\sqrt{-1}$

Q1.24.1 MCQ 2024 2M

If $\mathbf{A} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ is a matrix such that $\mathbf{A}^2 = \mathbf{I}$, where $\mathbf{I}$ is an identity matrix, then which of the following is TRUE?

A. $1 + a^2 + bc = 0$ **B.** $1 - a^2 + bc = 0$
C. $1 - a^2 - bc = 0$ **D.** $1 + a^2 - bc = 0$

Q1.23.1 MCQ 2023 1M

Given matrices $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix}$.

B is skew-symmetric matrix of A. B_{13} is

A. -3 **B.** -2 **C.** 2 **D.** 3

Q1.23.2 MCQ 2023 1M

Transformation matrix to translate a point P from $(10, 15)$ to $(15, 25)$ is

A. $\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q1.23.3 NAT 2023 2M

For the matrix $\begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$, eigenvalue corresponding to the eigenvector $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ is ______*(in integer)*.

Q1.22.1 MCQ 2022 2M

If a matrix is squared, then

A. both eigenvalues and eigenvectors must change

B. both eigenvalues and eigenvectors are retained

C. eigenvalues get squared but eigenvectors are retained

D. eigenvalues are retained but eigenvectors change

Q1.22.2 NAT 2022 1M

Matrix A as a product of two other matrices is given by

$$A = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \end{bmatrix}.$$

The value of $\det(A)$ is ______________. *[round off to nearest integer]*

Q1.21.1 MCQ 2021 2M

The eigenvalues of matrix $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$ are 5 and 10.

For matrix $B = A + \alpha I$, where α is a constant and I is 2×2 identity matrix, its eigenvalues are

A. 5, 10 **B.** $5 + \alpha, 10 + \alpha$

C. $5 - \alpha, 10 - \alpha$ **D.** $5\alpha, 10\alpha$

Q1.19.1 MCQ 2019 1M

For any real, square and non-singular matrix **B**, the det B^{-1} is

A. zero **B.** $(\det B)^{-1}$ **C.** $-(\det B)$ **D.** $\det B$

Q1.18.1 NAT 2018 1M

The diagonal elements of a 3x3 matrix are -10, 5 and 0, respectively. If two of its eigenvalues are -15 each, the third eigenvalue is _______

Q1.17.1 MCQ 2017 1M

For an orthogonal matrix Q, the valid equality is

A. $Q^T = Q^{-1}$ **B.** $Q = Q^{-1}$

C. $Q^T = Q$ **D.** $\det(Q) = 0$

Q1.16.1 MCQ 2016 2M

The number of solutions of the simultaneous algebraic equations $y = 3x + 3$ and $y = 3x + 5$ is

A. zero **B.** 1 **C.** 2 **D.** infinite

Q1.11.1 MCQ 2011 1M

If matrix $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ and matrix $B = \begin{bmatrix} 4 & 6 \\ 5 & 9 \end{bmatrix}$, the transpose of product of these two matrices, i.e. $(AB)^T$ is equal to

A. $\begin{bmatrix} 28 & 19 \\ 34 & 47 \end{bmatrix}$ **B.** $\begin{bmatrix} 19 & 34 \\ 47 & 28 \end{bmatrix}$

C. $\begin{bmatrix} 48 & 33 \\ 28 & 19 \end{bmatrix}$ **D.** $\begin{bmatrix} 28 & 19 \\ 48 & 33 \end{bmatrix}$

Q1.10.1 MCQ 2010 1M

If $\{1, 0, -1\}^T$ is an eigenvector of the following matrix,

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

then the corresponding eigenvalue is

A. 1 **B.** 2 **C.** 3 **D.** 5

Q1.10.2 MCQ 2010 1M

The value of q for which the following set of linear algebraic equations

$$2x + 3y = 0$$
$$6x + qy = 0$$

can have non-trivial solution is

A. 2 **B.** 7 **C.** 9 **D.** 11

Q1.09.1 MCQ 2009 2M

The value of x_3 obtained by solving the following system of linear equations is

$$x_1 + 2x_2 - 2x_3 = 4$$
$$2x_1 + x_2 + x_3 = -2$$
$$-x_1 + x_2 - x_3 = 2$$

A. -12 **B.** -2 **C.** 0 **D.** 12

Q1.08.1 MCQ 2008 2M

Inverse of the matrix $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is

A. $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ **B.** $\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

C. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ **D.** $\begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}$

Q1.07.1 MCQ 2007 1M

The determinant $\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix}$ evaluates to

A. 0 **B.** $2b(b-1)$

C. $2(1-b)(1+2b)$ **D.** $3b(1+b)$

Q1.07.2 MCQ 2007 2M

If A is square symmetric real valued matrix of dimension $2n$, the eigenvalues of A are

A. $2n$ distinct real values

B. $2n$ real values, not necessarily distinct

C. n distinct pairs of complex conjugate numbers

D. n pairs of complex conjugate numbers, not necessarily distinct

Q1.05.1 MCQ 2005 1M

The eigen values of the matrix M given are 15, 3, and 0. $M = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, the value of the determinant of a matrix is

A. 20 **B.** 10 **C.** 0 **D.** -10

Q1.95.1 MCQ 1995 1M

Given matrix $L = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 5 \end{bmatrix}$ and $M = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$ then $L \times M$ is

A. $\begin{bmatrix} 8 & 1 \\ 13 & 2 \\ 22 & 5 \end{bmatrix}$ **B.** $\begin{bmatrix} 6 & 5 \\ 9 & 8 \\ 12 & 13 \end{bmatrix}$ **C.** $\begin{bmatrix} 1 & 8 \\ 2 & 13 \\ 5 & 22 \end{bmatrix}$ **D.** $\begin{bmatrix} 6 & 2 \\ 9 & 4 \\ 0 & 5 \end{bmatrix}$

Q1.94.1 MCQ 1994 1M

For the following matrix $\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ the number of positive characteristic roots is

A. one **B.** two

C. four **D.** cannot be found

Q1.94.2 MCQ 1994 1M

If for a matrix, rank equals both the number of rows and number of columns, then the matrix is called

A. Non-singular **B.** singular

C. transpose **D.** minor

Q1.94.3 MCQ 1994 1M

The matrix $\begin{bmatrix} 1 & -4 \\ 1 & -5 \end{bmatrix}$ is an inverse of the matrix $\begin{bmatrix} 5 & -4 \\ 1 & -1 \end{bmatrix}$

A. True

B. False

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q1.26.1 B,D	Q1.25.1 A	Q1.24.1 5	Q1.24.2 A,B	Q1.23.1 C	Q1.23.2 A
Q1.22.1 C	Q1.22.2 A	Q1.21.1 8.0	Q1.21.2 10.0	Q1.20.1 C	Q1.20.2 A
Q1.19.1 3	Q1.18.1 C	Q1.18.2 2	Q1.17.1 4	Q1.17.2 C	Q1.17.3 C
Q1.16.1 C	Q1.16.2 1	Q1.16.3 3	Q1.16.4 D	Q1.16.5 D	Q1.15.1 2
Q1.15.2 16.5 to 17.5	Q1.15.3 B	Q1.15.4 C	Q1.14.1 1	Q1.14.2 D	Q1.14.3 1
Q1.14.4 200	Q1.14.5 B	Q1.14.6 49	Q1.14.7 B	Q1.13.1 A	Q1.13.2 B
Q1.12.1 B	Q1.11.1 B	Q1.10.1 C	Q1.09.1 D	Q1.08.1 C	Q1.08.2 B
Q1.06.1 C	Q1.06.2 A	Q1.06.3 C	Q1.05.1 C	Q1.05.2 C	Q1.00.1 B
Q1.98.1 D	Q1.94.1 A	Q1.94.2 A			
ELECTRICAL ENGINEERING					
Q1.26.1 A,B,D	Q1.26.2 0	Q1.26.3 A	Q1.26.4 A,B,D	Q1.26.5 A,B,C	Q1.25.1 B
Q1.25.2 B	Q1.25.3 A	Q1.24.1 C	Q1.24.2 29	Q1.23.1 A	Q1.22.1 B
Q1.22.2 C	Q1.22.3 A	Q1.21.1 B	Q1.21.2 1	Q1.20.1 *	Q1.19.1 D
Q1.19.2 C	Q1.18.1 5.5	Q1.18.2 0.9 to 1.1	Q1.17.1 C	Q1.17.2 A	Q1.16.1 3
Q1.16.2 A	Q1.16.3 B	Q1.16.4 D	Q1.16.5 D	Q1.15.1 9	Q1.15.2 B
Q1.15.3 A	Q1.14.1 0.33	Q1.14.2 C	Q1.14.3 A	Q1.13.1 D	Q1.11.1 B
Q1.11.2 D	Q1.10.1 B	Q1.09.1 C	Q1.08.1 D	Q1.08.2 A	Q1.08.3 B
Q1.08.4 D	Q1.07.1 B	Q1.07.2 B	Q1.07.3 D	Q1.07.4 C	Q1.07.5 A
Q1.05.1 A	Q1.05.2 B	Q1.02.1 C	Q1.98.1 C	Q1.97.1 *	Q1.95.1 *

Q1.94.1 C					
INSTRUMENTATION ENGINEERING					
Q1.26.1 C	Q1.26.2 A	Q1.25.1 A	Q1.25.2 2	Q1.24.1 C	Q1.24.2 -2
Q1.23.1 A	Q1.23.2 -8	Q1.22.1 A,D	Q1.22.2 A,C	Q1.21.1 A	Q1.21.2 4
Q1.20.1 B	Q1.20.2 D	Q1.19.1 10	Q1.18.1 A	Q1.18.2 C	Q1.17.1 1
Q1.17.2 D	Q1.17.3 C	Q1.16.1 -6	Q1.15.1 C	Q1.14.1 C	Q1.13.1 B
Q1.13.2 *	Q1.11.1 B	Q1.10.1 A	Q1.10.2 C	Q1.09.1 B	Q1.09.2 C
Q1.09.3 D	Q1.07.1 B	Q1.07.2 B	Q1.06.1 C	Q1.05.1 B	Q1.01.1 B
COMPUTER SCIENCE & IT					
Q1.26.1 A	Q1.26.2 A	Q1.26.3 B,D	Q1.26.4 24	Q1.26.5 48	Q1.25.1 A,D
Q1.25.2 D	Q1.25.3 C	Q1.25.4 A	Q1.25.5 A,B,C	Q1.24.1 B	Q1.24.2 A
Q1.24.3 A,B	Q1.23.1 B	Q1.22.1 D	Q1.22.2 A,C,D	Q1.21.1 3	Q1.21.2 4
Q1.20.1 C	Q1.19.1 12	Q1.18.1 3	Q1.18.2 D	Q1.17.1 C	Q1.17.2 B
Q1.17.3 2	Q1.17.4 5	Q1.16.1 15	Q1.16.2 C	Q1.16.3 0.124 to 0.126	Q1.16.4 1500
Q1.15.1 D	Q1.15.2 B	Q1.15.3 6	Q1.15.4 0	Q1.15.5 C	Q1.14.1 0
Q1.14.2 6	Q1.14.3 2	Q1.14.4 A	Q1.13.1 A	Q1.12.1 D	Q1.11.1 C
Q1.10.1 D	Q1.08.1 *	Q1.08.2 A	Q1.07.1 C	Q1.04.1 C	Q1.04.2 B
Q1.01.1 A	Q1.00.1 A	Q1.98.1 B	Q1.97.1 C	Q1.96.1 *	Q1.96.2 A
Q1.96.3 C	Q1.95.1 A	Q1.94.1 D			
MECHANICAL ENGINEERING					
Q1.25.1 C	Q1.24.1 D	Q1.24.2 A	Q1.22.1 A	Q1.22.2 A,B	Q1.22.3 2
Q1.22.4 B,C	Q1.21.1 B	Q1.21.2 D	Q1.21.3 B	Q1.20.1 A	Q1.20.2 B

Q1.20.3 1	Q1.19.1 C	Q1.19.2 D	Q1.18.1 0.25	Q1.17.1 D	Q1.17.2 5
Q1.17.3 0	Q1.16.1 D	Q1.16.2 A	Q1.16.3 C	Q1.16.4 2	Q1.15.1 B
Q1.15.2 A	Q1.15.3 A	Q1.14.1 A	Q1.14.2 B	Q1.14.3 D	Q1.14.4 D
Q1.14.5 D	Q1.13.1 C	Q1.13.2 C	Q1.12.1 B	Q1.10.1 A	Q1.09.1 A
Q1.08.1 C	Q1.08.2 B	Q1.07.1 A	Q1.07.2 B	Q1.06.1 A	Q1.06.2 A
Q1.06.3 C	Q1.05.1 B	Q1.95.1 C			
CIVIL ENGINEERING					
Q1.26.1 A	Q1.26.2 B	Q1.26.3 B	Q1.26.4 89.00 to 91.00	Q1.25.1 A,C	Q1.25.2 D
Q1.25.3 B	Q1.24.1 A	Q1.24.2 10	Q1.23.1 B,C	Q1.23.2 A,B,D	Q1.23.3 A
Q1.23.4 B	Q1.22.1 A,C	Q1.22.2 A,B	Q1.22.3 32	Q1.22.4 A,B	Q1.20.1 3
Q1.20.2 D	Q1.19.1 D	Q1.18.1 C	Q1.18.2 D	Q1.17.1 A	Q1.17.2 A
Q1.17.3 C	Q1.17.4 A	Q1.16.1 D	Q1.15.1 B	Q1.15.2 D	Q1.14.1 23
Q1.13.1 16	Q1.11.1 D	Q1.10.1 B	Q1.09.1 A	Q1.08.1 B	Q1.05.1 A
Q1.05.2 D	Q1.05.3 B	Q1.04.1 A	Q1.01.1 A	Q1.00.1 B	Q1.99.1 C
Q1.99.2 B	Q1.99.3 B	Q1.98.1 C	Q1.98.2 D	Q1.97.1 A	
CHEMICAL ENGINEERING					
Q1.26.1 B	Q1.24.1 A,C	Q1.24.2 -2	Q1.23.1 4160	Q1.22.1 B	Q1.21.1 3
Q1.21.2 D	Q1.19.1 C	Q1.18.1 1	Q1.16.1 15	Q1.15.1 D	Q1.15.2 7
Q1.13.1 B	Q1.12.1 D	Q1.11.1 D	Q1.10.1 A	Q1.10.2 C	Q1.09.1 C
Q1.09.2 A	Q1.07.1 B	Q1.05.1 D	Q1.02.1 *		
PRODUCTION & INDUSTRIAL ENGINEERING					
Q1.26.1 C	Q1.25.1 A	Q1.24.1 C	Q1.23.1 *	Q1.23.2 A	Q1.23.3 1

Q1.22.1 C	Q1.22.2 0	Q1.21.1 B	Q1.19.1 B	Q1.18.1 25	Q1.17.1 A
Q1.16.1 A	Q1.11.1 D	Q1.10.1 A	Q1.10.2 C	Q1.09.1 B	Q1.08.1 A
Q1.07.1 A	Q1.07.2 B	Q1.05.1 C	Q1.95.1 B	Q1.94.1 B	Q1.94.2 A
Q1.94.3 A					

UNIT

II

Calculus

Topics

1. Wavy Curve Method
2. Functions and Graphs
3. Limits
4. Continuity
5. Differentiability
6. Application of Derivatives
7. Tangent and Normal
8. Maxima and Minima
9. Taylor Series
10. Partial Derivatives
11. Integration
12. Multiple Integrals
13. Improper Integrals

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 51 **MCQ:** 32 **MSQ:** 1 **NAT:** 18
1M: 22 **2M:** 29 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2018)

EE — Electrical Engineering

Questions: 34 **MCQ:** 23 **MSQ:** 0 **NAT:** 11
1M: 17 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 2M(2026)–4M(2018)

IN — Instrumentation Engineering

Questions: 32 **MCQ:** 23 **MSQ:** 1 **NAT:** 8
1M: 18 **2M:** 14 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2020)

CS — Computer Science & IT

Questions: 38 **MCQ:** 24 **MSQ:** 3 **NAT:** 11
1M: 29 **2M:** 9 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2017)

ME — Mechanical Engineering

Questions: 48 **MCQ:** 42 **MSQ:** 0 **NAT:** 6
1M: 28 **2M:** 20 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–6M(2010)

CE — Civil Engineering

Questions: 57 **MCQ:** 48 **MSQ:** 2 **NAT:** 7
1M: 34 **2M:** 23 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–12M(2016)

CH — Chemical Engineering

Questions: 28 **MCQ:** 24 **MSQ:** 1 **NAT:** 3
1M: 16 **2M:** 12 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2022)

PI — Production & Industrial Engineering

Questions: 26 **MCQ:** 19 **MSQ:** 0 **NAT:** 7
1M: 18 **2M:** 8 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–6M(2016)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q2.26.1 NAT 2026 2M

Consider the square region R in the X-Y plane as shown with the dark shading in the Figure. The value of $\iint_R (x^2 + y^2 - 1)\, dx\, dy$ is ____.
(rounded off to two decimal places)

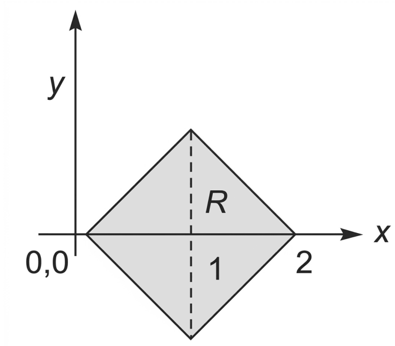


Fig. Q2.26.1

Q2.25.1 MSQ 2025 1M

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f(x) = 2x^3 - 3x^2 - 12x + 1$$

Which of the following statements is/are correct? (Here, $\mathbb{R}$ is the set of real numbers.)

- ☐ **A.** f has no global maximizer
- ☐ **B.** f has no global minimizer
- ☐ **C.** $x = -1$ is a local minimizer of f
- ☐ **D.** $x = 2$ is a local maximizer of f

Q2.23.1 MCQ 2023 2M

Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $v_1 = \alpha v_2 + e$, which minimizes the length of the error vector e, is

A. $\frac{7}{2}$ **B.** $\frac{-2}{7}$ **C.** $\frac{2}{7}$ **D.** $\frac{-7}{2}$

Q2.22.1 MCQ 2022 2M

The function $f(x) = 8\log_e x - x^2 + 3$ attains its minimum over the interval $[1, e]$ at $x =$ ______. (Here $\log_e x$ is the natural logarithm of x.)

A. 2 **B.** 1

C. e **D.** $\frac{1+e}{2}$

Q2.22.2 NAT 2022 2M

The value of the integral $\iint_D 3(x^2+y^2)dxdy$, where D is the shaded triangular region shown in the diagram, is ______ (rounded off to the nearest integer).

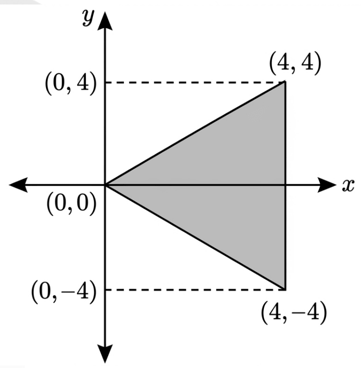


Fig. Q2.22.2

Q2.20.1 NAT 2020 2M

For the solid S shown below, the value of $\iiint_S x\,dxdydz$ (rounded off to two decimal places) is ______

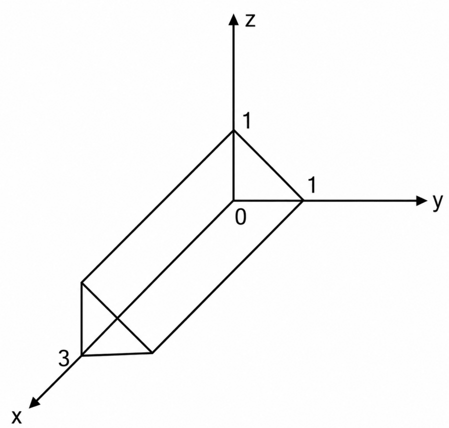


Fig. Q2.20.1

Q2.20.2 MCQ 2020 1M

The partial derivative of the following function $f(x,y,z) = e^{1-x\cos y} + xze^{-\frac{1}{(1+y^2)}}$ With respect to x at the point $(1,0,e)$ is

A. 1

B. 0

C. $\frac{1}{e}$

D. -1

Q2.19.1 NAT 2019 2M

The value of the integral $\int_0^{\pi}\int_y^{\pi} \frac{\sin x}{x} dx dy$, is equal to ______.

Q2.19.2 MCQ 2019 2M

Consider a differentiable function $f(x)$ on the set of real numbers such that $f(-1) = 0$ and $|f'(x)| \le 2$. Given these conditions, which one of the following inequalities is necessarily true for all $x \in [-2,2]$?

A. $f(x) \le \frac{1}{2}|x+1|$

B. $f(x) \le \frac{1}{2}|x|$

C. $f(x) \le 2|x+1|$

D. $f(x) \le 2|x|$

Q2.18.1 MCQ 2018 2M

Let $f(x,y) = \frac{ax^2+by^2}{xy}$, where a and b are constants. If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $x = 1$ and $y = 2$, then the relation between a and b is

A. $a = \frac{b}{4}$

B. $a = \frac{b}{2}$

C. a = 2b

D. a = 4b

Q2.18.2 NAT 2018 1M

Taylor series expansion of $f(x) = \int_0^x e^{-\left(\frac{t^2}{2}\right)} dt$ around $x = 0$ has the form $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots.$ The coefficient a_2 (correct to two decimal places) is equal to ______

Q2.18.3 NAT 2018 2M

Let $r = x^2 + y - z$ and $z^3 - xy + yz + y^3 = 1$. Assume that x and y are independent variables. At $(x,y,z) = (2,-1,1)$, the value (correct to two decimal places) of $\frac{\partial r}{\partial x}$ is ______

Q2.17.1 NAT 2017 2M

The minimum value of the function $f(x) = \frac{1}{3}x(x^2-3)$ in the interval $-100 \le x \le 100$ occurs at $x =$ ______.

Q2.17.2 MCQ 2017 2M

The values of the integrals $\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dy\right) dx$ and $\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dx\right) dy$ are

A. same and equal to 0.5

B. same and equal to -0.5

C. 0.5 and -0.5 respectively

D. -0.5 and 0.5 respectively

Q2.17.3 MCQ 2017 2M

Let $f(x) = e^{x+x^2}$ for real x. From among the following, choose the Taylor series approximation of $f(x)$ around $x = 0$, which includes all powers of x less than or equal to 3.

A. $1 + x + x^2 + x^3$

B. $1 + x + \frac{3}{2}x^2 + x^3$

C. $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$

D. $1 + x + 3x^2 + 7x^3$

Q2.17.4 NAT 2017 2M

A three dimensional region R of the finite volume is described by $x^2 + y^2 \le z^3, 0 \le z \le 1$. Where x, y, z are real. The volume of R (correct to two decimal places) is ______.

Q2.16.1 NAT 2016 1M

The integral $\frac{1}{2\pi}\iint_D (x+y+10) dx dy$ where D denotes the disc: $x^2 + y^2 \le 4$, evaluates to ______.

Q2.16.2 MCQ 2016 1M

As x varies from -1 to 3, which of the following describes the behaviour of the function $f(x) = x^3 - 3x^2 + 1$?

A. $f(x)$ increases monotonically

B. $f(x)$ increases, then decreases and increases again

C. $f(x)$ decreases, then increases and decreases again

D. $f(x)$ increases and then decreases

Q2.16.3 NAT 2016 1M

The region specified by $\{(\rho, \phi, z) : 3 \le \rho \le 5, \ \frac{\pi}{8} \le \phi \le \frac{\pi}{4}, \ 3 \le z \le 4.5\}$ in cylindrical coordinates has volume of ______.

Q2.16.4 MCQ 2016 1M

Given the following statements about a function $f : \mathbb{R} \to \mathbb{R}$, select the right option:

P: If $f(x)$ is continuous at $x = x_0$, then it is also differentiable at $x = x_0$
Q: If $f(x)$ is continuous at $x = x_0$, then it may not be differentiable at $x = x_0$
R: If $f(x)$ is differentiable at $x = x_0$, then it is also continuous at $x = x_0$

A. P is true, Q is false, R is false

B. P is false, Q is true, R is true

C. P is false, Q is true, R is false

D. P is true, Q is false, R is true

Q2.16.5 NAT 2016 2M

The integral $\int_0^1 \frac{dx}{\sqrt{(1-x)}}$ is equal to ______.

Q2.16.6 MCQ 2016 1M

How many distinct values of x satisfy the equation $\sin(x) = x/2$, where x is in radians?

A. 1 **B.** 2 **C.** 3 **D.** 4 or more

Q2.16.7 NAT 2016 1M

A triangle in the xy-plane is bounded by the straight lines $2x = 3y$, $y = 0$ and $x = 3$. The volume above the triangle and under the plane $x + y + z = 6$ is ______.

Q2.15.1 NAT 2015 2M

The maximum area (in sqare units) of a rectangle whose vertices lie on the ellipse $x^2 + 4y^2 = 1$ is ________.

Q2.15.2 NAT 2015 1M

The value of the integral $\int_{-\infty}^{\infty} 12\cos(2\pi t)\frac{\sin(4\pi t)}{4\pi t}dt$ is ________.

Q2.15.3 MCQ 2015 1M

A function $f(x) = 1 - x^2 + x^3$ is defined in the closed interval $[-1, 1]$. The value of x, in the open interval $(-1, 1)$ for which the mean value theorem is satisfied, is

A. $-\frac{1}{2}$ **B.** $-\frac{1}{3}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{2}$

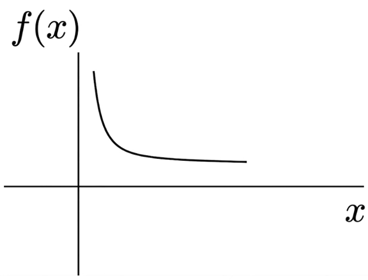
Q2.15.4 NAT 2015 1M

The value of $\sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n$ is ________.

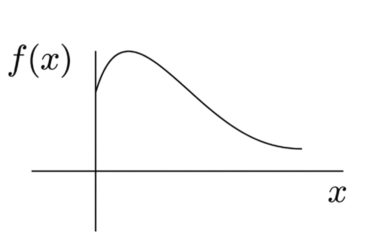
Q2.15.5 MCQ 2015 1M

Which one of the following graphs describes the following function? $f(x) = e^{-x}(x^2 + x + 1)$?

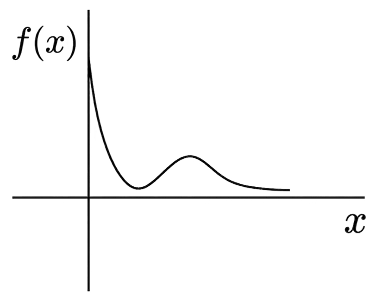
A.



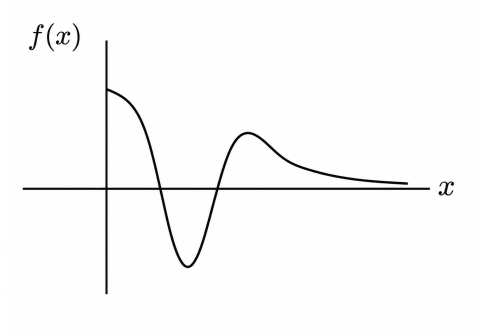
B.



C.



D.



Q2.15.6 MCQ 2015 1M

The contour on the $x-y$ plane, where the partial derivative of x^2+y^2 with respect to y is equal to the partial derivative of $6y+4x$ with respect to 'x', is

A. $y=2$

B. $x=2$

C. $x+y=4$

D. $x-y=0$

Q2.14.1 MCQ 2014 2M

The Taylor series expansion of $3\sin x + 2\cos x$ is

A. $2+3x-x^2-\dfrac{x^3}{2}+\dots$

B. $2-3x+x^2-\dfrac{x^3}{2}+\dots$

C. $2+3x+x^2+\dfrac{x^3}{2}+\dots$

D. $2-3x-x^2+\dfrac{x^3}{2}+\dots$

Q2.14.2 NAT 2014 2M

The volume under the surface $z(x,y)=x+y$ and above the triangle in the xy plane defined by $\{0\le y\le x \text{ and } 0\le x\le 12\}$ is ______.

Q2.14.3 MCQ 2014 1M

For $0\le t<\infty$, the maximum value of the function $f(t)=e^{-t}-2e^{-2t}$ occurs at

A. $t=\log_e 4$

B. $t=\log_e 2$

C. $t=0$

D. $t=\log_e 8$

Q2.14.4 MCQ 2014 1M

The value of $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$ is

A. $\ln 2$

B. 1.0

C. e

D. ∞

Q2.14.5 NAT 2014 1M

The maximum value of the function $f(x)=\ln(1+x)-x$ (where $x>-1$) occurs at $x=$ ______.

Q2.14.6 MCQ 2014 1M

The series $\sum_{n=0}^{\infty}\frac{1}{n!}$ converges to

A. $2\ln 2$

B. $\sqrt{2}$

C. 2

D. e

Q2.14.7 NAT 2014 2M

The maximum value of $f(x)=2x^3-9x^2+12x-3$ in the interval $0\le x\le 3$ is ______.

Q2.14.8 MCQ 2014 2M

For a right angled triangle, if the sum of the lengths of the hypotenuse and a side is kept constant, in order to have maximum area of the triangle, the angle between the hypotenuse and the side is

A. 12°

B. 36°

C. 60°

D. 45°

Q2.14.9 MCQ 2014 1M

If $z=xy\ln(xy)$, then

A. $x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=0$

B. $y\dfrac{\partial z}{\partial x}=x\dfrac{\partial z}{\partial y}$

C. $x\dfrac{\partial z}{\partial x}=y\dfrac{\partial z}{\partial y}$

D. $y\dfrac{\partial z}{\partial x}+x\dfrac{\partial z}{\partial y}=0$

Q2.12.1 MCQ 2012 2M

The maximum value of $f(x)=x^3-9x^2+24x+5$ in the interval $[1,6]$ is

A. 21

B. 25

C. 41

D. 46

Q2.10.1 MCQ 2010 2M

If $e^y=x^{1/x}$ then y has a

A. maximum at $x=e$

B. minimum at $x=e$

C. maximum at $x = e^{-1}$

D. minimum at $x = e^{-1}$

Q2.09.1 MCQ 2009 2M ○ ○ ○

The Taylor series expansion of $\dfrac{\sin x}{x-\pi}$ at $x = \pi$ is given by

A. $1 + \dfrac{(x-\pi)^2}{3!} + \dots$

B. $-1 - \dfrac{(x-\pi)^2}{3!} + \dots$

C. $1 - \dfrac{(x-\pi)^2}{3!} + \dots$

D. $-1 + \dfrac{(x-\pi)^2}{3!} + \dots$

Q2.08.1 MCQ 2008 1M ○ ○ ○

For real values of x, the minimum value of function $f(x) = e^x + e^{-x}$ is

A. 2 **B.** 1 **C.** 0.5 **D.** 0

Q2.08.2 MCQ 2008 1M ○ ○ ○

Which of the following function would have only odd powers of x in its Taylor series expansion about the point $x = 0$?

A. $\sin(x^3)$ **B.** $\sin(x^2)$

C. $\cos(x^3)$ **D.** $\cos(x^2)$

Q2.08.3 MCQ 2008 2M ○ ○ ○

In the Taylor series expansion of $e^x + \sin x$ about the point $x = \pi$, the coefficient of $(x-\pi)^2$ is

A. e^π **B.** $0.5e^\pi$

C. $e^\pi + 1$ **D.** $e^\pi - 1$

Q2.08.4 MCQ 2008 2M ○ ○ ○

The value of the integral of the function $g(x, y) = 4x^3 + 10y^4$ along the straight line segment from the point $(0, 0)$ to the point $(1, 2)$ in the xy-plane is

A. 33 **B.** 35 **C.** 40 **D.** 56

Q2.07.1 MCQ 2007 1M ○ ○ ○

$\lim_{\theta \to 0} \dfrac{\sin(\theta/2)}{\theta}$ is

A. 0.5 **B.** 1

C. 2 **D.** *not defined*

Q2.07.2 MCQ 2007 2M ○ ○ ○

The following plot shows a function y which varies linearly with x. The value of the integral $I = \int_1^2 y\,dx$ is

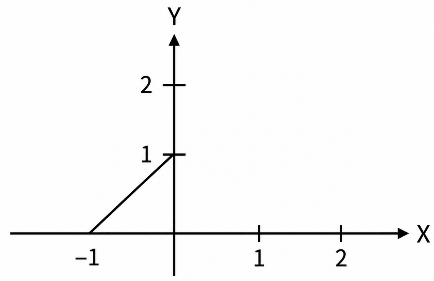


Fig. Q2.07.2

A. 1 **B.** 2.5

C. 4 **D.** 5

Q2.07.3 MCQ 2007 2M ○ ○ ○

For the function e^{-x}, the linear approximation around $x = 2$ is

A. $(3-x)e^{-2}$

B. $1 - x$

C. $\left[3 + 2\sqrt{2} - (1+\sqrt{2})x\right] e^{-2}$

D. e^{-2}

Q2.07.4 MCQ 2007 2M ○ ○ ○

For $|x| \ll 1$, $\coth(x)$ can be approximated as

A. x **B.** x^2

C. $\dfrac{1}{x}$ **D.** $\dfrac{1}{x^2}$

Q2.07.5 MCQ 2007 2M

Consider the function $f(x) = x^2 - x - 2$. The maximum value of $f(x)$ in the closed interval $[-4, 4]$ is

A. 18 **B.** 10

C. -2.25 **D.** indeterminate

Q2.05.1 MCQ 2005 2M

The value of the integral $I = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-x^2/8}\, dx$ is ________.

A. 1 **B.** π **C.** 2 **D.** 2π

Electrical Engineering

Q2.26.1 NAT 2026 2M

The integral $\frac{1}{\pi} \int_0^\infty \frac{x^{2026}}{(1+x^{2026})(1+x^2)}\, dx$ evaluates to ______.
(Round off to two decimal places)

Q2.22.1 MCQ 2022 2M

Let $f(x) = \int_0^x e^t (t-1)(t-2)dt$. Then $f(x)$ decreases in the interval

A. $x \in (1, 2)$ **B.** $x \in (2, 3)$

C. $x \in (0, 1)$ **D.** $x \in (0.5, 1)$

Q2.21.1 NAT 2021 1M

Suppose the circle $x^2 + y^2 = 1$ and $(x-1)^2 + (y-1)^2 = r^2$ intersect each other orthogonally at the point (u, v). Then $u + v =$ ________.

Q2.21.2 MCQ 2021 1M

Let $f(x)$ be a real-valued function such that $f'(x_0) = 0$ for some $x_0 \in (0, 1)$, and $f''(x) > 0$ for all $x \in (0, 1)$. Then $f(x)$ has

A. exactly one local minimum in $(0, 1)$

B. one local maximum in $(0, 1)$

C. no local minimum in $(0, 1)$

D. two distinct local minimum in $(0, 1)$

Q2.20.1 MCQ 2020 1M

Which of the following is true for all possible non-zero choices of integers m, n; $m \neq n$, or all possible non-zero choices of real numbers p, q; $p \neq q$ as applicable?

A. $\frac{1}{\pi} \int_0^\pi \sin m\theta \sin(n\theta) d\theta = 0$

B. $\frac{1}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin(p\theta) \sin(q\theta) d\theta = 0$

C. $\lim_{\alpha\to\infty} \frac{1}{2\alpha} \int_{-\alpha}^{\alpha} \sin p\theta \sin q\theta d\theta = 0$

D. $\frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(p\theta) \cos(q\theta) d\theta = 0$

Q2.20.2 MCQ 2020 2M

For real number x and y with $y = 3x^2+3x+1$, the maximum and minimum value of y for $x \in [-2, 0]$ are respectively

A. 1 and $\frac{1}{4}$ **B.** -2 and $\frac{-1}{2}$

C. 7 and $\frac{1}{4}$ **D.** 7 and 1

Q2.18.1 MCQ 2018 1M

Let f be a real-valued function of a real variable defined as $f(x) = x^2$ for $x \geq 0$, and $f(x) = -x^2$ for $x < 0$. Which one of the following statements is true?

A. $f(x)$ is discontinuous at $x = 0$

B. $f(x)$ is continuous but not differentiable at $x = 0$

C. $f(x)$ is differentiable but its first derivative is not continuous at $x = 0$

D. $f(x)$ is differentiable but its first derivative is not differentiable at $x = 0$

Q2.18.2 NAT 2018 2M

Let $f(x) = 3x^3 - 7x^2 + 5x + 6$. The maximum value of $f(x)$ over the interval $[0, 2]$ is ________ (up to 1 decimal place).

Q2.18.3 NAT 2018 1M

Let f be a real-valued function of a real variable defined as $f(x) = x - [x]$, where $[x]$ denotes the largest integer less than or equal to x. The value of $\int_{0.25}^{1.25} f(x)dx$ is ________ (up to 2 decimal places).

Q2.17.1 NAT 2017 1M

Let $I = c\iint_R xy^2 dxdy$, where R is the region shown in the figure and $c = 6 \times 10^{-4}$. The value of I equals ________. (Give the answer up to two decimal places)

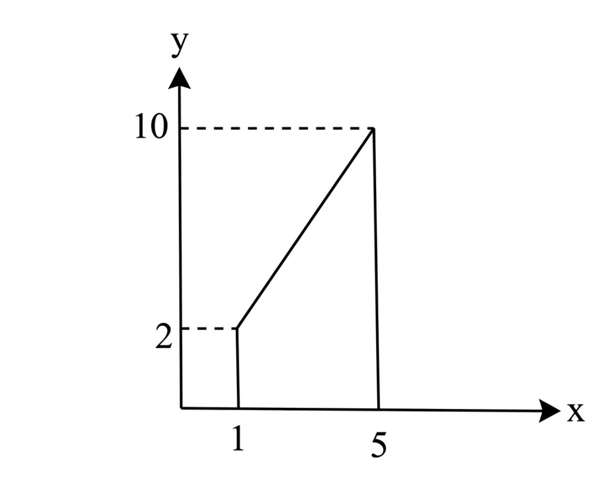


Fig. Q2.17.1

Q2.17.2 MCQ 2017 2M

A function $f(x)$ is defined as

$$f(x) = \begin{cases} e^x & ,x < 1 \\ \ln x + ax^2 + bx & ,x \geq 1 \end{cases} \quad \text{where } x \in \mathbb{R}.$$

Which one of the following statements is TRUE?

A. $f(x)$ is NOT differentiable at $x = 1$ for any values of a and b.

B. $f(x)$ is differentiable at $x = 1$ for the unique values of a and b.

C. $f(x)$ is differentiable at $x = 1$ for all values of a and b such that $a + b = e$.

D. $f(x)$ is differentiable at $x = 1$ for all values of a and b.

Q2.17.3 NAT 2017 1M

Let $y^2 - 2y + 1 = x$ and $\sqrt{x} + y = 5$. The value of $x + \sqrt{y}$ equals ________. (Given the answer up to three decimal places)

Q2.17.4 NAT 2017 1M

Consider a function $f(x, y, z)$ given by $f(x, y, z) = (x^2 + y^2 - 2z^2)(y^2 + z^2)$. The partial derivative of this function with respect to x at the point $x = 2, y = 1$ and $z = 3$ is ________.

Q2.17.5 NAT 2017 1M

Let x and y be integers satisfying the following equations

$$2x^2 + y^2 = 34$$
$$x + 2y = 11$$

The value of $(x + y)$ is ________.

Q2.17.6 MCQ 2017 2M

Let $g(x) = \begin{cases} -x, & x \leq 1 \\ x + 1 & x \geq 1 \end{cases}$ and $f(x) = \begin{cases} 1 - x, & x \leq 0 \\ x^2, & x > 0 \end{cases}$.

Consider the composition of f and g, i.e., $(f \circ g)(x) = f(g(x))$. The number of discontinuities in $(f \circ g)(x)$ present in the interval $(-\infty, 0)$ is

A. 0 **B.** 1

C. 2 **D.** 4

Q2.16.1 NAT 2016 2M

Let $S = \sum_{n=0}^{\infty} n\alpha^n$ where $|\alpha| < 1$. The value of α in the range $0 < \alpha < 1$, such that $S = 2\alpha$ is ________.

Q2.16.2 MCQ 2016 2M

The value of the integral $2\int_{-\infty}^{\infty} \left(\dfrac{\sin 2\pi t}{\pi t}\right) dt$ is equal to

A. 0 **B.** 0.5

C. 1 **D.** 2

Q2.16.3 NAT 2016 1M

The maximum value attained by the function $f(x) = x(x-1)(x-2)$ in the interval $[1, 2]$ is ________.

Q2.14.1 MCQ 2014 2M

Let $f(x) = xe^{-x}$. The maximum value of the function in the interval $(0, \infty)$ is

A. e^{-1} **B.** e

C. $1 - e^{-1}$ **D.** $1 + e^{-1}$

Q2.14.2 MCQ 2014 1M

Minimum of the real valued function $f(x) = (x-1)^{2/3}$ occurs at x equal to

A. $-\infty$ **B.** 0 **C.** 1 **D.** ∞

Q2.14.3 MCQ 2014 2M

To evaluate the double integral $\int_0^8 \left(\int_{y/2}^{(y/2)+1} \left(\frac{2x-y}{2}\right) dx\right) dy$, we make the substitution $u = \left(\frac{2x-y}{2}\right)$ and $v = \frac{y}{2}$. The integral will reduce to

A. $\int_0^4 \left(\int_0^2 2u du\right) dv$

B. $\int_0^4 \left(\int_0^1 2u du\right) dv$

C. $\int_0^4 \left(\int_0^1 u du\right) dv$

D. $\int_0^4 \left(\int_0^2 u du\right) dv$

Q2.14.4 MCQ 2014 2M

The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is

A. 20 **B.** 28 **C.** 16 **D.** 32

Q2.14.5 NAT 2014 1M

A particle, starting from origin at $t = 0$ s, is traveling along $x-$ axis with velocity

$$v = \frac{\pi}{2}\cos\left(\frac{\pi}{2}t\right) \text{ m/s}$$

At $t = 3$ s, the difference between the distance covered by the particle and the magnitude of displacement from the origin is _________.

Q2.13.1 MCQ 2013 2M

A function $y = 5x^2 + 10x$ is defined over an open interval $x = (1, 2)$. At least at one point in this interval, $\frac{dy}{dx}$ is exactly

A. 20 **B.** 25 **C.** 30 **D.** 35

Q2.11.1 MCQ 2011 1M

Roots of the algebraic equation $x^3 + x^2 + x + 1 = 0$ are

A. $(1, j, -j)$ **B.** $(1, -1, 1)$

C. $(0, 0, 0)$ **D.** $(-1, j, -j)$

Q2.11.2 MCQ 2011 2M

The function $f(x) = 2x - x^2 + 3$ has

A. a maxima at $x = 1$ and a minima at $x = 5$

B. a maxima at $x = 1$ and a minima at $x = -5$

C. only a maxima at $x = 1$

D. only a minima at $x = 1$

Q2.10.1 MCQ 2010 2M

At $t = 0$, the function $f(t) = \frac{\sin t}{t}$ has

A. a minimum **B.** a discontinuity

C. a point of inflection **D.** a maximum

Q2.10.2 MCQ 2010 1M

The value of the quantity, where

$$P = \int_0^1 xe^x \, dx$$

is

A. 0 **B.** 1 **C.** e **D.** 1/e

Q2.09.1 MCQ 2009 2M

If $f(x, y)$ is continuous function defined over $(x, y) \in [0, 1] \times [0, 1]$. Given two constraints, $x > y^2$ and $y > x^2$, the volume under $f(x, y)$ is

A. $\int_{y=0}^{y=1} \int_{x=y^2}^{x=\sqrt{y}} f(x, y) \, dx \, dy$

B. $\int_{y=x^2}^{y=1} \int_{x=y^2}^{x=1} f(x, y) \, dx \, dy$

C. $\int_{y=0}^{y=1} \int_{x=0}^{x=1} f(x, y) \, dx \, dy$

D. $\int_{x=0}^{y=\sqrt{x}} \int_{x=0}^{x=\sqrt{y}} f(x, y) \, dx \, dy$

Q2.07.1 MCQ 2007 1M

Consider the function $f(x) = (x^2 - 4)^2$ where x is a real number. Then the function has

A. Only one minimum **B.** Only two minima

C. Three minima **D.** Three maxima

Q2.07.2 MCQ 2007 2M

The integral $\frac{1}{2\pi}\int_0^{2\pi} \sin(t-\tau)\cos\tau\, d\tau$ equals

A. $\sin t \cos t$ **B.** 0

C. $\frac{1}{2}\cos t$ **D.** $\frac{1}{2}\sin t$

Q2.06.1 MCQ 2006 1M

The expression $V = \int_0^H \pi R^2 \left(1 - \frac{h}{H}\right)^2 dh$ for the volume of a cone is equal to ______.

A. $\int_0^R \pi R^2 \left(1 - \frac{h}{H}\right)^2 dr$ **B.** $\int_0^R \pi R^2 \left(1 - \frac{h}{H}\right)^2 dh$

C. $\int_0^H 2\pi r H \left(1 - \frac{r}{R}\right) dh$ **D.** $\int_0^R 2\pi r H \left(1 - \frac{r}{R}\right)^2 dr$

Q2.05.1 MCQ 2005 2M

If $S = \int_1^\infty x^{-3}\, dx$ then S has the value

A. $\frac{-1}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{2}$ **D.** 1

Q2.05.2 MCQ 2005 1M

For the function $f(x) = x^2 e^{-x}$, the maximum occurs when x is equal to

A. 2 **B.** 1 **C.** 0 **D.** −1

Instrumentation Engineering

Q2.26.1 MCQ 2026 1M

Which one among the following statements is true for the function $f(x) = x|x|$?

A. $f(x)$ is discontinuous but not differentiable at $x = 0$

B. $f(x)$ is discontinuous and differentiable at $x = 0$

C. $f(x)$ is continuous but not differentiable at $x = 0$

D. $f(x)$ is continuous and differentiable at $x = 0$

Q2.26.2 MSQ 2026 2M

Given $f(x, y) = x^2 - 2xy + y^2$

The complete contour of the equation $f(x, y) = 1$ is described by the option(s) ______.

- [ ] **A.** A line, $x - y = 1$
- [ ] **B.** A circle with radius 1
- [ ] **C.** An ellipse with length of major axis equal to 1
- [ ] **D.** A line, $x - y = -1$

Q2.25.1 MCQ 2025 2M

The value of the integral $\int_{-\pi}^{\pi} (\cos^6 x + \cos^4 x)\, dx$ is

A. $\frac{\pi}{2}$ **B.** $\frac{5\pi}{8}$ **C.** $\frac{11\pi}{8}$ **D.** $\frac{9\pi}{8}$

Q2.23.1 MCQ 2023 1M

What is $\lim_{x\to 0} f(x)$, where $f(x) = x \sin\frac{1}{x}$?

A. 0 **B.** 1 **C.** ∞ **D.** Limit does not exist

Q2.23.2 NAT 2023 2M

Consider the real-valued function $g(x) = \max\{(x-2)^2, -2x+7\}$, where $x \in (-\infty, \infty)$. The minimum value attained by $g(x)$ is ______ **(rounded off to one decimal place)**.

Q2.22.1 NAT 2022 1M

The global minimum of $x^3 e^{-|x|}$ for $x \in (-\infty, \infty)$ occurs at $x =$ ______ (round off to one decimal place)

Q2.21.1 NAT 2021 1M

Consider the function $f(x) = -x^2 + 10x + 100$. The minimum value of the function in the interval $[5, 10]$ is ______.

Q2.21.2 MCQ 2021 1M

Consider the sequence $x_n = 0.5x_{n-1} + 1, n = 1, 2, \ldots$ with $x_0 = 0$. Then $\lim_{n\to\infty} x_n$ is

A. 2 **B.** ∞ **C.** 1 **D.** 0

Q2.20.1 MCQ 2020 2M

A straight line drawn on an x-y plane intercepts the x-axis at -0.5 and the y-axis at 1. The equation that describes this line is _____.

A. $y = 2x + 1$ **B.** $y = x - 0.5$

C. $y = -0.5x + 1$ **D.** $y = 0.5x - 1$

Q2.20.2 NAT 2020 2M

Consider the function $f(x, y) = x^2 + y^2$. The minimum value the function attains on the line $x + y = 1$ (rounded off to one decimal place) is _____.

Q2.19.1 MCQ 2019 2M

The curve $y = f(x)$ is such that the tangent to the curve at every point (x, y) has a Y-axis intercept c, given by $c = -y$. then, $f(x)$ is proportional to

A. x^{-1} **B.** x^2 **C.** x^3 **D.** x^4

Q2.18.1 NAT 2018 1M

Consider two functions $f(x) = (x-2)^2$ and $g(x) = 2x-1$, where x is real. The smaller value of x for which $f(x) = g(x)$ is _____.

Q2.18.2 MCQ 2018 2M

Consider the following equations

$$\frac{\partial V(x,y)}{\partial x} = px^2 + y^2 + 2xy$$

$$\frac{\partial V(x,y)}{\partial y} = x^2 + qy^2 + 2xy$$

where p and q are constants, $V(x, y)$ that satisfies the above equations is

A. $p\frac{x^3}{3} + q\frac{y^3}{3} + 2xy + 6$

B. $p\frac{x^3}{3} + q\frac{y^3}{3} + 5$

C. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2 + xy$

D. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2$

Q2.18.3 MCQ 2018 1M

For $0 \le x \le 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.

A. $\left(0, \frac{\pi}{2}\right)$ **B.** $\left(\frac{\pi}{2}, \pi\right)$

C. $\left(\pi, \frac{3\pi}{2}\right)$ **D.** $\left(\frac{3\pi}{2}, 2\pi\right)$

Q2.16.1 NAT 2016 1M

A straight line of the form $y = mx + c$ passes through the origin and the point $(x, y) = (2, 6)$. The value of m is _____.

Q2.16.2 NAT 2016 1M

$\lim_{n\to\infty}\left(\sqrt{n^2+n} - \sqrt{n^2+1}\right)$ is _____.

Q2.16.3 NAT 2016 2M

Let $f : [-1, 1] \to \mathbb{R}$, where $f(x) = 2x^3 - x^4 - 10$. The minimum value of $f(x)$ is _____.

Q2.15.1 MCQ 2015 1M

The double integral $\int_0^a \int_0^y f(x, y)\, dx\, dy$ is equivalent to

A. $\int_0^x \int_0^y f(x, y)\, dx\, dy$

B. $\int_0^a \int_x^y f(x, y)\, dx\, dy$

C. $\int_0^a \int_x^a f(x, y)\, dy\, dx$

D. $\int_0^a \int_0^a f(x, y)\, dx\, dy$

Q2.14.1 MCQ 2014 1M

Given

$$x(t) = 3\sin(1000\pi t) \quad \text{and}$$
$$y(t) = 5\cos\left(1000\pi t + \frac{\pi}{4}\right)$$

The x-yplot will be

A. a circle

B. a mulit-loop closed curve

C. a hyperbola

D. an ellipse

Q2.13.1 MCQ 2013 1M

The type of the partial differential equation $\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$ is

A. Parabolic **B.** Elliptic

C. Hyperbolic **D.** Nonlinear

Q2.11.1 MCQ 2011 2M

The series $\sum_{m=0}^{\infty} \frac{1}{4^m}(x-1)^{2m}$ converges for

A. $-2 < x < 2$ **B.** $-1 < x < 3$

C. $-3 < x < 1$ **D.** $x < 3$

Q2.08.1 MCQ 2008 2M

The expression $e^{-\ln x}$ for $x > 0$ is equal to

A. $-x$ **B.** x **C.** x^{-1} **D.** $-x^{-1}$

Q2.08.2 MCQ 2008 2M

Consider the function $y = x^2 - 6x + 9$. The maximum value of y obtained when x varies over the interval 2 to 5 is

A. 1 **B.** 3 **C.** 4 **D.** 9

Q2.08.3 MCQ 2008 1M

Given $y = x^2 + 2x + 10$ the value of $\left.\frac{dy}{dx}\right|_{x=1}$ is equal to

A. 0 **B.** 4

C. 12 **D.** 13

Q2.08.4 MCQ 2008 1M

$\lim_{x\to 0} \frac{\sin x}{x}$ is

A. indeterminate **B.** 0

C. 1 **D.** ∞

Q2.07.1 MCQ 2007 2M

The value of $\int_0^{\infty}\int_0^{\infty} e^{-x^2} e^{-y^2}\, dx\, dy$ is

A. $\frac{\sqrt{\pi}}{2}$ **B.** $\sqrt{\pi}$ **C.** π **D.** $\frac{\pi}{4}$

Q2.07.2 MCQ 2007 2M

For real x, the maximum value of $\frac{e^{\sin x}}{e^{\cos x}}$ is

A. 1 **B.** e **C.** $e^{\sqrt{2}}$ **D.** ∞

Q2.07.3 MCQ 2007 2M

Consider the function $f(x) = |x|^3$, where x is real. Then the function $f(x)$ at $x = 0$ is

A. continuous but not differentiable

B. once differentiable but not twice.

C. twice differentiable but not thrice.

D. thrice differentiable

Q2.05.1 MCQ 2005 1M

$f = a_0x^n + a_1x^{n-1}y + \cdots + a_{n-1}xy^{n-1} + a_ny^n$ where a_i ($i = 0$ to n) are constants then $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}$ is

A. $\frac{f}{n}$ **B.** $\frac{n}{f}$

C. nf **D.** $n\sqrt{f}$

Q2.05.2 MCQ 2005 1M

The value of the integral $\int_{-1}^{1} \frac{1}{x^2}\, dx$ is

A. 2 **B.** does not exists

C. -2 **D.** ∞

Q2.01.1 MCQ 2001 1M

$$\lim_{x\to\frac{\pi}{4}} \frac{\sin 2\left(x - \frac{\pi}{4}\right)}{x - \frac{\pi}{4}} = \_\_\_\_$$

A. 0 **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Q2.99.1 MCQ 1999 1M

$$\lim_{x\to 0} \frac{1}{10}\frac{1 - e^{-j5x}}{1 - e^{-jx}} = \_\_\_\_$$

A. 0 **B.** 1.1 **C.** 0.5 **D.** 1

Computer Science & IT

Q2.26.1 MSQ 2026 2M

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as follows:

$$f(x) = \left(\frac{|x|}{2} - x\right)\left(x - \frac{|x|}{2}\right)$$

Which of the following statements is/are true?

- [ ] **A.** f has a local maximum
- [ ] **B.** f has a local minimum
- [ ] **C.** f' is continuous over $\mathbb{R}$
- [ ] **D.** f' is not differentiable over $\mathbb{R}$

Q2.26.2 MSQ 2026 1M

For a real number a, let $I(a) = \int_{-1}^{1}(3x^2 - ax + 1)\,dx$. Which of the following statements is/are true?

- [ ] **A.** The value of $I(a)$ is independent of the value of a
- [ ] **B.** The value of $I(a)$ can vary with the value of a
- [ ] **C.** There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a positive real number
- [ ] **D.** There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a negative real number

Q2.25.1 MCQ 2025 1M

The value of x such that $x > 1$, satisfying the equation $\int_1^x t \ln t \, dt = \frac{1}{4}$ is

A. $\sqrt{e}$ **B.** e **C.** e^2 **D.** $e - 1$

Q2.24.1 MCQ 2024 1M

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

A. $\{-1, 1, 2\}$ **B.** $\{-2, -1, 1\}$ **C.** $\{0, 1\}$ **D.** $\{-1, 0, 1\}$

Q2.24.2 MCQ 2024 1M

Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x)\,dx$?

A. 0 **B.** 1 **C.** 2 **D.** -1

Q2.23.1 MSQ 2023 1M

Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

- [ ] **A.** $f(x)$ does not have a local maximum.
- [ ] **B.** $f(x)$ has a local maximum.
- [ ] **C.** $f(x)$ does not have a local minimum.
- [ ] **D.** $f(x)$ has a local minimum.

Q2.23.2 NAT 2023 1M

The value of the definite integral

$$\int_{-3}^{3}\int_{-2}^{2}\int_{-1}^{1}(4x^2y - z^3)\,dz\,dy\,dx$$

is ________. (Rounded off to the nearest integer)

Q2.22.1 NAT 2022 1M

The value of the following limit is __________.

$$\lim_{x \to 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$$

Q2.21.1 NAT 2021 1M

Consider the following expression.

$$\lim_{x \to -3} \frac{\sqrt{2x + 22} - 4}{x + 3}$$

The value of the above expression (rounded to 2 decimal places) is __________.

Q2.21.2 NAT 2021 1M

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous function on the interval $[-3, 3]$ and a differentiable function in the interval $(-3, 3)$ such that for every x in the interval, $f'(x) \le 2$. If $f(-3) = 7$, then $f(3)$ is at most __________.

Q2.20.1 MCQ 2020 1M

Consider the functions

I. e^{-x}

II. $x^2 - \sin x$

III. $\sqrt{x^3 + 1}$

Which of the above functions is/are increasing everywhere in $[0, 1]$?

A. III only

B. II only

C. II and III only

D. I and III only

Q2.19.1 MCQ 2019 1M

Compute $\lim_{x\to3}\frac{x^4-81}{2x^2-5x-3}$

A. 1 **B.** 53/12

C. 108/7 **D.** Limit does not exist

Q2.18.1 MCQ 2018 1M

Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n+3$ for all $n = 0, 1, 2, \dots$?

A. $\frac{3}{(1-x)^2}$ **B.** $\frac{3x}{(1-x)^2}$ **C.** $\frac{2-x}{(1-x)^2}$ **D.** $\frac{3-x}{(1-x)^2}$

Q2.18.2 NAT 2018 1M

The value of $\int_0^{\pi/4} x\cos(x^2)\,dx$ correct to three decimal places (assuming that $\pi = 3.14$) is ________.

Q2.17.1 MCQ 2017 2M

The value of $\lim_{x\to1}\frac{x^7-2x^5+1}{x^3-3x^2+2}$

A. is 0 **B.** is -1

C. is 1 **D.** does not exist

Q2.17.2 MCQ 2017 2M

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to

A. the maximum of x and y

B. the minimum of x and y

C. 1

D. none of the above

Q2.17.3 MCQ 2017 1M

If $f(x) = R\sin\left(\frac{\pi x}{2}\right) + S$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2R}{\pi}$, then the constants R and S are, respectively

A. $\frac{2}{\pi}$ and $\frac{16}{\pi}$ **B.** $\frac{2}{\pi}$ and 0

C. $\frac{4}{\pi}$ and 0 **D.** $\frac{4}{\pi}$ and $\frac{16}{\pi}$

Q2.17.4 NAT 2017 1M

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b. The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ ________.

Q2.17.5 MCQ 2017 2M

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q2.16.1 NAT 2016 1M

$\lim_{x\to4}\frac{\sin(x-4)}{x-4} =$ ________.

Q2.16.2 NAT 2016 1M

Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x)+f(-x))$ is 10, then the degree of $(g(x)-g(-x))$ is ________.

Q2.15.1 MCQ 2015 1M

If $g(x) = 1-x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is:

A. $\frac{h(x)}{g(x)}$ **B.** $\frac{-1}{x}$ **C.** $\frac{g(x)}{h(x)}$ **D.** $\frac{x}{(1-x)^2}$

Q2.15.2 MCQ 2015 1M

$\lim_{x\to\infty} x^{1/x}$ is

A. ∞ **B.** 0 **C.** 1 **D.** Not defined

Q2.15.3 NAT 2015 2M

$\sum_{x=1}^{99}\frac{1}{x(x+1)} =$ ________.

Q2.15.4 NAT 2015 2M

$\int_{1/\pi}^{2/\pi}\frac{\cos(1/x)}{x^2}dx =$ ________.

Q2.15.5 MCQ 2015 1M

The value of $\lim_{x\to\infty}(1+x^2)^{e^{-x}}$ is

A. 0 **B.** $\frac{1}{2}$ **C.** 1 **D.** ∞

Q2.14.1 NAT 2014 1M

If $\int_0^{2\pi} |x\sin x| dx = k\pi$, then the value of k is equal to ________.

Q2.14.2 MCQ 2014 2M

The value of the integral given below is

$$\int_0^{\pi} x^2 \cos x dx$$

A. -2π **B.** π **C.** $-\pi$ **D.** 2π

Q2.14.3 MCQ 2014 1M

A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1, x = 2$ and $x = 3$. Which one of the following must be TRUE?

A. $f(0)f(4) < 0$ **B.** $f(0)f(4) > 0$

C. $f(0) + f(4) > 0$ **D.** $f(0) + f(4) < 0$

Q2.13.1 MCQ 2013 1M

Which one of the following functions is continuous at $x = 3$?

A. $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x-1, & \text{if } x > 3 \\ \frac{x+3}{3}, & \text{if } x < 3 \end{cases}$

B. $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8-x, & \text{if } x \neq 3 \end{cases}$

C. $f(x) = \begin{cases} x+3, & \text{if } x \leq 3 \\ x-4, & \text{if } x > 3 \end{cases}$

D. $f(x) = \frac{1}{x^3-27}$, if $x \neq 3$

Q2.12.1 MCQ 2012 1M

Consider the function $f(x) = \sin(x)$ in the interval $x \in [\pi/4, 7\pi/4]$. The number and location(s) of the local minima of this function are

A. One, at $\pi/2$

B. One, at $3\pi/2$

C. Two, at $\pi/2$ and $3\pi/2$

D. Two, at $\pi/4$ and $3\pi/2$

Q2.11.1 MCQ 2011 2M

Given $i = \sqrt{-1}$, what will be the evaluation of the definite integral $\int_0^{\pi/2} \frac{\cos x + i\sin x}{\cos x - i\sin x} dx$?

A. 0 **B.** 2 **C.** $-i$ **D.** i

Q2.10.1 MCQ 2010 1M

What is the value of $\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^{2n}$?

A. 0 **B.** e^{-2} **C.** $e^{-1/2}$ **D.** 1

Q2.09.1 MCQ 2009 1M

$\int_0^{\pi/4}(1-\tan x)/(1+\tan x)dx$ evaluates to

A. 0 **B.** 1 **C.** $\ln 2$ **D.** $\frac{1}{2}\ln 2$

Q2.05.1 MCQ 2005 1M

$\lim_{x\to 0} \frac{\sin^2 x}{x}$ is

A. 0 **B.** ∞ **C.** 1 **D.** Not defined

Q2.05.2 MCQ 2005 2M

The value of the integral $\int_0^{\pi} \theta\sin^3\theta d\theta$ is

A. $\frac{2\pi}{3}$ **B.** $\frac{\pi}{3}$ **C.** $\frac{\pi}{6}$ **D.** $\frac{\pi}{2}$

Q2.04.1 MCQ 2004 1M

The following integral $\int_1^e x\ln x dx$ is equal to

A. $\frac{e^2+1}{4}$ **B.** $\frac{e^2-1}{4}$ **C.** $\frac{e^2+1}{2}$ **D.** $\frac{e^2-1}{2}$

Q2.03.1 MCQ 2003 1M

The value of $\lim_{x\to 0}\left(\frac{-\sin x}{x}\right)$ is

A. 1 **B.** -1 **C.** 0 **D.** ∞

Mechanical Engineering

Q2.26.1 MCQ 2026 1M

Domain A is bounded by curve $x^2 = 4y$, ordinate $x = 2$ and x axis. The value of $\iint_A y\,dx\,dy$ is

A. $\frac{1}{5}$ **B.** $\frac{1}{3}$

C. $\frac{5}{12}$ **D.** $\frac{1}{2}$

Q2.25.1 MCQ 2025 2M

In the closed interval $[0, 3]$, the minimum value of the function f given below is

$$f(x) = 2x^3 - 9x^2 + 12x$$

A. 0 **B.** 4 **C.** 5 **D.** 9

Q2.23.1 MCQ 2023 1M

The figure shows the plot of a function over the interval $[-4, 4]$. Which one of the options given CORRECTLY identifies the function?

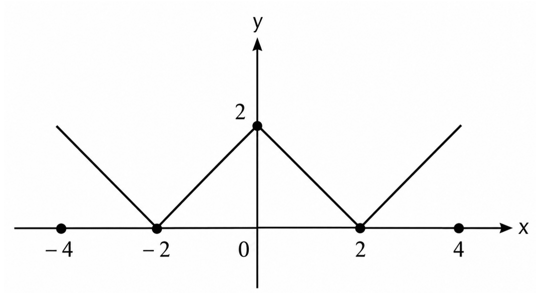


Fig. Q2.23.1

A. |2 - x| **B.** |2 - |x||

C. |2 + |x|| **D.** 2 - |x|

Q2.23.2 NAT 2023 2M

The smallest perimeter that a rectangle with area of 4 square units can have is ______ units. (Answer in integer)

Q2.22.1 MCQ 2022 1M

The limit $p = \lim_{x\to\pi}\left(\frac{x^2 + \alpha x + 2\pi^2}{x - \pi + 2\sin x}\right)$ has a finite value for a real α. The value of α and the corresponding limit p are

A. $\alpha = -3\pi$, and $p = \pi$

B. $\alpha = -2\pi$, and $p = 2\pi$

C. $\alpha = \pi$, and $p = \pi$

D. $\alpha = 2\pi$, and $p = 3\pi$

Q2.22.2 MCQ 2022 1M

Given $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$. If a,b are positive integers, the value of $\int_{-\infty}^{\infty} e^{-a(x+b)^2}dx$ is ______.

A. $\sqrt{\pi a}$ **B.** $\sqrt{\frac{\pi}{a}}$

C. $b\sqrt{\pi a}$ **D.** $b\sqrt{\frac{\pi}{a}}$

Q2.21.1 MCQ 2021 2M

The value of $\int_0^{\pi/2}\int_0^{\cos\theta} r\sin\theta\,dr\,d\theta$ is

A. π **B.** $\frac{1}{6}$ **C.** 0 **D.** $\frac{4}{3}$

Q2.21.2 MCQ 2021 1M

The value of $\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)$ is

A. 1 **B.** $\frac{1}{3}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{2}$

Q2.21.3 MCQ 2021 2M

Let $f(x) = x^2 - 2x + 2$ be a continuous function defined on $x \in [1, 3]$. The point x at which the tangent of $f(x)$ becomes parallel to the straight line joining $f(1)$ and $f(3)$ is

A. 3 **B.** 1 **C.** 0 **D.** 2

Q2.20.1 MCQ 2020 1M

Let $I = \int_{x=0}^{1}\int_{y=0}^{x^2} xy^2\,dy\,dx$ .Then here I may also be expressed as

A. $\int_{y=0}^{1}\int_{x=0}^{\sqrt{y}} yx^2\,dx\,dy$ **B.** $\int_{y=0}^{1}\int_{x=\sqrt{y}}^{1} yx^2\,dx\,dy$

C. $\int_{y=0}^{1}\int_{x=0}^{\sqrt{y}} xy^2\,dx\,dy$ **D.** $\int_{y=0}^{1}\int_{x=\sqrt{y}}^{1} xy^2\,dx\,dy$

Q2.20.2 MCQ 2020 1M

The value of $\lim_{x\to 1}\left(\frac{1-e^{-c(1-x)}}{1-xe^{-c(1-x)}}\right)$ is

A. c **B.** $\frac{c+1}{c}$

C. $c+1$ **D.** $\frac{c}{c+1}$

Q2.19.1 MCQ 2019 1M

A parabola $x = y^2$ with $0 \le x \le 1$ is shown in the figure. The volume of the solid of rotation obtained by rotating the shaded area by 360° around the x-axis is

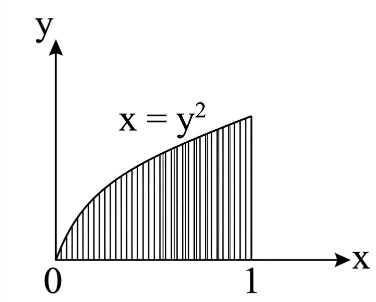


Fig. Q2.19.1

A. π **B.** $\frac{\pi}{4}$

C. 2π **D.** $\frac{\pi}{2}$

Q2.19.2 NAT 2019 2M

The value of the following definite integral $\int_1^e (x\ln x)dx$ is ______. (round off to three decimal places)

Q2.19.3 MCQ 2019 2M

The derivative of $f(x) = \cos(x)$ can be estimated using the approximation

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

The percentage error is calculated as

$$\left(\frac{\text{Exact value} - \text{approximate value}}{\text{Exact value}}\right) \times 100.$$

The percentage error in the derivative of $f(x)$ at $x = \pi/6$ radian, choosing $h = 0.1$ radian, is

A. $> 5\%$

B. $> 0.1\%$ and $< 1\%$

C. $> 1\%$ and $< 5\%$

D. $< 0.1\%$

Q2.18.1 MCQ 2018 1M

According to Mean Value Theorem, for a continuous function $f(x)$ in the interval $[a, b]$, there exists a value ξ in this interval such that $\int_a^b f(x)dx =$

A. $f(\xi)(b-a)$ **B.** $f(b)(\xi - a)$

C. $f(a)(b-\xi)$ **D.** 0

Q2.17.1 MCQ 2017 1M

The value of $\lim_{x\to 0}\left(\frac{x^3 - \sin(x)}{x}\right)$ is

A. 0 **B.** 3 **C.** 1 **D.** -1

Q2.17.2 MCQ 2017 2M

A parametric curve defined by $x = \cos\left(\frac{\pi u}{2}\right)$ and $y = \sin\left(\frac{\pi u}{2}\right)$ in the range $0 \le u \le 1$ is rotated about the X-axis by 360 degrees. Area of the surface generated is

A. $\frac{\pi}{2}$ **B.** π **C.** 2π **D.** 4π

Q2.16.1 NAT 2016 2M

Consider the function $f(x) = 2x^3 - 3x^2$ in the domain $[-1, 2]$. The global minimum of $f(x)$ is ______.

Q2.16.2 MCQ 2016 1M

The values of x for which the function $f(x) = \dfrac{x^2 - 3x - 4}{x^2 + 3x - 4}$ is NOT continuous are

A. 4 and -1 **B.** 4 and 1

C. -4 and 1 **D.** -4 and -1

Q2.16.3 MCQ 2016 1M

$\lim_{x\to 0} \dfrac{\log_e(1+4x)}{e^{3x}-1}$ is equal to

A. 0 **B.** $\dfrac{1}{12}$ **C.** $\dfrac{4}{3}$ **D.** 1

Q2.16.4 MCQ 2016 2M

$\lim_{x\to\infty} \left(\sqrt{x^2+x-1} - x\right)$ is

A. 0 **B.** ∞ **C.** $1/2$ **D.** $-\infty$

Q2.15.1 NAT 2015 1M

The value of $\lim_{x\to 0} \left(\dfrac{-\sin x}{2\sin x + x\cos x}\right)$ is ______.

Q2.15.2 MCQ 2015 1M

The value of $\lim_{x\to 0} \dfrac{1-\cos(x^2)}{2x^4}$ is

A. 0 **B.** $\dfrac{1}{2}$ **C.** $\dfrac{1}{4}$ **D.** undefined

Q2.15.3 NAT 2015 2M

Consider a spatial curve in three-dimensional space given in parametric form by $x(t) = \cos t$, $y(t) = \sin t$, $z(t) = \dfrac{2}{\pi}t$, $0 \le t \le \dfrac{\pi}{2}$. The length of the curve is ______.

Q2.15.4 NAT 2015 1M

Consider an ant crawling along the curve $(x-2)^2 + y^2 = 4$, where x and y are in meters. The ant starts at the point $(4, 0)$ and moves counter-clockwise with a speed of 1.57 meters per second. The time taken by the ant to reach the point $(2, 2)$ is ______ (in seconds).

Q2.15.5 MCQ 2015 1M

At $x = 0$, the function $f(x) = |x|$ has

A. a minimum

B. a maximum

C. a point of inflexion

D. neither a maximum nor minimum

Q2.14.1 MCQ 2014 1M

$\lim_{x\to 0} \dfrac{x - \sin x}{1 - \cos x}$ is

A. 0 **B.** 1 **C.** 3 **D.** not defined

Q2.14.2 MCQ 2014 1M

$\lim_{x\to 0} \left(\dfrac{e^{2x}-1}{\sin(4x)}\right)$ is equal to

A. 0 **B.** 0.5 **C.** 1 **D.** 2

Q2.14.3 MCQ 2014 1M

If a function is continuous at a point,

A. the limit of the function may not exist at the point

B. The function must be derivable at the point

C. the limit of the function at the point tends to infinity

D. the limit must exist at the point and the value of limit should be same as the value of the function at the point.

Q2.14.4 MCQ 2014 1M

The value of the integral $\displaystyle\int_0^2 \frac{(x-1)^2 \sin(x-1)}{(x-1)^2 + \cos(x-1)}\,dx$ is

A. 3 **B.** 0 **C.** -1 **D.** -2

Q2.14.5 MCQ 2014 2M

The value of the integral $\displaystyle\int_0^2 \int_0^x e^{x+y}\,dy\,dx$ is

A. $\dfrac{1}{2}(e-1)$ **B.** $\dfrac{1}{2}(e^2-1)^2$

C. $\dfrac{1}{2}(e^2-e)$ **D.** $\dfrac{1}{2}\left(e - \dfrac{1}{e}\right)^2$

Q2.13.1 MCQ 2013 2M

The value of the definite integral $\int_1^e \sqrt{x}\ln(x)dx$ is

A. $\frac{4}{9}\sqrt{e^3}+\frac{2}{9}$

B. $\frac{2}{9}\sqrt{e^3}-\frac{4}{9}$

C. $\frac{2}{9}\sqrt{e^3}+\frac{4}{9}$

D. $\frac{4}{9}\sqrt{e^3}-\frac{2}{9}$

Q2.12.1 MCQ 2012 1M

At $x=0$, the function $f(x)=x^3+1$ has

A. a maximum value

B. a minimum value

C. a singularity

D. a point of inflection

Q2.12.2 MCQ 2012 2M

A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y=2x-0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

A. 8 meters

B. 10 meters

C. 12 meters

D. 14 meters

Q2.11.1 MCQ 2011 1M

If $f(x)$ is even function and a is a positive real number, then $\int_{-a}^{a} f(x)dx$ equals _____.

A. 0

B. a

C. $2a$

D. $2\int_0^a f(x)dx$

Q2.10.1 MCQ 2010 1M

The infinite series $f(x)=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+-\ldots$ Converges to

A. $\cos(x)$

B. $\sin(x)$

C. $\sin\text{ h}(x)$

D. e^x

Q2.10.2 MCQ 2010 2M

The value of the integral $\int_{-\alpha}^{\alpha}\frac{dx}{1+x^2}$

A. $-\pi$

B. $-\frac{\pi}{2}$

C. $\frac{\pi}{2}$

D. π

Q2.10.3 MCQ 2010 2M

The function $y=|2-3x|$

A. is continuous $\forall\, x\in R$ and differentiable $\forall\, x\in R$

B. is continuous $\forall\, x\in R$ and differentiable $\forall\, x\in R$ except at $x=\frac{3}{2}$

C. is continuous $\forall\, x\in R$ and differentiable $\forall\, x\in R$ except at $x=\frac{2}{3}$

D. is continuous $\forall\, x\in R$ and except at $x=3$ and differentiable $\forall\, x\in R$

Q2.10.4 MCQ 2010 1M

The parabolic arc $y=\sqrt{x}, 1\le x\le 2$ is revolved around the x-axis. The volume of the solid of revolution is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{3\pi}{4}$

D. $\frac{3\pi}{2}$

Q2.09.1 MCQ 2009 2M

The distance between the origin and the point nearest to it on the surface $z^2=1+xy$ is

A. 1

B. $\frac{\sqrt{3}}{2}$

C. $\sqrt{3}$

D. 2

Q2.08.1 MCQ 2008 2M

Consider the shaded triangular region P shown in the figure. What is $\iint_P xy\,dx\,dy$?

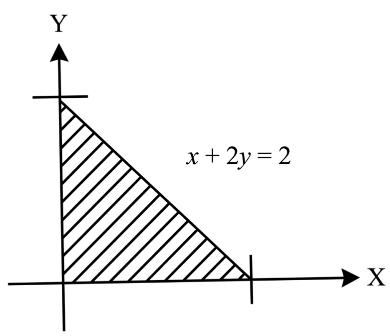


Fig. Q2.08.1

A. $\frac{1}{6}$ B. $\frac{2}{9}$ C. $\frac{7}{16}$ D. 1

Q2.08.2 MCQ 2008 1M

In the Taylor series expansion of e^x about $x = 2$, the coefficient of $(x-2)^4$ is

A. $\frac{1}{4!}$ B. $\frac{2^4}{4!}$ C. $\frac{e^2}{4!}$ D. $\frac{e^4}{4!}$

Q2.08.3 MCQ 2008 1M

The value of $\lim_{x\to 8} \frac{x^{1/3}-2}{x-8}$ is

A. $\frac{1}{16}$ B. $\frac{1}{12}$ C. $\frac{1}{8}$ D. $\frac{1}{4}$

Q2.08.4 MCQ 2008 2M

Which of the following integrals is unbounded?

A. $\int_0^{\pi/4} \tan x\, dx$ B. $\int_0^{\infty} \frac{1}{1+x^2} dx$

C. $\int_0^{\infty} x \cdot e^{-x}\, dx$ D. $\int_0^1 \frac{1}{1-x} dx$

Q2.08.5 MCQ 2008 1M

The length of the curve $y = \frac{2}{3}x^{3/2}$ between $x = 0$ & $x = 1$ is

A. 0.27 B. 0.67 C. 1 D. 1.22

Q2.08.6 MCQ 2008 2M

Let $f = y^x$. What is $\frac{\partial^2 f}{\partial x\, \partial y}$ at $x = 2, y = 1$?

A. 0 B. ln 2 C. 1 D. $\frac{1}{\ln 2}$

Q2.07.1 MCQ 2007 1M

$\lim_{x\to 0} \frac{e^x - \left(1 + x + \frac{x^2}{2}\right)}{x^3} =$ _____.

A. 0 B. $\frac{1}{6}$ C. $\frac{1}{3}$ D. 1

Q2.07.2 MCQ 2007 2M

If $y = x + \sqrt{x + \sqrt{x + \sqrt{x + \ldots \infty}}}$, then $y(2) =$ _____.

A. 4 (or) 1 B. 4 only

C. 1 only D. Undefined

Civil Engineering

Q2.26.1 MCQ 2026 2M

Let $f(x)$ be a continuous function defined in $[0,2] \to \mathbb{R}$ and satisfying the equation $\int_0^2 f(x)[x - f(x)]dx = \frac{2}{3}$. The value of $f(1)$ is

A. 1 B. 2 C. $\frac{1}{2}$ D. 0

Q2.25.1 MCQ 2025 1M

Integration of ln(x) with x i.e $\int \ln(x)\, dx =$ ________.

A. $x \cdot \ln(x) - x +$ Constant B. $x - \ln(x) +$ Constant

C. $x \cdot \ln(x) + x +$ Constant D. $\ln(x) - x +$ Constant

Q2.25.2 MCQ 2025 2M

The value of $\lim_{x\to\infty}(x - \sqrt{x^2 + x})$ is equal to

A. -1 B. -0.5 C. -2 D. 0

Q2.25.3 MSQ 2025 2M

Consider the function given below and pick one or more **CORRECT** statement(s) from the following choices. $f(x) = x^3 - \frac{15}{2}x^2 + 18x + 20$

- ☐ A. $f(x)$ has a local minimum at $x = 3$.
- ☐ B. $f(x)$ has a local maximum at $x = 3$.
- ☐ C. $f(x)$ has a local minimum at $x = 2$.
- ☐ D. $f(x)$ has a local maximum at $x = 2$.

Q2.25.4 NAT 2025 2M

The maximum value of the function $h(x) = -x^3 + 2x^2$ in the interval $[-1, 1.5]$ is equal to ______________ (rounded off to 1 decimal place).

Q2.24.1 MCQ 2024 1M

The smallest positive root of the equation

$$x^5 - 5x^4 - 10x^3 + 50x^2 + 9x - 45 = 0$$

lies in the range

A. $0 < x \le 2$ **B.** $2 < x \le 4$

C. $6 \le x \le 8$ **D.** $10 \le x \le 100$

Q2.24.2 MCQ 2024 1M

The function $f(x) = x^3 - 27x + 4$, $1 \le x \le 6$ has

A. Maxima point **B.** Minima point

C. Saddle point **D.** Inflection point

Q2.24.3 NAT 2024 2M

The expression for computing the effective interest rate (i_{eff}) using continuous compounding for a nominal interest rate of 5% is

$$i_{eff} = \lim_{m\to\infty}\left(1 + \frac{0.05}{m}\right)^m - 1$$

The effective interest rate (in percentage) is ________

Q2.23.1 MCQ 2023 1M

For the integral $I = \int_{-1}^{1} \frac{1}{x^2}\, dx$ which of the following statements is TRUE?

A. $I = 0$ **B.** $I = 2$

C. $I = -2$ **D.** The integral does not converge

Q2.23.2 MCQ 2023 2M

For the function $f(x) = e^x|\sin x|$; $x \in \mathbb{R}$, which of the following statements is/are TRUE?

A. The function is continuous at all x

B. The function is differentiable at all x

C. The function is periodic

D. The function is bounded

Q2.22.1 MSQ 2022 2M

Let $\max\{a, b\}$ denote the maximum of two real numbers a and b. Which of the following statement(s) is/are TURE about the function $f(x) = \max\{3 - x, x - 1\}$?

- [ ] **A.** It is continuous on its domain
- [ ] **B.** It has a local minimum at $x = 2$
- [ ] **C.** It has a local maximum at $x = 2$
- [ ] **D.** It is differentiable on its domain

Q2.22.2 MCQ 2022 1M

$\int \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots\right) dx$ is equal to

A. $\frac{1}{1+x}$ + Constant **B.** $\frac{1}{1+x^2}$ + Constant

C. $-\frac{1}{1-x}$ + Constant **D.** $-\frac{1}{1-x^2}$ + Constant

Q2.22.3 MCQ 2022 2M

Consider the polynomial $f(x) = x^3 - 6x^2 + 11x - 6$ on the domain S given by $1 \le x \le 3$. The first and second derivatives are $f'(x) and f''(x)$. Consider the following statements.

I. The given polynomial is zero at the boundary points $x = 1$ and $x = 3$

II. There exists one local maxima of $f(x)$ within the domain S.

III. The second derivative $f''(x) > 0$ throughout the domain S

IV. There exists one local minima of $f(x)$ within the domain S.

The correct option is:

A. Only statements I, II and III are correct

B. Only statements I, II and IV are correct

C. Only statements I and IV are correct

D. Only statements II and IV are correct

Q2.21.1 NAT 2021 1M

The volume determined from $\iiint_V 8xyz\, dV$ for $V = [2, 3] \times [1, 2] \times [0, 1]$ will be (in integer) ________.

Q2.21.2 NAT 2021 1M

Consider the limit:

$$\lim_{x\to 1}\left(\frac{1}{\ln x} - \frac{1}{x-1}\right)$$

The limit (correct up to one decimal place) is ________.

Q2.21.3 NAT 2021 2M

The value (round off to one decimal place) of $\int_{-1}^{1} xe^{|x|}\,dx$ is ________.

Q2.21.4 MCQ 2021 1M

The value of $\lim_{x\to\infty} \frac{x\ln(x)}{1+x^2}$ is

A. ∞ **B.** 0.5 **C.** 0 **D.** 1.0

Q2.20.1 MCQ 2020 1M

The value of $\lim_{x\to\infty} \frac{x^2-5x+4}{4x^2+2x}$ is

A. 1 **B.** $\frac{1}{2}$ **C.** 0 **D.** $\frac{1}{4}$

Q2.20.2 MCQ 2020 1M

The value of $\lim_{x\to\infty} \frac{\sqrt{9x^2+2020}}{x+7}$ is

A. indeterminable **B.** 1 **C.** $\frac{7}{9}$ **D.** 3

Q2.20.3 MCQ 2020 1M

The area of an ellipse represented by an equation $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ is

A. $\frac{\pi ab}{4}$

B. $\frac{4\pi ab}{3}$

C. $\frac{\pi ab}{2}$

D. πab

Q2.19.1 MCQ 2019 1M

Which one of the following is correct?

A. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = 2$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = 1$

B. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = \infty$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = 1$

C. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = 1$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = 1$

D. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = 2$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = \infty$

Q2.19.2 MCQ 2019 1M

For a small value of h, the Taylor series expansion for $f(x+h)$ is

A. $f(x)+hf'(x)+\frac{h^2}{2!}f''(x)+\frac{h^3}{3!}f'''(x)+\ldots\infty$

B. $f(x)-hf'(x)+\frac{h^2}{2}f''(x)-\frac{h^3}{3}f'''(x)+\ldots\infty$

C. $f(x)+hf'(x)+\frac{h^2}{2}f''(x)+\frac{h^3}{3}f'''(x)+\ldots\infty$

D. $f(x)-hf'(x)+\frac{h^2}{2!}f''(x)-\frac{h^3}{3!}f'''(x)+\ldots\infty$

Q2.19.3 MCQ 2019 2M

Consider two functions: $x = \psi\ln\phi$ and $y = \phi\ln\psi$. Which one of the following is the correct expression for $\frac{\partial\psi}{\partial x}$?

A. $\frac{x\ln\phi}{\ln\phi\ln\psi-1}$

B. $\frac{x\ln\psi}{\ln\phi\ln\psi-1}$

C. $\frac{\ln\phi}{\ln\phi\ln\psi-1}$

D. $\frac{\ln\psi}{\ln\phi\ln\psi-1}$

Q2.19.4 MCQ 2019 2M

Which one of the following is NOT a correct statement?

A. The function $\sqrt[x]{x}$, $(x>0)$, has the global minimum at $x=e$

B. The function $|x|$ has the global minima at $x=0$

C. The function x^3 has neither global minima nor global maxima

D. The function $\sqrt[x]{x}$, $(x>0)$, has the global maxima at $x=e$

Q2.19.5 MCQ 2019 1M

The following inequality is true for all x close to 0.

$$\left(2-\frac{x^2}{3}\right) < \frac{x\sin x}{1-\cos x} < 2$$

What is the value $\lim_{x\to 0}\frac{x\sin x}{1-\cos x}$?

A. 2 **B.** 1 **C.** 0 **D.** $\frac{1}{2}$

Q2.18.1 MCQ 2018 1M

At the point $x = 0$, the function $f(x) = x^3$ has

A. local maximum
B. local minimum
C. both local maximum and minimum
D. neither local maximum nor local minimum

Q2.18.2 MCQ 2018 1M

The value of the integral $\int_0^{\pi} x\cos^2 x\,dx$ is

A. $\frac{\pi^2}{8}$ B. $\frac{\pi^2}{4}$ C. $\frac{\pi^2}{2}$ D. π^2

Q2.17.1 MCQ 2017 1M

The tangent to the curve represented by $y = x\ln x$ is required to have 45° inclination with the x-axis. The coordinates of the tangent point would be

A. $(1,0)$ B. $(0,1)$
C. $(1,1)$ D. $(\sqrt{2},\sqrt{2})$

Q2.17.2 MCQ 2017 2M

Consider the following definite integral

$$I = \int_0^1 \frac{(\sin^{-1}x)^2}{\sqrt{1-x^2}}\,dx$$

The value of the integral is

A. $\frac{\pi^3}{24}$ B. $\frac{\pi^3}{12}$ C. $\frac{\pi^3}{48}$ D. $\frac{\pi^3}{64}$

Q2.17.3 NAT 2017 1M

$\lim_{x\to 0}\left(\frac{\tan x}{x^2-x}\right)$ is equal to ________.

Q2.17.4 MCQ 2017 1M

Let x be a continuous variable defined over the interval $(-\infty,\infty)$ and $f(x) = e^{-x-e^{-x}}$. The integral $g(x) = \int f(x)\,dx$ is equal to

A. $e^{e^{-x}}$ B. $e^{-e^{-x}}$ C. $e^{-e^{x}}$ D. e^{-x}

Q2.17.5 MCQ 2017 1M

Let $W = f(x,y)$, where x and y are functions of t. Then, according to the chain rule, $\frac{dw}{dt}$ is equal to

A. $\frac{dw}{dx}\frac{dx}{dt}+\frac{dw}{dy}\frac{dt}{dt}$ B. $\frac{\partial w}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial t}$

C. $\frac{\partial w}{\partial x}\frac{dx}{dt}+\frac{\partial w}{\partial y}\frac{dy}{dt}$ D. $\frac{dw}{dx}\frac{\partial x}{\partial t}+\frac{dw}{dy}\frac{\partial y}{\partial t}$

Q2.16.1 MCQ 2016 2M

The quadratic approximation of $f(x) = x^3 - 3x^2 - 5$ at the point $x = 0$ is

A. $3x^2 - 6x - 5$ B. $-3x^2 - 5$
C. $-3x^2 + 6x - 5$ D. $3x^2 - 5$

Q2.16.2 MCQ 2016 2M

The value of $\int_0^{\infty}\frac{1}{1+x^2}dx + \int_0^{\infty}\frac{\sin x}{x}dx$ is

A. $\frac{\pi}{2}$ B. π C. $\frac{3\pi}{2}$ D. 1

Q2.16.3 MCQ 2016 2M

The area of the region bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is

A. $\frac{59}{6}$ B. $\frac{9}{2}$ C. $\frac{10}{3}$ D. $\frac{7}{6}$

Q2.16.4 MCQ 2016 1M

The optimum value of the function $f(x) = x^2 - 4x + 2$ is

A. 2 (maximum) B. 2 (minimum)
C. −2 (maximum) D. −2 (minimum)

Q2.16.5 MCQ 2016 1M

What is the value of $\lim_{\substack{x\to 0\\ y\to 0}}\frac{xy}{x^2+y^2}$?

A. 1 B. −1 C. 0 D. Limit does not exist

Q2.16.6 MCQ 2016 2M

The angle of intersection of the curves $x^2 = 4y$ and $y^2 = 4x$ at point $(0,0)$ is

A. 0° B. 30° C. 45° D. 90°

Q2.16.7 NAT 2016 2M

The area between the parabola $x^2 = 8y$ and the straight line $y = 8$ is ______________.

Q2.15.1 MCQ 2015 1M

While minimizing the function $f(x)$, necessary and sufficient conditions for a point x_0 to be a minima are:

A. $f'(x_0) > 0$ and $f''(x_0) = 0$

B. $f'(x_0) < 0$ and $f''(x_0) = 0$

C. $f'(x_0) = 0$ and $f''(x_0) < 0$

D. $f'(x_0) = 0$ and $f''(x_0) > 0$

Q2.15.2 MCQ 2015 1M

Given, $i = \sqrt{-1}$, the value of the definite integral, $I = \int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$ is:

A. 1 **B.** -1 **C.** i **D.** $-i$

Q2.15.3 MCQ 2015 1M

$\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^{2x}$ is equal to

A. e^{-2} **B.** e **C.** 1 **D.** e^2

Q2.14.1 MCQ 2014 1M

$\lim_{x\to\infty} \left(\frac{x + \sin x}{x}\right)$ equals

A. $-\infty$ **B.** 1 **C.** 0 **D.** ∞

Q2.14.2 MCQ 2014 1M

With reference to the conventional Cartesian (x, y) coordinate system, the vertices of a triangle have the following coordinates: $(x_1, y_1) = (1, 0)$; $(x_2, y_2) = (2, 2)$; and $(x_3, y_3) = (4, 3)$. The area of the triangle is equal to

A. $\frac{3}{2}$ **B.** $\frac{4}{5}$ **C.** $\frac{3}{4}$ **D.** $\frac{5}{2}$

Q2.14.3 MCQ 2014 2M

The expression $\lim_{a\to 0} \frac{x^a - 1}{a}$ is equal to

A. $\log x$ **B.** $x \log x$

C. 0 **D.** ∞

Q2.13.1 MCQ 2013 2M

The solution for $\int_0^{\pi/6} \cos^4 3\theta \sin^3 6\theta \, d\theta$ is:

A. 0 **B.** 1 **C.** $\frac{1}{15}$ **D.** $\frac{8}{3}$

Q2.12.1 MCQ 2012 1M

The infinite series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$ corresponds to

A. $\sec x$ **B.** $\cos x$

C. e^x **D.** $1 + \sin^2 x$

Q2.11.1 MCQ 2011 2M

What should be the value of λ such that the function defined below is continuous at $x = \frac{\pi}{2}$?

$$f(x) = \begin{cases} \dfrac{\lambda \cos x}{\frac{\pi}{2} - x}, & \text{if } x \neq \frac{\pi}{2} \\ 1, & \text{if } x = \frac{\pi}{2} \end{cases}$$

A. 0 **B.** 2π **C.** 1 **D.** $\frac{\pi}{2}$

Q2.11.2 MCQ 2011 2M

What is the value of the definite integral?

$$\int_0^a \frac{\sqrt{x}}{\sqrt{x} + \sqrt{a - x}} dx?$$

A. 0 **B.** $\frac{a}{2}$ **C.** a **D.** $2a$

Q2.10.1 MCQ 2010 1M

The $\lim_{x\to 0} \frac{\sin\left(\frac{2}{3}x\right)}{x}$ is

A. $\frac{2}{3}$ **B.** 1 **C.** $\frac{3}{2}$ **D.** ∞

Q2.10.2 MCQ 2010 2M

Given a function $f(x, y) = 4x^2 + 6y^2 - 8x - 4y + 8$, the optimal values of $f(x, y)$ is

A. a minimum equal to $\frac{10}{3}$

B. a maximum equal to $\frac{10}{3}$

C. a minimum equal to $\frac{8}{3}$

D. a maximum equal to $\frac{8}{3}$

Q2.10.3 MCQ 2010 2M

A parabolic cable is held between two supports at the same level. The horizontal span between the supports is L. The sag at the mid-span is h. The equation of the parabola is $y = 4h\frac{x^2}{L^2}$, where x is the horizontal coordinate and y is the vertical coordinate with the origin at the centre of the cable. The expression for the total length of the cable is

A. $\int_0^L \sqrt{1+64\frac{h^2x^2}{L^4}}dx$ **B.** $2\int_0^{L/2} \sqrt{1+64\frac{h^3x^2}{L^4}}dx$

C. $\int_0^{L/2} \sqrt{1+64\frac{h^2x^2}{L^4}}dx$ **D.** $2\int_0^{L/2} \sqrt{1+64\frac{h^2x^2}{L^4}}dx$

Q2.02.1 MCQ 2002 1M

The following function has local minima at which value of x, $f(x) = x\sqrt{5-x^2}$

A. $\frac{-\sqrt{5}}{2}$ **B.** $\sqrt{5}$ **C.** $\sqrt{\frac{5}{2}}$ **D.** $-\sqrt{\frac{5}{2}}$

Q2.02.2 MCQ 2002 1M

Limit of the following sequence as $n \to \infty$ is ______
$x_n = n^{\frac{1}{n}}$

A. 0 **B.** 1 **C.** ∞ **D.** $-\infty$

Q2.01.3 MCQ 2001 1M

Limit of the following series as x approaches $\frac{\pi}{2}$ is

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

A. $\frac{2\pi}{3}$ **B.** $\frac{\pi}{2}$ **C.** $\frac{\pi}{3}$ **D.** 1

Q2.00.1 MCQ 2000 1M

The Taylor series expansion of $\sin x$ about $x = \frac{\pi}{6}$ is given by

A. $\frac{1}{2} + \frac{\sqrt{3}}{2}\left(x-\frac{\pi}{6}\right) - \frac{1}{4}\left(x-\frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12}\left(x-\frac{\pi}{6}\right)^3 + \ldots$

B. $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$

C. $\frac{x-\frac{\pi}{6}}{1!} - \frac{\left(x-\frac{\pi}{6}\right)^3}{3!} + \frac{\left(x-\frac{\pi}{6}\right)^5}{5!} - \frac{\left(x-\frac{\pi}{6}\right)^7}{7!} + \ldots$

D. $\frac{1}{2}$

Q2.99.1 MCQ 1999 1M

Number of inflection points for the curve $y = x + 2x^4$ is ______.

A. 3 **B.** 1 **C.** 0 **D.** 2

Chemical Engineering

Q2.26.1 MCQ 2026 1M

Which of the following is the value of $\lim_{x\to 0} \frac{e^x - x - 1}{\cos x - 1}$

A. -1 **B.** 0 **C.** 1 **D.** ∞

Q2.24.1 MCQ 2024 1M

The first non-zero term in the Taylor series expansion of $(1-x) - e^{-x}$ about $x = 0$ is

A. 1 **B.** -1 **C.** $\frac{x^2}{2}$ **D.** $-\frac{x^2}{2}$

Q2.23.1 MCQ 2023 1M

Which one of the following is the CORRECT value of y, as defined by the expression given below?

$$y = \lim_{x\to 0} \frac{2x}{e^x - 1}$$

A. 1 **B.** 2 **C.** 0 **D.** ∞

Q2.22.1 MCQ 2022 1M

Let $f(x) = e^{-|x|}$, where x is real. The value of $\frac{df}{dx}$ at $x = -1$ is

A. $-e$ **B.** e **C.** $1/e$ **D.** $-1/e$

Q2.22.2 MCQ 2022 1M

The value of the real variable $x \geq 0$, which maximizes the function $f(x) = x^e e^{-x}$ is

A. e **B.** 0 **C.** $1/e$ **D.** 1

Q2.22.3 MCQ 2022 2M

The partial differential equation

$$\frac{\partial u}{\partial t} = \frac{1}{\pi^2} \frac{\partial^2 u}{\partial x^2}$$

where, $t \geq 0$ and $x \in [0, 1]$, is subjected to the following initial and boundary conditions:

$$u(x, 0) = \sin(\pi x)$$

$$u(0, t) = 0$$

$$u(1, t) = 0$$

The value of t at which $\dfrac{u(0.5, t)}{u(0.5, 0)} = \dfrac{1}{e}$ is

A. 1 **B.** e **C.** π **D.** $1/e$

Q2.21.1 MCQ 2021 1M

The function $\cos(x)$ is approximated using Taylor series around $x = 0$ as $\cos(x) \approx 1 + ax + bx^2 + cx^3 + dx^4$. The values of a, b, c and d are

A. $a = 1,\ b = -0.5,\ c = -1,\ d = -0.25$

B. $a = 0,\ b = -0.5,\ c = 0,\ d = 0.042$

C. $a = 0,\ b = 0.5,\ c = 0,\ d = 0.042$

D. $a = -0.5,\ b = 0,\ c = 0.042,\ d = 0$

Q2.21.2 MSQ 2021 1M

For the function $f(x) = \begin{cases} -x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ the CORRECT statement(s) is/are

A. $f(x)$ is continuous at $x = 1$

B. $f(x)$ is differentiable at $x = 1$

C. $f(x)$ is continuous at $x = 0$

D. $f(x)$ is differentiable at $x = 0$

Q2.20.1 MCQ 2020 1M

Consider the hyperbolic functions in **Group – 1** and their definitions in **Group – 2**.

	Group – 1		Group – 2
P	$\tanh x$	**I**	$\dfrac{e^x + e^{-x}}{e^x - e^{-x}}$
Q	$\coth x$	**II**	$\dfrac{2}{e^x + e^{-x}}$
R	$\operatorname{sech} x$	**III**	$\dfrac{2}{e^x - e^{-x}}$
S	$\operatorname{cosech} x$	**IV**	$\dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

The correct combination is

A. P – IV, Q – I, R – III, S – II

B. P – II, Q – III, R – I, S – IV

C. P – IV, Q – I, R – II, S – III

D. P – I, Q – II, R – IV, S – III

Q2.20.2 MCQ 2020 2M

The maximum value of the function $f(x) = -\frac{5}{3}x^3 + 10x^2 - 15x + 16$ in the interval $(0.5, 3.5)$ is

A. 0 **B.** 8 **C.** 16 **D.** 48

Q2.19.1 MCQ 2019 1M

The value of the expression $\lim_{x \to \frac{\pi}{2}} \left| \frac{\tan x}{x} \right|$ is

A. ∞ **B.** 0 **C.** 1 **D.** -1

Q2.18.1 NAT 2018 2M

If $y = e^{-x^2}$ then the value of $\lim_{x \to \infty} \frac{1}{x} \frac{dy}{dx}$ is ________

Q2.17.1 NAT 2017 1M

The value of $\lim_{x \to 0} \frac{\tan(x)}{x}$ is ________.

Q2.16.1 NAT 2016 2M ◯ ◯ ◯

The Lagrange mean-value theorem is satisfied for $f(x) = x^3 + 5$, in the interval $(1, 4)$ at a value (rounded off to the second decimal place) of x equal to ________

Q2.14.4 MCQ 2014 1M ◯ ◯ ◯

If $f(x)$ is a real and continuous function of x, the Taylor series expansion of $f(x)$ about its minima will **NEVER** have a term containing

A. first derivative

B. second derivative

C. third derivative

D. any higher derivative

Q2.13.1 MCQ 2013 1M ◯ ◯ ◯

Evaluate

$$\int \frac{dx}{e^x - 1}$$

(Note: C is a constant of integration.)

A. $\frac{e^x}{e^x - 1} + C$ **B.** $\frac{\ln(e^x - 1)}{e^x} + C$

C. $\ln\left(\frac{e^x}{e^x - 1}\right) + C$ **D.** $\ln(1 - e^{-x}) + C$

Q2.12.1 MCQ 2012 2M ◯ ◯ ◯

If a is a constant, then value of the integral $a^2 \int_0^\infty xe^{-ax}dx$

A. $1/a$ **B.** a **C.** 1 **D.** 0

Q2.11.1 MCQ 2011 1M ◯ ◯ ◯

Which **ONE** of the following functions $y(x)$ has the slope of its tangent equal to $\frac{ax}{y}$?

A. $y = \frac{x + b}{a}$ **B.** $y = ax + b$

C. $y = \sqrt{\frac{x^2 + b}{a}}$ **D.** $y = \sqrt{ax^2 + b}$

Q2.11.2 MCQ 2011 2M ◯ ◯ ◯

The value of the improper integral $\int_{-\infty}^{\infty} \frac{dx}{(1 + x^2)}$ is

A. -2π **B.** 0 **C.** π **D.** 2π

Q2.10.1 MCQ 2010 2M ◯ ◯ ◯

For a function $g(x)$, if $g(0) = 0$ and $g'(0) = 2$, then

$$\lim_{x \to 0} \int_0^{g(x)} \frac{2t}{x} dt$$

is equal to

A. ∞ **B.** 2 **C.** 0 **D.** $-\infty$

Q2.09.1 MCQ 2009 2M ◯ ◯ ◯

The value of the limit

$$\lim_{x \to \pi/2} \frac{\cos x}{(x - \pi/2)^3}$$

is

A. $-\infty$ **B.** 0 **C.** 1 **D.** ∞

Q2.08.1 MCQ 2008 1M ◯ ◯ ◯

The limit of $\frac{\sin x}{x}$ as $x \to \infty$ is

A. -1 **B.** 0 **C.** 1 **D.** ∞

Q2.08.2 MCQ 2008 2M ◯ ◯ ◯

The first four terms of the Taylor series expansion of $\cos x$ about the point $x = 0$ are

A. $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$ **B.** $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!}$

C. $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$ **D.** $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$

Q2.07.1 MCQ 2007 2M ◯ ◯ ◯

The family of curves that is orthogonal to

$$xy = c$$

is

A. $y = c_1 x$ **B.** $y = c_1/x$

C. $y^2 + x^2 = c_1$ **D.** $y^2 - x^2 = c_1$

Q2.06.1 MCQ 2006 1M

The derivative of $|x|$ with respect to x when $x \neq 0$ is

A. $|x|/x$ **B.** -1 **C.** 1 **D.** Undefined

Q2.05.1 MCQ 2005 2M

In the domain $-\infty < x < \infty$, the function $y(x) = x^3 e^{-x}$ has

A. no maximum and no minimum

B. one maximum and no minimum

C. no maximum and one minimum

D. one maximum and one minimum

Q2.05.2 MCQ 2005 2M

In the limit $x \to 0$, what is the limiting value of the function $F(x)$ given below?

$$F(x) = \frac{1 - \cos(2x)}{x^2}$$

A. 0 **B.** 1 **C.** 2 **D.** ∞

Q2.04.1 MCQ 2004 1M

If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, then $\frac{dy}{dx}$ will be equal to

A. $\sin\left(\frac{\theta}{2}\right)$ **B.** $\cos\left(\frac{\theta}{2}\right)$ **C.** $\tan\left(\frac{\theta}{2}\right)$ **D.** $\cot\left(\frac{\theta}{2}\right)$

Production & Industrial Engineering

Q2.26.1 MCQ 2026 1M

Which ONE of the following options is the closest approximation to the value of the given definite integral?

$$\int_0^1 e^{-x^2}\, dx$$

Note: Use numerical integration

A. 1 **B.** $\frac{1}{e}$

C. $\frac{e+1}{2e}$ **D.** $\frac{1+e}{2}$

Q2.24.1 NAT 2024 1M

If

$$\lim_{x\to 1}\left(\frac{x^2 - 2ax + b}{x - 1}\right) = 8$$

then $(a - b)$ is ________. (Answer in integer)

Q2.23.1 MCQ 2023 2M

Given, $z(x, y) = e^{x-2y}$, where $x(t) = e^t$ and $y(t) = e^{-t}$. All the variables are real. The total differential $\frac{dz}{dt}$ is

A. $-z(x + 2y)$ **B.** $-z(x - 2y)$

C. $z(x + 2y)$ **D.** $z(x - 2y)$

Q2.23.2 MCQ 2023 1M

The power series expansion of a function is given

$$\frac{1}{x}\ln(1 + x) = 1 + bx + cx^2 + \ldots$$

for $0 < x \leq 1$.

The values of constants b and c, respectively, are

A. $-\frac{1}{2}$ and $\frac{1}{3}$ **B.** $\frac{1}{2}$ and $-\frac{1}{3}$

C. -1 and $\frac{1}{2}$ **D.** 1 and $-\frac{1}{2}$

Q2.23.3 MCQ 2023 1M

What is $\lim_{x\to 0} f(x)$, where $f(x) = x \sin\frac{1}{x}$?

A. 0 **B.** 1

C. ∞ **D.** Limit does not exist

Q2.23.4 NAT 2023 1M

Consider the real-valued function $g(x) = \max\{(x-2)^2, -2x+7\}$, where $x \in (-\infty, \infty)$. The minimum value attained by $g(x)$ is _______ (rounded off to one decimal place).

Q2.22.1 NAT 2022 2M

The value of $\lim_{x\to 1} \frac{x^3 - 3x + 2}{x^3 - x^2 - x + 1}$ is _______.

Q2.21.1 NAT 2021 2M

The minimum value of function f defined by $f(x, y, z) = x^2 + 5y^2 + 5z^2 - 4x + 40y - 40z + 300$ is ______.
[in integer]

Q2.21.2 NAT 2021 2M

If $(3i + 1)x + (4i + 4)y + 5 = 0$ with x, y being real and $i = \sqrt{-1}$, then $x =$ ______.
[correct upto one decimal place]

Q2.19.1 MCQ 2019 1M

The solution of $\int_1^a \int_1^b \frac{dxdy}{xy}$ is

A. $\ln(ab)$ **B.** $\ln(a/b)$
C. $\ln(a) + \ln(b)$ **D.** $\ln(a)\ln(b)$

Q2.18.1 MCQ 2018 1M

A real-valued function y of real variable x is such that $y = 5|x|$. At $x = 0$, the function is

A. discontinuous but differentiable
B. both continuous and differentiable
C. discontinuous and not differentiable
D. continuous but not differentiable

Q2.16.1 MCQ 2016 1M

At $x = 0$, the function is

$$f(x) = \left|\sin\frac{2\pi x}{L}\right| (-\infty < x < \infty, L > 0)$$

A. continuous and differentiable.
B. not continuous and not differentiable.
C. not continuous but differentiable.
D. continuous but not differentiable.

Q2.16.2 MCQ 2016 1M

For the two functions

$$f(x, y) = x^3 - 3xy^2 \text{ and } g(x, y) = 3x^2y - y^3$$

Which one of the following options is correct?

A. $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial x}$ **B.** $\frac{\partial f}{\partial x} = -\frac{\partial g}{\partial y}$
C. $\frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}$ **D.** $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$

Q2.16.3 MCQ 2016 2M

The range of values of k for which the function $f(x) = (k^2 - 4)x^2 + 6x^3 + 8x^4$ has a local maxima at point $x = 0$ is

A. $k < -2$ or $k > 2$ **B.** $k \leq -2$ or $k \geq 2$
C. $-2 < k < 2$ **D.** $-2 \leq k \leq 2$

Q2.16.4 NAT 2016 2M

$\lim_{x\to 0} \left(\frac{e^{5x} - 1}{x}\right)^2$ is equal to ______.

Q2.15.1 MCQ 2015 1M

The function $f(x) = x^2 = x + x + x + \ldots x$ times, is defined

A. at all real values of x
B. only at positive integer values of x
C. only at negative integer values of x
D. only at rational values of x

Q2.15.2 MCQ 2015 2M

The value of $\lim_{(x,y)\to(0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ is

A. 0 **B.** 1 **C.** $\frac{1}{2}$ **D.** ∞

Q2.15.3 MCQ 2015 2M

The curve $y = x^4$ is

A. concave up for all values of x
B. concave down for all values of x
C. concave up only for positive values of x
D. concave up only for negative values of x

Q2.10.1 MCQ 2010 1M

If $f(x) = \sin|x|$ then the value of $\frac{df}{dx}$ at $x = \frac{-\pi}{4}$ is

A. 0 **B.** $-\frac{1}{\sqrt{2}}$ **C.** $\frac{1}{\sqrt{2}}$ **D.** 1

Q2.10.2 MCQ 2010 1M

The integral $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-x^2}{2}} dx$ is equal to

A. $\frac{1}{2}$ **B.** 1 **C.** $\frac{1}{\sqrt{2}}$ **D.** ∞

Q2.09.1 MCQ 2009 1M

The total derivative of the function 'xy' is

A. $x\,dy + y\,dx$ **B.** $x\,dx + y\,dy$

C. $dx + dy$ **D.** $dx\,dy$

Q2.08.1 MCQ 2008 1M

The value of the integral $\int_{-\pi/2}^{\pi/2}(x\cos x)\,dx$ is

A. 0 **B.** π

C. $\pi - 2$ **D.** $\pi + 2$

Q2.08.2 MCQ 2008 1M

The value of the expression $\lim_{x\to 0}\left[\frac{\sin(x)}{e^x x}\right]$ is

A. 0 **B.** 1

C. $\frac{1}{2}$ **D.** $\frac{1}{1+e}$

Q2.07.1 MCQ 2007 1M

What is the value of $\lim_{x\to\pi/4}\frac{\cos x - \sin x}{x - \pi/4}$

A. $\sqrt{2}$ **B.** $-\sqrt{2}$

C. 0 **D.** Limit does not exist

Q2.07.2 MCQ 2007 1M

For the function $f(x, y) = x^2 - y^2$ defined on $\mathbb{R}^2$, the point $(0, 0)$ is

A. a local minimum

B. Neither a local minimum (nor) a local maximum

C. a local maximum

D. Both a local minimum and a local maximum

Q2.95.1 NAT 1995 1M

Given $y = \int_1^{x^2}\cos t\,dt$, then $\frac{dy}{dx} =$ ______.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q2.26.1 0.67	Q2.25.1 A,B	Q2.23.1 C	Q2.22.1 B	Q2.22.2 512	Q2.20.1 2.25
Q2.20.2 B	Q2.19.1 2	Q2.19.2 C	Q2.18.1 D	Q2.18.2 0	Q2.18.3 4.50
Q2.17.1 -100	Q2.17.2 C	Q2.17.3 C	Q2.17.4 0.79	Q2.16.1 20	Q2.16.2 B
Q2.16.3 4.714	Q2.16.4 B	Q2.16.5 2	Q2.16.6 C	Q2.16.7 10	Q2.15.1 1
Q2.15.2 3	Q2.15.3 B	Q2.15.4 2	Q2.15.5 B	Q2.15.6 A	Q2.14.1 A
Q2.14.2 864	Q2.14.3 A	Q2.14.4 C	Q2.14.5 0	Q2.14.6 D	Q2.14.7 6
Q2.14.8 C	Q2.14.9 C	Q2.12.1 C	Q2.10.1 A	Q2.09.1 D	Q2.08.1 A
Q2.08.2 A	Q2.08.3 B	Q2.08.4 A	Q2.07.1 A	Q2.07.2 B	Q2.07.3 A
Q2.07.4 C	Q2.07.5 A	Q2.05.1 A			
ELECTRICAL ENGINEERING					
Q2.26.1 0.25	Q2.22.1 A	Q2.21.1 1	Q2.21.2 A	Q2.20.1 D	Q2.20.2 C
Q2.18.1 D	Q2.18.2 12	Q2.18.3 0.50	Q2.17.1 1	Q2.17.2 B	Q2.17.3 5.732
Q2.17.4 40	Q2.17.5 7	Q2.17.6 A	Q2.16.1 0.2928	Q2.16.2 D	Q2.16.3 0
Q2.14.1 A	Q2.14.2 C	Q2.14.3 B	Q2.14.4 B	Q2.14.5 2	Q2.13.1 B
Q2.11.1 D	Q2.11.2 C	Q2.10.1 B	Q2.10.2 B	Q2.09.1 A	Q2.07.1 B
Q2.07.2 D	Q2.06.1 D	Q2.05.1 C	Q2.05.2 A		
INSTRUMENTATION ENGINEERING					
Q2.26.1 D	Q2.26.2 A,D	Q2.25.1 C	Q2.23.1 A	Q2.23.2 1.0	Q2.22.1 -3.0
Q2.21.1 100.0	Q2.21.2 A	Q2.20.1 A	Q2.20.2 0.5	Q2.19.1 B	Q2.18.1 1.0
Q2.18.2 D	Q2.18.3 B	Q2.16.1 3.0	Q2.16.2 0.5	Q2.16.3 -13.0	Q2.15.1 C

Q2.14.1 D	Q2.13.1 A	Q2.11.1 B	Q2.08.1 C	Q2.08.2 C	Q2.08.3 B
Q2.08.4 C	Q2.07.1 D	Q2.07.2 C	Q2.07.3 C	Q2.05.1 C	Q2.05.2 D
Q2.01.1 D	Q2.99.1 C				
COMPUTER SCIENCE & IT					
Q2.26.1 A,C,D	Q2.26.2 A,C	Q2.25.1 A	Q2.24.1 D	Q2.24.2 B	Q2.23.1 B,D
Q2.23.2 0	Q2.22.1 -0.5	Q2.21.1 0.25	Q2.21.2 19	Q2.20.1 A	Q2.19.1 C
Q2.18.1 D	Q2.18.2 0.29	Q2.17.1 C	Q2.17.2 B	Q2.17.3 C	Q2.17.4 8
Q2.17.5 C	Q2.16.1 1	Q2.16.2 9	Q2.15.1 A	Q2.15.2 C	Q2.15.3 0.99
Q2.15.4 -1	Q2.15.5 A	Q2.14.1 4	Q2.14.2 A	Q2.14.3 A	Q2.13.1 A
Q2.12.1 B	Q2.11.1 D	Q2.10.1 B	Q2.09.1 D	Q2.05.1 A	Q2.05.2 A
Q2.04.1 A	Q2.03.1 B				
MECHANICAL ENGINEERING					
Q2.26.1 A	Q2.25.1 A	Q2.23.1 B	Q2.23.2 8	Q2.22.1 A	Q2.22.2 B
Q2.21.1 B	Q2.21.2 D	Q2.21.3 D	Q2.20.1 D	Q2.20.2 D	Q2.19.1 D
Q2.19.2 2.097	Q2.19.3 B	Q2.18.1 A	Q2.17.1 D	Q2.17.2 C	Q2.16.1 -5
Q2.16.2 C	Q2.16.3 C	Q2.16.4 C	Q2.15.1 -0.333	Q2.15.2 C	Q2.15.3 1.8614
Q2.15.4 2	Q2.15.5 A	Q2.14.1 A	Q2.14.2 B	Q2.14.3 D	Q2.14.4 B
Q2.14.5 B	Q2.13.1 C	Q2.12.1 D	Q2.12.2 B	Q2.11.1 D	Q2.10.1 B
Q2.10.2 D	Q2.10.3 C	Q2.10.4 D	Q2.09.1 A	Q2.08.1 A	Q2.08.2 C
Q2.08.3 B	Q2.08.4 D	Q2.08.5 D	Q2.08.6 C	Q2.07.1 B	Q2.07.2 B
CIVIL ENGINEERING					
Q2.26.1 C	Q2.25.1 A	Q2.25.2 B	Q2.25.3 A,D	Q2.25.4 3	Q2.24.1 A

Q2.24.2 B	Q2.24.3 5.13	Q2.23.1 D	Q2.23.2 A	Q2.22.1 A, B	Q2.22.2 *
Q2.22.3 B	Q2.21.1 15	Q2.21.2 0.5	Q2.21.3 0.0	Q2.21.4 C	Q2.20.1 D
Q2.20.2 D	Q2.20.3 D	Q2.19.1 A	Q2.19.2 A	Q2.19.3 D	Q2.19.4 A
Q2.19.5 A	Q2.18.1 D	Q2.18.2 B	Q2.17.1 A	Q2.17.2 A	Q2.17.3 -1
Q2.17.4 B	Q2.17.5 C	Q2.16.1 B	Q2.16.2 B	Q2.16.3 B	Q2.16.4 D
Q2.16.5 D	Q2.16.6 D	Q2.16.7 85.33	Q2.15.1 D	Q2.15.2 C	Q2.15.3 D
Q2.14.1 B	Q2.14.2 A	Q2.14.3 A	Q2.13.1 C	Q2.12.1 C	Q2.11.1 C
Q2.11.2 B	Q2.10.1 A	Q2.10.2 A	Q2.10.3 D	Q2.02.1 D	Q2.02.2 B
Q2.01.3 D	Q2.00.1 A	Q2.99.1 C			
CHEMICAL ENGINEERING					
Q2.26.1 A	Q2.24.1 D	Q2.23.1 B	Q2.22.1 C	Q2.22.2 A	Q2.22.3 A
Q2.21.1 B	Q2.21.2 A,B,C	Q2.20.1 C	Q2.20.2 C	Q2.19.1 A	Q2.18.1 0
Q2.17.1 1	Q2.16.1 2.65	Q2.14.4 A	Q2.13.1 D	Q2.12.1 C	Q2.11.1 D
Q2.11.2 C	Q2.10.1 C	Q2.09.1 A	Q2.08.1 B	Q2.08.2 C	Q2.07.1 D
Q2.06.1 A	Q2.05.1 B	Q2.05.2 C	Q2.04.1 C		
PRODUCTION & INDUSTRIAL ENGINEERING					
Q2.26.1 C	Q2.24.1 4	Q2.23.1 C	Q2.23.2 A	Q2.23.3 A	Q2.23.4 1
Q2.22.1 1.4 to 1.6	Q2.21.1 136	Q2.21.2 2.5	Q2.19.1 D	Q2.18.1 D	Q2.16.1 D
Q2.16.2 C	Q2.16.3 C	Q2.16.4 25	Q2.15.1 B	Q2.15.2 A	Q2.15.3 A
Q2.10.1 B	Q2.10.2 B	Q2.09.1 A	Q2.08.1 A	Q2.08.2 B	Q2.07.1 B
Q2.07.2 B	Q2.95.1 $2x\cos(x^2)$				

UNIT

III

Probability and Statistics

Topics

1. Probability
2. Random Variables
3. Probability Distributions
4. Mean and Variance
5. Bayes Theorem
6. Conditional Probability
7. Correlation
8. Regression
9. Estimation
10. Hypothesis Testing

Branches: EC,EE,IN,CS,ME,CE,CH,PI

PrepFusion

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 53 **MCQ:** 29 **MSQ:** 1 **NAT:** 23
1M: 23 **2M:** 29 **Desc:** 1 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2013)

EE — Electrical Engineering

Questions: 34 **MCQ:** 23 **MSQ:** 1 **NAT:** 10
1M: 16 **2M:** 18 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2013)–3M(2025)

IN — Instrumentation Engineering

Questions: 33 **MCQ:** 21 **MSQ:** 0 **NAT:** 12
1M: 16 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2018)

CS — Computer Science & IT

Questions: 58 **MCQ:** 31 **MSQ:** 2 **NAT:** 25
1M: 25 **2M:** 33 **Desc:** 0 **Avg Weightage** ~4M **Range:** 1M(2013)–5M(2026)

ME — Mechanical Engineering

Questions: 67 **MCQ:** 50 **MSQ:** 0 **NAT:** 17
1M: 29 **2M:** 38 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2013)

CE — Civil Engineering

Questions: 58 **MCQ:** 32 **MSQ:** 1 **NAT:** 25
1M: 31 **2M:** 27 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2020)–6M(2016)

CH — Chemical Engineering

Questions: 26 **MCQ:** 18 **MSQ:** 1 **NAT:** 7
1M: 11 **2M:** 15 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–4M(2014)

PI — Production & Industrial Engineering

Questions: 42 **MCQ:** 29 **MSQ:** 0 **NAT:** 13
1M: 20 **2M:** 22 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q3.26.1 MCQ 2026 2M

Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows Normal distribution with zero mean and unity variance. Y and Z are defined as, $Y = X + N$ and $Z = X^2 + N$. Which of the following pairs represents the values of correlation between X and Y and that between X and Z?

A. $\frac{1}{3}$ and 0
B. $\frac{1}{3}$ and $\frac{1}{9}$
C. $\frac{1}{3}$ and $\frac{1}{3}$
D. 1 and 0

Q3.25.1 MCQ 2025 1M

A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement. What is the probability that the two balls drawn have different colours?

A. $\frac{2}{3}$
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. 1

Q3.25.2 NAT 2025 2M

Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable X denote the sum of the outcomes obtained. The expectation of X is ________ *(rounded off to two decimal places).*

Q3.24.1 NAT 2024 2M

Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval [0,1]. The probability that $X \geq Y$ is ________.

Q3.22.1 NAT 2022 1M

The bar graph shows the frequency of the number of wickets taken in a match by a bowler in her career. For example, in 17 of her matches, the bowler has taken 5 wickets each. The median number of wickets taken by the bowler in a match is ________. *(rounded off to one decimal place).*

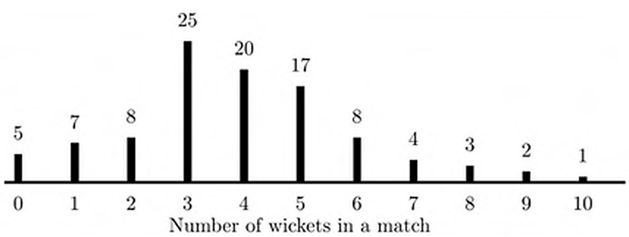


Fig. Q3.22.1

Q3.21.1 MCQ 2021 1M

Two continuous random variables X and Y are related as

$$Y = 2X + 3$$

Let σ_X^2 and σ_Y^2 denote the variances of X and Y, respectively. The variances are related as

A. $\sigma_Y^2 = 2\ \sigma_X^2$
B. $\sigma_Y^2 = 4\ \sigma_X^2$
C. $\sigma_Y^2 = 5\ \sigma_X^2$
D. $\sigma_Y^2 = 25\ \sigma_X^2$

Q3.21.2 MCQ 2021 2M

A box contains the following three coins.

I. A fair coin with head on one face and tail on the other face.

II. A coin with heads on both the faces.

III. A coin with tails on both the faces.

A coin is picked randomly from the box and tossed. Out of the two remaining coins in the box, one coin is then picked randomly and tossed. If the first toss results in a head, the probability of getting a head in the second toss is

A. $\frac{2}{5}$
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. $\frac{2}{3}$

Q3.20.1 NAT 2020 1M

The two sides of a fair coin are labelled as 0 and 1. The coin is tossed two times independently. Let M and N denote the labels corresponding to the outcomes of those tosses. For a random variable X, defined as $X = min(M, N)$, the expected value of $E(X)$ **(rounded off to two decimal places)** is ________.

Q3.20.2 NAT 2020 2M

X is a random variable with uniform probability density function in the interval $[-2, 10]$. For $Y = 2X - 6$, the conditional probability $P(Y \leq 7 | X \geq 5)$ **(rounded off to three decimal places)** is ________.

Q3.19.1 NAT 2019 1M

If X and Y are random variables such that $E[2X + Y] = 0$ and $E[X + 2Y] = 33$, then $E[X] + E[Y] =$ ________.

Q3.19.2 NAT 2019 1M

Let Z be an exponential random variable with mean 1. That is, the cumulative distribution function of Z is given by

$$F_Z(x) = \begin{cases} 1 - e^{-x} & if\ x \geq 0 \\ 0 & if\ x < 0 \end{cases}$$

Then $Pr(Z > 2|Z > 1)$, rounded off to two decimal places, is equal to ______.

Q3.18.1 NAT 2018 1M

Let X_1, X_2, X_3 and X_4 be independent normal random variables with zero mean and unit variance. The probability that X_4 is the smallest among the four is ______.

Q3.17.1 NAT 2017 1M

Three fair cubical dice are thrown simultaneously. The probability that all three dice have the same number of dots on the faces showing up is (up to third decimal place) ______.

Q3.17.2 NAT 2017 2M

Passengers try repeatedly to get a seat reservation in any train running between two stations until they are successful. If there is 40% chance of getting reservation in any attempt by a passenger. Then the average number of attempts that passengers need to get a seat reserved is ______.

Q3.16.1 NAT 2016 1M

The second moment of a Poisson-distributed random variable is 2. The mean of the random variable is ______

Q3.16.2 NAT 2016 2M

Two random variables X and Y are distributed according to

$$f_{X,Y}(x, y) = \begin{cases} (x + y), & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(X + Y \leq 1)$ is ______.

Q3.16.3 NAT 2016 1M

The probability of getting a "head" in a single toss of a biased coin is 0.3. The coin is tossed repeatedly till a "head" is obtained. If the tosses are independent, then the probability of getting "head" for the first time in the fifth toss is ______

Q3.15.1 MCQ 2015 1M

Suppose A and B are two independent events with probabilities $P(A) \neq 0$ and $P(B) \neq 0$. Let $\overline{A}$ and $\overline{B}$ be their complements. Which one of the following statements is FALSE?

A. $P(A \cap B) = P(A)P(B)$

B. $P(A|B) = P(A)$

C. $P(A \cup B) = P(A) + P(B)$

D. $P(\overline{A} \cap \overline{B}) = P(\overline{A})P(\overline{B})$

Q3.15.2 NAT 2015 2M

Let the random variable X represent the number of times a fair coin needs to be tossed till two consecutive heads appear for the first time. The expectation of X is ______

Q3.15.3 MCQ 2015 2M

Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables. If $P(X = 0) = p$ and $P(Y = 0) = q$, then $P(X + Y \geq 1)$ is equal to

A. $pq + (1 - p)(1 - q)$

B. pq

C. $p(1 - q)$

D. $1 - pq$

Q3.15.4 NAT 2015 2M

A fair die with faces $\{1, 2, 3, 4, 5, 6\}$ is thrown repeatedly till '3' is observed for the first time. Let X denote the number of times the die is thrown. The expected value of X is ______.

Q3.15.5 NAT 2015 2M

The variance of the random variable X with probability density function $f(x) = \frac{1}{2}|x|e^{-|x|}$ is ______

Q3.14.1 NAT 2014 1M

In a housing society, half of the families have a single child per family, while the remaining half have two children per family. The probability that a child picked at random, has a sibling is ______

Q3.14.2 MCQ 2014 2M

Let X be a real-valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2 respectively. The relation which always holds true is

A. $(E[X])^2 > E[X^2]$ **B.** $E[X^2] \geq (E[X])^2$

C. $E[X^2] = (E[X])^2$ **D.** $E[X^2] > (E[X])^2$

Q3.14.3 NAT 2014 1M

Let X be a random variable which is uniformly chosen from the set of positive numbers less than 100. The expectation, $E[X]$, is ______

Q3.14.4 MCQ 2014 1M

An unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is

A. 0.067 **B.** 0.073 **C.** 0.082 **D.** 0.091

Q3.14.5 NAT 2014 2M

A fair coin is tossed repeatedly till both head and tail appear at least once. The average number of tosses required is ______

Q3.14.6 NAT 2014 2M

Let X_1, X_2 and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability that $P\{X_1 + X_2 \leq X_3\}$ is ______.

Q3.14.7 NAT 2014 1M

Let X be a zero mean unit variance Gaussian random variable. $E[|X|]$ is equal to ______

Q3.14.8 MCQ 2014 1M

If calls arrive at a telephone exchange such that the time of arrival of any call is independent of the time of arrival of earlier or future calls, the probability distribution function of the total number of calls in a fixed time interval will be

A. Poisson **B.** Gaussian

C. Exponential **D.** Gamma

Q3.14.9 NAT 2014 2M

Parcels from sender S to receiver R pass sequentially through two post-offices. Each post-office has a probability $\frac{1}{5}$ of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by the second post-office is ______

Q3.13.1 MCQ 2013 2M

Let U and V be two independent zero mean Gaussian random variables of variances $\frac{1}{4}$ and $\frac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

A. $\frac{4}{9}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{5}{9}$

Q3.13.2 MCQ 2013 2M

Consider two identically distributed zero-mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then, for all values of x

A. $F(x) - G(x) \leq 0$ **B.** $F(x) - G(x) \geq 0$

C. $(F(x) - G(x)) \cdot x \leq 0$ **D.** $(F(x) - G(x)) \cdot x \geq 0$

Q3.12.1 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that max $[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$ **B.** $\frac{9}{16}$ **C.** $\frac{1}{4}$ **D.** $\frac{2}{3}$

Q3.12.2 MCQ 2012 2M

A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

A. $\frac{1}{3}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{4}$

Q3.11.1 MCQ 2011 2M

A fair dice is tossed two times. The probability that the second toss results in a value that is higher than the first toss is

A. $\frac{2}{36}$ **B.** $\frac{2}{6}$ **C.** $\frac{5}{12}$ **D.** $\frac{1}{2}$

Q3.10.1 MCQ 2010 2M

A fair coin is tossed independently four times. The probability of the event "the number of times heads show up is more than the number of times tails show up" is

A. $\frac{1}{16}$ **B.** $\frac{1}{8}$ **C.** $\frac{1}{4}$ **D.** $\frac{5}{16}$

Q3.09.1 MCQ 2009 1M

A fair coin is tossed 10 times. What is the probability that ONLY the first two tosses will yield heads?

A. $\left(\frac{1}{2}\right)^2$ **B.** ${}^{10}C_2\left(\frac{1}{2}\right)^2$ **C.** $\left(\frac{1}{2}\right)^{10}$ **D.** ${}^{10}C_2\left(\frac{1}{2}\right)^{10}$

Q3.09.2 MCQ 2009 2M

Consider two independent random variables X and Y with identical distributions. The variables X and Y take values 0, 1 and 2 with probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$ respectively. What is the conditional probability $P(X+Y=2|X-Y=0)$?

A. 0 **B.** $\frac{1}{16}$ **C.** $\frac{1}{6}$ **D.** 1

Q3.09.3 MCQ 2009 2M

A discrete random variable X takes values from 1 to 5 with probabilities as shown in the table. A student calculates the mean of X as 3.5 and her teacher calculates the variance of X as 1.5. Which of the following statements is true?

k	1	2	3	4	5
$P(X=k)$	0.1	0.2	0.4	0.2	0.1

A. Both the student and the teacher are right

B. Both the student and the teacher are wrong

C. The student is wrong but the teacher is right

D. The student is right but the teacher is wrong

Q3.08.1 MCQ 2008 2M

$P_X(x) = Me^{-2|x|} + Ne^{-3|x|}$ is the probability density function for the real random variable X, over the entire x axis. M and N are both positive real numbers. The equation relating M and N is

A. $M+\frac{2}{3}N=1$ **B.** $2M+\frac{1}{3}N=1$

C. $M+N=1$ **D.** $M+N=3$

Q3.07.1 MCQ 2007 1M

If E denotes expectation, the variance of a random variable X is given by

A. $E[X^2]-E^2[X]$ **B.** $E[X^2]+E^2[X]$

C. $E[X^2]$ **D.** $E^2[X]$

Q3.07.2 MCQ 2007 2M

An examination consists of two papers, Paper 1 and Paper 2. The probability of failing in Paper 1 is 0.3 and that in Paper 2 is 0.2. Given that a student has failed in Paper 2, the probability of failing in Paper 1 is 0.6. The probability of a student failing in both the papers is

A. 0.5 **B.** 0.18 **C.** 0.12 **D.** 0.06

Q3.06.1 MCQ 2006 1M

A probability density function is of the form

$$p(x) = Ke^{-\alpha|x|}, x \in (-\infty, \infty)$$

The value of K is

A. 0.5 **B.** 1 **C.** 0.5α **D.** α

Q3.06.2 MCQ 2006 2M

Three companies X, Y, and Z supply computers to a university. The percentage of computers supplied by them and the probability of those being defective are tabulated below

Company	*% of Computer Supplied*	*Probability of being supplied defective*
X	60%	0.01
Y	30%	0.02
Z	10%	0.03

Given that a computer is defective, the probability that was supplied by Y is

A. 0.1 **B.** 0.2 **C.** 0.3 **D.** 0.4

Q3.06.3 MCQ 2006 2M

A uniformly distributed random variable X with probability density function

$$f_x(x) = \frac{1}{10}[u(x+5)-u(x-5)]$$

where $u(\cdot)$ is the unit step function is passed through a transformation given in the figure below. The probability density function of the transformed random variable Y would be

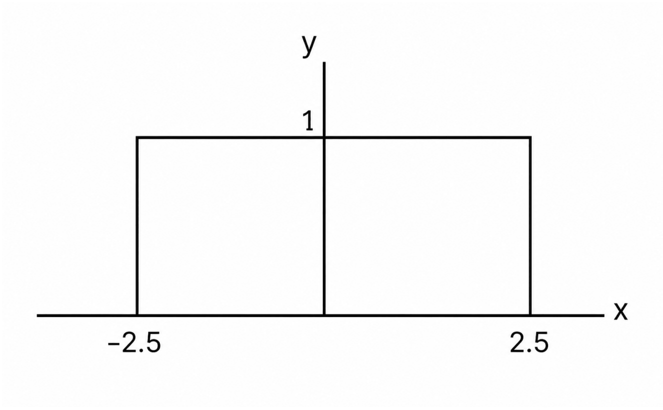


Fig. Q3.06.3

A. $f_Y(y) = \frac{1}{5}[u(y+2.5) - u(y-2.5)]$

B. $f_Y(y) = 0.5\delta(y) + 0.5\delta(y-1)$

C. $f_Y(y) = 0.25\delta(y+2.5) + 0.25\delta(y-2.5) + 0.5\delta(y)$

D. $f_Y(y) = 0.25\delta(y+2.5) + 0.25\delta(y-2.5)] + \frac{1}{10}[u(y+2.5) - u(y-2.5)$

Q3.05.1 MCQ 2005 1M ○○○

A fair dice is rolled twice. The probability that an odd number will follow an even number is

A. $\frac{1}{2}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{4}$

Q3.04.1 MCQ 2004 1M ○○○

The distribution function $F_x(x)$ of a random variable X is shown in the figure. The probability that $X = 1$ is

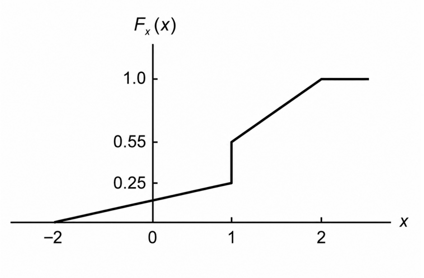


Fig. Q3.04.1

A. zero **B.** 0.25 **C.** 0.55 **D.** 0.30

Q3.03.1 MCQ 2003 2M ○○○

Let X and Y be two statistically independent random variables uniformly distributed in the ranges (-1,1) and (-2,1) respectively. Let $Z = X + Y$. Then the probability that $(z \le -1)$ is

A. zero **B.** $\frac{1}{6}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{12}$

Q3.01.1 MCQ 2001 1M ○○○

The PDF of a Gaussian random variable X is given by $P_x(x) = \frac{1}{3\sqrt{2\pi}} e^{\frac{-(x-4)^2}{18}}$. The probability of the event $(X = 4)$ is

A. $\frac{1}{2}$ **B.** $\frac{1}{3\sqrt{2\pi}}$ **C.** 0 **D.** $\frac{1}{4}$

Q3.93.1 NAT 1993 2M ○○○

The function shown in figure, can represent a probability density function for A ________

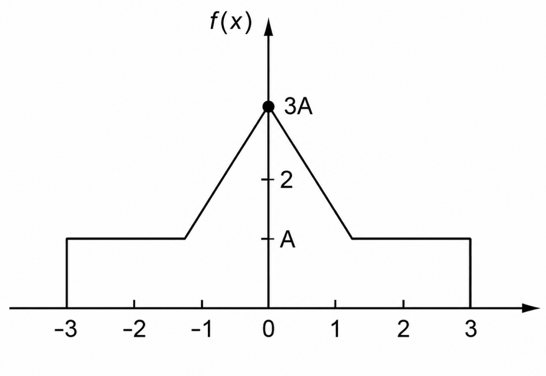


Fig. Q3.93.1

Q3.92.1 MCQ 1992 2M ○○○

For a random variable X following the probability density function $p(x)$, shown in given figure the mean and the variance are, respectively,

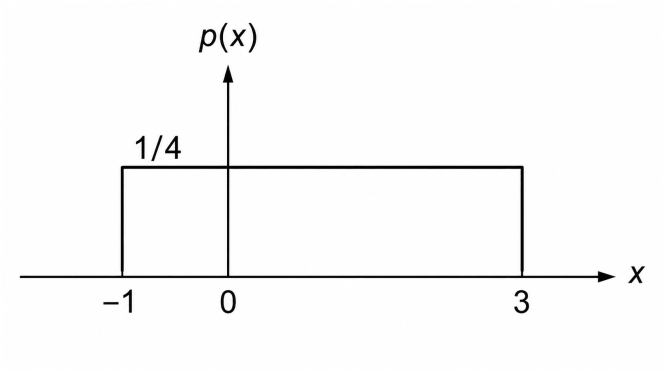


Fig. Q3.92.1

A. $\frac{1}{2}$ and $\frac{2}{3}$

B. 1 and $\frac{4}{3}$

C. 1 and $\frac{2}{3}$

D. 2 and $\frac{4}{3}$

Q3.88.1 MSQ 1988 ○ ○ ○

Events A and B are mutually exclusive and have nonzero probability. Which of the following statement(s) are true?

☐ **A.** $P(A \cup B) = P(A) + P(B)$

☐ **B.** $P(B^C) > P(A)$

☐ **C.** $P(A \cap B) = P(A)P(B)$

☐ **D.** $P(B^C) < P(A)$

Electrical Engineering

Q3.26.1 NAT 2026 2M ○ ○ ○

Let X and Y be two real-valued random variables with $E(X) = 1$, $E(Y) = 2$, $E(X^2) = 4$, $E(Y^2) = 9$ and $E(XY) = 0.9$, where E denotes the expectation operator.The value of α that minimizes $E\left((X - \alpha Y)^2\right)$ is ________.

(Round off to one decimal place)

Q3.25.1 MCQ 2025 2M ○ ○ ○

Let X and Y be continuous random variables with probability density functions $P_X(x)$ and $P_Y(y)$, respectively. Further, let $Y = X^2$ and

$$P_X(x) = \begin{cases} 1, & x \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$$

Which one of the following options is correct?

A. $P_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

B. $P_Y(y) = \begin{cases} 1, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

C. $P_Y(y) = \begin{cases} 1.5\sqrt{y}, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

D. $P_Y(y) = \begin{cases} 2y, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

Q3.25.2 MCQ 2025 1M ○ ○ ○

Consider discrete random variables X and Y with probabilities as follows:

$$P(X = 0 \text{ and } Y = 0) = \frac{1}{4}$$

$$P(X = 1 \text{ and } Y = 0) = \frac{1}{8}$$

$$P(X = 0 \text{ and } Y = 1) = \frac{1}{2}$$

$$P(X = 1 \text{ and } Y = 1) = \frac{1}{8}$$

Given $X = 1$, the expected value of Y is

A. $\frac{1}{4}$ **B.** $\frac{1}{2}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{3}$

Q3.24.1 MSQ 2024 2M ○ ○ ○

Let X be a discrete random variable that is uniformly distributed over the set $\{-10, -9, \ldots, 0, \ldots, 9, 10\}$.
Which of the following random variables is/are uniformly distributed?

A. X^2

B. X^3

C. $(X - 5)^2$

D. $(X + 10)^2$

Q3.23.1 NAT 2023 2M ○ ○ ○

The expected number of trials for first occurrence of a "head" in a biased coin is known to be 4. The probability of first occurrence of a "head" in the second trial is ________.

(Round off to 3 decimal places).

Q3.23.2 MCQ | 2023 | 1M

One million random numbers are generated from a statistically stationary process with a Gaussian distribution with mean zero and standard deviation σ_0.
The σ_0 is estimated by randomly drawing out 10,000 numbers of samples (x_n). The estimates $\hat{\sigma}_1, \hat{\sigma}_2$ are computed in the following two ways.

$$\hat{\sigma}_1^2 = \frac{1}{10000}\sum_{n=1}^{10000} x_n^2$$

$$\hat{\sigma}_2^2 = \frac{1}{9999}\sum_{n=1}^{10000} x_n^2$$

Which of the following statements is true?

A. $E(\hat{\sigma}_2^2) = \sigma_0^2$ **B.** $E(\hat{\sigma}_2) = \sigma_0$

C. $E(\hat{\sigma}_1^2) = \sigma_0^2$ **D.** $E(\hat{\sigma}_1) = E(\hat{\sigma}_2)$

Q3.22.1 MCQ | 2022 | 1M

Consider two dices having faces as P, Q, R, S, T, U, if both the dices are rolled together then what is the probability of getting any combination of P, Q, R on upper faces?

A. $\frac{1}{4}$ **B.** $\frac{1}{6}$ **C.** $\frac{5}{36}$ **D.** 0

Q3.22.2 NAT | 2022 | 1M

Let the probability density function of a random variable x be given as

$$f(x) = ae^{-2|x|}$$

The value of a is ________.

Q3.21.1 MCQ | 2021 | 2M

Suppose the probability that a coin toss shows "head" is p, where $0 < p < 1$. The coin is tossed repeatedly until the first "head" appears. The expected number of tosses required is

A. $\frac{p}{1-p}$ **B.** $\frac{1}{p}$

C. $\frac{1}{p^2}$ **D.** $\frac{1-p}{p}$

Q3.19.1 NAT | 2019 | 2M

The probability of a resistor being defective is 0.02. There are 50 such resistors in a circuit. The probability of two or more defective resistors in the circuit (round off to two decimal places) is ________.

Q3.19.2 MCQ | 2019 | 1M

The mean-square of a zero-mean random process is kT/C, where k is Boltzman's constant, T is the absolute temperature, and C is a capacitance. The standard deviation of the random process is

A. $\frac{\sqrt{kT}}{C}$ **B.** $\sqrt{\frac{kT}{C}}$

C. $\frac{C}{kT}$ **D.** $\frac{kT}{C}$

Q3.17.1 MCQ | 2017 | 2M

A person decides to toss a fair coin repeatedly until he gets a head. He will make at most 3 tosses. Let the random variable Y denote the number of heads. The value of $\text{var}\{Y\}$, where $\text{var}[\cdot]$ denotes the variance, equals

A. $\frac{7}{8}$ **B.** $\frac{49}{64}$ **C.** $\frac{7}{64}$ **D.** $\frac{105}{64}$

Q3.17.2 MCQ | 2017 | 1M

A urn contains 5 red ball and 5 black balls. In the first draw, one ball is picked at random and discarded without noticing its colour. The probability to get a red ball in the second draw is

A. $\frac{1}{2}$ **B.** $\frac{4}{9}$ **C.** $\frac{5}{9}$ **D.** $\frac{6}{9}$

Q3.17.3 NAT | 2017 | 1M

Assume that in a traffic junction, the cycle of the traffic signal lights is 2 minutes of green (vehicle does not stop) and 3 minutes of red (vehicle stops). Consider that the arrival time of vehicles at the junction is uniformly distributed over 5 minute cycle. The expected waiting time (in minutes) for the vehicle at the junction is ________

Q3.16.1 MCQ | 2016 | 2M

Let the probability density function of a variable, X, be given as

$$f_X(x) = \frac{3}{2}e^{-3x}u(x) + ae^{4x}u(-x)$$

Where $u(x)$ is the unit step function.
Then the value of 'a' and $\text{Prob}\{X \leq 0\}$, respectively, are

A. $2, \frac{1}{2}$ **B.** $4, \frac{1}{2}$ **C.** $2, \frac{1}{4}$ **D.** $4, \frac{1}{4}$

Q3.16.2 NAT 2016 2M

Candidates were asked to come to an interview with 3 pens each. Black, blue, green and red were the permitted pen colours that the candidate could bring. The probability that a candidate comes with all 3 pens having the same colour is ___________

Q3.15.1 MCQ 2015 2M

Two coins R and S are tossed. The 4 joint events $H_R H_S$, $T_R T_S$, $H_R T_S$, $T_R H_S$ have probabilities 0.28, 0.18, 0.30, 0.24 respectively, where H represents head and T represents tail. Which one of the following is TRUE?

A. The coin tosses are independent.

B. R is fair, S is not.

C. S is fair, R is not.

D. The coin tosses are dependent.

Q3.15.2 MCQ 2015 2M

Two players, A and B, alternately keep rolling a fair dice. The person to get a six first wins the game. Given that player A starts the game, the probability that A wins the game is

A. $\frac{5}{11}$ **B.** $\frac{1}{2}$ **C.** $\frac{7}{13}$ **D.** $\frac{6}{11}$

Q3.15.3 NAT 2015 1M

A random variable X has probability density function $f(x)$ as given below:

$$f(x) = \begin{cases} a + bx & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If the expected value $E[X] = \frac{2}{3}$, then $P[X < 0.5]$ is ___________

Q3.14.1 NAT 2014 1M

Lifetime of an electric bulb is a random variable with density $f(x) = kx^2$, where x is measured in years. If the minimum and maximum lifetimes of bulb are 1 and 2 years respectively, then the value of k is ___________

Q3.14.2 NAT 2014 2M

Let X be a random variable with probability density function

$$f(x) = \begin{cases} 0.2, & \text{for } |x| \leq 1 \\ 0.1, & \text{for } 1 < |x| \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(0.5 < X < 5)$ is ___________

Q3.14.3 NAT 2014 1M

Consider a dice with the property that the probability of a face with n dots showing up is proportional to n. The probability of the face with three dots showing up is ___________.

Q3.14.4 MCQ 2014 2M

A fair coin is tossed n times. The probability that the difference between the number of heads and tails is $(n-3)$ is

A. 2^{-n} **B.** 0

C. ${}^nC_{n-3}\, 2^{-n}$ **D.** 2^{-n+3}

Q3.13.1 MCQ 2013 1M

A continuous random variable X has a probability density function

$$f(x) = e^{-x}, \quad 0 < x < \infty.$$

Then $P\{X > 1\}$ is

A. 0.368 **B.** 0.5 **C.** 0.632 **D.** 1.0

Q3.12.1 MCQ 2012 1M

A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

A. $\frac{1}{3}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{4}$

Q3.12.2 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$ **B.** $\frac{9}{16}$ **C.** $\frac{1}{4}$ **D.** $\frac{2}{3}$

Q3.11.1 MCQ 2011 2M

A zero mean random signal is uniformly distributed between limits $-a$ and $+a$ and its mean square value is equal to its variance. Then the r.m.s value of the signal is

A. $\frac{a}{\sqrt{3}}$ **B.** $\frac{a}{\sqrt{2}}$ **C.** $a\sqrt{2}$ **D.** $a\sqrt{3}$

Q3.10.1 MCQ 2010 2M

A box contains 4 white balls and 3 red balls. In succession, two balls are randomly selected and removed from the box. Given that the first removed ball is white, the probability that the second removed ball is red is

A. $\frac{1}{3}$ **B.** $\frac{3}{7}$ **C.** $\frac{1}{2}$ **D.** $\frac{4}{7}$

Q3.09.1 MCQ 2009 2M

Assume for simplicity that N people, all born in April (a month of 30 days) are collected in a room, consider the event of at least two people in the room being born on the same date of the month even if in different years e.g. 1980 and 1985. What is the smallest N so that the probability of this exceeds 0.5 is?

A. 20 **B.** 7 **C.** 15 **D.** 16

Q3.08.1 MCQ 2008 1M

X is uniformly distributed random variable that take values between 0 and 1. The value of $E(X^3)$ will be,

A. 0 **B.** $\frac{1}{8}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{2}$

Q3.07.1 MCQ 2007 1M

A loaded dice has following probability distribution of occurrences

Dice value	1	2	3	4	5	6
Probability	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$

If three identical dice as the above are thrown, the probability of occurrence of values 1, 5 and 6 on the three dice is

A. Same as that of occurrence of 3, 4, 5.

B. Same as that of occurrence of 1, 2, 5.

C. $\frac{1}{128}$

D. $\frac{5}{8}$

Q3.06.1 MCQ 2006 2M

Two fair dice are rolled and the sum r of the numbers turned up is considered

A. $Pr(r > 6) = \left(\frac{1}{6}\right)$

B. $Pr(r/3 \text{ is an integer}) = \left(\frac{5}{6}\right)$

C. $Pr(r = 8 \mid r/4 \text{ is an integer}) = \left(\frac{5}{9}\right)$

D. $Pr(r = 6 \mid r/5 \text{ is an integer}) = \left(\frac{1}{18}\right)$

Q3.05.1 MCQ 2005 1M

If P and Q are two random events, then which of the following is true?

A. Independence of P and Q implies that Probability $(P \cap Q) = 0$

B. Probability $(P \cap Q) \geq$ Probability (P)+Probability (Q)

C. If P and Q are mutually exclusive then they must be independent

D. Probability $(P \cap Q) \leq$ Probability (P)

Q3.05.2 MCQ 2005 2M

A fair coin is tossed three times in succession. If the first toss produces a head, then the probability of getting exactly two heads in three tosses is

A. $\frac{1}{8}$ **B.** $\frac{1}{2}$ **C.** $\frac{3}{8}$ **D.** $\frac{3}{4}$

Instrumentation Engineering

Q3.26.1 NAT 2026 1M

A data set is given to be $[1, 2, 0, -1, -3, 1, 2, 0, 1]$. The median of the data set is_______. (rounded off to the nearest integer)

Q3.26.2 MCQ 2026 2M

A box contains three apples and two oranges. Two fruits are removed randomly in succession. The probability that the first is an apple and the second is an orange is_______.

A. $\frac{3}{5}$ **B.** $\frac{3}{20}$ **C.** $\frac{3}{10}$ **D.** $\frac{2}{5}$

Q3.25.1 NAT 2025 2M

The probability of a student missing a class is 0.1. In a total number of 10 classes, the probability that the student will not miss more than one class is _____. (rounded off to two decimal places).

Q3.24.1 NAT 2024 1M

Indian Premier League has divided the sixteen cricket teams into two equal pools : Pool-A and Pool-B. Four teams of Pool-A have blue logo jerseys while the rest four have red logo jerseys. Five teams of Pool-B have blue logo jerseys while the rest three have red logo jerseys. If one team from each pool reaches the final, the probability that one team has a blue logo jersey and another has a red logo jersey is _______ (rounded off to one decimal place).

Q3.24.2 NAT 2024 2M

A random variable X has a probability density function

$$f_X(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

The probability of $X > 2$ is _______ (rounded off to three decimal places).

Q3.23.1 NAT 2023 1M

X is a discrete random variable which takes values 0, 1 and 2. The probabilities are $P(X = 0) = 0.25$ and $P(X = 1) = 0.5$. With $E[\cdot]$ denoting the expectation operator, the value of $E[X] - E[X^2]$ is _______ (rounded off to one decimal places).

Q3.23.2 NAT 2023 2M

How many five digit numbers can be formed using the integers 3, 4, 5 and 6 with exactly one digit appearing twice?

Q3.22.1 NAT 2022 1M

A box contains 6 red and 4 blue balls. If three ball are drawn in succession without replacement. Then, the probability of 2$^{\text{nd}}$ and 3$^{\text{rd}}$ ball of having color red is _______.

Q3.20.1 NAT 2020 1M

A player throws a ball at a basket kept at a distance. The probability that the ball falls into the basket in a single attempt is 0.1. The player attempts to throw the ball twice. Considering each attempt to be independent, the probability that this player puts the ball into the basket only in the second attempt (rounded off to two decimal places) is _______.

Q3.20.2 NAT 2020 2M

Consider two identical bags B_1 and B_2, each containing 10 balls of identical shapes and sizes. Bag B_1 contains 7 Red and 3 Green balls, while bag B_2 contains 3 Red and 7 Green balls. A bag is picked at random and a ball is drawn from it, which was found to be Red. The probability that the Red ball came from bag B_1 (rounded off to one decimal place) is _______.

Q3.19.1 MCQ 2019 2M

The function $p(x)$ is given by

$$p(x) = \frac{A}{x^\mu}$$

where A and μ are constants with $\mu > 1$ and $1 \leq x < \infty$ and $p(x) = 0$ for $-\infty < x < 1$. For $p(x)$ to be probability density function, the value of A should be equal to

A. $\mu - 1$ **B.** $\mu + 1$

C. $\frac{1}{(\mu - 1)}$ **D.** $\frac{1}{(\mu + 1)}$

Q3.19.2 MCQ 2019 1M

A box has 8 red balls and 8 green balls. Two balls are drawn randomly in succession from the box without replacement. The probability that the first ball drawn is red and the second ball drawn is green is

A. $\frac{4}{15}$ **B.** $\frac{7}{16}$ **C.** $\frac{1}{2}$ **D.** $\frac{8}{15}$

Q3.18.1 MCQ 2018 1M

X and Y are two independent random variables with variances 1 and 2, respectively. Let $Z = X - Y$. The variance of Z is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q3.18.2 MCQ 2018 1M

Consider a sequence of tossing of a fair coin where the outcomes of tosses are independent. The probability of getting the head for the third time in the fifth toss is

A. $\frac{5}{16}$ **B.** $\frac{3}{16}$ **C.** $\frac{3}{5}$ **D.** $\frac{9}{16}$

Q3.18.3 MCQ 2018 2M

Two bags A and B have equal number of balls. Bag A has 20% red balls and 80% green balls. Bag B has 30% red balls, 60% green balls and 10% yellow balls. Contents of Bags A and B are mixed thoroughly and a ball is randomly picked from the mixture. What is the chance that the ball picked is red?

A. 20% **B.** 25% **C.** 30% **D.** 40%

Q3.17.1 MCQ 2017 2M

The probability that a communication system will have high fidelity is 0.81. The probability that the system will have both high fidelity and high selectivity is 0.18. The probability that a given system with high fidelity will have high selectivity is

A. 0.181 **B.** 0.191 **C.** 0.222 **D.** 0.826

Q3.16.1 MCQ 2016 2M

An urn contains 5 red and 7 green balls. A ball is drawn at random and its colour is noted. The ball is placed back into the urn along with another ball of the same colour. The probability of getting a red ball in the next draw is

A. $\frac{65}{156}$ **B.** $\frac{67}{156}$ **C.** $\frac{79}{156}$ **D.** $\frac{89}{156}$

Q3.15.1 MCQ 2015 2M

The probability that a thermistor randomly picked up from a production unit is defective is 0.1. The probability that out of 10 thermistors, randomly picked up 3 are defective is

A. 0.001 **B.** 0.057 **C.** 0.107 **D.** 0.3

Q3.15.2 NAT 2015 2M

The probability density function of a random variable X is $P_X(x) = e^{-x}$ for $x \geq 0$ and 0 otherwise. The expected value of the function $g_2(x) = e^{3x/4}$ is __________.

Q3.14.1 NAT 2014 1M

The figure shows the schematic of a production process with machines A, B and C. An input job needs to be pre-processed either by A or by B before it is fed to C, from which the final finished product comes out. The probabilities of failure of the machines are given as

$$P_A = 0.15, \qquad P_B = 0.05, \qquad P_C = 0.1.$$

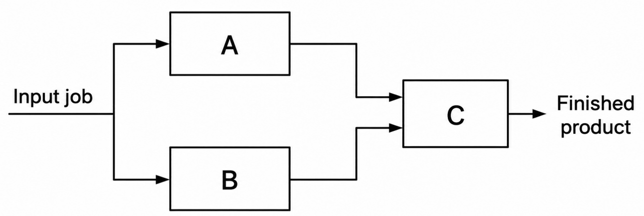


Fig. Q3.14.1

Assuming independence of failures of the machines, the probability that a given job is successfully processed (up to the third decimal place) is __________.

Q3.14.2 NAT 2014 1M

Given that x is a random variable in the range $[0, \infty]$ with a probability density function

$$\frac{e^{-x/2}}{K},$$

the value of the constant K is __________.

Q3.13.1 MCQ 2013 1M

A continuous random variable X has a probability density function

$$f(x) = e^{-x}, \ 0 < x < \infty$$

Then $P(X > 1)$ is

A. 0.368 **B.** 0.5 **C.** 0.632 **D.** 1.0

Q3.12.1 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$ **B.** $\frac{9}{16}$ **C.** $\frac{1}{4}$ **D.** $\frac{2}{3}$

Q3.11.1 MCQ 2011 2M

The box 1 contains chips numbered 3, 6, 9, 12 and 15. The box 2 contains chips numbered 6, 11, 16, 21 and 26. Two chips, one from each box, are drawn at random. The numbers written on these chips are multiplied. The probability for the product to be an even number is

A. $\frac{6}{25}$ **B.** $\frac{2}{5}$ **C.** $\frac{3}{5}$ **D.** $\frac{19}{25}$

Q3.10.1 MCQ 2010 1M

The diameters of 10000 ball bearings were measured. The mean diameter and standard deviation were found to be 10 mm and 0.05 mm respectively. Assuming Gaussian distribution of measurements, it can be expected that the number of measurements more than 10.15 mm will be

A. 230 **B.** 115 **C.** 15 **D.** 2

Q3.09.1 MCQ 2009 2M

A screening test is carried out to detect a certain disease. It is found that 12% of the positive reports and 15% of the negative reports are incorrect. Assuming that the probability of a person getting a positive report is 0.01, the probability that a person tested gets incorrect report is

A. 0.0027 **B.** 0.0173 **C.** 0.1497 **D.** 0.2100

Q3.08.1 MCQ 2008 2M

A random variable is uniformly distributed over the interval 2 to 10. Its variance will be

A. $\frac{16}{3}$ **B.** 6 **C.** $\frac{256}{9}$ **D.** 36

Q3.08.2 MCQ 2008 2M

Consider a Gaussian distributed random variable with zero mean and standard deviation σ. The value of its cumulative distribution function at the origin will be

A. 0 **B.** 0.5 **C.** 1 **D.** 10

Q3.07.1 MCQ 2007 1M

Assume that the duration in minutes of a telephone conversation follows the exponential distribution

$$f(x) = \frac{1}{5}e^{-x/5}, \qquad x \geq 0.$$

The probability that the conversation will exceed five minutes is

A. $\frac{1}{e}$ **B.** $1 - \frac{1}{e}$ **C.** $\frac{1}{e^2}$ **D.** $1 - \frac{1}{e^2}$

Q3.06.1 MCQ 2006 2M

You have gone to a cyber-cafe with a friend. You found that the cyber-cafe has only three terminals. All terminals are unoccupied. You and your friend have to make a random choice of selecting a terminal. What is the probability that both of you will NOT select the same terminal?

A. $\frac{1}{9}$ **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** 1

Q3.06.2 MCQ 2006 2M

Probability density function $p(x)$ of a random variable x is as shown below. The value of α is

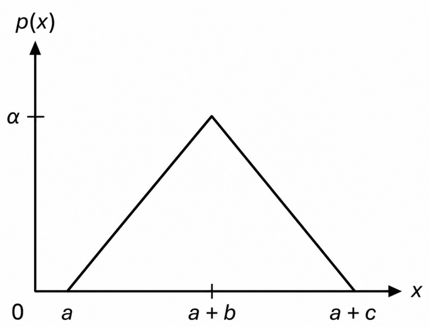


Fig. Q3.06.2

A. $\frac{2}{c}$ **B.** $\frac{1}{c}$
C. $\frac{2}{(b+c)}$ **D.** $\frac{1}{(b+c)}$

Q3.06.3 MCQ 2006 1M

Two dices are rolled simultaneously. The probability that the sum of digits on the top surface of the two dices is even, is

A. 0.5 **B.** 0.25 **C.** 0.167 **D.** 0.125

Q3.05.1 MCQ 2005 1M

The probability that there are 53 Sundays in a randomly chosen leap year is

A. $\frac{1}{7}$ **B.** $\frac{1}{14}$ **C.** $\frac{1}{28}$ **D.** $\frac{2}{7}$

Computer Science & IT

Q3.26.1 MCQ 2026 2M

An unbiased six-faced dice whose faces are marked with numbers $1, 2, 3, 4, 5$, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is __________

A. $\frac{1}{6}$ **B.** $\frac{5}{18}$ **C.** $\frac{7}{18}$ **D.** $\frac{5}{6}$

Q3.26.2 MCQ 2026 1M

An urn contains one red ball and one blue ball. At each step, a ball is picked uniformly at random from the urn, and this ball together with another ball of the same color is put back in the urn. The probability that there are equal number of red and blue balls after two steps is

A. $\frac{1}{4}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{2}$ **D.** $\frac{2}{3}$

Q3.26.3 NAT 2026 2M

Let X be a random variable which takes values in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Further,

$$\Pr(X = 1) = \Pr(X = 2) = \Pr(X = 5) = \Pr(X = 7) = \frac{1}{6}$$

and

$$\Pr(X = 3) = \Pr(X = 4) = \Pr(X = 6) = \Pr(X = 8) = \frac{1}{12}.$$

The expected value of X, denoted by $E[X]$, is equal to __________. (rounded off to two decimal places)

Q3.26.4 MCQ 2026 1M

A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

A. $\frac{1}{4}$ **B.** $\frac{3}{4}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{8}$

Q3.26.5 MCQ 2026 1M

The probability density function $f(x)$ of a random variable X which takes real values is

$$f(x) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{x^2}{18}\right), \qquad x \in (-\infty, +\infty)$$

Which one of the following statements is correct about the random variable X?

A. X is an exponential random variable

B. X is a normal random variable

C. X is a Poisson random variable

D. X is a uniform random variable

Q3.26.6 NAT 2026 2M

Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E_1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E_2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails.
The conditional probability $P(E_1 \mid E_2)$ is equal to __________. (rounded off to one decimal place)

Q3.25.1 NAT 2025 2M

A quadratic polynomial $(x-\alpha)(x-\beta)$ over complex numbers is said to be square invariant if $(x-\alpha)(x-\beta) = (x-\alpha^2)(x-\beta^2)$.. Suppose from the set of all square invariant quadratic polynomials we choose one at random.The probability that the roots of the chosen polynomial are equal is __________(rounded off to one decimal place)

Q3.25.2 NAT 2025 2M

The unit interval $(0, 1)$ is divided at a point chosen uniformly distributed over $(0, 1)$ in $\mathbb{R}$ into two disjoint subintervals.
The expected length of the subinterval that contains 0.4 is __________. (rounded off to two decimal places)

Q3.25.3 NAT 2025 1M

A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(\text{head}) = 0.5$ and for a fake coin, $P(\text{head}) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is __________. (Rounded off to two decimal places)

Q3.25.4 NAT 2025 2M

Consider a probability distribution given by the density function $P(x)$.

$$P(x) = \begin{cases} Cx^2, & \text{for } 1 \le x \le 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$

The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is ________. (Rounded off to three decimal places)

Q3.24.1 MCQ 2024 1M

When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (i.e. $1, 2, 3, 4, 5$ and 6) is

A. $\frac{5}{324}$ **B.** $\frac{11}{324}$ **C.** $\frac{1}{324}$ **D.** $\frac{7}{324}$

Q3.24.2 MCQ 2024 2M

Let x and y be random variables, not necessarily independent, that take real values in the interval $[0, 1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$ respectively. Which one of the following statements is TRUE?

A. $\bar{z} = \bar{x}\bar{y}$ **B.** $\bar{z} \geq \bar{x}\bar{y}$

C. $\bar{z} \leq \bar{x}\bar{y}$ **D.** $\bar{z} \leq \bar{x}$

Q3.24.3 NAT 2024 2M

A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is ________.

Q3.24.4 MSQ 2024 1M

Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statement is/are TRUE?

- [ ] **A.** $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
- [ ] **B.** The two events A and B are independent
- [ ] **C.** $P(A \cup B) = 0.7$
- [ ] **D.** $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B, respectively

Q3.24.5 MCQ 2024 1M

Consider a permutation sampled uniformly at random from the set of all permutations of $1, 2, 3, \ldots, n$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

A. Event X is more likely than event Y

B. The events X and Y are independent

C. Either event X or Y must occur

D. The events X and Y are mutually exclusive

Q3.23.1 MSQ 2023 2M

Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

- [ ] **A.** A and B are independent.
- [ ] **B.** A and C are independent.
- [ ] **C.** B and C are independent.
- [ ] **D.** $\text{Prob}(B \mid C) = \text{Prob}(B)$

Q3.21.1 MCQ 2021 1M

The shape of the cumulative distribution function of Gaussian distribution is

A. Bell-shaped **B.** S-shaped

C. Horizontal line **D.** Straight line at 45 degree angle

Q3.21.2 MCQ 2021 2M

A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

A. $\frac{r}{r+b}$

B. $\frac{r}{r+b+3}$

C. $\left(\frac{r}{r+b}\right)\left(\frac{r+1}{r+b+1}\right)\left(\frac{r+2}{r+b+2}\right)\left(\frac{r+3}{r+b+3}\right)$

D. $\frac{r+3}{r+b+3}$

Q3.21.3 MCQ 2021 2M

In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted.

- If the first question is answered wrong, the student gets zero marks.

- If the first questions is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

Question	Probability of answering correctly	Marks
QuesA	0.8	10
QuesB	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

A. First QuesA and then QuesB. Expected marks 14.

B. First QuesB and then QuesA. Expected marks 22.

C. First QuesB and then QuesA. Expected marks 14.

D. First QuesA and then QuesB. Expected marks 16.

Q3.21.4 NAT 2021 1M

For a given biased coin, the probability that the outcome of a toss is a head is 0.4. This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal places) is ________.

Q3.21.5 NAT 2021 2M

A sender (S) transmits a signal, which can be one of the two kinds: H and L with probability 0.1 and 0.9 respectively, to a receiver (R). In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H, is 0.7.

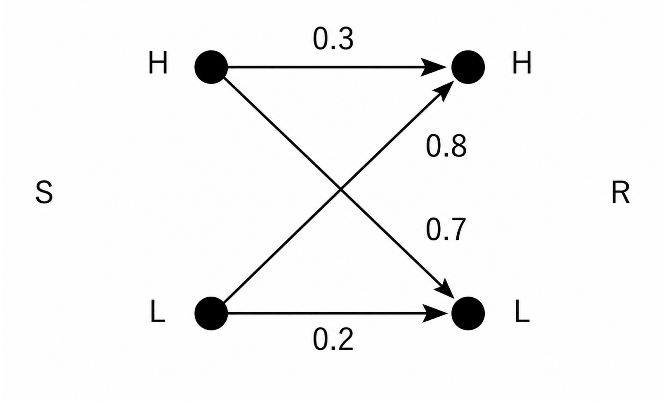


Fig. Q3.21.5

If the received signal is H, the probability that the transmitted signal was H (rounded to 2 decimal places) is ________.

Q3.20.1 NAT 2020 2M

For $n > 2$, let $a \in \{0, 1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0, 1\}^n$. Then, the probability that $\sum_{i=1}^{n} a_i x_i$ is an odd number is ________.

Q3.19.1 NAT 2019 1M

Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \ldots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is ________.

Q3.19.2 NAT 2019 2M

Suppose Y is distributed uniformly in the open interval $(1, 6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) ________.

Q3.18.1 NAT 2018 1M

Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equiprobable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is ________.

Q3.18.2 NAT 2018 2M

Consider Guwahati (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D \mid H_G) = 0.40$. Similarly, the next two entries are $P(M_D \mid H_G) = 0.48$ and $P(L_D \mid H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$ and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is ________.

Q3.17.1 NAT 2017 2M

If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(X+2)^2]$ equals ________.

Q3.17.2 NAT 2017 1M

Let X be a Gaussian random variable with mean 0 and variance σ^2. Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b. The median of Y is ________.

Q3.17.3 MCQ 2017 1M

The probability that a k-digit number does NOT contain the digits 0, 5, or 9 is

A. 0.3^k **B.** 0.6^k **C.** 0.7^k **D.** 0.9^k

Q3.16.1 NAT 2016 1M

A probability density function on the interval $[a, 1]$ is given by $\frac{1}{x^2}$ and outside this interval the value of the function is zero. The value of a is ____.

Q3.16.2 NAT 2016 2M

Consider the following experiment.

Step 1. Flip a fair coin twice.

Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places) ________ .

Q3.16.3 NAT 2016 1M

Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is ________ .

Q3.14.1 NAT 2014 1M

The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p. Then $100p =$ ________.

Q3.14.2 NAT 2014 2M

Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)\,P(B)$ is _____.

Q3.14.3 NAT 2014 2M

The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is ____.

Q3.14.4 NAT 2014 2M

Four fair six-sided dice are rolled. The probability that the sum of the results being 22 is $X/1296$.

The value of X is ____.

Q3.14.5 NAT 2014 1M

Each of the nine words in the sentence "The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is __________. (The answer should be rounded to one decimal place.)

Q3.14.6 NAT 2014 1M

Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is ______.

Q3.13.1 MCQ 2013 1M

Suppose p is the number of cars per minute passing through a certain road junction between 5 PM, and p has Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

A. $8/(2e^3)$ **B.** $9/(2e^3)$

C. $17/(2e^3)$ **D.** $26/(2e^3)$

Q3.12.1 MCQ 2012 1M

Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

A. 0 and 0.5 **B.** 0 and 1

C. 0.5 and 1 **D.** 0.25 and 0.75

Q3.12.2 MCQ 2012 2M

Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2 or 3 the die is rolled a second time. What is the probability that the sum of total values that turn up is at least 6?

A. $10/21$ **B.** $5/12$ **C.** $2/3$ **D.** $1/6$

Q3.11.1 MCQ 2011 1M

If the difference between the expectation of the square of a random variable ($E[x^2]$) and the square of the expectation of the random variable $(E[x])^2$ is denoted by R, then

A. $R = 0$ **B.** $R < 0$ **C.** $R \geq 0$ **D.** $R > 0$

Q3.11.2 MCQ 2011 2M

Consider the finite sequence of random values $X = [x_1, x_2, \ldots, x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X. Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constant. Let μ_y be the mean and σ_y be the standard deviation of this sequence. Which one of the following statements INCORRECT?

A. Index position of mode of X in X is the same as the index position of mode of Y in Y.

B. Index position of median of X in X is the same as the index position of median of Y in Y.

C. $\mu_y = a\mu_x + b$

D. $\sigma_y = a\sigma_x + b$

Q3.11.3 MCQ 2011 2M

A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

A. $1/5$ **B.** $4/25$ **C.** 1.4 **D.** $2/5$

Q3.10.1 MCQ 2010 2M

Consider a company that assembles computers. The probability of a faulty assembly of any computer is p. The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q. What is the probability of a computer being declared faulty?

A. $pq + (1-p)(1-q)$ **B.** $(1-q)p$

C. $(1-p)q$ **D.** pq

Q3.09.1 MCQ 2009 2M

An unbalanced dice (with 6 faces, numbered from 1 to 6) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same.
If the probability that the face is even given that it is greater than 3 is 0.75, which one of the following options is closest to the probability that the face value exceeds 3?

A. 0.453 **B.** 0.468 **C.** 0.485 **D.** 0.492

Q3.09.2 MCQ 2009 1M

If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

A. $1/3$ **B.** $1/4$ **C.** $1/2$ **D.** $2/3$

Q3.08.1 MCQ 2008 2M

Let X be a random variable following normal distribution with mean $+1$ and variance 4. Let Y be another normal variable with mean -1 and variance unknown,. If $P(X \le -1) = P(Y \ge 2)$ the standard deviation of Y is

A. 3 **B.** 2 **C.** $\sqrt{2}$ **D.** 1

Q3.08.2 MCQ 2008 2M

Aishwarya studies either computer science or mathematics everyday. if she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

A. 0.24 **B.** 0.36 **C.** 0.4 **D.** 0.6

Q3.07.1 MCQ 2007 2M

Suppose we uniformly and randomly select a permutation from the 20! permutations of $1, 2, 3, \ldots, 20$. What is the probability that 2 appears at an earlier position that any other even number in the selected permutation?

A. $\frac{1}{2}$ **B.** $\frac{1}{10}$

C. $\frac{9!}{20!}$ **D.** None of these

Q3.06.1 MCQ 2006 2M

For each element in a set of size $2n$, an unbiased coin is tossed. All the $2n$ coin tossed are independent. An element is chosen if the corresponding coin toss were head. The probability that exactly n elements are chosen is

A. $\binom{2n}{n}/4^n$ **B.** $\binom{2n}{n}/2^n$ **C.** $1/\binom{2n}{n}$ **D.** $\frac{1}{2}$

Q3.05.1 MCQ 2005 1M

Let $f(x)$ be the continuous probability density function of a random variable X. The probability that $a < X \le b$, is

A. $f(b-a)$ **B.** $f(b) - f(a)$

C. $\int_a^b f(x)\,dx$ **D.** $\int_a^b x f(x)\,dx$

Q3.04.1 MCQ 2004 2M

A point is randomly selected with uniform probability in the X-Y plane within the rectangle with corners at $(0,0)$, $(1,0)$, $(1,2)$ and $(0,2)$. If p is the length of the position vector of the point, the expected value of p^2 is

A. $2/3$ **B.** 1 **C.** $4/3$ **D.** $5/3$

Q3.04.2 MCQ 2004 2M

An examination paper has 150 multiple-choice questions of one mark each, with each question having four choices. Each incorrect answer fetches -0.25 mark. Suppose 1000 students choose all their answers randomly with uniform probability. The sum total of the expected marks obtained by all these students is

A. 0 **B.** 2550 **C.** 7525 **D.** 9375

Q3.04.3 MCQ 2004 1M

If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

A. $3/8$ **B.** $1/2$ **C.** $5/8$ **D.** $3/4$

Q3.04.4 MCQ 2004 2M

Two n bit binary strings, S_1 and S_2 are chosen randomly with uniform probability. The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d is

A. ${}^nC_d/2^n$ **B.** ${}^nC_d/2^d$

C. $d/2^n$ **D.** $1/2^d$

Q3.03.1 MCQ 2003 2M

A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

A. $f_1(t) + f_2(t)$

B. $\int_0^t f_1(x) f_2(x)\, dx$

C. $\int_0^t f_1(x) f_2(t-x)\, dx$

D. $\max\{f_1(t),\ f_2(t)\}$

Q3.03.2 MCQ 2003 1M

Let $P(E)$ denote the probability of the event E. Given $P(A) = 1$, $P(B) = 1/2$, the values of $P(A/B)$ and $P(B/A)$ respectively are

A. $1/4,\ 1/2$

B. $1/2,\ 1/4$

C. $1/2,\ 1$

D. $1,\ 1/2$

Mechanical Engineering

Q3.26.1 MCQ 2026 2M

A student council of 10 members consists of two students from engineering school, three students from science school and five students from arts school. The university administration selects three students from the council at random. What is the chance that out of the selected students, two belong to the same school and the third belongs to different school?

A. $\frac{79}{120}$ **B.** $\frac{2}{120}$ **C.** $\frac{89}{120}$ **D.** $\frac{69}{120}$

Q3.25.1 MCQ 2025 1M

If two unbiased coins are tossed, then what is the probability of having at least one head?

A. 0.25 **B.** 0.5 **C.** 0.675 **D.** 0.75

Q3.24.1 NAT 2024 2M

Let X be a continuous random variable defined on $[0, 1]$ such that its probability density function $f(x) = 1$ for $0 \le x \le 1$ and 0 otherwise. Let $Y = \log_e(X + 1)$. Then the expected value of Y is ________ (rounded off to 2 decimal places).

Q3.23.1 MCQ 2023 1M

A machine produces a defective component with a probability of 0.015. The number of defective components in a packed box containing 200 components produced by the machine follows a Poisson distribution. The mean and the variance of the distribution are

A. 3 and 3, respectively

B. $\sqrt{3}$ and $\sqrt{3}$ respectively

C. 0.015 and 0.015, respectively

D. 3 and 9, respectively

Q3.22.1 NAT 2022 2M

Let a random variable x follows poisson's distribution such that $\text{prob}(x = 1) = \text{prob}(x = 2)$. The value of $\text{prob}(x = 3)$ is ________. (Round off to 2 decimal places)

Q3.21.1 MCQ 2021 1M

The Mean and variance, respectively, of a binomial distribution for n independent trials with the probability of success as p, are [Set-02]

A. $\sqrt{np},\ np(1-2p)$

B. $np,\ np$

C. $np,\ np(1-p)$

D. $\sqrt{np},\ \sqrt{np(1-p)}$

Q3.20.1 NAT 2020 1M

A fair coin is tossed 20 times. The probability that 'head' will appear exactly 4 times in the first ten tosses, and 'tail' will appear exactly 4 times in the next ten tosses is ________ (round off to 3 decimal places)

Q3.20.2 NAT 2020 1M

A company is hiring to fill four managerial vacancies. The candidates are five men and three women. If every candidate is equally likely to be chosen then the probability that at least one women will be selected is ________ (round off to 2 decimal places).

Q3.20.3 NAT 2020 2M

Consider two exponentially distributed random variables X and Y, both having a mean of 0.50. Let $Z = X + Y$ and r be the correlation coefficient between X and Y. If the variance of Z equals 0, then the value of r is ________ (round off to 2 decimal places).

Q3.19.1 MCQ 2019 2M

The variable X takes a value between 0 and 10 with uniform probability distribution. The variable Y takes a value between 0 and 20 with uniform probability distribution. The probability of the sum of variable $(X + Y)$ being greater than 20 is

A. 0.25 **B.** 0.5 **C.** 0 **D.** 0.33

Q3.19.2 NAT 2019 1M

If x is the mean of data $3, x, 2, 4$ then the mode is ________.

Q3.19.3 NAT 2019 2M

The probability that a part manufactured by a company will be defective is 0.05. If 15 such parts are selected randomly and inspected, then the probability that atleast two parts will be defective is ________.

Q3.18.1 MCQ 2018 1M

Four red balls, four green balls and four blue balls are put in a box. Three balls are pulled out of the box at random one after another without replacement. The probability that all the three balls are red is

A. $\frac{1}{72}$ **B.** $\frac{1}{55}$ **C.** $\frac{1}{36}$ **D.** $\frac{1}{27}$

Q3.18.2 MCQ 2018 1M

A six-faced fair dice is rolled five times. Then probability (in %) of obtaining 'ONE' at least four times is

A. 33.3 **B.** 3.33 **C.** 0.33 **D.** 0.0033

Q3.18.3 MCQ 2018 2M

Let X_1, X_2 be two normal random variable with means μ_1, μ_2 and standard deviation σ_1, σ_2 respectively. Consider $Y = X_1 - X_2$; $\mu_1 = \mu_2 = 1, \sigma_1 = 1, \sigma_2 = 2$ then,

A. Y is normal distributed random with mean 0 and variance 1.

B. Y is normal distributed with mean 0 and variance 5.

C. Y has mean 0 and variance 5, but is NOT normally distributed.

D. Y has mean 0 and variance 1, but is NOT normally distributed.

Q3.18.4 MCQ 2018 2M

Let X_1 and X_2 be two independent exponentially distributed random variables with means 0.5 and 0.25 respectively. Then $Y = \min(X_1, X_2)$ is

A. Exponentially distributed with mean 1/6.

B. Exponentially distributed with mean 2.

C. Normally distributed with mean 3/4.

D. Normally distributed with mean 1/6.

Q3.17.1 NAT 2017 1M

Two coins are tossed simultaneously. The probability (upto two decimal points accuracy) of getting at least one head is ____________.

Q3.17.2 MCQ 2017 1M

A sample of 15 data is as follows: 17, 18, 17, 17, 13, 18, 5, 5, 6, 7, 8, 9, 20, 17, 3. The mode of the data is

A. 4 **B.** 13 **C.** 17 **D.** 20

Q3.17.3 NAT 2017 1M

A six-face fair dice is rolled a large number of times. The mean value of the outcomes is ________.

Q3.16.1 MCQ 2016 1M

Consider a Poisson distribution for the tossing of a biased coin. The mean for this distribution is μ. The standard deviation for this distribution is given by

A. $\sqrt{\mu}$ **B.** μ^2 **C.** μ **D.** $\frac{1}{\mu}$

Q3.16.2 NAT 2016 2M

The probability that a screw manufactured by a company is defective is 0.1. The company sells screws in packets containing 5 screws and gives a guarantee of replacement if one or more screws in the packet are found to be defective. The probability that a packet would have to be replaced is ________.

Q3.16.3 NAT 2016 1M

The area (in percentage) under standard normal distribution curve of random variable Z within limits from -3 to $+3$ is __________.

Q3.16.4 MCQ 2016 1M

Three cards were drawn from a pack of 52 cards. The probability that they are a king, a queen, and a jack is

A. $\frac{16}{5525}$ **B.** $\frac{64}{2197}$ **C.** $\frac{3}{13}$ **D.** $\frac{8}{16575}$

Q3.15.1 MCQ 2015 2M

The probability of obtaining at least two "SIX" in throwing a fair dice 4 times is

A. $\frac{425}{432}$ **B.** $\frac{19}{144}$ **C.** $\frac{13}{144}$ **D.** $\frac{125}{432}$

Q3.15.2 MCQ 2015 1M

If $P(X) = \frac{1}{4}$, $P(Y) = \frac{1}{3}$, and $P(X \cap Y) = \frac{1}{12}$, the value of $P(Y|X)$ is

A. $\frac{1}{4}$ **B.** $\frac{4}{25}$ **C.** $\frac{1}{3}$ **D.** $\frac{29}{50}$

Q3.15.3 MCQ 2015 1M

Among the four normal distribution with probability density functions as shown below, which one has the lowest variance?

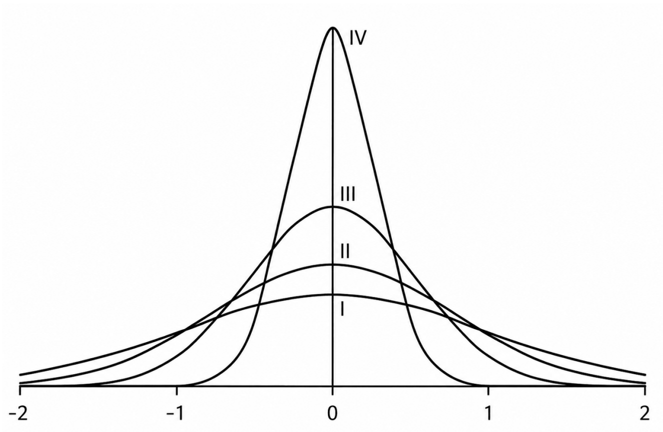


Fig. Q3.15.3

A. I **B.** II **C.** III **D.** IV

Q3.15.4 NAT 2015 1M

Three vendors were asked to supply a very high precision component. The respective probabilities of their meeting the strict design specifications are 0.8, 0.7 and 0.5. Each vendor supplies one component. The probability that out of total three components supplied by the vendors, at least one will meet the design specification is ___________.

Q3.15.5 MCQ 2015 2M

The chance of a student passing an exam is 20%. The chance of a student passing the exam and getting above 90% marks in it is 5%. Given that a student passes the examination, the probability that the student gets above 90% marks is

A. $\frac{1}{18}$ **B.** $\frac{1}{4}$ **C.** $\frac{2}{9}$ **D.** $\frac{5}{18}$

Q3.14.1 MCQ 2014 2M

In the following table, x is a discrete random variable and $p(x)$ is the probability density. The standard deviation of x is

x	1	2	3
$p(x)$	0.3	0.6	0.1

A. 0.18 **B.** 0.36 **C.** 0.54 **D.** 0.6

Q3.14.2 MCQ 2014 2M

A machine produces 0, 1 or 2 defective pieces in a day with associated probability of $\frac{1}{6}$, $\frac{2}{3}$ and $\frac{1}{6}$, respectively. The mean value and the variance of the number of defective pieces produced by the machine in a day, respectively, are

A. 1 and $\frac{1}{3}$ **B.** $\frac{1}{3}$ and 1

C. 1 and $\frac{4}{3}$ **D.** $\frac{1}{3}$ and $\frac{4}{3}$

Q3.14.3 NAT 2014 2M

Consider an unbiased cubic dice with opposite faces coloured identically and each face coloured red, blue or green such that each colour appears only two times on the dice. If the dice is thrown thrice, the probability of obtaining red colour on top face of the dice at least two times is ___________.

Q3.14.4 MCQ 2014 2M

The number of accidents occurring in a plant in a month follows Poisson distribution with mean as 5.2. The probability of occurrence of less than 2 accidents in the plant during a randomly selected month is

A. 0.029 **B.** 0.034 **C.** 0.039 **D.** 0.044

Q3.14.5 MCQ 2014 1M

A box contains 25 parts of which 10 are defective. Two parts are being drawn simultaneously in a random manner from the box. The probability of both the parts being good is

A. $\frac{7}{20}$ **B.** $\frac{42}{125}$ **C.** $\frac{25}{29}$ **D.** $\frac{5}{9}$

Q3.14.6 NAT 2014 2M

A group consists of equal number of men and women. Of this group 20% of the men and 50% of the women are unemployed. If a person is selected at random from this group, the probability of the selected person being employed is ___________.

Q3.14.7 NAT 2014 1M

Demand during lead time with associated probabilities is shown below

Demand	50	70	75	80	85
Probability	0.15	0.14	0.21	0.20	0.30

Expected demand during lead time is ___________

Q3.14.8 NAT 2014 2M

A nationalized bank has found that the daily balance available in its savings accounts follows a normal distribution with a mean of Rs. 500 and a standard deviation of Rs. 50. The percentage of savings account holders, who maintain an average daily balance more than Rs. 500 is ___________

Q3.14.9 NAT 2014 1M

A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch accepted is ___________

Q3.13.1 MCQ 2013 1M

Let X be a normal random variable with mean 1 and variance 4. The probability $P\{X < 0\}$ is

A. 0.5

B. greater than zero and less than 0.5

C. greater than 0.5 and less than 1.0

D. 1.0

Q3.13.2 MCQ 2013 2M

Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

A. $\frac{13}{90}$ **B.** $\frac{12}{90}$ **C.** $\frac{78}{90}$ **D.** $\frac{77}{90}$

Q3.13.3 MCQ 2013 2M

The probability that a student knows the correct answer to a multiple choice question is $\frac{2}{3}$. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is $\frac{1}{4}$. Given that the student has answered the question correctly, the conditional probability that the student knows the correct answer is

A. $\frac{2}{3}$ **B.** $\frac{3}{4}$ **C.** $\frac{5}{6}$ **D.** $\frac{8}{9}$

Q3.12.1 MCQ 2012 2M

A box contains 4 red balls and 6 black balls. Three balls are selected randomly from the box one after another, without replacement. The probability that the selected set contains one red ball and two black balls is

A. $\frac{1}{20}$ **B.** $\frac{1}{12}$ **C.** $\frac{3}{10}$ **D.** $\frac{1}{2}$

Q3.12.2 MCQ 2012 2M

An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 **B.** 0.334 **C.** 0.667 **D.** 0.720

Q3.11.1 MCQ 2011 2M

An unbiased coin is tossed five times. The outcome of each toss is either a head or a tail. The probability of getting at least one head is

A. $\frac{1}{32}$ **B.** $\frac{13}{32}$ **C.** $\frac{16}{32}$ **D.** $\frac{31}{32}$

Q3.10.1 MCQ 2010 2M

A box contains 2 washers, 3 nuts and 4 bolts. Items are drawn from the box at random one at a time without replacement. The probability of drawing 2 washers first followed by 3 nuts and subsequently the 4 bolts is

A. $\frac{2}{315}$ **B.** $\frac{1}{630}$ **C.** $\frac{1}{1260}$ **D.** $\frac{1}{2520}$

Q3.09.1 MCQ 2009 1M

If three coins are tossed simultaneously, the probability of getting at least one head is

A. $\frac{1}{8}$ **B.** $\frac{3}{8}$ **C.** $\frac{1}{2}$ **D.** $\frac{7}{8}$

Q3.09.2 MCQ 2009 2M

The standard deviation of a uniformly distributed random variable between 0 and 1 is

A. $\frac{1}{\sqrt{12}}$ **B.** $\frac{1}{\sqrt{3}}$ **C.** $\frac{5}{\sqrt{12}}$ **D.** $\frac{7}{\sqrt{12}}$

Q3.08.1 MCQ 2008 1M

A coin is tossed 4 times. What is the probability of getting heads exactly 3 times?

A. $\frac{1}{4}$ **B.** $\frac{3}{8}$ **C.** $\frac{1}{2}$ **D.** $\frac{3}{4}$

Q3.07.1 MCQ 2007 2M

Let X and Y be two independent random variables. Which one of the relations between expectation (E), variance (Var) and covariance (Cov) given below is False?

A. $E(XY) = E(X)\,E(Y)$

B. $\mathrm{Cov}(X, Y) = 0$

C. $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

D. $E(X^2Y^2) = (E(X))^2\,(E(Y))^2$

Q3.06.1 MCQ 2006 1M

A box contains 20 defective items and 80 non-defective items. If two items are selected at random without replacement, what will be the probability that both the items are defective?

A. $\frac{1}{5}$ **B.** $\frac{1}{25}$ **C.** $\frac{20}{99}$ **D.** $\frac{19}{495}$

Q3.06.2 MCQ 2006 2M

Consider the continuous random variable with probability density function

$$f(t) = 1 + t \quad \text{for } -1 \le t \le 0$$

$$= 1 - t \quad \text{for } 0 \le t \le 1$$

The standard deviation of the random variable is

A. $\frac{1}{\sqrt{3}}$ **B.** $\frac{1}{\sqrt{6}}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{6}$

Q3.05.1 MCQ 2005 1M

A lot has 10% defective items. Ten items are chosen randomly from this lot. The probability that exactly 2 of the chosen items are defective is

A. 0.0036 **B.** 0.1937 **C.** 0.2234 **D.** 0.3874

Q3.05.2 MCQ 2005 2M

A single die is thrown twice. What is the probability that the sum is neither 8 nor 9?

A. $\frac{1}{9}$ **B.** $\frac{5}{36}$ **C.** $\frac{1}{4}$ **D.** $\frac{3}{4}$

Q3.04.1 MCQ 2004 2M

The following data about the flow of liquid was observed in a continuous chemical process plant

Flow rate (litres/sec)	7.5 to 7.7	7.7 to 7.9	7.9 to 8.1	8.1 to 8.3	8.3 to 8.5	8.5 to 8.7
Frequency	1	5	35	17	12	10

Mean flow rate of the liquid is

A. 8.00 litres/sec **B.** 8.06 litres/sec

C. 8.16 litres/sec **D.** 8.26 litres/sec

Q3.04.2 MCQ 2004 2M

From a pack of regular playing cards, two cards are drawn at random. What is the probability that both cards will be Kings, if first card is NOT replaced?

A. $\frac{1}{26}$ **B.** $\frac{1}{52}$ **C.** $\frac{1}{169}$ **D.** $\frac{1}{221}$

Q3.03.1 MCQ 2003 2M

A box contains 5 black and 5 red balls. Two balls are randomly picked one after another from the box, without replacement. The probability for both balls being red is

A. $\frac{1}{90}$ **B.** $\frac{1}{2}$ **C.** $\frac{19}{90}$ **D.** $\frac{2}{9}$

Q3.03.2 MCQ 2003 2M

Let $P(E)$ denote the probability of an event E. Given $P(A) = 1$, $P(B) = \frac{1}{2}$, the values of $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$ respectively are

A. $\frac{1}{4}, \frac{1}{2}$ **B.** $\frac{1}{2}, \frac{1}{4}$ **C.** $\frac{1}{2}, 1$ **D.** $1, \frac{1}{2}$

Q3.02.1 MCQ 2002 1M

Two dice are thrown. What is the probability that the sum of the numbers on the two dice is eight?

A. $\frac{5}{36}$ **B.** $\frac{5}{18}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{3}$

Q3.02.2 MCQ 2002 2M

Manish has to travel from A to D changing buses at stops B and C enroute. The maximum waiting time at either stop can be 8 minutes each, but any time of waiting up to 8 minutes is equally likely at both places. He can afford up to 13 minutes of total waiting time if he is to arrive at D on time. What is the probability that Manish will arrive late at D?

A. $\frac{8}{13}$ **B.** $\frac{13}{64}$ **C.** $\frac{119}{128}$ **D.** $\frac{9}{128}$

Q3.02.3 MCQ 2002 2M

Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between successive arrivals. The length of a phone call is distributed exponentially with mean 3 minutes. The probability that an arrival does not have to wait before service is

A. 0.3 **B.** 0.5 **C.** 0.7 **D.** 0.9

Q3.01.1 MCQ 2001 2M

An unbiased coin is tossed three times. The probability that the head turns up in exactly two cases is

A. $\frac{1}{9}$ **B.** $\frac{1}{8}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{8}$

Q3.00.1 MCQ 2000 1M

E_1 and E_2 are events in a probability space satisfying the following constants $P(E_1) = P(E_2)$, $P(E_1 \cup E_2) = 1$, E_1 and E_2 are independent then $P(E_1) =$ __________.

A. 0 **B.** $\frac{1}{4}$ **C.** $\frac{1}{2}$ **D.** 1

Q3.00.2 MCQ 2000 2M

In a manufacturing plant, the probability of making a defective bolt is 0.1. The mean and standard deviation of defective bolts in a total of 900 bolts are respectively

A. 90 and 9 **B.** 9 and 90

C. 18 and 9 **D.** 9 and 81

Q3.99.1 MCQ 1999 2M

At a production machine, parts arrive according to a Poisson process at the rate of 0.35 parts per minute. Processing time for parts have exponential distribution with mean of 2 minutes. What is the probability that a random part arrival finds that there are already 8 parts in the system (in machine + in queue)?

A. 0.0247 **B.** 0.0576 **C.** 0.0173 **D.** 0.082

Q3.98.1 MCQ 1998 1M

The probability that two friends share the same birth-month is:

A. $\frac{1}{6}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{144}$ **D.** $\frac{1}{24}$

Q3.97.1 MCQ 1997 2M

A box contains 5 black balls and 3 red balls. A total of three balls are picked from the box one after another, without replacing them back. The probability of getting two black balls and one red ball is

A. $\frac{3}{8}$ **B.** $\frac{2}{15}$ **C.** $\frac{15}{28}$ **D.** $\frac{1}{2}$

Q3.96.1 MCQ 1996 2M

The probability of a defective piece being produced in a manufacturing process is 0.01. The probability that out of 5 successive pieces, only one is defective, is

A. $(0.99)^4(0.01)$ **B.** $(0.99)(0.01)^4$

C. $5 \times (0.99)(0.01)^4$ **D.** $5 \times (0.99)^4(0.01)$

Q3.93.1 MCQ 1993 1M

If 20 percent managers are technocrats, the probability that a random committee of 5 managers consists of exactly 2 technocrats is:

A. 0.2048 **B.** 0.4000

C. 0.4096 **D.** 0.9421

Civil Engineering

Q3.26.1 NAT 2026 1M

Bag I contains 4 white and 6 black balls.
Bag II contains 4 white and 3 black balls.
One ball is drawn at random from any one of the two bags and it is found to be a black ball. The probability that the black ball was drawn from Bag I is ________ (rounded off to two decimal places).

Q3.26.2 NAT 2026 2M

The age (in years) of a population is normally distributed with a mean of 36 and standard deviation of 12. The height (in cm) of the same population is also normally distributed with a mean of 160 and standard deviation of 10.
If the probability of age greater than 50 years is equal to the probability of height greater than h, the value of h (in cm) is ________ (rounded off to two decimal places).

Q3.26.3 NAT 2026 1M

The probability (in %) that a storm having return period of 15 years may occur in the next 10 years is ________ (rounded off to two decimal places).

Q3.25.1 NAT 2025 2M

A one-way, single lane road has traffic that consists of 30% trucks and 70% cars. The speed of trucks (in km/h) is a uniform random variable on the interval $(30, 60)$, and the speed of cars (in km/h) is a uniform random variable on the interval $(40, 80)$. The speed limit on the road is 50 km/h. The percentage of vehicles that exceed the speed limit is ________ (rounded off to 1 decimal place).
Note: X is a uniform random variable on the interval (α, β), if its probability density function is given by

$$f(x) = \begin{cases} \dfrac{1}{\beta - \alpha} & \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases}$$

Q3.25.2 MCQ 2025 1M

X is a random variable that can take any one of the values 0, 1, 7, 11, and 12. The probability mass function for X is
$P(X = 0) = 0.4$; $P(X = 1) = 0.3$; $P(X = 7) = 0.1$;
$P(X = 11) = 0.1$; $P(X = 12) = 0.1$
Then, the variance of X is

A. 20.81 **B.** 28.40 **C.** 31.70 **D.** 10.89

Q3.25.3 MSQ 2025 1M

Given that A and B are not null sets, which of the following statements regarding probability is/are CORRECT?

- [ ] **A.** $P(A \cap B) = P(A)\,P(B)$, if A and B are mutually exclusive.
- [ ] **B.** Conditional probability, $P(A|B) = 1$ if $B \subset A$.
- [ ] **C.** $P(A \cup B) = P(A) + P(B)$, if A and B are mutually exclusive.
- [ ] **D.** $P(A \cap B) = 0$, if A and B are independent.

Q3.25.4 NAT 2025 2M

Consider a discrete random variable X whose probabilities are given below. The standard deviation of the random variable is ________ (round off to one decimal place).

x_i	1	2	4	8
$P(X = x_i)$	0.3	0.1	0.3	0.3

Q3.24.1 MCQ 2024 2M

In a sample of 100 heart patients, each patient has 80% chance of having a heart attack without medicine X. It is clinically known that medicine X reduces the probability of having a heart attack by 50%. Medicine X is taken by 50 of these 100 patients. The probability that a randomly selected patient, out of the 100 patients, takes medicine X and has a heart attack is

A. 40% **B.** 60% **C.** 20% **D.** 30%

Q3.24.2 MCQ 2024 1M

The probability that a student passes only in Mathematics is $\frac{1}{3}$. The probability that the student passes only in English is $\frac{4}{9}$. The probability that the student passes in both of these subjects is $\frac{1}{6}$. The probability that the student will pass in at least one of these two subjects is

A. $\frac{17}{18}$ **B.** $\frac{11}{18}$ **C.** $\frac{14}{18}$ **D.** $\frac{1}{18}$

Q3.23.1 MCQ 2023 1M

A remote village has exactly 1000 vehicles with sequential registration numbers starting from 1000. Out of the total vehicles, 30% are without pollution clearance certificate. Further, even- and odd-numbered vehicles are operated on even- and odd-numbered dates, respectively.
If 100 vehicles are chosen at random on an even-numbered date, the number of vehicles expected without pollution clearance certificate is ________.

A. 15 **B.** 30 **C.** 50 **D.** 70

Q3.23.2 NAT 2023 1M

The probabilities of occurrences of two independent events A and B are 0.5 and 0.8, respectively. What is the probability of occurrence of at least A or B (rounded off to one decimal place)? ________

Q3.23.3 MCQ 2023 1M

Which of the following probability distribution functions (PDFs) has the mean greater than the median?

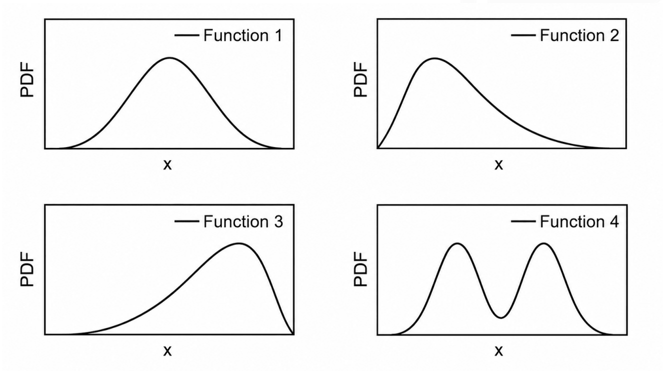


Fig. Q3.23.3

A. Function 1 **B.** Function 2

C. Function 3 **D.** Function 4

Q3.22.1 NAT 2022 2M

At a traffic intersection, cars and buses arrive randomly according to independent Poisson processes at an average rate of 4 vehicles per hour and 2 vehicles per hour, respectively. The probability of observing at least 2 vehicles in 30 minutes is ________. (*round off to two decimal places*)

Q3.22.2 NAT 2022 2M

A pair of six-faced dice is rolled thrice. The probability that the sum of the outcomes in each roll equals 4 in exactly two of the three attempts is ________. (*round off to three decimal places*)

Q3.22.3 NAT 2022 2M

A set of observations of independent variable (x) and the corresponding dependent variable (y) is given below.

x	5	2	4	3
y	16	10	13	12

Based on the data, the coefficient a of the linear regression model

$$y = a + bx$$

is estimated as 6.1.
The coefficient b is ________. (*round off to one decimal place*)

Q3.21.1 NAT 2021 2M

Vehicular arrival at an isolated intersection follows the Poisson distribution. The mean of vehicular arrival rate is 2 vehicle per minute. The probability (*round off to two decimal places*) that at least 2 vehicles will arrive in any given 1-minute interval is ________

Q3.21.2 MCQ 2021 1M

The shape of the cumulative distribution function of Gaussian distribution is

A. Bell-shaped

B. S-shaped

C. Horizontal line

D. Straight line at 45 degree angle

Q3.20.1 NAT 2020 1M

The probability that a 50 year flood may NOT occur at all during 25 years life of a project (*round off to two decimal places*), is ________.

Q3.20.2 NAT 2020 1M

A fair (unbiased) coin is tossed 15 times. The probability of getting exactly 8 Heads (*round off to three decimal places*), is ________.

Q3.19.1 MCQ 2019 2M

The probability density function of a continuous random variable distributed uniformly between x and y (for $y > x$) is

A. $\frac{1}{x-y}$ **B.** $\frac{1}{y-x}$ **C.** $x-y$ **D.** $y-x$

Q3.19.2 NAT 2019 2M

Traffic on a highway is moving at a rate of 360 vehicles per hour at a location. If the number of vehicles arriving on this highway follows Poisson distribution, the probability (round off to 2 decimal places) that the headway between successive vehicles lies between 6 and 10 seconds is ______.

Q3.18.1 NAT 2018 1M

Probability (up to one decimal place) of consecutively picking 3 red balls without replacement from a box containing 5 red balls and 1 white ball is ______.

Q3.18.2 MCQ 2018 1M

A probability distribution with right skew is shown in the figure.

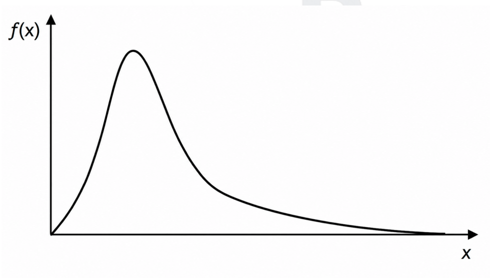


Fig. Q3.18.2

The correct statement for the probability distribution is

A. Mean is equal to mode

B. Mean is greater than median but less than mode

C. Mean is greater than median and mode

D. Mode is greater than median

Q3.18.3 MCQ 2018 1M

The graph of a function $f(x)$ is shown in the figure.

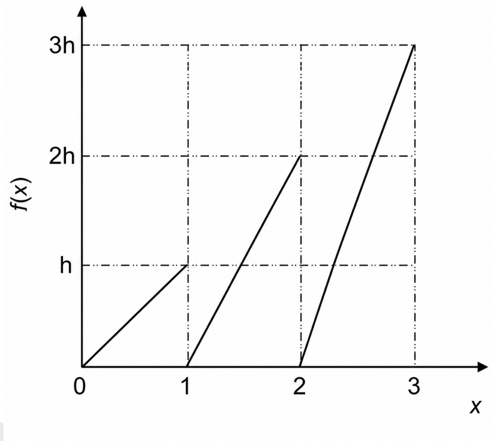


Fig. Q3.18.3

For $f(x)$ to be a valid probability density function, the value of h is

A. 1/3 **B.** 2/3 **C.** 1 **D.** 3

Q3.18.4 NAT 2018 1M

The frequency distribution of the compressive strength of 20 concrete cube specimens is given in the table.

f (MPa)	Number of specimens with compressive strength equal to f
23	4
28	2
22.5	5
31	5
29	4

If μ is the mean strength of the specimens and σ is the standard deviation, the number of specimens (out of 20) with compressive strength less than $\mu - 3\sigma$ is ______.

Q3.17.1 NAT 2017 1M

For a construction project, the mean and standard deviation of the completion time are 200 days and 6.1 days, respectively. Assume normal distribution and use the value of standard normal deviate $z = 1.64$ for the 95% confidence level. The maximum time required (in days) for the completion of the project would be ________.

Q3.17.2 NAT 2017 1M

A two-faced fair coin has its faces designated as head (H) and tail (T). This coin is tossed three times in succession to record the following outcomes: H, H, H. If the coin is tossed one more time, the probability (up to one decimal place) of obtaining H again, given the previous realizations of H, H and H, would be ________.

Q3.17.3 MCQ 2017 2M

For the function $f(x) = a + bx$, $0 \le x \le 1$, to be a valid probability density function, which one of the following statements is correct?

A. $a = 1,\ b = 4$ **B.** $a = 0.5,\ b = 1$

C. $a = 0,\ b = 1$ **D.** $a = 1,\ b = -1$

Q3.17.4 NAT 2017 1M

Vehicles arriving at an intersection from one of the approach road follow the Poisson distribution. The mean rate of arrival is 900 vehicles per hour. If a gap is defined as the time difference between two successive vehicle arrivals (with vehicles assumed to be points), the probability (up to four decimal places) that the gap is greater than 8 seconds is ________.

Q3.16.1 MCQ 2016 2M

If $f(x)$ and $g(x)$ are two probability density functions,

$$f(x) = \begin{cases} \frac{x}{a} + 1, & -a \le x < 0 \\ -\frac{x}{a} + 1, & 0 \le x \le a \\ 0, & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} -\frac{x}{a}, & -a \le x < 0 \\ \frac{x}{a}, & 0 \le x \le a \\ 0, & \text{otherwise} \end{cases}$$

Which one of the following statements is true?

A. Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are same

B. Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are different

C. Mean of $f(x)$ and $g(x)$ are different; Variance of $f(x)$ and $g(x)$ are same

D. Mean of $f(x)$ and $g(x)$ are different; Variance of $f(x)$ and $g(x)$ are different

Q3.16.2 MCQ 2016 2M

Probability density function of a random variable X is given below

$$f(x) = \begin{cases} 0.25, & \text{if } 1 \le x \le 5 \\ 0, & \text{otherwise} \end{cases}$$

$P(X \le 4)$ is

A. $\frac{3}{4}$ **B.** $\frac{1}{2}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{8}$

Q3.16.3 MCQ 2016 1M

X and Y are two random independent events. It is known that $P(X) = 0.40$ and $P(X \cup Y^c) = 0.7$. Which one of the following is the value of $P(X \cup Y)$?

A. 0.7 **B.** 0.5 **C.** 0.4 **D.** 0.3

Q3.16.4 NAT 2016 1M

The spot speeds (expressed in km/hr) observed at a road section are 66, 62, 45, 79, 32, 51, 56, 60, 53, and 49. The median speed (expressed in km/hr) is ________.

(Note: answer with one decimal accuracy)

Q3.15.1 NAT 2015 1M

Consider the following probability mass function (p.m.f.) of a random variable X:

$$p(x, q) = \begin{cases} q, & \text{if } X = 0, \\ 1 - q, & \text{if } X = 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $q = 0.4$, the variance of X is ________.

Q3.15.2 NAT 2015 2M

The probability density function of a random variable, x is

$$f(x) = \begin{cases} \frac{x}{4}(4 - x^2), & \text{for } 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

The mean, μ_x of the random variable is ________.

Q3.14.1 NAT 2014 1M ○○○

The probability density function of evaporation E on any day during a year in a watershed is given by

$$f(E) = \begin{cases} \frac{1}{5} & 0 \le E \le 5 \text{ mm/day} \\ 0 & \text{otherwise} \end{cases}$$

The probability that E lies in between 2 and 4 mm/day in a day in the watershed is (in decimal) __________

Q3.14.2 NAT 2014 2M ○○○

An observer counts 240 veh/h at a specific highway location. Assume that the vehicle arrival at the location is Poisson distributed, the probability of having one vehicle arriving over a 30-second time interval is __________

Q3.14.3 MCQ 2014 1M ○○○

A fair (unbiased) coin was tossed four times in succession and resulted in the following outcomes: (i) Head, (ii) Head, (iii) Head, (iv) Head. The probability of obtaining a 'Tail' when the coin is tossed again is

A. 0 **B.** $\frac{1}{2}$ **C.** $\frac{4}{5}$ **D.** $\frac{1}{5}$

Q3.14.4 NAT 2014 2M ○○○

A traffic office imposes on an average 5 number of penalties daily on traffic violators. Assume that the number of penalties on different days is independent and follows a Poisson distribution. The probability that there will be less than 4 penalties in a day is __________

Q3.14.5 MCQ 2014 1M ○○○

If $\{x\}$ is a continuous, real valued random variable defined over the interval $(-\infty, +\infty)$ and its occurrence is defined by the density function given as:

$$f(x) = \frac{1}{\sqrt{2\pi}\, b} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2}$$

where 'a' and 'b' are the statistical attributes of the random variable $\{x\}$. The value of the integral

$$\int_{-\infty}^{a} \frac{1}{\sqrt{2\pi}\, b} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2} dx$$

is

A. 1 **B.** 0.5 **C.** π **D.** $\frac{\pi}{2}$

Q3.13.1 MCQ 2013 1M ○○○

A 1 hour rainfall of 10 cm has return period of 50 year. The probability that 1 hour of rainfall 10 cm or more will occur in each of two successive years is

A. 0.04 **B.** 0.2 **C.** 0.2 **D.** 0.0004

Q3.13.2 NAT 2013 2M ○○○

Find the value of λ such that function $f(x)$ is valid probability density function?

$$f(x) = \begin{cases} \lambda(x-1)(2-x) & \text{for } 1 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Q3.12.1 MCQ 2012 2M ○○○

An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.
The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 **B.** 0.334 **C.** 0.667 **D.** 0.720

Q3.12.2 MCQ 2012 2M ○○○

In an experiment, positive and negative values are equally likely to occur. The probability of obtaining atmost one negative value in five trials is

A. $\frac{1}{32}$ **B.** $\frac{2}{32}$ **C.** $\frac{3}{32}$ **D.** $\frac{6}{32}$

Q3.12.3 MCQ 2012 1M ○○○

The annual precipitation data of a city is normally distributed with mean and standard deviation as 1000mm and 200mm, respectively. The probability that the annual precipitation will be more than that of 1200mm is

A. < 50% **B.** 50% **C.** 75% **D.** 100%

Q3.11.1 MCQ 2011 1M ○○○

There are two containers, with one containing 4 Red and 3 Green balls and the other containing 3 blue and 4 Green balls. One ball is drawn at random from each container. The probability that one of the balls is Red and the other is Blue will be

A. $\frac{1}{7}$ **B.** $\frac{9}{49}$ **C.** $\frac{12}{49}$ **D.** $\frac{3}{7}$

Q3.11.2 MCQ 2011 1M

The probability that k number of vehicles arrive (i.e., cross a predefined line) in time t is given as $(\lambda t)^k e^{-\lambda t}/k!$. where λ is the average vehicle arrival rate. What is the probability that the time headway is greater than or equal to time t_1?

A. $\lambda e^{\lambda t_1}$ **B.** λe^{-t_1} **C.** $e^{\lambda t_1}$ **D.** $e^{-\lambda t_1}$

Q3.10.1 MCQ 2010 1M

Two coins are simultaneously tossed. The probability of two heads simultaneously appearing is

A. $\frac{1}{8}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{2}$

Q3.09.1 MCQ 2009 2M

The standard normal probability function can be approximated as

$$F(x_N) = \frac{1}{1 + \exp\left(-1.7255 x_N |x_N|^{0.12}\right)}$$

where x_N = standard normal deviate. If mean and standard deviation of annual precipitation are 102 cm and 27 cm respectively, the probability that the annual precipitation will be between 90 cm and 102 cm is

A. 66.7% **B.** 50.0% **C.** 33.3% **D.** 16.7%

Q3.08.1 MCQ 2008 2M

A person on a trip has a choice between private car and public transport. The probability of using a private car is 0.45. While using the public transport, further choices available are bus and metro, out of which the probability of commuting by a bus is 0.55. In such a situation, the probability (rounded up to two decimals) of using a car, bus and metro, respectively would be

A. 0.45, 0.30 and 0.25 **B.** 0.45, 0.25 and 0.30

C. 0.45, 0.55 and 0.00 **D.** 0.45, 0.35 and 0.20

Q3.08.2 MCQ 2008 2M

If probability density function of a random variable X is

$$f(x) = x^2 \text{ for } -1 \le x \le 1, \text{ and}$$

$$= 0 \text{ for any other value of } x$$

Then, the percentage probability

$$P\left(-\frac{1}{3} \le x \le \frac{1}{3}\right)$$

is

A. 0.247 **B.** 2.47 **C.** 24.7 **D.** 247

Q3.08.3 MCQ 2008 2M

Three values of x and y are to be fitted in a straight line in the form $y = a + bx$ by the method of least squares. Given: $\sum x = 6$, $\sum y = 21$, $\sum x^2 = 14$ and $\sum xy = 46$, the values of a and b are respectively

A. 2 and 3 **B.** 1 and 2

C. 2 and 1 **D.** 3 and 2

Q3.07.1 MCQ 2007 2M

If the standard deviation of the spot speed of vehicles in a highway is 8.8 kmph and the mean speed of the vehicles is 33 kmph, the coefficient of variation in speed is

A. 0.1517 **B.** 0.1867 **C.** 0.2666 **D.** 0.3646

Q3.06.1 MCQ 2006 2M

There are 25 calculators in a box. Two of them are defective. Suppose 5 calculators are randomly picked for inspection (i.e., each has the same chance of being selected), what is the probability that only one of the defective calculators will be included in the inspection?

A. $\frac{1}{2}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{5}$

Q3.06.2 MCQ 2006 2M

A class of first year B. Tech. students is composed of four batches A, B, C and D, each consisting of 30 students. It is found that the sessional marks of students in Engineering Drawing in batch C have a mean of 6.6 and standard deviation of 2.3. The mean and standard deviation for the entire class are 5.5 and 4.2, respectively. It is decided by the course instructor to normalize the marks of the students of all batches to have the same mean and standard deviation as that of the entire class. Due to this, the marks of a student in batch C are changed from 8.5 to

A. 6.0 **B.** 7.0 **C.** 8.0 **D.** 9.0

Q3.05.1 MCQ 2005 1M

Which one of the following statement is NOT true?

A. The measure of skewness is dependent upon the amount of dispersion.

B. In a symmetric distribution, the values of mean, mode and median are the same.

C. In a positively skewed distribution: mean > median > mode.

D. In a negatively skewed distribution: mode > mean > median.

Q3.04.1 MCQ 2004 2M

A hydraulic structure has four gates which operate independently. The probability of failure of each gate is 0.2. Given that gate 1 has failed, the probability that both gates 2 and 3 will fail is

A. 0.240 **B.** 0.200 **C.** 0.040 **D.** 0.008

Q3.03.1 MCQ 2003 1M

A box contains 10 screws, 3 of which are defective. Two screws are drawn at random with replacement. The probability that none of the two screws is defective will be

A. 100%

B. 50%

C. 49%

D. None of these

Chemical Engineering

Q3.26.1 NAT 2026 1M

An experiment is performed by rolling an unbiased die once. Events A and B are defined as follows:
Event A: The outcome of the rolled die is a prime number.
Event B: The outcome of the rolled die is an odd number less than 4.
The conditional probability of A given B, denoted by $P(A \mid B)$, is ________ (rounded off to one decimal place).

Q3.25.1 MCQ 2025 1M

A box contains 3 identical green balls and 7 identical blue balls. Two balls are randomly drawn without replacement from the box. The probability of drawing 1 green and 1 blue ball is

A. $\dfrac{{}^3P_1 \times {}^7P_1}{{}^{10}P_2}$

B. $\dfrac{{}^{10}P_3 \times {}^{10}P_7}{{}^{10}P_2}$

C. $\dfrac{{}^{10}C_3 \times {}^{10}C_7}{{}^{10}C_2}$

D. $\dfrac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2}$

Q3.25.2 MCQ 2025 2M

A probability distribution function is given as

$$p(x) = \begin{cases} \dfrac{1}{a}, & x \in (0, a) \\ 0, & \text{otherwise} \end{cases}$$

where a is a positive constant. For a function $f(x) = x^2$, the expectation of $f(x)$ is

A. $\dfrac{a^2}{3}$ **B.** $\dfrac{a^3}{3}$ **C.** $\dfrac{2a^2}{3}$ **D.** $\dfrac{2a^3}{3}$

Q3.24.1 MCQ 2024 1M

Consider the normal probability distribution function

$$f(x) = \frac{4}{\sqrt{2\pi}} e^{-8(x+3)^2}$$

If μ and σ are the mean and standard deviation of $f(x)$ respectively, then the ordered pair (μ, σ) is

A. $\left(3, \dfrac{1}{4}\right)$ **B.** $\left(-3, \dfrac{1}{4}\right)$ **C.** $(3, 4)$ **D.** $(-3, 4)$

Q3.23.1 MCQ 2023 2M

An exhibition was held in a hall on 15 August 2022 between 3 PM and 4 PM during which any person was allowed to enter only once. Visitors who entered before 3:40 PM exited the hall exactly after 20 minutes from their time of entry. Visitors who entered at or after 3:40 PM, exited exactly at 4 PM. The probability distribution of the arrival time of any visitor is uniform between 3 PM and 4 PM. Two persons X and Y entered the exhibition hall independent of each other. Which one of the following values is the probability that their visits to the exhibition overlapped with each other?

A. $\dfrac{5}{9}$ **B.** $\dfrac{4}{9}$ **C.** $\dfrac{2}{9}$ **D.** $\dfrac{7}{9}$

Q3.21.1 NAT 2021 2M

The probability distribution function of a random variable X is shown in the following figure.

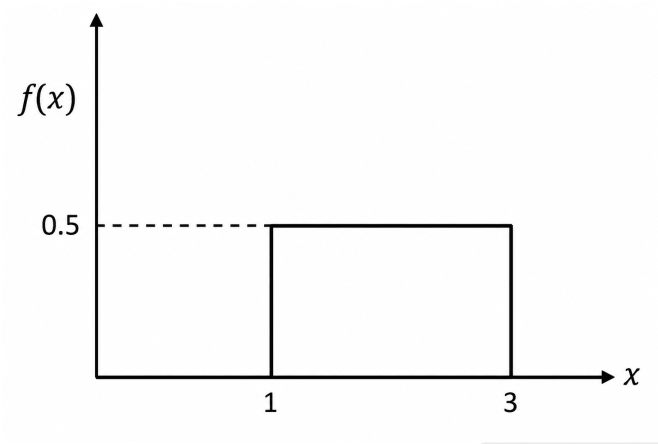


Fig. Q3.21.1

From this distribution, random samples with sample size $n = 68$ are taken. If $\overline{X}$ is the sample mean, the standard deviation of the probability distribution of $\overline{X}$, i.e. $\sigma_{\overline{X}}$ is ________ (round off to 3 decimal places).

Q3.20.1 MCQ 2020 2M

In a box, there are 5 green balls and 10 blue balls. A person picks 6 balls randomly. The probability that the person has picked 4 green balls and 2 blue balls is

A. $\frac{42}{1001}$ **B.** $\frac{45}{1001}$ **C.** $\frac{240}{1001}$ **D.** $\frac{420}{1001}$

Q3.19.1 NAT 2019 2M

Two unbiased dice are thrown. Each dice can show any number between 1 and 6. The probability that the sum of the outcomes of the two dice is divisible by 4 is ________ (rounded off to two decimal places).

Q3.18.1 MCQ 2018 1M

A watch uses two electronic circuits (ECs). Each EC has a failure probability of 0.1 in one year of operation. Both ECs are required for functioning of the watch. The probability of the watch functioning for one year without failure is

A. 0.99 **B.** 0.90 **C.** 0.81 **D.** 0.80

Q3.17.1 MCQ 2017 1M

The marks obtained by a set of students are: 38, 84, 45, 70, 75, 60, 48.
The mean and median marks, respectively, are

A. 45 and 75 **B.** 55 and 48

C. 60 and 60 **D.** 60 and 70

Q3.17.2 NAT 2017 2M

A box has 6 red balls and 4 white balls. A ball is picked at random and replaced in the box, after which a second ball is picked.
The probability of both the balls being red, rounded to 2 decimal places, is ________.

Q3.16.1 NAT 2016 2M

The model $y = mx^2$ is to be fit to the data given below.

x	1	$\sqrt{2}$	$\sqrt{3}$
y	2	5	8

Using linear regression, the value (rounded off to the second decimal place) of m is ________.

Q3.14.1 NAT 2014 2M

In rolling of two fair dice, the outcome of an experiment is considered to be the sum of the numbers appearing on the dice. The probability is highest for the outcome of ________.

Q3.14.2 MCQ 2014 2M

Consider the following two normal distributions:

$$f_1(x) = \exp(-\pi x^2)$$

$$f_2(x) = \frac{1}{2\pi}\exp\left\{-\frac{1}{4\pi}(x^2 + 2x + 1)\right\}$$

If μ and σ denote the mean and standard deviation, respectively, then

A. $\mu_1 < \mu_2$ and $\sigma_1^2 < \sigma_2^2$ **B.** $\mu_1 < \mu_2$ and $\sigma_1^2 > \sigma_2^2$

C. $\mu_1 > \mu_2$ and $\sigma_1^2 < \sigma_2^2$ **D.** $\mu_1 > \mu_2$ and $\sigma_1^2 > \sigma_2^2$

Q3.13.1 NAT 2013 1M

For two rolls of a fair die, the probability of getting a 4 in the first roll and a number less than 4 in the second roll, up to 3 digits after the decimal point, is ________.

Q3.13.2 MCQ 2013 1M

The number of emails received on six consecutive days is 11, 9, 18, 18, 4 and 15, respectively. What are the median and the mode for these data?

A. 18 and 11, respectively **B.** 13 and 18, respectively

C. 13 and 12.5, respectively **D.** 12.5 and 18, respectively

Q3.12.1 MCQ 2012 1M

A box containing 10 identical compartments has 6 red balls and 2 blue balls. If each compartment can hold only one ball, then the number of different possible arrangements are

A. 1026 **B.** 1062 **C.** 1260 **D.** 1620

Q3.10.1 MCQ 2010 2M

X and Y are independent random variables. X follows a binomial distribution, with $N = 5$ and $p = \frac{1}{2}$. Y takes integer values 1 and 2, with equal probability. Then the probability that $X = Y$ is

A. $\frac{15}{64}$ **B.** $\frac{15}{32}$ **C.** $\frac{1}{2}$ **D.** $\frac{15}{16}$

Q3.10.2 MCQ 2010 2M

A box contains three red and two black balls. Four balls are removed from the box one by one, without replacement. The probability of the ball remaining in the box being red, is

A. $\frac{609}{625}$ **B.** $\frac{3}{5}$ **C.** $\frac{2}{5}$ **D.** $\frac{81}{625}$

Q3.09.1 MCQ 2009 2M

A fair die is rolled. Let R denote the event of obtaining a number less than or equal to 5 and S denote the event of obtaining an odd number. Then which ONE of the following about the probability (P) is TRUE?

A. $P(R \mid S) = 1$ **B.** $P(R \mid S) = 0$

C. $P(S \mid R) = 1$ **D.** $P(S \mid R) = 0$

Q3.08.1 MCQ 2008 2M

The Poisson distribution is given by

$$P(r) = \frac{m^r}{r!}\exp(-m).$$

The first moment about the origin for this distribution is

A. 0 **B.** m **C.** $\frac{1}{m}$ **D.** m^2

Q3.06.1 MSQ 2006 2M

A pair of fair dice is rolled three times. What is the probability that 10 (sum of the number on the two faces) will show up exactly once?

A. $\frac{121}{1728}$ **B.** $\frac{363}{1728}$ **C.** $\frac{121}{576}$ **D.** $\frac{363}{576}$

Q3.06.2 MCQ 2006 2M

A company purchased components from the firms P, Q, and R as shown in the table below.

Firm	Total number of components purchased	Number of components likely to be defective
P	1000	5
Q	2500	5
R	500	2

The components are stored together. One of the component is selected at random and found to be defective. What is the probability that it was supplied by firm R?

A. $\frac{1}{250}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{6}$

Q3.05.1 MCQ 2005 1M

Two bags contain ten coins each and the coins in each bag are numbered from 1 to 10. One coin is drawn at random from each bag. The probability that one of the coins has value 1, 2, 3, or 4, while the other has value 7, 8, 9 or 10 is

A. $\frac{2}{5}$ **B.** $\frac{4}{25}$ **C.** $\frac{8}{25}$ **D.** $\frac{16}{25}$

Q3.04.1 MCQ 2004 1M ◯◯◯

A box contains three blue balls and four red balls. Another identical box contains two blue balls and five red balls. One ball is picked at random from one of the two boxes and it is red. The probability that it came from the first box is

A. $\frac{2}{3}$ **B.** $\frac{4}{9}$ **C.** $\frac{4}{7}$ **D.** $\frac{2}{7}$

Q3.02.1 MCQ 2002 1M ◯◯◯

A fair die is rolled four times. Find the probability that six shows up twice.

A. $\frac{1}{2}$ **B.** $\frac{16}{325}$ **C.** $\frac{1}{36}$ **D.** $\frac{25}{216}$

Production & Industrial Engineering

Q3.26.1 NAT 2026 2M ◯◯◯

The number of soldering defects that occur in a semiconductor device follows a discrete Poisson distribution with a probability mass function

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!}.$$

The average number of soldering defects per semiconductor device is 3. The probability that a randomly selected semiconductor device will have at least two soldering defects is ________ (rounded off to two decimal places).

Q3.26.2 MCQ 2026 2M ◯◯◯

The function, $r(t)$, signifies the likelihood that a component has survived up to time t and fails in the next instant of time.

$$r(t) = \frac{f(t)}{1 - F(t)}$$

where $f(t)$ denotes the probability density function for time to failure with corresponding cumulative distribution function given by,

$$F(t) = 1 - e^{-at^b}, \qquad a > 0,\ b > 0$$

Which ONE of the following represents $r(t)$?

A. $a \times b \times t^{b-1} \times e^{-at^b}$

B. $a \times b \times t^{b-1}$

C. $1 - (a \times b \times t^{b-1})$

D. $1 - \left(a \times b \times t^{b-1} \times e^{-at^b}\right)$

Q3.26.3 NAT 2026 1M ◯◯◯

A box in a machine shop consists of 5 coated and 10 uncoated cutting inserts which are of otherwise similar characteristics. The inserts got mixed up randomly in the box. An operator has taken 4 of them at once without noticing the differences to mount on a 4-tooth face milling cutter that uses inserts.

The probability of the milling cutter having all uncoated inserts is ________ (rounded off to two decimal places).

Q3.25.1 MCQ 2025 1M ◯◯◯

A bag contains 5 red, 7 green and 3 blue balls. Two balls are drawn at random from the bag one-by-one. The probability of the second drawn ball being red is

A. $\frac{1}{5}$ **B.** $\frac{1}{3}$ **C.** $\frac{2}{5}$ **D.** $\frac{2}{3}$

Q3.24.1 NAT 2024 2M ◯◯◯

Seven cards numbered 1 to 7 are placed in a box. After thoroughly mixing all the cards, one card is drawn at random.

If it is known that the number on the card drawn is odd, then the probability that the number on the card drawn is greater than 4 is ________ %. (Answer in integer)

Q3.24.2 NAT 2024 1M ◯◯◯

If X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{K}{4}, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

then the value of K is ________. (Answer in integer)

Q3.23.1 MCQ 2023 1M ◯◯◯

Three unbiased coins are tossed. Provided that at least two outcomes are tails, the probability of having all three outcomes as tails is

A. $\frac{1}{8}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{2}$

Q3.23.2 MCQ 2023 2M

Two cards are drawn one after the other from a regular deck of 52 playing cards without replacement. The probability that the drawn cards are of different suits is

A. $\frac{39}{51}$ **B.** $\frac{13}{52}$ **C.** $\frac{2}{52}$ **D.** $\frac{2}{51}$

Q3.22.1 NAT 2022 1M

If the interarrival time is exponential and 8 customers per hour arrive in a bank, then the probability of no arrival of customer during a period of 15 minutes is __________. [round off to two decimal places]

Q3.22.2 NAT 2022 1M

An operator manufactures 10 identical spur gears in a lot. One spur gear is defective in the lot. Three spur gears are drawn at random without replacement. The probability of getting all three gears as non-defective is __________. [round off to two decimal places]

Q3.21.1 NAT 2021 2M

The probability mass function $P(x)$ of a discrete random variable X is given by $P(x) = \frac{1}{2^x}$, where $x = 1, 2, \cdots, \infty$. The expected value of X is __________. *[in integer]*

Q3.21.2 NAT 2021 2M

The time to pass through a security screening at an airport follows an exponential distribution. The mean time to pass through the security screening is 15 minutes. To catch the flight, a passenger must clear the security screening within 15 minutes. The probability that the passenger will miss the flight is __________. *[round off to 3 decimal places]*

Q3.20.1 MCQ 2020 1M

If x is a random variable with the expected value of 5 and the variance of 1, then the expected value of x^2 is

A. 24 **B.** 25 **C.** 26 **D.** 36

Q3.20.2 NAT 2020 2M

If the probability density function of a random variable x is given by

$$f(x) = \begin{cases} \frac{kx^2}{2}, & -1 \le x \le 1, \\ 0, & \text{elsewhere,} \end{cases}$$

the value of k is __________.

Q3.19.1 NAT 2019 2M

A box contains 2 red and 3 black balls. Three balls are randomly chosen from the box and are placed in a bag. Then the probability that there are 1 red and 2 black balls in the bag, is __________.

Q3.18.1 NAT 2018 1M

The probabilities of occurrence of events F and G are $P(F) = 0.3$ and $P(G) = 0.4$, respectively. The probability that both events occur simultaneously is $P(F \cap G) = 0.2$. The probability of occurrence of at least one event $P(F \cup G)$ is __________.

Q3.18.2 NAT 2018 1M

Weights (in kg) of six products are 3, 7, 6, 2, 3 and 4. The median weight (in kg, up to one decimal place) is __________.

Q3.17.1 NAT 2017 2M

Two machines are defective in a lot of 10. A combination of four machines is to be picked at a time from the lot. The maximum number of combinations that can be obtained without any defective machine is __________.

Q3.16.1 MCQ 2016 1M

A fair coin is tossed N times. The probability that head does not turn up in any of the tosses is

A. $\left(\frac{1}{2}\right)^{N-1}$ **B.** $1 - \left(\frac{1}{2}\right)^{N-1}$ **C.** $\left(\frac{1}{2}\right)^{N}$ **D.** $1 - \left(\frac{1}{2}\right)^{N}$

Q3.16.2 MCQ 2016 1M

A normal random variable X has the following probability density function

$$f_X(x) = \frac{1}{\sqrt{8\pi}} e^{-\left\{\frac{(x-1)^2}{8}\right\}}, \qquad -\infty < x < \infty$$

Then $\int_1^{\infty} f_X(x)\,dx =$

A. 0 **B.** $\frac{1}{2}$ **C.** $1 - \frac{1}{e}$ **D.** 1

Q3.15.1 MCQ 2015 1M

A product is an assembly of 5 different components. The product can be sequentially assembled in two possible ways. If the 5 components are placed in a box and these are drawn at random from the box, then the probability of getting a correct sequence is

A. $\frac{2}{5!}$ **B.** $\frac{2}{5}$ **C.** $\frac{2}{(5-2)!}$ **D.** $\frac{2}{(5-3)!}$

Q3.14.1 MCQ 2014 2M

Marks obtained by 100 students in an examination are given in the table.

Sl. No.	Marks Obtained	Number of Students
1	25	20
2	30	20
3	35	40
4	40	20

What would be the mean, median, and mode of the marks obtained by the students?

A. Mean 33; Median 35; Mode 40

B. Mean 35; Median 32.5; Mode 40

C. Mean 33; Median 35; Mode 35

D. Mean 35; Median 32.5; Mode 35

Q3.14.2 MCQ 2014 2M

In a given day in the rainy season, it may rain 70% of the time. If it rains, chance that a village fair will make a loss on that day is 80%. However, if it does not rain, chance that the fair will make a loss on that day is only 10%. If the fair has not made a loss on a given day in the rainy season, what is the probability that it has not rained on that day?

A. $\frac{3}{10}$ **B.** $\frac{9}{11}$ **C.** $\frac{14}{17}$ **D.** $\frac{27}{41}$

Q3.13.1 MCQ 2013 1M

Customers arrive at a ticket counter at a rate of 50 per *hr* and tickets are issued in the order of their arrival. The average time taken for issuing a ticket is 1 *min*. Assuming that customer arrivals form a Poisson process and service times are exponentially distributed, the average waiting time in queue in *min* is

A. 3 **B.** 4 **C.** 5 **D.** 6

Q3.13.2 MCQ 2013 1M

Let X be a normal random variable with mean 1 and variance 4. The probability $P\{X < 0\}$ is

A. 0.5

B. greater than zero and less than 0.5

C. greater than 0.5 and less than 1.0

D. 1.0

Q3.13.3 MCQ 2013 2M

The probability that a student knows the correct answer to a multiple choice question is $\frac{2}{3}$. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is $\frac{1}{4}$. Given that the student has answered the question correctly, the conditional probability that the student knows the correct answer is

A. $\frac{2}{3}$ **B.** $\frac{3}{4}$ **C.** $\frac{5}{6}$ **D.** $\frac{8}{9}$

Q3.13.4 MCQ 2013 1M

A company manufactures 1000 toys every day. On an average, 10% of the toys are defective and 40% of the defective toys can be reworked into defect-free ones. The average number of defect-free toys manufactured daily is

A. 900 **B.** 920 **C.** 940 **D.** 960

Q3.12.1 MCQ 2012 2M

A box contains 4 red balls and 6 black balls. Three balls are selected randomly from the box one after another, without replacement. The probability that the selected set contains one red ball and two black balls is

A. $\frac{1}{20}$ **B.** $\frac{1}{12}$ **C.** $\frac{3}{10}$ **D.** $\frac{1}{2}$

Q3.11.1 MCQ 2011 1M

It is estimated that the average number of events during a year is three. What is the probability of occurrence of not more than two events over a two-year duration? Assume that the number of events follows a Poisson distribution.

A. 0.052 **B.** 0.062 **C.** 0.072 **D.** 0.082

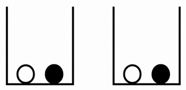
Q3.10.1 MCQ 2010 1M

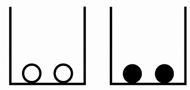
If a random variable X satisfies the Poisson's distribution with a mean value of 2, then the probability that $X \geq 2$ is

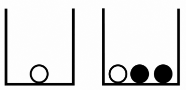
A. $2e^{-2}$ **B.** $1 - 2e^{-2}$ **C.** $3e^{-2}$ **D.** $1 - 3e^{-2}$

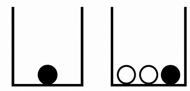
Q3.10.2 MCQ 2010 2M

Two white and two black balls, kept in two bins, are arranged in four ways as shown below. In each arrangement, a bin has to be chosen randomly and only one ball needs to be picked randomly from the chosen bin. Which one of the following arrangements has the highest probability for getting a white ball picked?

A. 

B. 

C. 

D. 

Q3.09.1 MCQ 2009 2M

Consider the joint probability mass function of random variables X and Y as shown in table below: For instance, $P\{X = 1, Y = 2\} = 0.3$

	$X = 1$	$X = 2$
$Y = 1$	0.2	0.3
$Y = 2$	0.3	0.1
$Y = 3$	0.1	0

The value of $P\{X = 2 \mid Y = 2\}$ is

A. 0.10 **B.** 0.25 **C.** 0.40 **D.** 0.75

Q3.08.1 MCQ 2008 1M

For a random variable x $(-\infty < x < \infty)$ following normal distribution, the mean is $\mu = 100$. If the probability is $P = \alpha$ for $x \geq 110$, then the probability of x lying between 90 and 110, i.e., $P(90 \leq x \leq 110)$ will be equal to

A. $1 - 2\alpha$ **B.** $1 - \alpha$ **C.** $1 - \frac{\alpha}{2}$ **D.** 2α

Q3.08.2 MCQ 2008 2M

In a game, two players X and Y toss a coin alternately. Whosoever gets a 'head' first, wins the game and the game is terminated. Assuming that player X starts the game, the probability of player X winning the game is

A. $\frac{1}{3}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{4}$

Q3.08.3 MCQ 2008 2M

Customers arrive at a service counter manned by a single person according to a Poisson distribution with a mean arrival rate of 30 per hour. The time required to serve a customer follows an exponential distribution with a mean of 100 seconds. The average waiting time (in hours) of a customer in the system will be

A. 0.138 **B.** 0.166 **C.** 0.276 **D.** 0.332

Q3.07.1 MCQ 2007 1M

Two cards are drawn at random in succession, with replacement, from a deck of 52 well shuffled cards. Probability of getting both 'Aces' is

A. $\frac{1}{169}$ **B.** $\frac{2}{169}$ **C.** $\frac{1}{13}$ **D.** $\frac{2}{13}$

Q3.07.2 MCQ 2007 2M

If X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} K(5x - 2x^2), & 0 \leq x \leq 2, \\ 0, & \text{otherwise} \end{cases}$$

Then $P(X > 1)$ is:

A. $\frac{3}{14}$ **B.** $\frac{4}{5}$ **C.** $\frac{14}{17}$ **D.** $\frac{17}{28}$

Q3.07.3 MCQ 2007 2M

The random variable X takes on the values 1, 2, or 3 with probabilities $(2 + 5P)/5$, $(1 + 3P)/5$, and $(1.5 + 2P)/5$, respectively. The values of P and $E[X]$ are respectively

A. 0.05, 1.87 **B.** 1.90, 5.87

C. 0.05, 1.10 **D.** 0.25, 1.40

Q3.07.4 MCQ 2007 2M

$X_1, \ldots, X_{100}$ are Bernoulli random variables with a probability of success equal to 0.6. By the Central Limit Theorem, the random variable

$$Y = \sum_{i=1,\ldots,100} X_i$$

is approximately normally distributed. Then Y has mean and variance respectively equal to

A. 40 and 24 **B.** 60 and 24

C. 40 and 12 **D.** 60 and 12

Q3.07.5 MCQ 2007 2M

The average number of accidents occurring monthly on an assembly shop floor is 2. The probability that there will be at least one accident in this month is estimated to be

A. 0.055 **B.** 0.456 **C.** 0.865 **D.** 0.950

Q3.05.1 MCQ 2005 1M

Two dice are thrown simultaneously. The probability that the sum of numbers on both exceeds 8 is,

A. $\frac{4}{36}$ **B.** $\frac{7}{36}$ **C.** $\frac{9}{36}$ **D.** $\frac{10}{36}$

Q3.05.2 MCQ 2005 2M

The life of a bulb (in hours) is a random variable with an exponential distribution

$$f(t) = \alpha e^{-\alpha t}, \qquad 0 \le t \le \infty.$$

The probability that its value lies between 100 and 200 hours is

A. $e^{-100\alpha} - e^{-200\alpha}$ **B.** $e^{-100} - e^{-200}$

C. $e^{-100\alpha} + e^{-200\alpha}$ **D.** $e^{-200\alpha} - e^{-100\alpha}$

Answer Key

ELECTRONICS & COMMUNICATION ENGINEERING					
Q3.26.1 A	Q3.25.1 A	Q3.25.2 7.00	Q3.24.1 0.5	Q3.22.1 4.0	Q3.21.1 B
Q3.21.2 B	Q3.20.1 0.25	Q3.20.2 0.299 to 0.301	Q3.19.1 11	Q3.19.2 0.36 to 0.38	Q3.18.1 0.25
Q3.17.1 0.026 to 0.028	Q3.17.2 2.4 to 2.6	Q3.16.1 0.9 to 1.1	Q3.16.2 0.32 to 0.34	Q3.16.3 0.07 to 0.08	Q3.15.1 C
Q3.15.2 6	Q3.15.3 D	Q3.15.4 6	Q3.15.5 6	Q3.14.1 0.65 to 0.68	Q3.14.2 B
Q3.14.3 49.9 to 50.1	Q3.14.4 C	Q3.14.5 2.9 to 3.1	Q3.14.6 0.15 to 0.18	Q3.14.7 0.79 to 0.81	Q3.14.8 A
Q3.14.9 0.43 to 0.45	Q3.13.1 B	Q3.13.2 D	Q3.12.1 B	Q3.12.2 C	Q3.11.1 C
Q3.10.1 D	Q3.09.1 C	Q3.09.2 C	Q3.09.3 B	Q3.08.1 A	Q3.07.1 A
Q3.07.2 C	Q3.06.1 C	Q3.06.2 D	Q3.06.3 B	Q3.05.1 D	Q3.04.1 D
Q3.03.1 C	Q3.01.1 C	Q3.93.1 0.125	Q3.92.1 B	Q3.88.1 A,B	
ELECTRICAL ENGINEERING					
Q3.26.1 0.1	Q3.25.1 A	Q3.25.2 B	Q3.24.1 B,D	Q3.23.1 0.188	Q3.23.2 C
Q3.22.1 A	Q3.22.2 1	Q3.21.1 B	Q3.19.1 0.26	Q3.19.2 B	Q3.17.1 C
Q3.17.2 A	Q3.17.3 0.9	Q3.16.1 A	Q3.16.2 0.2	Q3.15.1 D	Q3.15.2 D
Q3.15.3 0.25	Q3.14.1 0.42	Q3.14.2 0.4	Q3.14.3 0.142	Q3.14.4 B	Q3.13.1 A
Q3.12.1 C	Q3.12.2 B	Q3.11.1 A	Q3.10.1 C	Q3.09.1 B	Q3.08.1 C
Q3.07.1 C	Q3.06.1 C	Q3.05.1 D	Q3.05.2 B		
INSTRUMENTATION ENGINEERING					
Q3.26.1 1	Q3.26.2 C	Q3.25.1 0.74	Q3.24.1 0.5	Q3.24.2 0.135	Q3.23.1 -0.5
Q3.23.2 240	Q3.22.1 0.33	Q3.20.1 0.09	Q3.20.2 0.7	Q3.19.1 A	Q3.19.2 A
Q3.18.1 D	Q3.18.2 B	Q3.18.3 B	Q3.17.1 C	Q3.16.1 A	Q3.15.1 B

Q3.15.2 4	Q3.14.1 0.890 to 0.899	Q3.14.2 2	Q3.13.1 A	Q3.12.1 B	Q3.11.1 D
Q3.10.1 C	Q3.09.1 C	Q3.08.1 A	Q3.08.2 B	Q3.07.1 A	Q3.06.1 C
Q3.06.2 A	Q3.06.3 A	Q3.05.1 D			
COMPUTER SCIENCE & IT					
Q3.26.1 C	Q3.26.2 B	Q3.26.3 4.24 to 4.26	Q3.26.4 A	Q3.26.5 B	Q3.26.6 0.5
Q3.25.1 0.5	Q3.25.2 0.70 to 0.80	Q3.25.3 0.49 to 0.51	Q3.25.4 0.300 to 0.302	Q3.24.1 A	Q3.24.2 D
Q3.24.3 0.370 to 0.380	Q3.24.4 A,C	Q3.24.5 B	Q3.23.1 C,D	Q3.21.1 B	Q3.21.2 A
Q3.21.3 D	Q3.21.4 15 to 16	Q3.21.5 0.04	Q3.20.1 0.5	Q3.19.1 0.502 to 0.504	Q3.19.2 0.8
Q3.18.1 0.021 to 0.024	Q3.18.2 0.60 to 0.62	Q3.17.1 54	Q3.17.2 0	Q3.17.3 C	Q3.16.1 0.5
Q3.16.2 0.33 to 0.34	Q3.16.3 0.55	Q3.14.1 11.85 to 11.95	Q3.14.2 0.25	Q3.14.3 0.259 to 0.261	Q3.14.4 10
Q3.14.5 3.8 to 3.9	Q3.14.6 0.24 to 0.27	Q3.13.1 C	Q3.12.1 C	Q3.12.2 B	Q3.11.1 C
Q3.11.2 D	Q3.11.3 A	Q3.10.1 A	Q3.09.1 B	Q3.09.2 A	Q3.08.1 A
Q3.08.2 C	Q3.07.1 B	Q3.06.1 A	Q3.05.1 C	Q3.04.1 D	Q3.04.2 D
Q3.04.3 A	Q3.04.4 A	Q3.03.1 C	Q3.03.2 D		
MECHANICAL ENGINEERING					
Q3.26.1 A	Q3.25.1 D	Q3.24.1 0.38 to 0.39	Q3.23.1 A	Q3.22.1 0.17 to 0.19	Q3.21.1 C
Q3.20.1 0.041 to 0.043	Q3.20.2 0.90 to 0.95	Q3.20.3 -1 to 0.98	Q3.19.1 A	Q3.19.2 3	Q3.19.3 0.16 to 0.18
Q3.18.1 B	Q3.18.2 C	Q3.18.3 B	Q3.18.4 A	Q3.17.1 0.75	Q3.17.2 C
Q3.17.3 3.5	Q3.16.1 A	Q3.16.2 0.41	Q3.16.3 99.73	Q3.16.4 A	Q3.15.1 B
Q3.15.2 C	Q3.15.3 D	Q3.15.4 0.97	Q3.15.5 B	Q3.14.1 D	Q3.14.2 A
Q3.14.3 0.26	Q3.14.4 B	Q3.14.5 A	Q3.14.6 0.65	Q3.14.7 74.55	Q3.14.8 50
Q3.14.9 0.8145	Q3.13.1 B	Q3.13.2 D	Q3.13.3 D	Q3.12.1 D	Q3.12.2 B

Q3.11.1 D	Q3.10.1 C	Q3.09.1 D	Q3.09.2 A	Q3.08.1 A	Q3.07.1 D
Q3.06.1 D	Q3.06.2 B	Q3.05.1 B	Q3.05.2 D	Q3.04.1 C	Q3.04.2 D
Q3.03.1 D	Q3.03.2 D	Q3.02.1 A	Q3.02.2 D	Q3.02.3 C	Q3.01.1 D
Q3.00.1 D	Q3.00.2 A	Q3.99.1 C	Q3.98.1 B	Q3.97.1 C	Q3.96.1 D
Q3.93.1 A					
CIVIL ENGINEERING					
Q3.26.1 0.50 to 0.66	Q3.26.2 170.00 to 173.00	Q3.26.3 45.00 to 55.00	Q3.25.1 61.0 to 63.0	Q3.25.2 A	Q3.25.3 B,C
Q3.25.4 2.7 to 2.9	Q3.24.1 C	Q3.24.2 A	Q3.23.1 B	Q3.23.2 0.9	Q3.23.3 B
Q3.22.1 0.78 to 0.82	Q3.22.2 0.018 to 0.020	Q3.22.3 1.9	Q3.21.1 0.58 to 0.60	Q3.21.2 B	Q3.20.1 0.59 to 0.61
Q3.20.2 0.196	Q3.19.1 B	Q3.19.2 0.17 to 0.19	Q3.18.1 0.5	Q3.18.2 C	Q3.18.3 A
Q3.18.4 0	Q3.17.1 209 to 219	Q3.17.2 0.5	Q3.17.3 B	Q3.17.4 0.1276 to 0.1372	Q3.16.1 B
Q3.16.2 A	Q3.16.3 A	Q3.16.4 54.49 to 54.51	Q3.15.1 0.23 to 0.25	Q3.15.2 1.06 to 1.07	Q3.14.1 0.4
Q3.14.2 0.25 to 0.28	Q3.14.3 B	Q3.14.4 0.26 to 0.27	Q3.14.5 B	Q3.13.1 D	Q3.13.2 6
Q3.12.1 B	Q3.12.2 D	Q3.12.3 A	Q3.11.1 C	Q3.11.2 D	Q3.10.1 C
Q3.09.1 D	Q3.08.1 A	Q3.08.2 B	Q3.08.3 D	Q3.07.1 C	Q3.06.1 B
Q3.06.2 D	Q3.05.1 D	Q3.04.1 C	Q3.03.1 C		
CHEMICAL ENGINEERING					
Q3.26.1 0.5	Q3.25.1 D	Q3.25.2 A	Q3.24.1 B	Q3.23.1 A	Q3.21.1 0.069 to 0.071
Q3.20.1 B	Q3.19.1 0.24 to 0.26	Q3.18.1 C	Q3.17.1 C	Q3.17.2 0.35 to 0.37	Q3.16.1 2.25 to 2.72
Q3.14.1 6.99 to 7.01	Q3.14.2 C	Q3.13.1 0.083	Q3.13.2 B	Q3.12.1 C	Q3.10.1 B
Q3.10.2 B	Q3.09.1 A	Q3.08.1 B	Q3.06.1 B,C	Q3.06.2 D	Q3.05.1 C

Q3.04.1 B	Q3.02.1 D				
PRODUCTION & INDUSTRIAL ENGINEERING					
Q3.26.1 0.78 to 0.82	Q3.26.2 B	Q3.26.3 0.14 to 0.16	Q3.25.1 B	Q3.24.1 50	Q3.24.2 4
Q3.23.1 B	Q3.23.2 A	Q3.22.1 0.13 to 0.15	Q3.22.2 0.70	Q3.21.1 2	Q3.21.2 0.365 to 0.370
Q3.20.1 C	Q3.20.2 3	Q3.19.1 0.59 to 0.61	Q3.18.1 0.49 to 0.51	Q3.18.2 3.49 to 3.51	Q3.17.1 70
Q3.16.1 C	Q3.16.2 B	Q3.15.1 A	Q3.14.1 C	Q3.14.2 D	Q3.13.1 C
Q3.13.2 B	Q3.13.3 D	Q3.13.4 C	Q3.12.1 D	Q3.11.1 B	Q3.10.1 D
Q3.10.2 C	Q3.09.1 B	Q3.08.1 A	Q3.08.2 C	Q3.08.3 B	Q3.07.1 A
Q3.07.2 D	Q3.07.3 A	Q3.07.4 B	Q3.07.5 C	Q3.05.1 D	Q3.05.2 A

UNIT

IV

Vector Calculus

Topics

1. Gradient
2. Divergence
3. Curl
4. Line Integrals
5. Surface Integrals
6. Volume Integrals
7. Gauss Divergence Theorem
8. Stokes' Theorem
9. Green's Theorem

Note to Reader

Vector Calculus is present in branches with paper codes CE, ME, EE, EC, IN, CH, PI, AE, AG, MT, MN according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 28 **MCQ:** 18 **MSQ:** 2 **NAT:** 8
1M: 12 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2023)

EE — Electrical Engineering

Questions: 27 **MCQ:** 18 **MSQ:** 0 **NAT:** 9
1M: 10 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–6M(2022)

IN — Instrumentation Engineering

Questions: 11 **MCQ:** 9 **MSQ:** 0 **NAT:** 2
1M: 7 **2M:** 4 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–2M(2025)

CS — Computer Science & IT

Questions: 0 **MCQ:** 0 **MSQ:** 0 **NAT:** 0
1M: 0 **2M:** 0 **Desc:** 0 **Avg Weightage** ~0M **Range:** 0M(0)–0M(0)

ME — Mechanical Engineering

Questions: 44 **MCQ:** 32 **MSQ:** 0 **NAT:** 12
1M: 20 **2M:** 24 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–5M(2025)

CE — Civil Engineering

Questions: 17 **MCQ:** 11 **MSQ:** 2 **NAT:** 4
1M: 6 **2M:** 11 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–2M(2024)

CH — Chemical Engineering

Questions: 6 **MCQ:** 4 **MSQ:** 0 **NAT:** 2
1M: 2 **2M:** 4 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2023)–3M(2026)

PI — Production & Industrial Engineering

Questions: 12 **MCQ:** 9 **MSQ:** 0 **NAT:** 3
1M: 9 **2M:** 3 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–3M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q4.26.1 MCQ 2026 1M

A surface is given by $z^2 = 2x^2 - y^2$ and $\vec{n}$ and $-\vec{n}$ are unit normal vectors to the surface at the point $\vec{P} = \hat{\imath} + \sqrt{2}\,\hat{k}$. Which of the following vectors can be $\vec{n}$, where $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are the unit vectors along x, y and z axes, respectively?

A. $\hat{\imath} - \sqrt{2}\,\hat{k}$

B. $\frac{2}{3}\hat{\imath} - \frac{1}{3}\hat{k}$

C. $\sqrt{2}\hat{\imath} - \sqrt{3}\hat{k}$

D. $\frac{\sqrt{2}\hat{\imath} - \hat{k}}{\sqrt{3}}$

Q4.24.1 MSQ 2024 1M

Let $\rho(x, y, z, t)$ and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t. Assume $\frac{\partial \rho}{\partial t}$ is continuous. Let V be an arbitrary volume in space enclosed by the closed surface S and $\hat{n}$ be the outward unit normal of S.

Which of the following equations is/are equivalent to $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$?

A. $\int_V \frac{\partial \rho}{\partial t}\, dv = -\oint_S \rho u \cdot \hat{n}\, ds$

B. $\int_V \frac{\partial \rho}{\partial t}\, dv = \oint_S \rho u \cdot \hat{n}\, ds$

C. $\int_V \frac{\partial \rho}{\partial t}\, dv = -\int_V \nabla \cdot (\rho u)\, dv$

D. $\int_V \frac{\partial \rho}{\partial t}\, dv = \int_V \nabla \cdot (\rho u)\, dv$

Q4.24.2 MSQ 2024 2M

Let F_1, F_2, and F_3 be functions of (x, y, z). Suppose that for every given pair of points A and B in space, the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$ evaluates to the same value along any path C that starts at A and ends at B. Then which of the following is/are true?

A. For every closed path Γ, we have $\oint_\Gamma (F_1 dx + F_2 dy + F_3 dz) = 0$.

B. There exists a differentiable scalar function $f(x, y, z)$ such that

$$F_1 = \frac{\partial f}{\partial x},\ F_2 = \frac{\partial f}{\partial y},\ F_3 = \frac{\partial f}{\partial z}.$$

C. $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0.$

D. $\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x},\ \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}.$

Q4.23.1 MCQ 2023 1M

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $\mathbf{v}_1 = \alpha \mathbf{v}_2 + \mathbf{e}$, which minimizes the length of the error vector $\mathbf{e}$, is

A. $\frac{7}{2}$ **B.** $-\frac{2}{7}$ **C.** $\frac{2}{7}$ **D.** $-\frac{7}{2}$

Q4.23.2 MCQ 2023 1M

The rate of increase, of a scalar field $f(x, y, z) = xyz$, in the direction $\mathbf{v} = (2, 1, 2)$ at a point $(0, 2, 1)$ is

A. $\frac{2}{3}$ **B.** $\frac{4}{3}$ **C.** 2 **D.** 4

Q4.23.3 MCQ 2023 2M

The value of the line integral

$$\int_P^Q \left(z^2\, dx + 3y^2\, dy + 2xz\, dz\right)$$

along the straight line joining the points $P(1, 1, 2)$ and $Q(2, 3, 1)$ is

A. 20 **B.** 24 **C.** 29 **D.** -5

Q4.22.1 MCQ 2022 1M

Consider the two-dimensional vector field $\vec{F}(x, y) = x\,\vec{i} + y\,\vec{j}$, where $\vec{i}$ and $\vec{j}$ denote the unit vectors along the x-axis and the y-axis, respectively. A contour C in the x-y plane, as shown in the figure, is composed of two horizontal lines connected at the two ends by two semicircular arcs of unit radius. The contour is traversed in the counter-clockwise sense. The value of the closed path integral

$$\oint_C \vec{F}(x, y) \cdot (dx\,\vec{i} + dy\,\vec{j})$$

is ___________.

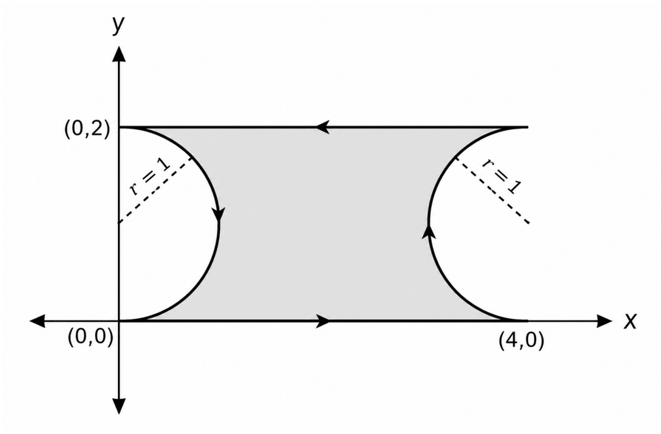


Fig. Q4.22.1

A. 0 **B.** 1 **C.** $8 + 2\pi$ **D.** -1

Q4.21.1 MCQ 2021 1M

The vector function $\mathbf{F}(\mathbf{r}) = -x\,\hat{i} + y\,\hat{j}$ is defined over a circular arc C shown in the figure.

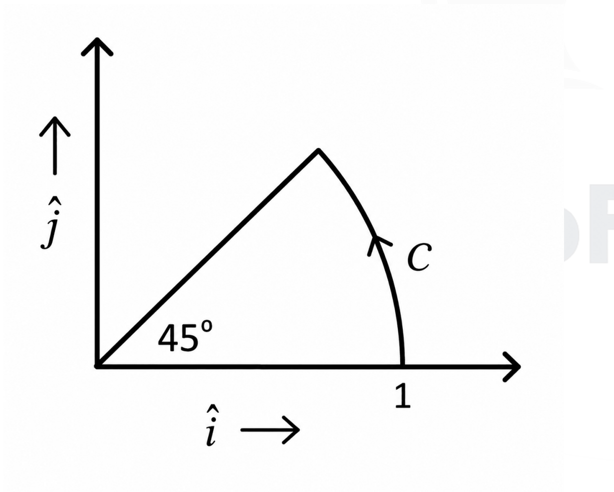


Fig. Q4.21.1

The line integral of $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$ is

A. $\frac{1}{2}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{6}$ **D.** $\frac{1}{3}$

Q4.20.1 MCQ 2020 1M

For a vector field $\vec{A}$, which one of the following is FALSE?

A. $\vec{A}$ is solenoidal if $\nabla \cdot \vec{A} = 0$.

B. $\nabla \times \vec{A}$ is another vector field.

C. $\vec{A}$ is irrotational if $\nabla^2 \vec{A} = 0$.

D. $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$.

Q4.19.1 MCQ 2019 2M

Consider the line integral

$$\int_C (x\,dy - y\,dx)$$

the integral being taken in a counterclockwise direction over the closed curve C that forms the boundary of the region R shown in the figure below. The region R is the area enclosed by the union of a 2×3 rectangle and a semi-circle of radius 1. The line integral evaluates to

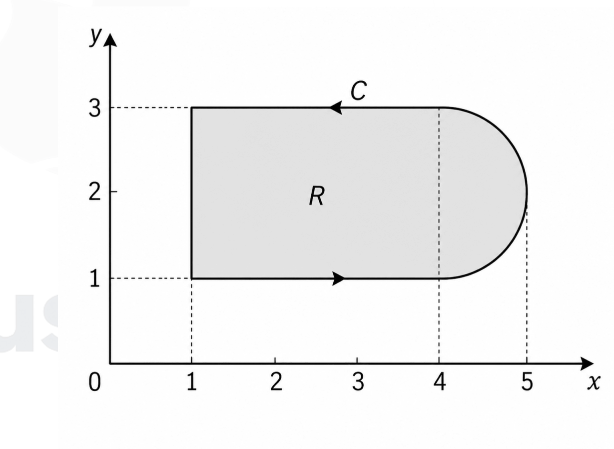


Fig. Q4.19.1

A. $6 + \pi/2$ **B.** $8 + \pi$ **C.** $12 + \pi$ **D.** $16 + 2\pi$

Q4.17.1 NAT 2017 2M

Let $I = \int_C (2z\,dx + 2y\,dy + 2x\,dz)$ where x, y, z are real, and let C be the straight line segment from point $A : (0, 2, 1)$ to point $B : (4, 1, -1)$. The value of I is ___________.

Q4.17.2 MCQ 2017 2M

If the vector function

$$\vec{F} = \hat{a}_x(3y - k_1 z) + \hat{a}_y(k_2 x - 2z) - \hat{a}_z(k_3 y + z)$$

is irrotational, then the values of the constants k_1, k_2 and k_3, respectively, are

A. 0.3, -2.5, 0.5 **B.** 0.0, 3.0, 2.0

C. 0.3, 0.33, 0.5 **D.** 4.0, 3.0, 2.0

Q4.17.3 NAT 2017 1M

The smaller angle (in degrees) between the planes $x+y+z=1$ and $2x-y+2z=0$ is __________.

Q4.16.1 NAT 2016 2M

The integral $\frac{1}{2\pi}\iint_D (x+y+10)\,dx\,dy$, where D denotes the disc: $x^2+y^2 \le 4$ evaluates to ______.

Q4.16.2 NAT 2016 2M

Suppose C is the closed curve defined as the circle $x^2+y^2=1$ with C oriented anti-clockwise. The value of $\oint (xy^2\,dx + x^2 y\,dy)$ over the curve C equals __________.

Q4.15.1 MCQ 2015 2M

A vector $\vec{P}$ is given by

$$\vec{P} = x^3 y\,\hat{a}_x - x^2 y^2\,\hat{a}_y - x^2 yz\,\hat{a}_z$$

Which of the following statements is TRUE?

A. $\vec{P}$ is solenoidal but not irrotational.

B. $\vec{P}$ is irrotational but not solenoidal.

C. $\vec{P}$ is neither solenoidal nor irrotational.

D. $\vec{P}$ is both solenoidal and irrotational.

Q4.14.1 NAT 2014 2M

If $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$, and $|\vec{r}| = r$, then

$$\operatorname{div}\left(r^2\nabla(\ln r)\right) = \_\_\_\_\_\_\_\_\_\_.$$

Q4.14.2 NAT 2014 1M

The directional derivative of

$$f(x,y) = \frac{xy}{\sqrt{2}}(x+y)$$

at $(1,1)$ in the direction of the unit vector at an angle of $\frac{\pi}{4}$ with y-axis, is given by __________.

Q4.14.3 NAT 2014 2M

Given $\vec{F} = z\hat{a}_x + x\hat{a}_y + y\hat{a}_z$. If S represents the portion of the sphere

$$x^2+y^2+z^2=1$$

for $z \ge 0$, then

$$\int_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

is __________.

Q4.13.1 MCQ 2013 1M

Consider a vector field $\vec{A}(\vec{r})$. The closed loop line integral $\oint \vec{A} \bullet d\vec{l}$ can be expressed as

A. $\oiint (\nabla \times \vec{A}) \bullet d\vec{s}$ over the closed surface bounded by the loop

B. $\oiiint (\nabla \bullet \vec{A})\,dv$ over the closed volume bounded by the loop

C. $\iiint (\nabla \bullet \vec{A})\,dv$ over the open volume bounded by the loop

D. $\iint (\nabla \times \vec{A}) \bullet d\vec{s}$ over the open surface bounded by the loop

Q4.13.2 MCQ 2013 1M

The divergence of the vector field $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is

A. 0 **B.** 1/3 **C.** 1 **D.** 3

Q4.12.1 MCQ 2012 2M

The direction of vector $\vec{A}$ is radially outward from the origin, with $|\vec{A}| = Kr^n$, where $r^2 = x^2+y^2+z^2$ and K is constant. The value of n for which $\nabla \cdot \vec{A} = 0$ is

A. -2 **B.** 2

C. 1 **D.** 0

Q4.11.1 MCQ 2011 2M

Consider a closed surface S surrounding a volume V. If $\vec{r}$ is the position vector of a point inside S, with $\hat{n}$ the unit normal on S, the value of the integral

$$\oiint_S 5\vec{r}\cdot\hat{n}\,ds$$

is

A. 3V

B. 5V

C. 10V

D. 15V

Q4.10.1 MCQ 2010 2M

If $\vec{A} = xy\,\hat{a}_x + x^2\,\hat{a}_y$, then $\oint_C \vec{A}\bullet d\vec{l}$ over the path shown in the figure is

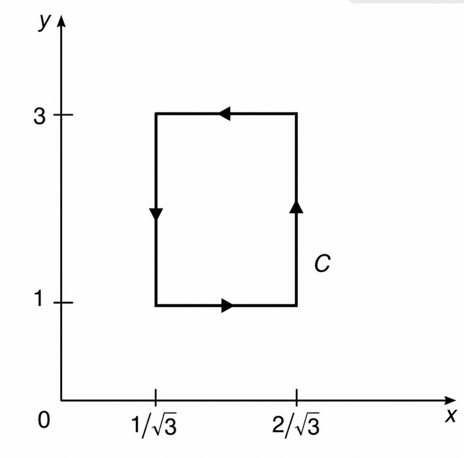


Fig. Q4.10.1

A. 0

B. $\dfrac{2}{\sqrt{3}}$

C. 1

D. $2\sqrt{3}$

Q4.08.1 MCQ 2008 2M

Consider points P and Q in the x-y plane, with $P = (1,0)$ and $Q = (0,1)$. The line integral

$$2\int_P^Q (x\,dx + y\,dy)$$

along the semicircle with the line segment PQ as its diameter

A. is -1

B. is 0

C. is 1

D. depends on the direction (clockwise or anti-clockwise) of the semicircle

Q4.08.2 MCQ 2008 2M

The value of the integral of the function $g(x,y) = 4x^3 + 10y^4$ along the straight line segment from the point $(0,0)$ to the point $(1,2)$ in the x-y plane is

A. 33

B. 35

C. 40

D. 56

Q4.06.1 MCQ 2006 1M

$\iint (\nabla\times P)\bullet d\vec{s}$, where P is a vector, is equal to

A. $\oint P\bullet d\vec{l}$

B. $\oint \nabla\times\nabla\times P\bullet d\vec{l}$

C. $\oint \nabla\times P\bullet d\vec{l}$

D. $\iiint \nabla\bullet P\,dv$

Q4.93.1 NAT 1993 2M

Given, $\vec{V} = x\cos^2 y\,\hat{i} + x^2 e^z\,\hat{j} + z\sin^2 y\,\hat{k}$ and S the surface of a unit cube with one corner at the origin and edges parallel to the coordinate axes, the value of the integral $\iint_S \vec{V}\cdot\hat{n}\,dS$ is ________

Electrical Engineering

Q4.26.1 NAT 2026 1M

Given that $\vec{F}(x,y,z) = \sin(y)\,\hat{x} + \cos(x)\,\hat{y} + 5\,\hat{z}$, the integral

$$\oiint_S \vec{F}(x,y,z)\cdot d\vec{s}$$

over the unit sphere S centered at the origin evaluates to ________ (Round off to one decimal place)

Q4.25.1 NAT 2025 2M

Let $(x,y)\in\mathbb{R}^2$. The rate of change of the real valued function,

$$V(x,y) = x^2 + x + y^2 + 1$$

at the origin in the direction of the point $(1,2)$ is ________ (round off to the nearest integer).

Q4.24.1 MCQ 2024 2M

Consider a vector $\vec{u} = 2\hat{x} + \hat{y} + 2\hat{z}$, where $\hat{x}, \hat{y}, \hat{z}$ represent unit vector along the coordinate axes x, y, z respectively. The directional derivative of the function

$$f(x,y,z) = 2\ln(xy) + \ln(yz) + 3\ln(xz)$$

at the point $(x, y, z) = (1, 1, 1)$ in the direction of $\vec{u}$ is

A. 0 **B.** $\frac{7}{5\sqrt{2}}$

C. 7 **D.** 21

Q4.23.1 NAT 2023 2M

A quadratic function of two variables is given as

$$f(x_1, x_2) = x_1^2 + 2x_2^2 + 3x_1 + 3x_2 + x_1x_2 + 1$$

The magnitude of the maximum rate of change of the function at the point $(1, 1)$ is __________(Round off to the nearest integer).

Q4.23.2 NAT 2023 2M

The closed curve shown in the figure is described by $r = 1+\cos\theta$, where $r = \sqrt{x^2 + y^2}$; $x = r\cos\theta$, $y = r\sin\theta$. The magnitude of the line integral of the vector field $\mathbf{F} = -y\hat{i} + x\hat{j}$ around the closed curve is __________ (Round off to 2 decimal places).

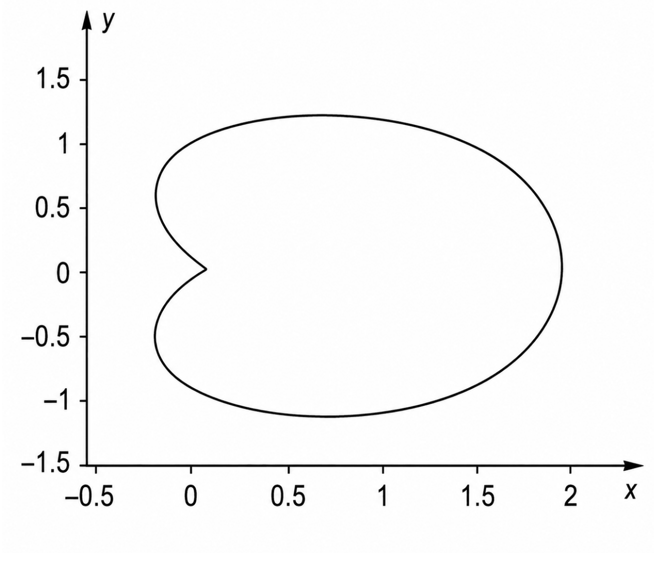


Fig. Q4.23.2

Q4.23.3 MCQ 2023 1M

For a given vector $W = [1, 2, 3]^T$, the vector normal to the plane defined by $W^T X = 1$ is

A. $[-2, -2, 2]^T$ **B.** $[3, 0, -1]^T$

C. $[3, 2, 1]^T$ **D.** $[1, 2, 3]^T$

Q4.22.1 MCQ 2022 2M

Let R be a region in the first quadrant of the xy plane enclosed by a closed curve C considered in counter-clockwise direction. Which of the following expressions does not represent the area of the region R?

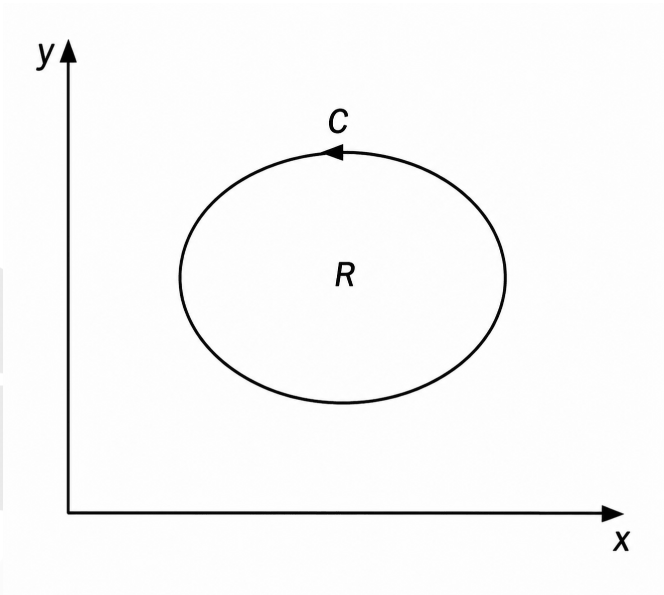


Fig. Q4.22.1

A. $\iint_R dxdy$ **B.** $\oint_C x\,dy$

C. $\oint_C y\,dx$ **D.** $\frac{1}{2}\oint_C (x\,dy - y\,dx)$

Q4.22.2 MCQ 2022 2M

Let $\vec{E}(x, y, z) = 2x^2\hat{i} + 5y\hat{j} + 3z\hat{k}$. The value of

$$\iiint_V (\vec{\nabla}\cdot\vec{E})\,dV,$$

where V is the volume enclosed by the unit cube defined by $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$, is

A. 3 **B.** 8

C. 10 **D.** 5

Q4.22.3 MCQ 2022 2M

Let $f(x, y, z) = 4x^2 + 7xy + 3xz^2$. The direction in which the function $f(x, y, z)$ increases most rapidly at point $P = (1, 0, 2)$ is

A. $20\hat{i}+7\hat{j}$

B. $20\hat{i}+7\hat{j}+12\hat{k}$

C. $20\hat{i}+12\hat{k}$

D. $20\hat{i}$

Q4.20.1 NAT 2020 2M

Let a_x and a_y be unit vectors along x and y directions, respectively. A vector function is given by $F = a_x y - a_y x$. The line integral of the above function $\int_C F \cdot dl$ along the curve C, which follows the parabola $y = x^2$, as shown below is ________ (rounded off to 2 decimal places).

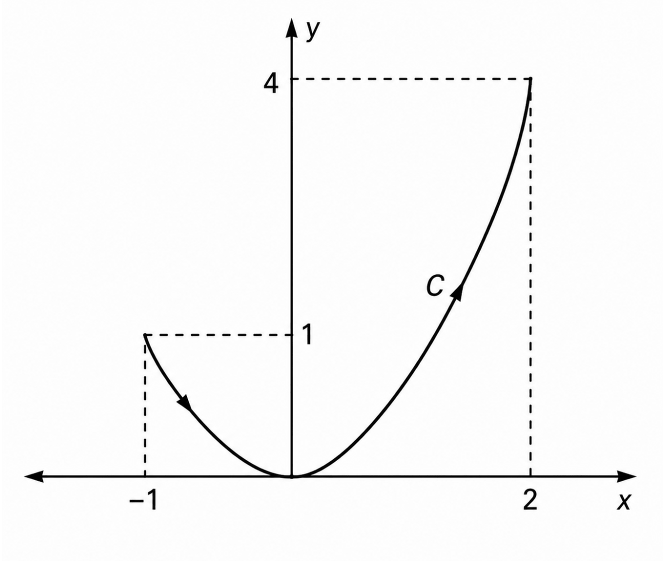


Fig. Q4.20.1

Q4.20.2 MCQ 2020 2M

The vector function expressed by

$$F = a_x(5y - k_1 z) + a_y(3z + k_2 x) + a_z(k_3 y - 4x)$$

represents a conservative field, where a_x, a_y, a_z are unit vectors along x, y and z directions, respectively. The values of constants k_1, k_2, k_3 are given by:

A. $k_1 = 3,\ k_2 = 3,\ k_3 = 7$

B. $k_1 = 4,\ k_2 = 5,\ k_3 = 3$

C. $k_1 = 3,\ k_2 = 8,\ k_3 = 5$

D. $k_1 = 0,\ k_2 = 0,\ k_3 = 0$

Q4.19.1 NAT 2019 1M

If $f = 2x^3 + 3y^2 + 4z$, the value of line integral $\int_C \nabla f \cdot d\mathbf{r}$ evaluated over contour C formed by the segments $(-3,-3,2) \to (2,-3,2) \to (2,6,2) \to (2,6,-1)$ is ________.

Q4.18.1 NAT 2018 2M

As shown in the figure, C is the arc from the point $(3,0)$ to the point $(0,3)$ on the circle $x^2+y^2=9$. The value of the integral

$$\int_C \left[(y^2+2yx)\,dx + (2xy+x^2)\,dy\right]$$

is ________ (upto 2 decimal places).

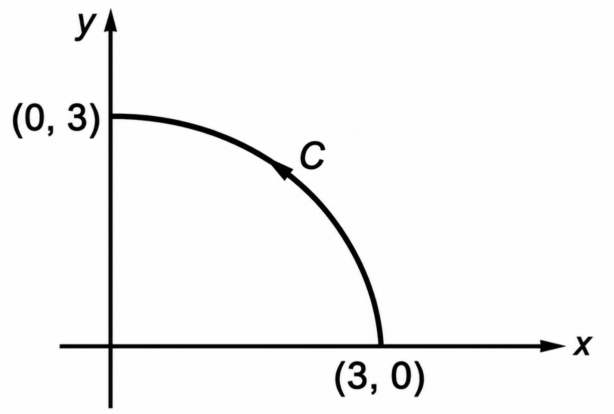


Fig. Q4.18.1

Q4.18.2 MCQ 2018 1M

The value of the directional derivative of the function $\phi(x,y,z) = xy^2 + yz^2 + zx^2$ at the point $(2,-1,1)$ in the direction of the vector $\mathbf{p} = \hat{i} + 2\hat{j} + 2\hat{k}$ is

A. 1

B. 0.95

C. 0.93

D. 0.9

Q4.16.1 NAT 2016 2M

The line integral of the vector field $F = 5xz\hat{i} + (3x^2+2y)\hat{j} + x^2z\hat{k}$ along a path from $(0,0,0)$ to $(1,1,1)$ parameterized by (t,t^2,t) is ________.

Q4.16.2 MCQ 2016 1M

The value of the line integral

$$\int_C (2xy^2\,dx + 2x^2y\,dy + dz)$$

along a path joining the origin $(0,0,0)$ and the point $(1,1,1)$ is

A. 0

B. 2

C. 4

D. 6

Q4.15.1 NAT 2015 2M

The volume enclosed by the surface $f(x,y) = e^x$ over the triangle bounded by the lines $x = y$, $x = 0$ and $y = 1$ in the xy plane is ________.

Q4.14.1 MCQ 2014 1M

Let $\nabla \cdot (f\vec{v}) = x^2y + y^2z + z^2x$, where f and $\vec{v}$ are scalar and vector fields respectively. If $\vec{v} = y\hat{i} + z\hat{j} + x\hat{k}$, then $\vec{v} \cdot \nabla f$ is

A. $x^2y + y^2z + z^2x$ **B.** $2xy + 2yz + 2zx$

C. $x + y + z$ **D.** 0

Q4.14.2 MCQ 2014 2M

The line integral of function $F = yz\hat{i}$, in the counter clockwise direction, along the circle $x^2 + y^2 = 1$ at $z = 1$ is

A. -2π **B.** $-\pi$

C. π **D.** 2π

Q4.13.1 MCQ 2013 1M

Given a vector field $\vec{F} = y^2x\hat{a}_x - yz\hat{a}_y - x^2\hat{a}_z$, the line integral $\int \vec{F} \cdot d\vec{l}$ evaluated along a segment on the x-axis from $x = 1$ to $x = 2$ is

A. -2.33 **B.** 0

C. 2.33 **D.** 7

Q4.13.2 MCQ 2013 1M

The curl of the gradient of the scalar field defined by $V = 2x^2y + 3y^2z + 4z^2x$ is

A. $4xy\,a_x + 6yz\,a_y + 8zx\,a_z$

B. $4a_x + 6a_y + 8a_z$

C. $(4xy + 4z^2)\,a_x + (2x^2 + 6yz)\,a_y + (3y^2 + 8zx)\,a_z$

D. 0

Q4.12.1 MCQ 2012 2M

The direction of vector $\vec{A}$ is radially outward from the origin, with $\left|\vec{A}\right| = kr^n$, where $r^2 = x^2 + y^2 + z^2$ and k is constant. The value of n for which $\nabla \cdot \vec{A} = 0$ is

A. -2 **B.** 2

C. 1 **D.** 0

Q4.11.1 MCQ 2011 1M

The two vectors $[1, 1, 1]$ and $[1, a, a^2]$ where $a = \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$, are

A. orthonormal. **B.** orthogonal.

C. parallel. **D.** collinear.

Q4.10.1 MCQ 2010 1M

Divergence of the three-dimensional radial vector field $\vec{r}$ is

A. 3 **B.** $1/r$

C. $\hat{i} + \hat{j} + \hat{k}$ **D.** $3(\hat{i} + \hat{j} + \hat{k})$

Q4.09.1 MCQ 2009 2M

$\vec{F}(x,y) = (x^2 + xy)\hat{a}_x + (y^2 + xy)\hat{a}_y$. Its line integral over the straight line from $(x,y) = (0,2)$ to $(x,y) = (2,0)$ evaluates to

A. -8 **B.** 4

C. 8 **D.** 0

Q4.06.1 MCQ 2006 2M

A surface $S(x,y) = 2x+5y-3$ is integrated once over a path consisting of the points that satisfy $(x+1)^2+(y-1)^2 = \sqrt{2}$. The integral evaluates to

A. $17\sqrt{2}$ **B.** $\frac{17}{\sqrt{2}}$

C. $\frac{\sqrt{2}}{17}$ **D.** 0

Q4.05.1 MCQ 2005 2M

For the scalar field $u = \frac{x^2}{2} + \frac{y^2}{3}$, magnitude of the gradient at the point $(1,3)$ is

A. $\sqrt{\frac{13}{9}}$ **B.** $\sqrt{\frac{9}{2}}$

C. $\sqrt{5}$ **D.** $\frac{9}{2}$

Instrumentation Engineering

Q4.25.1 MCQ 2025 2M

The value of the surface integral

$$\iint_S [(2x+z)\,dy\,dz + (2x+z)\,dx\,dz + (2z+y)\,dx\,dy]$$

over the sphere $S : x^2 + y^2 + z^2 = 9$ is

A. 72π **B.** 144π

C. 36π **D.** 432π

Q4.20.1 MCQ 2020 1M

The unit vectors along the mutually perpendicular x, y and z axes are $\hat{i}$, $\hat{j}$ and $\hat{k}$ respectively.Consider the plane $z = 0$ and two vector $\vec{a}$ and $\vec{b}$ on that plane such that $\vec{a} \neq \alpha\vec{b}$ for any scalar α. A vector perpendicular to both $\vec{a}$ and $\vec{b}$ is ______.

A. $\hat{i} - \hat{j}$ **B.** $-\hat{j}$

C. $\hat{k}$ **D.** $\hat{i}$

Q4.19.1 MCQ 2019 1M

$\vec{a}, \vec{b}, \vec{c}$ are three orthogonal vectors, Given that $\vec{a} = \hat{i} + 2\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, the vector $\vec{c}$ is parallel to

A. $\hat{i} + 2\hat{j} + 3\hat{k}$ **B.** $2\hat{i} + \hat{j}$

C. $2\hat{i} - \hat{j}$ **D.** $4\hat{k}$

Q4.18.1 MCQ 2018 2M

Given $\vec{F} = (x^2 - 2y)\,\hat{i} - 4yz\,\hat{j} + 4xz^2\,\hat{k}$, the value of the line integral

$$\int_C \vec{F} \cdot d\vec{l}$$

along the straight line C from $(0, 0, 0)$ to $(1, 1, 1)$ is

A. $\frac{3}{16}$ **B.** 0

C. $-\frac{5}{12}$ **D.** -1

Q4.17.1 NAT 2017 2M

The angle between two vectors $X_1 = [2\ 6\ 14]^T$ and $X_2 = [-12\ 8\ 16]^T$ in radian is ______.

Q4.16.1 MCQ 2016 1M

The vector that is NOT perpendicular to the vectors $(i + j + k)$ and $(i + 2j + 3k)$ is ____.

A. $(i - 2j + k)$ **B.** $(-i + 2j - k)$

C. $(0i + 0j + 0k)$ **D.** $(4i + 3j + 5k)$

Q4.15.1 MCQ 2015 1M

The magnitude of the directional derivative of the function $f(x, y) = x^2 + 3y^2$ in a direction normal to the circle $x^2 + y^2 = 2$, at the point $(1, 1)$, is

A. $4\sqrt{2}$ **B.** $5\sqrt{2}$

C. $7\sqrt{2}$ **D.** $9\sqrt{2}$

Q4.14.1 NAT 2014 1M

A vector is defined as

$$f = y\hat{i} + x\hat{j} + z\hat{k}$$

where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors in Cartesian (x, y, z) coordinate system.The surface integral $\oiint f \cdot ds$ over the closed surface S of a cube with vertices having the following coordinates: $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(1, 1, 1)$, $(0, 1, 1)$ is ____

Q4.13.1 MCQ 2013 1M

For a vector E, which one of the following statements is NOT TRUE?

A. If $\nabla \cdot E = 0$, E is called solenoidal.

B. If $\nabla \times E = 0$, E is called conservative.

C. If $\nabla \times E = 0$, E is called irrotational.

D. If $\nabla \cdot E = 0$, E is called irrotational.

Q4.12.1 MCQ 2012 2M

The direction of vector A is radially outward from origin, with $|A| = kr^n$ where $r^2 = x^2 + y^2 + z^2$ and k is a constant. The value of n for which $\nabla \cdot A = 0$ is

A. -2 **B.** 1

C. 2 **D.** 0

Q4.09.1 MCQ 2009 1M

A sphere of unit radius is centered at the origin. The unit normal at a point (x, y, z) on the surface of the sphere is the vector

A. (x, y, z)

B. $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

C. $\left(\frac{x}{\sqrt{3}}, \frac{y}{\sqrt{3}}, \frac{z}{\sqrt{3}}\right)$

D. $\left(\frac{x}{\sqrt{2}}, \frac{y}{\sqrt{2}}, \frac{z}{\sqrt{2}}\right)$

Mechanical Engineering

Q4.26.1 MCQ 2026 1M

Domain A is bounded by curve $x^2 = 4y$, ordinate $x = 2$, and x axis. The value of $\iint_A y\, dxdy$ is

A. $\frac{1}{5}$

B. $\frac{1}{3}$

C. $\frac{5}{12}$

D. $\frac{1}{2}$

Q4.25.1 NAT 2025 2M

Consider a velocity field $\vec{V} = 3z\hat{i} + 0\hat{j} + Cx\hat{k}$, where C is a constant. if the flow is irrotational, the value of C is __________ (rounded off to 1 decimal place).

Q4.25.2 NAT 2025 2M

The directional derivative of the function f given below at the point $(1, 0)$ in the direction of $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ is __________ (Rounded off to 1 decimal place).

$$f(x, y) = x^2 + xy^2$$

Q4.25.3 MCQ 2025 1M

The divergence of the curl of a vector field is

A. the magnitude of this vector field

B. the argument of this vector field

C. the magnitude of the curl of this vector field

D. Zero

Q4.24.1 MCQ 2024 1M

The value of the surface integral

$$\oiint_S z\, dxdy$$

where S is the external surface of the sphere $x^2 + y^2 + z^2 = R^2$ is

A. $4\pi R^3$

B. $\frac{4\pi R^3}{3}$

C. 0

D. πR^3

Q4.23.1 NAT 2023 1M

A vector field

$$B(x, y, z) = x\hat{i} + y\hat{j} - 2z\hat{k}$$

is defined over a conical region having height $h = 2$, base radius $r = 3$ and axis along z, as shown in the figure. The base of the cone lies in the x-y plane and is centered at the origin. If n denotes the unit outward normal to the curved surface S of the cone, the value of the integral,

$$\int_S B \cdot n\, dS$$

equals __________. [Answer in integer]

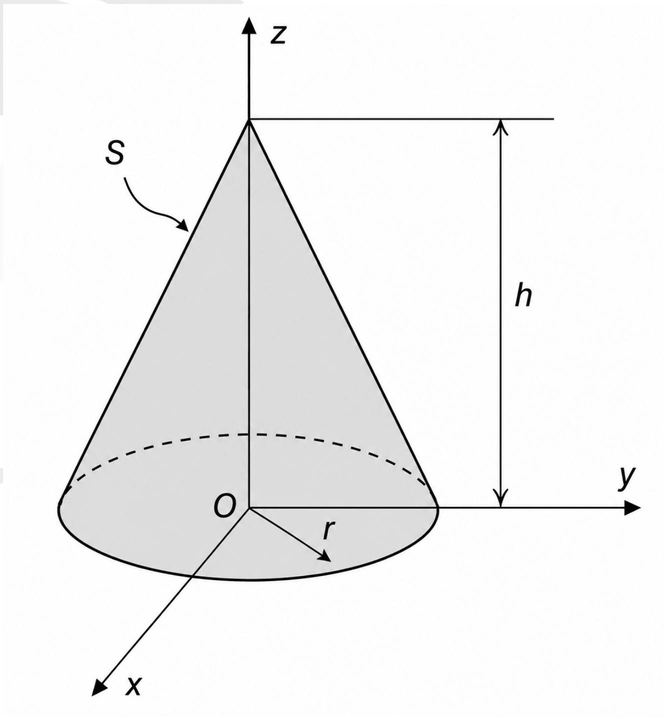


Fig. Q4.23.1

Q4.22.1 MCQ 2022 1M

Consider a cube of unit edge length and sides parallel to coordinate axes with its centroid at the point $(1, 2, 3)$. The surface integral $\int_A \vec{F} \cdot d\vec{A}$ of a vector field, $\vec{F} = 3x\hat{i} + 5y\hat{j} + 6z\hat{k}$ over the entire surface A area of the cube is _____.

A. 14

B. 27

C. 28

D. 31

Q4.22.2 NAT 2022 2M ◯ ◯ ◯

Consider two vectors:

$$\vec{a} = 5i + 7j + 2k$$

$$\vec{b} = 3i - j + 6k$$

Magnitude of the component of $\vec{a}$ orthogonal to $\vec{b}$ in the plane containing the vectors $\vec{a}$ and $\vec{b}$ is __________ (round off to 2 decimal places).

Q4.22.3 MCQ 2022 1M ◯ ◯ ◯

Given a function $\phi = \frac{1}{2}(x^2 + y^2 + z^2)$ in three-dimensional Cartesian space, the value of the surface integral

$$\oiint_S \hat{n} \cdot \nabla\phi \, dS,$$

where S is the surface of a sphere of unit radius and $\hat{n}$ is the outward unit normal vector on S, is

A. 4π **B.** 3π

C. $\frac{4\pi}{3}$ **D.** 0

Q4.20.1 MCQ 2020 2M ◯ ◯ ◯

The directional derivative $f(x, y, z) = xyz$ at point $(-1, 1, 3)$ in the direction of vector $\hat{\imath} - 2\hat{\jmath} + 2\hat{k}$ is

A. $\frac{7}{3}$ **B.** 7

C. $-\frac{7}{3}$ **D.** $3\hat{\imath} - 3\hat{\jmath} - \hat{k}$

Q4.20.2 MCQ 2020 2M ◯ ◯ ◯

A vector field is defined as

$$\vec{F}(x, y, z) = \frac{x}{[x^2 + y^2 + z^2]^{3/2}}\hat{\imath} + \frac{y}{[x^2 + y^2 + z^2]^{3/2}}\hat{\jmath} + \frac{z}{[x^2 + y^2 + z^2]^{3/2}}\hat{k}$$

where, $\hat{\imath}, \hat{\jmath}, \hat{k}$ are unit vectors along the axes of a right-handed rectangular/Cartesian coordinate system. The surface integral $\iint \vec{F} d\vec{S}$ (where $d\vec{S}$ is an elemental surface area vector) evaluated over the inner and outer surfaces of a spherical shell formed by two concentric spheres with origin as the center, and internal and external radii of 1 and 2, respectively, is

A. 0 **B.** 2π

C. 8π **D.** 4π

Q4.20.3 NAT 2020 1M ◯ ◯ ◯

For three vectors $\vec{A} = 2\hat{\jmath} - 3\hat{k}$, $\vec{B} = -2\hat{\imath} + \hat{k}$ and $\vec{C} = 3\hat{\imath} - \hat{\jmath}$, where $\hat{\imath}, \hat{\jmath}$ and $\hat{k}$ are unit vectors along the axes of a right-handed rectangular/Cartesian coordinate system, the value of

$$\left(\vec{A} \cdot (\vec{B} \times \vec{C}) + 6\right)$$

is __________.

Q4.19.1 MCQ 2019 2M ◯ ◯ ◯

Given a vector $\vec{U} = \frac{1}{3}\left(-y^3\hat{\imath} + x^3\hat{\jmath} + z^3\hat{k}\right)$ and $\hat{n}$ as the unit normal vector to the surface of the hemisphere $(x^2 + y^2 + z^2 = 1;\ z \geq 0)$, the value of integral $\int(\nabla \times \vec{U}) \cdot \hat{n}\, dS$ evaluated on the curved surface of the hemisphere S is

A. π **B.** $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$ **D.** $\frac{\pi}{3}$

Q4.19.2 MCQ 2019 1M ◯ ◯ ◯

The directional derivative of the function $f(x, y) = x^2 + y^2$ along a line directed from $(0, 0)$ to $(1, 1)$, evaluated at point $x = 1,\ y = 1$ is

A. $4\sqrt{2}$ **B.** 2

C. $2\sqrt{2}$ **D.** $\sqrt{2}$

Q4.19.3 MCQ 2019 1M ◯ ◯ ◯

The position vector OP of point $P(20, 10)$ is rotated anti-clockwise in $X - Y$ plane by an angle of 30° such that the point P occupies position Q as shown in the figure. The coordinates (x, y) of Q are

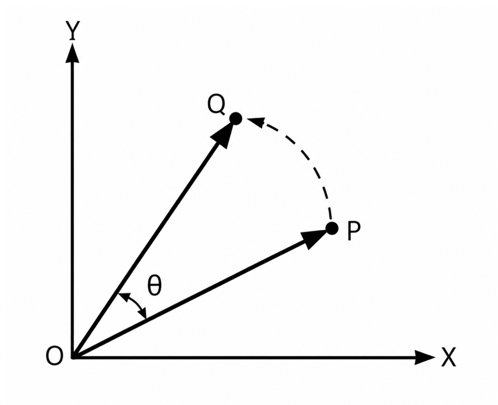


Fig. Q4.19.3

A. (13.40, 22.32) **B.** (22.32, 8.26)

C. (12.32, 18.66) **D.** (18.66, 12.32)

Q4.18.1 MCQ 2018 2M

For a position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ the norm of the vector can be defined as

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}.$$

Given a function $\phi = \ln|\vec{r}|$, its gradient $\nabla\phi$ is

A. $\vec{r}$ **B.** $\dfrac{\vec{r}}{|\vec{r}|}$

C. $\dfrac{\vec{r}}{\vec{r}\cdot\vec{r}}$ **D.** $\dfrac{\vec{r}}{|\vec{r}|^3}$

Q4.18.2 MCQ 2018 1M

The divergence of the vector field $\vec{u} = e^x(\cos y\,\hat{i} + \sin y\,\hat{j})$ is

A. 0 **B.** $e^x \cos y + e^x \sin y$

C. $2e^x \cos y$ **D.** $2e^x \sin y$

Q4.18.3 MCQ 2018 2M

The value of integral $\oiint_S \vec{r}\cdot\vec{n}\,ds$ over the closed surface S bounding a volume V, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector and $\vec{n}$ is the normal to the surface S, is

A. V **B.** $2V$

C. $3V$ **D.** $4V$

Q4.17.1 NAT 2017 2M

The surface integral $\iint_S \vec{F}\cdot\hat{n}\,dS$ over the surface S of the sphere $x^2 + y^2 + z^2 = 9$, where $\vec{F} = (x+y)\hat{i} + (x+z)\hat{j} + (y+z)\hat{k}$ and $\hat{n}$ is the unit outward surface normal, yields _____.

Q4.17.2 NAT 2017 2M

For the vector $\vec{V} = 2yz\hat{i} + 3xz\hat{j} + 4xy\hat{k}$, the value of $\nabla\cdot(\nabla\times\vec{V})$ is _____.

Q4.17.3 MCQ 2017 2M

A parametric curve defined by

$$x = \cos\left(\frac{\pi u}{2}\right), \qquad y = \sin\left(\frac{\pi u}{2}\right)$$

in the range $0 \le u \le 1$ is rotated about the x-axis by 360 degrees. Area of the surface generated is

A. $\dfrac{\pi}{2}$ **B.** π

C. 2π **D.** 4π

Q4.17.4 MCQ 2017 1M

Consider the two-dimensional velocity field given by

$$\vec{V} = (5 + a_1x + b_1y)\hat{i} + (4 + a_2x + b_2y)\hat{j},$$

where a_1, b_1, a_2 and b_2 are constants. Which one of the following conditions needs to be satisfied for the flow to be incompressible?

A. $a_1 + b_1 = 0$ **B.** $a_1 + b_2 = 0$

C. $a_2 + b_2 = 0$ **D.** $a_2 + b_1 = 0$

Q4.16.1 NAT 2016 2M

The value of the line integral $\oint_C \vec{F}\cdot\vec{r}'\,ds$, where C is a circle of radius $\dfrac{4}{\sqrt{\pi}}$ units is _____.

Here, $\vec{F}(x,y) = y\hat{i} + 2x\hat{j}$ and $\vec{r}'$ is the UNIT tangent vector on the curve C at an arc length s from a reference point on the curve. $\hat{i}$ and $\hat{j}$ are the basis vectors in the x-y Cartesian reference. In evaluating the line integral, the curve has to be traversed in the counter-clockwise direction.

Q4.16.2 NAT 2016 2M

A scalar potential ϕ has the following gradient : $\nabla\phi = yz\hat{i} + xz\hat{j} + xy\hat{k}$. Consider the integral $\int_C \nabla\phi\cdot d\vec{r}$ on the curve $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. The curve C is parameterized as follows :

$$\begin{cases} x = t \\ y = t^2 \\ z = 3t^2 \end{cases} \quad \text{and } 1 \le t \le 3$$

The value of the integral is _____

Q4.15.1 NAT 2015 2M

The surface integral $\iint_S \frac{1}{\pi}(9xi - 3yj) \cdot n\, dS$ over the sphere given by $x^2 + y^2 + z^2 = 9$ is ______.

Q4.15.2 NAT 2015 2M

The value of $\int_C \left[(3x - 8y^2)\, dx + (4y - 6xy)\, dy\right]$, (where C is the boundary of the region boundary by $x = 0$, $y = 0$ and $x + y = 1$) is ______.

Q4.15.3 MCQ 2015 2M

Curl of vector $V(x, y, z) = 2x^2\hat{i} + 3z^2\hat{j} + y^3\hat{k}$ at $x = y = z = 1$ is

A. $-3\hat{i}$

B. $3\hat{i}$

C. $3\hat{i} - 4\hat{j}$

D. $3\hat{i} - 6\hat{k}$

Q4.15.4 NAT 2015 2M

Consider a spatial curve in three-dimensional space given in parametric form by

$$x(t) = \cos t, \qquad y(t) = \sin t, \qquad z(t) = \frac{2}{\pi}t, 0 \le t \le \frac{\pi}{2}.$$

The length of the curve is ________.

Q4.14.1 MCQ 2014 1M

Divergence of the vector field $x^2z\hat{i} + xy\hat{j} - yz^2\hat{k}$ at $(1, -1, 1)$ is

A. 0

B. 3

C. 5

D. 6

Q4.14.2 MCQ 2014 1M

Curl of vector $\vec{F} = x^2z^2\hat{i} - 2xy^2z\hat{j} + 2y^2z^3\hat{k}$ is

A. $(4yz^3 + 2xy^2)\hat{i} + 2x^2z\hat{j} - 2y^2z\hat{k}$

B. $(4yz^3 + 2xy^2)\hat{i} - 2x^2z\hat{j} - 2y^2z\hat{k}$

C. $2xz^2\hat{i} - 4xyz\hat{j} + 6y^2z^2\hat{k}$

D. $2xz^2\hat{i} + 4xyz\hat{j} + 6y^2z^2\hat{k}$

Q4.14.3 MCQ 2014 1M

Which one of the following describes the relationship among the three vectors, $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $5\hat{i} + 6\hat{j} + 4\hat{k}$?

A. The vectors are mutually perpendicular

B. The vectors are linearly dependent

C. The vectors are linearly independent

D. The vectors are unit vectors

Q4.14.4 MCQ 2014 2M

The integral

$$\oint_C (y\, dx - x\, dy)$$

is evaluated along the circle $x^2 + y^2 = \frac{1}{4}$ traversed in counter clockwise direction. The integral is equal to

A. 0

B. $-\frac{\pi}{4}$

C. $-\frac{\pi}{2}$

D. $\frac{\pi}{4}$

Q4.13.1 MCQ 2013 2M

The following surface integral is to be evaluated over a sphere for the given steady velocity vector field $F = x\hat{i} + y\hat{j} + z\hat{k}$ defined with respect to a Cartesian coordinate system having $\hat{i}$, $\hat{j}$ and $\hat{k}$ as unit base vectors.

$$\iint_S \frac{1}{4}(F \cdot n)\, dA$$

where S is the sphere, $x^2 + y^2 + z^2 = 1$ and n is the outward unit normal vector to the sphere. The value of the surface integral is

A. π

B. 2π

C. $3\pi/4$

D. 4π

Q4.12.1 MCQ 2012 1M

For the spherical surface $x^2 + y^2 + z^2 = 1$, the unit outward normal vector at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ is given by

A. $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

B. $\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

C. $\hat{k}$

D. $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}$

Q4.10.1 MCQ 2010 2M ◯ ◯ ◯

Velocity vector of a flow field is given as $\vec{V} = 2xy\hat{i} - x^2z\hat{j}$. The vorticity vector at $(1, 1, 1)$ is

A. $4\hat{i} - \hat{j}$ **B.** $4\hat{i} - \hat{k}$

C. $\hat{i} - 4\hat{j}$ **D.** $\hat{i} - 4\hat{k}$

Q4.09.1 MCQ 2009 1M ◯ ◯ ◯

The divergence of the vector field $3xz\hat{i} + 2xy\hat{j} - yz^2\hat{k}$ at a point $(1, 1, 1)$ is equal to

A. 7 **B.** 4

C. 3 **D.** 0

Q4.09.2 MCQ 2009 2M ◯ ◯ ◯

A path AB in the form of one quarter of a circle of unit radius is shown in the figure. Integration of $(x + y)^2$ on path AB traversed in a counter-clockwise sense is

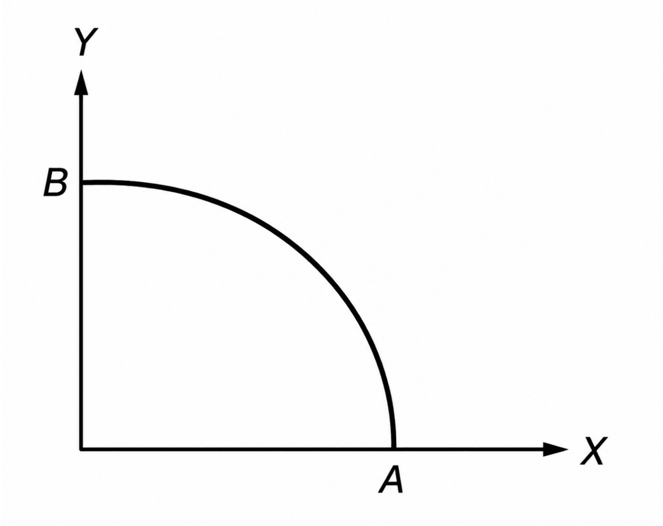


Fig. Q4.09.2

A. $\frac{\pi}{2} - 1$ **B.** $\frac{\pi}{2} + 1$

C. $\frac{\pi}{2}$ **D.** 1

Q4.08.1 MCQ 2008 2M ◯ ◯ ◯

The directional derivative of the scalar function $f(x, y, z) = x^2 + 2y^2 + z$ at the point $P = (1, 1, 2)$ in the direction of the vector $\vec{a} = 3\hat{i} - 4\hat{j}$ is

A. -4 **B.** -2

C. -1 **D.** 1

Q4.07.1 MCQ 2007 2M ◯ ◯ ◯

The area of a triangle formed by the tips of vectors $\overline{a}$, $\overline{b}$ and $\overline{c}$ is

A. $\frac{1}{2}(\overline{a} - \overline{b}) \cdot (\overline{a} - \overline{c})$ **B.** $\frac{1}{2}\left|(\overline{a} - \overline{b}) \times (\overline{a} - \overline{c})\right|$

C. $\frac{1}{2}\left|\overline{a} \times \overline{b} \times \overline{c}\right|$ **D.** $\frac{1}{2}(\overline{a} \times \overline{b}) \cdot \overline{c}$

Q4.05.1 MCQ 2005 2M ◯ ◯ ◯

The line integral $\int \vec{V} \cdot d\vec{r}$ of the vector $\vec{V}(\vec{r}) = 2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$ from the origin to the point $P(1, 1, 1)$

A. is 1

B. is zero

C. is -1

D. cannot be determined without specifying path

Q4.05.2 MCQ 2005 1M ◯ ◯ ◯

By a change of variables

$$x(u, v) = uv, \qquad y(u, v) = \frac{v}{u}$$

in a double integral, the integrand $f(x, y)$ changes to

$$f\left(uv, \frac{v}{u}\right)\phi(u, v).$$

Then, $\phi(u, v)$ is

A. $2\frac{v}{u}$ **B.** $2uv$

C. v^2 **D.** 1

Q4.04.1 MCQ 2004 1M ◯ ◯ ◯

The angle between two unit-magnitude coplanar vectors $P(0.866, 0.500, 0)$ and $Q(0.259, 0.966, 0)$ will be

A. $0°$ **B.** $30°$

C. $45°$ **D.** $60°$

Q4.99.1 MCQ 1999 1M

For the function $\phi = ax^2y - y^3$ to represent the velocity potential of an ideal fluid, $\nabla^2\phi$ should be equal to zero. In that case, the value of 'a' has to be:

A. -1

B. 1

C. -3

D. 3

Q4.95.1 MCQ 1995 1M

If $\overline{V}$ is a differentiable vector function and f is sufficiently differentiable scalar function then curl$(f\overline{V})$ = __________.

A. $(\text{grad } f) \times \overline{V} + (f \text{ curl } \overline{V})$

B. $\overline{o}$

C. $f \text{ curl } \overline{V}$

D. $(\text{grad } f) \times \overline{V}$

Civil Engineering

Q4.26.1 MCQ 2026 2M

Vector field $\vec{V}$ is defined as

$$\vec{V} = 3x^2yz\,\hat{\imath} - 5xy\,\hat{\jmath} + 6yz^2\,\hat{k}$$

The curl of $\vec{V}$ at point $(2, -1, 1)$ is

A. $6\hat{\imath} - 12\hat{\jmath} - 7\hat{k}$

B. $-12\hat{\imath} - 10\hat{\jmath} - 12\hat{k}$

C. -34

D. $\begin{bmatrix} -12 & 12 & -12 \\ 5 & -10 & 0 \\ 0 & 6 & -12 \end{bmatrix}$

Q4.25.1 MSQ 2025 1M

Consider a velocity vector, $\vec{V}$ in (x, y, z) coordinates given below. Pick one or more CORRECT statement(s) from the choices given below:

$$\vec{V} = u\hat{x} + v\hat{y}$$

☐ **A.** z-component of Curl of velocity;

$$\nabla \times \vec{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{z}$$

☐ **B.** z-component of Curl of velocity;

$$\nabla \times \vec{V} = \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)\hat{z}$$

☐ **C.** Divergence of velocity;

$$\nabla \cdot \vec{V} = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

☐ **D.** Divergence of velocity;

$$\nabla \cdot \vec{V} = \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

Q4.24.1 MSQ 2024 2M

Three vectors $\vec{p}$, $\vec{q}$, and $\vec{r}$ are given as

$$\vec{p} = \hat{\imath} + \hat{\jmath} + \hat{k}$$

$$\vec{q} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$$

$$\vec{r} = 2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}$$

Which of the following is/are CORRECT?

☐ **A.** $\vec{p} \times (\vec{q} \times \vec{r}) = (\vec{p} \times \vec{q}) \times \vec{r}$

☐ **B.** $\vec{r} \cdot (\vec{p} \times \vec{q}) = (\vec{q} \times \vec{p}) \cdot \vec{r}$

☐ **C.** $\vec{p} \times (\vec{q} \times \vec{r}) + \vec{q} \times (\vec{r} \times \vec{p}) + \vec{r} \times (\vec{p} \times \vec{q}) = \vec{0}$

☐ **D.** $\vec{p} \times (\vec{q} \times \vec{r}) = (\vec{p} \cdot \vec{r})\vec{q} - (\vec{p} \cdot \vec{q})\vec{r}$

Q4.24.2 MCQ 2024 2M

A vector field $\vec{p}$ and a scalar field r are given by

$$\vec{p} = (2x^2 - 3xy + z^2)\hat{\imath} + (2y^2 - 3yz + x^2)\hat{\jmath} + (2z^2 - 3zx + x^2)\hat{k}$$

$$r = 6x^2 + 4y^2 - z^2 - 9xyz - 2xy + 3xz - yz$$

Consider the statement P and Q:
P : Curl of the gradient of the scalar field r is a null vector.
Q : Divergence of curl of the vector field $\vec{p}$ is zero.
Which one of the following options is CORRECT?

A. Both P and Q are TRUE.

B. Both P and Q are FALSE

C. P is TRUE and Q FALSE.

D. P is FALSE and Q is TRUE.

Q4.23.1 MCQ 2023 1M

Let ϕ be a scalar field, and u be a vector field.
Which of the following identities is true for $\mathrm{div}(\phi u)$?

A. $\mathrm{div}(\phi u) = \phi\,\mathrm{div}(u) + u \cdot \mathrm{grad}(\phi)$

B. $\mathrm{div}(\phi u) = \phi\,\mathrm{div}(u) + u \times \mathrm{grad}(\phi)$

C. $\mathrm{div}(\phi u) = \phi\,\mathrm{grad}(u) + u \cdot \mathrm{grad}(\phi)$

D. $\mathrm{div}(\phi u) = \phi\,\mathrm{grad}(u) + u \times \mathrm{grad}(\phi)$

Q4.21.1 MCQ 2021 1M

The unit normal vector to the surface $X^2+Y^2+Z^2-48=0$ at the point $(4,4,4)$ is

A. $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ **B.** $\frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}$

C. $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ **D.** $\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}$

Q4.21.2 NAT 2021 2M

A function is defined in Cartesian coordinate system as $f(x,y) = xe^y$. The value of the directional derivative of the function (in integer) at the point $(2,0)$ along the direction of the straight line segment from point $(2,0)$ to point $\left(\frac{1}{2}, 2\right)$ is ________.

Q4.20.1 NAT 2020 2M

If C represents a line segment between $(0,0,0)$ and $(1,1,1)$ in Cartesian coordinate system, the value (expressed as integer) of the line integral

$$\int_C [(y+z)\,dx + (x+z)\,dy + (x+y)\,dz]$$

is ________.

Q4.18.1 NAT 2018 2M

The value (up to two decimal places) of a line integral $\int_C \vec{F}(\vec{r}) \cdot d\vec{r}$, for $\vec{F}(\vec{r}) = x^2\hat{i} + y^2\hat{j}$ along C which is a straight line joining $(0,0)$ to $(1,1)$ is ________.

Q4.15.1 NAT 2015 2M

The directional derivative of the field

$$u(x,y,z) = x^2 - 3yz$$

in the direction of the vector $(\hat{i} + \hat{j} - 2\hat{k})$ at point $(2,-1,4)$ is ________.

Q4.12.1 MCQ 2012 2M

For the parallelogram $OPQR$ shown in the sketch, $\overrightarrow{OP} = a\hat{i} + b\hat{j}$ and $\overrightarrow{OR} = c\hat{i} + d\hat{j}$. The area of the parallelogram is

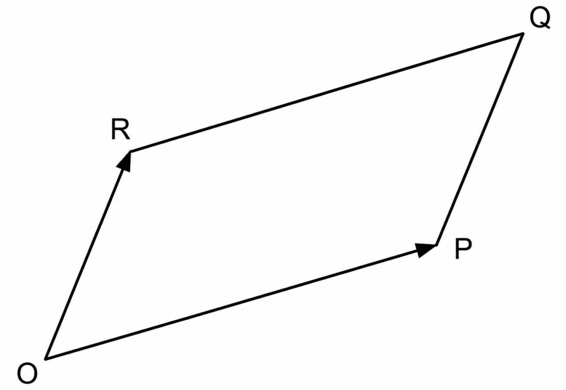


Fig. Q4.12.1

A. $ad - bc$ **B.** $ac + bd$

C. $ad + bc$ **D.** $ab - cd$

Q4.11.1 MCQ 2011 1M

If $\vec{a}$ and $\vec{b}$ are two arbitrary vectors with magnitudes a and b, respectively, $\left|\vec{a} \times \vec{b}\right|^2$ will be equal to

A. $a^2b^2 - (\vec{a} \cdot \vec{b})^2$ **B.** $ab - \vec{a} \cdot \vec{b}$

C. $a^2b^2 + (\vec{a} \cdot \vec{b})^2$ **D.** $ab + \vec{a} \cdot \vec{b}$

Q4.09.1 MCQ 2009 2M

For a scalar function $f(x,y,z) = x^2 + 3y^2 + 2z^2$, the directional derivative at the point $P(1,2,-1)$ in the direction of a vector $\hat{i} - \hat{j} + 2\hat{k}$ is

A. -18 **B.** $-3\sqrt{6}$

C. $3\sqrt{6}$ **D.** 18

Q4.09.2 MCQ 2009 1M

For a scalar function $f(x,y,z) = x^2+3y^2+2z^2$, the gradient at the point $P(1,2,-1)$ is

A. $2\hat{i} + 6\hat{j} + 4\hat{k}$ **B.** $2\hat{i} + 12\hat{j} - 4\hat{k}$

C. $2\hat{i} + 12\hat{j} + 4\hat{k}$ **D.** $\sqrt{56}$

Q4.06.1 MCQ 2006 2M

The directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at the point $P(2, 1, 3)$ in the direction of the vector $\vec{a} = \hat{i} - 2\hat{k}$ is

A. -2.785 **B.** -2.145

C. -1.789 **D.** 1.000

Q4.05.1 MCQ 2005 2M

Value of the integral $\oint_C (xydy - y^2dx)$, where, C is the square cut from the first quadrant by the lines $x = 1$ and $y = 1$ will be (Use Green's theorem to change the line integral into double integral)

A. $\frac{1}{2}$ **B.** 1

C. $\frac{3}{2}$ **D.** $\frac{5}{3}$

Q4.03.1 MCQ 2003 1M

If P, Q and R are three points having coordinates $(3, -2, -1)$, $(1, 3, 4)$, $(2, 1, -2)$ in XYZ space, then the distance from point P to plane OQR (O being the origin of the coordinate system) is given by

A. 3 **B.** 5

C. 7 **D.** 9

Chemical Engineering

Q4.26.1 MCQ 2026 1M

A two-dimensional temperature profile is given as $T = 2x^2 + 3xy + y^2$. If $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y directions, respectively, which one of the following is the directional derivative of T at the location $x = 2, y = 2$?

A. $8\hat{i} + 12\hat{j}$ **B.** $14\hat{i} + 10\hat{j}$

C. $8\hat{i} - 12\hat{j}$ **D.** $14\hat{i} - 10\hat{j}$

Q4.26.2 MCQ 2026 2M

Consider a flow with the following velocity field

$$\vec{V} = (x + y)\hat{i} + (y + z)\hat{j} + (z + x)\hat{k}$$

where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors in the x, y and z directions, respectively. Which one of the following is CORRECT?

A. The flow is incompressible and irrotational.

B. The flow is incompressible and rotational.

C. The flow is compressible and rotational.

D. The flow is compressible and irrotational.

Q4.25.1 MCQ 2025 2M

Consider a Cartesian coordinate system with orthogonal unit basis vectors $\hat{i}$, $\hat{j}$ defined over a domain: $x, y \in [0, 1]$. Choose the condition for which the divergence of the vector field $v = ax\hat{i} - by\hat{j}$ is zero.

A. $a - b = 0$ **B.** $a < b$

C. $a > b$ **D.** $a + b = 0$

Q4.24.1 NAT 2024 2M

Consider the line integral $\int_C F(r) \cdot dr$, with $F(r) = x\hat{i} + y\hat{j} + z\hat{k}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors in the (x, y, z) Cartesian coordinate system. The path C is given by $r(t) = \cos(t)\hat{i} + \sin(t)\hat{j} + t\hat{k}$, where $0 \le t \le \pi$. The value of the integral, rounded off to 2 decimal places, is ________.

Q4.23.1 MCQ 2023 1M

For a two-dimensional plane, the unit vectors, $(\hat{e}_r, \hat{e}_\theta)$ of the polar coordinate system and $(\hat{i}, \hat{j})$ of the Cartesian coordinate system, are related by the following two equations.

$$\hat{e}_r = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$$

$$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

Which one of the following is the CORRECT value of $\frac{\partial(\hat{e}_r + \hat{e}_\theta)}{\partial\theta}$?

A. 1 **B.** $\hat{e}_\theta$

C. $\hat{e}_r + \hat{e}_\theta$ **D.** $-\hat{e}_r + \hat{e}_\theta$

Q4.22.1 NAT 2022 2M

Consider a sphere of radius 4, centered at the origin, with outward unit normal $\hat{n}$ on its surface S. The value of the surface integral

$$\iint_S \left(\frac{2x\hat{i} + 3y\hat{j} + 4z\hat{k}}{4\pi}\right) \cdot \hat{n}\, dA$$

is ________.

(rounded off to one decimal place)

Production & Industrial Engineering

Q4.26.1 NAT 2026 1M

Consider the line integral

$$\int_C (2x\,dx + 2y\,dy + 2z\,dz)$$

where C is a semi-circle in the $z = 0$ plane with start point at $(0, 0, 0)$ and end point at $(1, 0, 0)$.
The value of the line integral is ______ (in integer).

Q4.25.1 NAT 2025 2M

If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the orthogonal unit vectors in Cartesian x-y-z coordinate system, the rate of change of the function $f(x, y, z) = x^2 + 2y^2 + z$ at point $(1, 1, 1)$ in the direction of $3\hat{i} + 4\hat{k}$ is ________. (Answer in integer)

Q4.24.1 MCQ 2024 1M

A vector field is given as $F(x, y) = (100x + 100y)\hat{i} + (-50x + 200y)\hat{j}$, where $\hat{i}$ and $\hat{j}$ are the unit vectors along the x and y axes in the Cartesian frame, respectively. Then the value of

$$\oint_C F(x, y) \cdot dl$$

where $dl = dx\hat{i} + dy\hat{j}$ is an elemental path taken over an anticlockwise circular contour C of radius $r = 2$ is

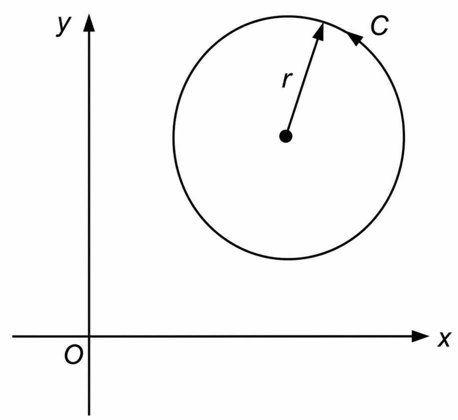


Fig. Q4.24.1

A. -100π **B.** -800π

C. -400π **D.** 400π

Q4.22.1 MCQ 2022 1M

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors, the vector triple product $(\vec{a} \times \vec{b}) \times \vec{c}$ is given by

A. $(\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b}$ **B.** $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$

C. $(\vec{b} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{c})\vec{b}$ **D.** $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

Q4.18.1 NAT 2018 1M

The value of surface integral

$$\iint_S (9x\hat{i} - 2y\hat{j} - z\hat{k}) \cdot \hat{n}\, dS$$

over the surface of sphere S, $x^2 + y^2 + z^2 = 9$ where $\hat{n}$ is unit outward normal to surface element dS is ________.

Q4.17.1 MCQ 2017 1M

For two non-zero vectors $\vec{A}$ and $\vec{B}$, if $\vec{A} + \vec{B}$ is perpendicular to $\vec{A} - \vec{B}$ then,

A. Magnitude of $\vec{A}$ is twice magnitude of $\vec{B}$.

B. Magnitude of $\vec{A}$ is half the magnitude of $\vec{B}$.

C. $\vec{A}$ and $\vec{B}$ cannot be orthogonal.

D. The magnitudes of $\vec{A}$ and $\vec{B}$ are equal.

Q4.14.1 MCQ 2014 2M

If $\phi = 2x^3y^2z^4$ then $\nabla^2\phi$ is

A. $12xy^2z^4 + 4x^2z^2 + 20x^3y^2z^3$

B. $2x^2y^2z + 4x^3z^4 + 24x^3y^2z^2$

C. $12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2$

D. $4xyz^2 + 4x^2z^2 + 24x^3y^2z^2$

Q4.14.2 MCQ 2014 1M

Directional derivative of $\phi = 2xz - y^2$ at the point $(1, 3, 2)$ becomes maximum in the direction of:

A. $4\hat{i} + 2\hat{j} - 3\hat{k}$ **B.** $4\hat{i} - 6\hat{j} + 2\hat{k}$

C. $2\hat{i} - 6\hat{j} + 2\hat{k}$ **D.** $4\hat{i} - 6\hat{j} - 2\hat{k}$

Q4.11.1 MCQ 2011 1M

If $A(0,4,3)$, $B(0,0,0)$ and $C(3,0,4)$ are three points defined in x, y, z co-ordinate system, then which of the following vector is perpendicular to both vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$.

A. $16\hat{i} + 9\hat{j} - 12\hat{k}$ **B.** $16\hat{i} - 9\hat{j} + 12\hat{k}$

C. $16\hat{i} - 9\hat{j} - 12\hat{k}$ **D.** $16\hat{i} + 9\hat{j} + 12\hat{k}$

Q4.11.2 MCQ 2011 1M

The line integral

$$\int_{P_1}^{P_2} (y\,dx + x\,dy)$$

for $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$ along the semicircle P_1, P_2 shown in the figure is

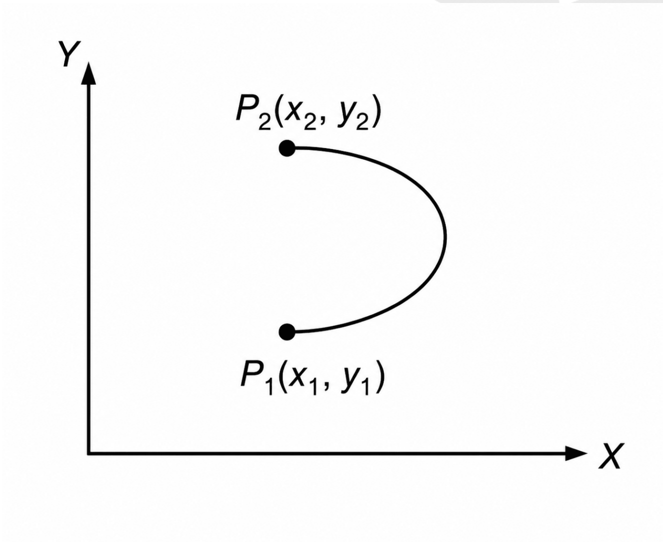


Fig. Q4.11.2

A. $x_2y_2 - x_1y_1$ **B.** $(y_2^2 - y_1^2) + (x_2^2 - x_1^2)$

C. $(x_2 - x_1)(y_2 - y_1)$ **D.** $(y_2 - y_1)^2 + (x_2 - x_1)^2$

Q4.08.1 MCQ 2008 1M

If $\vec{r}$ is the position vector of any point on a closed surface S that encloses volume V then $\iint_S \vec{r} \cdot d\vec{S}$ is equal to

A. $\frac{1}{2}V$ **B.** V

C. $2V$ **D.** $3V$

Q4.07.1 MCQ 2007 2M

Divergence of vector field $\vec{V}(x,y,z) = -(x\cos xy + y)\hat{i} + (y\cos xy)\hat{j} + [(\sin z^2) + x^2 + y^2]\hat{k}$ is

A. $2z\cos z^2$ **B.** $\sin xy + 2z\cos z^2$

C. $x\sin xy - \cos z$ **D.** None of these

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q4.26.1 D	Q4.24.1 A,C	Q4.24.2 A,B,D	Q4.23.1 C	Q4.23.2 B	Q4.23.3 B
Q4.22.1 A	Q4.21.1 A	Q4.20.1 C	Q4.19.1 C	Q4.17.1 -11	Q4.17.2 B
Q4.17.3 54 to 55	Q4.16.1 20	Q4.16.2 0	Q4.15.1 A	Q4.14.1 3	Q4.14.2 3
Q4.14.3 3.13 to 3.15	Q4.13.1 D	Q4.13.2 D	Q4.12.1 A	Q4.11.1 D	Q4.10.1 C
Q4.08.1 B	Q4.08.2 A	Q4.06.1 A	Q4.93.1 1		
ELECTRICAL ENGINEERING					
Q4.26.1 0	Q4.25.1 0 to 1	Q4.24.1 C	Q4.23.1 10	Q4.23.2 9 to 10	Q4.23.3 D
Q4.22.1 C	Q4.22.2 C	Q4.22.3 B	Q4.20.1 -3	Q4.20.2 B	Q4.19.1 139
Q4.18.1 0	Q4.18.2 A	Q4.16.1 4 to 5	Q4.16.2 B	Q4.15.1 0 to 1	Q4.14.1 A
Q4.14.2 B	Q4.13.1 B	Q4.13.2 D	Q4.12.1 A	Q4.11.1 B	Q4.10.1 A
Q4.09.1 D	Q4.06.1 D	Q4.05.1 C			
INSTRUMENTATION ENGINEERING					
Q4.25.1 B	Q4.20.1 C	Q4.19.1 C	Q4.18.1 D	Q4.17.1 0.65 to 0.80	Q4.16.1 D
Q4.15.1 A	Q4.14.1 1	Q4.13.1 D	Q4.12.1 A	Q4.09.1 A	
MECHANICAL ENGINEERING					
Q4.26.1 A	Q4.25.1 3	Q4.25.2 1	Q4.25.3 D	Q4.24.1 B	Q4.23.1 0
Q4.22.1 A	Q4.22.2 8 to 9	Q4.22.3 A	Q4.20.1 A	Q4.20.2 A	Q4.20.3 6
Q4.19.1 B	Q4.19.2 C	Q4.19.3 C	Q4.18.1 C	Q4.18.2 C	Q4.18.3 C
Q4.17.1 226	Q4.17.2 0	Q4.17.3 C	Q4.17.4 B	Q4.16.1 16	Q4.16.2 726
Q4.15.1 216	Q4.15.2 1 to 2	Q4.15.3 A	Q4.15.4 1.86	Q4.14.1 C	Q4.14.2 A

Q4.14.3 B	Q4.14.4 C	Q4.13.1 A	Q4.12.1 A	Q4.10.1 D	Q4.09.1 C
Q4.09.2 B	Q4.08.1 B	Q4.07.1 B	Q4.05.1 A	Q4.05.2 A	Q4.04.1 C
Q4.99.1 D	Q4.95.1 A				
CIVIL ENGINEERING					
Q4.26.1 A	Q4.25.1 A,C	Q4.24.1 B,C,D	Q4.24.2 A	Q4.23.1 A	Q4.21.1 A
Q4.21.2 1	Q4.20.1 3	Q4.18.1 0 to 1	Q4.15.1 -5.72 to -5.70	Q4.12.1 A	Q4.11.1 A
Q4.09.1 B	Q4.09.2 B	Q4.06.1 C	Q4.05.1 C	Q4.03.1 A	
CHEMICAL ENGINEERING					
Q4.26.1 B	Q4.26.2 C	Q4.25.1 A	Q4.24.1 4.5 to 5	Q4.23.1 D	Q4.22.1 192
PRODUCTION & INDUSTRIAL ENGINEERING					
Q4.26.1 1	Q4.25.1 2	Q4.24.1 *	Q4.22.1 B	Q4.18.1 677 to 679	Q4.17.1 D
Q4.14.1 C	Q4.14.2 B	Q4.11.1 A	Q4.11.2 A	Q4.08.1 D	Q4.07.1 A

UNIT

V

Complex Variable

Topics

1. Complex Numbers
2. Analytic Functions
3. Cauchy–Riemann Equations
4. Harmonic Functions
5. Complex Integration
6. Cauchy's Integral Theorem
7. Taylor Series
8. Laurent Series
9. Residue Theorem
10. Conformal Mapping

Note to Reader

Complex Variable Analysis is present in branches with paper codes AE, BM, CH, EC, EE, IN, MA, MN, ME, NM, PE, XE according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 31 **MCQ:** 22 **MSQ:** 1 **NAT:** 8
1M: 17 **2M:** 14 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2022)–2M(2025)

EE — Electrical Engineering

Questions: 27 **MCQ:** 23 **MSQ:** 1 **NAT:** 3
1M: 17 **2M:** 10 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–3M(2021)

IN — Instrumentation Engineering

Questions: 30 **MCQ:** 23 **MSQ:** 3 **NAT:** 4
1M: 17 **2M:** 13 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2022)

ME — Mechanical Engineering

Questions: 26 **MCQ:** 21 **MSQ:** 0 **NAT:** 5
1M: 12 **2M:** 14 **Desc:** 0 **Avg Weightage** ~2M **Range:** 0M(2015)–3M(2020)

CE — Civil Engineering

Questions: 8 **MCQ:** 8 **MSQ:** 0 **NAT:** 0
1M: 5 **2M:** 3 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2011)–1M(2011)

CH — Chemical Engineering

Questions: 14 **MCQ:** 12 **MSQ:** 0 **NAT:** 2
1M: 8 **2M:** 6 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–3M(2015)

PI — Production & Industrial Engineering

Questions: 15 **MCQ:** 13 **MSQ:** 1 **NAT:** 1
1M: 11 **2M:** 4 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q5.25.1 MSQ | 2025 | 2M

Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?

- [ ] **A.** $\oint_C e^z\,dz = 0$
- [ ] **B.** $\oint_C z^n\,dz = 0$, where n is an even integer
- [ ] **C.** $\oint_C \cos z\,dz \neq 0$
- [ ] **D.** $\oint_C \sec z\,dz \neq 0$

Q5.24.1 MCQ | 2024 | 2M

Let z be a complex variable. If $f(z) = \frac{\sin(\pi z)}{z^2(z-2)}$ and C is the circle in the complex plane with $|z| = 3$ then $\oint_C f(z)\,dz$ is ________.

A. $\pi^2 j$ **B.** $j\pi\left(\frac{1}{2}-\pi\right)$ **C.** $j\pi\left(\frac{1}{2}+\pi\right)$ **D.** $-\pi^2 j$

Q5.23.1 MCQ | 2023 | 1M

Let $w^4 = 16j$. Which of the following cannot be a value of w?

A. $2e^{\frac{j2\pi}{8}}$ **B.** $2e^{\frac{j\pi}{8}}$ **C.** $2e^{\frac{j5\pi}{8}}$ **D.** $2e^{\frac{j9\pi}{8}}$

Q5.23.2 MCQ | 2023 | 1M

The value of the contour integral, $\oint_C \left(\frac{z+2}{z^2+2z+2}\right) dz$, where the contour C is $\left\{z : \left|z+1-\frac{3}{2}j\right| = 1\right\}$, taken in the counter clockwise direction, is

A. $-\pi(1+j)$ **B.** $\pi(1+j)$

C. $\pi(1-j)$ **D.** $-\pi(1-j)$

Q5.22.1 NAT | 2022 | 1M

A simple closed path C in the complex plane is shown in the figure. If

$$\oint_C \frac{2^z}{z^2-1}\,dz = -i\pi A,$$

where $i = \sqrt{-1}$, then the value of A is ________ (rounded off to two decimal places).

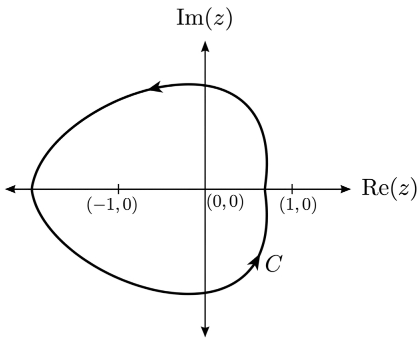


Fig. Q5.22.1

Q5.21.1 MCQ | 2021 | 2M

Consider the integral

$$\oint_C \frac{\sin(x)}{x^2(x^2+4)}\,dx$$

where C is a counter-clockwise oriented circle defined as $|x-i| = 2$. The value of the integral is

A. $-\frac{\pi}{8}\sin(2i)$ **B.** $\frac{\pi}{8}\sin(2i)$

C. $-\frac{\pi}{4}\sin(2i)$ **D.** $\frac{\pi}{4}\sin(2i)$

Q5.19.1 MCQ | 2019 | 1M

Which one of the following functions is analytic over the entire complex plane?

A. $\ln(z)$ **B.** $e^{1/z}$ **C.** $\dfrac{1}{1-z}$ **D.** $\cos(z)$

Q5.19.2 NAT | 2019 | 1M

The value of the contour integral

$$\frac{1}{2\pi j}\oint\left(z+\frac{1}{z}\right)^2 dz$$

evaluated over the unit circle $|z| = 1$ is __________.

Q5.18.1 NAT | 2018 | 2M

The contour C given below is on the complex plane $z = x + jy$, where $j = \sqrt{-1}$. The value of the integral $\frac{1}{\pi j}\oint_C \frac{dz}{z^2-1}$ is ________.

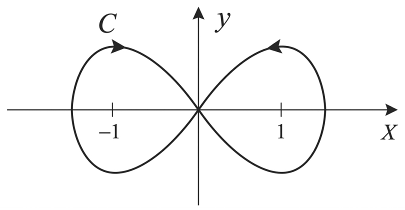


Fig. Q5.18.1

Q5.17.1 | MCQ | 2017 | 1M | ○ ○ ○

The residues of a function

$$f(z) = \frac{1}{(z-4)(z+1)^3}$$

are

A. $\frac{-1}{27}$ and $\frac{-1}{125}$ **B.** $\frac{1}{125}$ and $\frac{-1}{125}$

C. $\frac{-1}{27}$ and $\frac{1}{5}$ **D.** $\frac{1}{125}$ and $\frac{-1}{5}$

Q5.17.2 | MCQ | 2017 | 1M | ○ ○ ○

An integral I over a counterclockwise circle C is given by

$$I = \oint_C \frac{z^2-1}{z^2+1} e^z dz.$$

If C is defined as $|z| = 3$, then the value of I is

A. $-\pi i \sin(1)$ **B.** $-2\pi i \sin(1)$

C. $-3\pi i \sin(1)$ **D.** $-4\pi i \sin(1)$

Q5.16.1 | NAT | 2016 | 2M | ○ ○ ○

In the following integral, the contour C encloses the points $2\pi j$ and $-2\pi j$

$$-\frac{1}{2\pi} \oint_C \frac{\sin z}{(z-2\pi j)^3} dz$$

The value of the integral is ___________

Q5.16.2 | NAT | 2016 | 1M | ○ ○ ○

Consider the complex valued function $f(z) = 2z^3 + b|z|^3$ where z is a complex variable. The value of b for which the function $f(z)$ is analytic is ___________.

Q5.16.3 | MCQ | 2016 | 2M | ○ ○ ○

The values of the integral $\frac{1}{2\pi j} \oint_c \frac{e^z}{z-2} dz$ along a closed contour c in anti-clockwise direction for

(i) The point $z_0 = 2$ inside the contour c, and

(ii) The point $z_0 = 2$ outside the contour c,

Respectively, are

A. (i) 2.72 (ii) 0 **B.** (i) 7.39 (ii) 0

C. (i) 0 (ii) 2.72 **D.** (i) 0 (ii) 7.39

Q5.15.1 | MCQ | 2015 | 1M | ○ ○ ○

Let $z = x + iy$ be a complex variable. Consider that contour integration is performed along the unit circle in anticlockwise direction. Which one of the following statements is NOT TRUE?

A. The residue of $\frac{z}{z^2-1}$ at $z = 1$ is $1/2$

B. $\oint_C z^2 dz = 0$

C. $\frac{1}{2\pi i} \oint_C \frac{1}{z} dz = 1$

D. $\bar{z}$ (complex conjugate of z) is an analytical function

Q5.15.2 | MCQ | 2015 | 1M | ○ ○ ○

The value of x for which all the eigen-values of the matrix given below are real is

$$\begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

A. $5 + j$ **B.** $5 - j$ **C.** $1 - 5j$ **D.** $1 + 5j$

Q5.15.3 | NAT | 2015 | 1M | ○ ○ ○

Let $f(z) = \frac{az+b}{cz+d}$. If $f(z_1) = f(z_2)$ for all $z_1 \neq z_2$, $a = 2$, $b = 4$ and $c = 5$, then d should be equal to ___________.

Q5.15.4 | NAT | 2015 | 2M | ○ ○ ○

If C denotes the counterclockwise unit circle, the value of the contour integral

$$\frac{1}{2\pi j} \oint_C Re\{z\} dz$$

is ___________.

Q5.15.5 MCQ 2015 1M

If C is a circle of radius r with centre z_0, in the complex z-plane and if n is a non-zero integer, then $\oint_C \frac{dz}{(z-z_0)^{n+1}}$ equals

A. $2\pi nj$ **B.** 0 **C.** $\frac{nj}{2\pi}$ **D.** $2\pi n$

Q5.14.1 MCQ 2014 2M

The real part of an analytic function $f(z)$ where $z = x + jy$ is given by $e^{-y}\cos(x)$. The imaginary part of $f(z)$ is

A. $e^{y}\cos(x)$ **B.** $e^{-y}\sin(x)$

C. $-e^{y}\sin(x)$ **D.** $-e^{-y}\sin(x)$

Q5.14.2 MCQ 2014 1M

For an all-pass system $H(z) = \frac{(z^{-1}-b)}{(1-az^{-1})}$, where $|H(e^{-j\omega})| = 1$, for all ω. If $\text{Re}(a) \neq 0$, $\text{Im}(a) \neq 0$, then b equals

A. a **B.** a^* **C.** $1/a^*$ **D.** $1/a$

Q5.14.3 NAT 2014 2M

Let $H_1(z) = (1-pz^{-1})^{-1}$, $H_2(z) = (1-qz^{-1})^{-1}$, $H(z) = H_1(z) + rH_2(z)$. The quantities p, q, r are real numbers. Consider $p = \frac{1}{2}$, $q = -\frac{1}{4}$, $|r| < 1$. If the zero of $H(z)$ lies on the unit circle, then r = ______

Q5.12.1 MCQ 2012 1M

Given $f(z) = \frac{1}{z+1} - \frac{2}{z+3}$. If C is a counterclockwise path in the z-plane such that $|z+1| = 1$, the value of $\frac{1}{2\pi j}\oint_C f(z)\,dz$ is

A. -2 **B.** -1 **C.** 1 **D.** 2

Q5.11.1 MCQ 2011 1M

The value of the integral $\oint_C \frac{-3z+4}{(z^2+4z+5)}dz$ where C is the circle $|z| = 1$ is given by

A. 0 **B.** $1/10$ **C.** $4/5$ **D.** 1

Q5.10.1 MCQ 2010 2M

The residues of a complex function $X(z) = \frac{1-2z}{z(z-1)(z-2)}$ at its poles are

A. $\frac{1}{2}, -\frac{1}{2}$ and 1 **B.** $\frac{1}{2}, \frac{1}{2}$ and -1

C. $\frac{1}{2}$, 1 and $-\frac{3}{2}$ **D.** $\frac{1}{2}$, -1 and $\frac{3}{2}$

Q5.09.1 MCQ 2009 1M

If $f(z) = c_0 + c_1 z^{-1}$, then $\oint_{\substack{\text{unit}\\ \text{circle}}} \frac{1+f(z)}{z}dz$ is given by

A. $2\pi c_1$ **B.** $2\pi(1+c_0)$ **C.** $2\pi j c_1$ **D.** $2\pi j(1+c_0)$

Q5.08.1 MCQ 2008 1M

The equation $\sin(z) = 10$ has

A. no real or complex solution

B. exactly two distinct complex solutions

C. a unique solution

D. an infinite number of complex solutions

Q5.08.2 MCQ 2008 2M

The residue of the function $f(z) = \frac{1}{(z+2)^2(z-2)^2}$ at $z = 2$ is

A. $-\frac{1}{32}$ **B.** $-\frac{1}{16}$ **C.** $\frac{1}{16}$ **D.** $\frac{1}{32}$

Q5.07.1 MCQ 2007 2M

If the semi-circular contour D of radius 2 is as shown in the figure, then the value of the integral $\oint_D \frac{1}{(s^2-1)}ds$ is

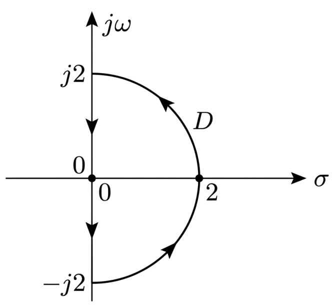


Fig. Q5.07.1

A. $j\pi$ **B.** $-j\pi$ **C.** $-\pi$ **D.** π

Q5.06.1 MCQ 2006 2M

For the function of a complex variable $W = \ln Z$ (where, $W = u + jv$ and $Z = x + jy$), then u = constant lines get mapped in Z-plane as

A. Set of radial straight lines **B.** Set of concentric circles

C. Set of confocal hyperbolas **D.** Set of confocal ellipses

Q5.06.2 MCQ 2006 2M

The value of the constant integral $\oint_{|z-j|=2} \frac{1}{z^2+4} dz$ in positive sense is

A. $\frac{j\pi}{2}$ **B.** $\frac{-\pi}{2}$ **C.** $\frac{-j\pi}{2}$ **D.** $\frac{\pi}{2}$

Electrical Engineering

Q5.26.1 NAT 2026 2M

The magnitude of the contour integral

$$\oint_C \left(\frac{(z+1)^2}{(z-i)(z-2)} \right) dz$$

over the contour $C : |z - 2 - i| = 3/2$ is _____ (Round off to two decimal places) Note: z is a complex variable and $i = \sqrt{-1}$

Q5.25.1 NAT 2025 2M

Let C be a clockwise oriented closed curve in the complex plane defined by $|z| = 1$. Further, let $f(z) = jz$ be a complex function, where $j = \sqrt{-1}$. Then, $\oint_C f(z)dz$ = _____ (round off to the nearest integer).

Q5.24.1 MSQ 2024 1M

Which of the following complex functions is/are analytic on the complex plane?

- ☐ **A.** $f(z) = j\mathrm{Re}(z)$
- ☐ **B.** $f(z) = \mathrm{Im}(z)$
- ☐ **C.** $f(z) = e^{|z|}$
- ☐ **D.** $f(z) = z^2 - z$

Q5.24.2 NAT 2024 1M

Consider the complex function $f(z) = \cos z + e^{z^2}$. The coefficient of z^5 in the Taylor series expansion of $f(z)$ about the origin is _____ (rounded off to 1 decimal place).

Q5.21.1 MCQ 2021 1M

Let $p(z) = z^3 + (1 + j)z^2 + (2 + j)z + 3$, where z is a complex number. Which one of the following is true?

A. conjugate$\{p(z)\} = p($conjugate$\{z\})$ for all z

B. The sum of the roots of $p(z) = 0$ is a real number

C. The complex roots of the equation $p(z) = 0$ come in conjugate pairs

D. All the roots cannot be real

Q5.21.2 MCQ 2021 2M

Let $(-1 - j)$, $(3 - j)$, $(3 + j)$ and $(-1 + j)$ be the vertices of a rectangle C in the complex plane. Assuming that C is traversed in counter-clockwise direction, the value of the contour integral $\oint_C \frac{dz}{z^2(z-4)}$ is

A. $j\pi/2$ **B.** 0 **C.** $-j\pi/8$ **D.** $j\pi/16$

Q5.20.1 MCQ 2020 1M

The value of the following complex integral, with C representing the unit circle centered at origin in the counter-clockwise sense, is:

$$\int_C \frac{z^2+1}{z^2-2z} dz$$

A. $8\pi i$ **B.** $-8\pi i$ **C.** $-\pi i$ **D.** πi

Q5.19.1 MCQ 2019 1M

Which one of the following functions is analytic in the region $|z| \leq 1$?

A. $\frac{z^2-1}{z}$ **B.** $\frac{z^2-1}{z+2}$ **C.** $\frac{z^2-1}{z-0.5}$ **D.** $\frac{z^2-1}{z+j0.5}$

Q5.19.2 MCQ 2019 2M

The closed loop line integral

$$\oint_{|z|=5} \frac{z^3+z^2+8}{z+2} dz$$

evaluated counter-clockwise, is

A. $+8j\pi$ **B.** $-8j\pi$ **C.** $-4j\pi$ **D.** $+4j\pi$

Q5.18.1 MCQ 2018 1M

The value of the integral $\oint_C \frac{z+1}{z^2-4} dz$ in counter clockwise direction around a circle C of radius 1 with center at the point $z = -2$ is

A. $\frac{\pi i}{2}$ **B.** $2\pi i$ **C.** $-\frac{\pi i}{2}$ **D.** $-2\pi i$

Q5.18.2 MCQ 2018 2M

If C is a circle $|z| = 4$ and $f(z) = \frac{z^2}{(z^2-3z+2)^2}$, then $\oint_C f(z)dz$ is

A. 1 **B.** 0 **C.** -1 **D.** -2

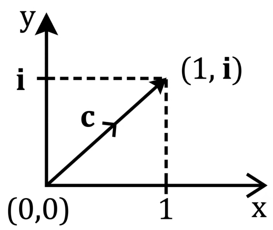
Q5.17.1 MCQ 2017 1M

For a complex number z , $\lim_{z\to i} \frac{z^2+1}{z^3+2z-i(z^2+2)}$ is

A. $-2i$ **B.** $-i$ **C.** i **D.** $2i$

Q5.17.2 MCQ 2017 2M

Consider the line integral $I = \int_c (x^2 + iy^2)\, dz$, where $z = x + iy$. The line c is shown in the figure below.



The value of I is

A. $\frac{1}{2}i$ **B.** $\frac{2}{3}i$ **C.** $\frac{3}{4}i$ **D.** $\frac{4}{5}i$

Q5.16.1 MCQ 2016 1M

The value of the integral

$$\oint_C \frac{2z+5}{\left(z-\frac{1}{2}\right)(z^2-4z+5)} dz$$

over the contour $|z| = 1$, taken in the anti-clockwise direction, would be

A. $\frac{24\pi i}{13}$ **B.** $\frac{48\pi i}{13}$ **C.** $\frac{24}{13}$ **D.** $\frac{12}{13}$

Q5.16.2 MCQ 2016 1M

Consider the function $f(z) = z + z^*$ where z is a complex variable and z^* denotes it complex conjugate. Which one of the following is TRUE?

A. $f(z)$ is both continuous and analytic

B. $f(z)$ is continuous but not analytic

C. $f(z)$ is not continuous but is analytic

D. $f(z)$ is neither continuous nor analytic

Q5.15.1 MCQ 2015 1M

Given $f(z) = g(z) + h(z)$, where f, g, h are complex valued functions of a complex variable z. Which one of the following statements is TRUE?

A. If $f(z)$ is differentiable at z_0, then $g(z)$ and $h(z)$ are also differentiable at z_0.

B. If $g(z)$ and $h(z)$ are differentiable at z_0, then $f(z)$ is also differentiable at z_0.

C. If $f(z)$ is continuous at z_0, then it is differentiable at z_0.

D. If $f(z)$ is differentiable at z_0, then so are its real and imaginary parts.

Q5.14.1 MCQ 2014 1M

Let S be the set of points in the complex plane corresponding to the unit circle. (That is, $S = \{z : |z| = 1\}$). Consider the function $f(z) = zz^*$ where z^* denotes the complex conjugate of z. The $f(z)$ maps S to which one of the following in the complex plane

A. unit circle

B. horizontal axis line segment from origin to $(1, 0)$

C. the point $(1, 0)$

D. the entire horizontal axis

Q5.14.2 MCQ 2014 1M

All the values of the multi-valued complex function 1^i, where $i = \sqrt{-1}$, are

A. purely imaginary. **B.** real and non-negative.

C. on the unit circle. **D.** equal in real and imaginary parts.

Q5.14.3 MCQ 2014 2M

Integration of the complex function $f(z) = \frac{z^2}{z^2-1}$, in the counterclockwise direction, around $|z-1| = 1$, is

A. $-\pi i$ **B.** 0 **C.** πi **D.** $2\pi i$

Q5.13.1 MCQ 2013 1M

Square roots of $-i$, where $i = \sqrt{-1}$, are

A. $i, -i$

B. $\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}), \cos(\frac{3\pi}{4}) + i\sin(\frac{3\pi}{4})$

C. $\cos(\frac{\pi}{4}) + i\sin(\frac{3\pi}{4}), \cos(\frac{3\pi}{4}) + i\sin(\frac{\pi}{4})$

D. $\cos(\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4}), \cos(-\frac{3\pi}{4}) + i\sin(\frac{3\pi}{4})$

Q5.13.2 MCQ 2013 2M

$\oint \frac{z^2-4}{z^2+4}\,dz$ evaluated anticlockwise around the circle $|z-i| = 2$, where $i = \sqrt{-1}$, is

A. -4π **B.** 0 **C.** $2+\pi$ **D.** $2+2i$

Q5.12.1 MCQ 2012 1M

If $x = \sqrt{-1}$, then the value of x^x is

A. $e^{-\pi/2}$ **B.** $e^{\pi/2}$ **C.** x **D.** 1

Q5.11.1 MCQ 2011 1M

Roots of the algebraic equation $x^3 + x^2 + x + 1 = 0$ are

A. $(+1, +j, -j)$ **B.** $(+1, -1, +1)$

C. $(0, 0, 0)$ **D.** $(-1, +j, -j)$

Q5.11.2 MCQ 2011 1M

A point z has been plotted in the complex plane, as shown in figure below. The plot of the complex number $y = \frac{1}{z}$ is

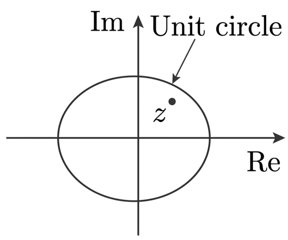
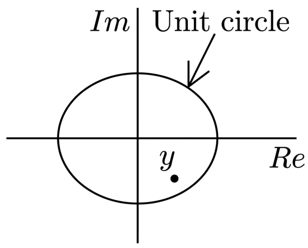
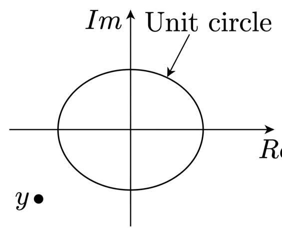


Fig. Q5.11.2

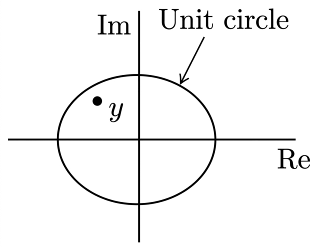
A.



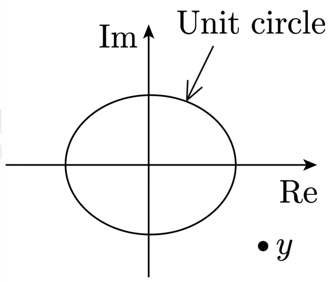
B.



C.



D.



Q5.08.1 MCQ 2008 2M

Given $X(z) = \frac{z}{(z-a)^2}$ with $|z| > a$, the residue of $X(z)z^{n-1}$ at $z = a$ for $n \geq 0$ will be

A. a^{n-1} **B.** a^n **C.** na^n **D.** na^{n-1}

Q5.07.1 MCQ 2007 2M

The value of $\oint_C \frac{dz}{(1+z^2)}$ where C is the contour $|z - i/2| = 1$ is

A. $2\pi i$ **B.** π **C.** $\tan^{-1} z$ **D.** $\pi i \tan^{-1} z$

Q5.97.1 MCQ 1997 1M

e^z is a periodic with a period of

A. 2π **B.** $2\pi i$ **C.** π **D.** $i\pi$

Instrumentation Engineering

Q5.26.1 MCQ 2026 1M

Consider a function $f(z) = z^2 + z + 1$ where $z \in \mathbb{C}$ is a complex variable. A simple closed contour γ in z-plane encloses the point $z = 1 + 0j$. The value of integral $\oint_\gamma \frac{f(z)}{z-1}\,dz =$ ________.

A. $6\pi j$ **B.** $3\pi j$ **C.** $12\pi j$ **D.** πj

Q5.25.1 NAT 2025 1M

Consider the function $f(z) = \frac{2z+1}{z^2-z}$, where z is a complex variable. The sum of the residues at singular points of $f(z)$ is _____ (in integer).

Q5.25.2 MSQ 2025 2M

Choose the correct statement(s) from the following options, regarding Cauchy's theorem on complex integration $\oint_C f(z)\,dz$ where C is a simple closed path in a simply connected domain D.

- ☐ **A.** Cauchy's theorem cannot be directly applied to conclude that $\oint_C f(z)\,dz = 0$ when $f(z) = \frac{1}{z^2}$, and C is the unit circle
- ☐ **B.** If $f(z)$ is analytic in D, then it can be concluded that $\oint_C f(z)\,dz = 0$ for any simple closed path C in D
- ☐ **C.** The function $f(z)$ must be analytic in D to conclude $\oint_C f(z)\,dz = 0$ for any simple closed path C in D
- ☐ **D.** $\oint_C f(z)\,dz \neq 0$ when $f(z) = \frac{1}{z^2}$, since the function is not analytic at $z = 0$

Q5.24.1 MCQ 2024 1M

Let $z = x + iy$ be a complex variable and $\bar{z}$ be its complex conjugate. The equation $\bar{z}^2 + z^2 = 2$ represents a

A. parabola **B.** hyperbola **C.** ellipse **D.** circle

Q5.24.2 MSQ 2024 2M

The complex functions $f(z) = u(x,y) + i\,v(x,y)$ and $\overline{f(z)} = u(x,y) - i\,v(x,y)$ are both analytic in a given domain. Choose the correct option(s) from the following.

A. $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0$ **B.** $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \neq 0$

C. $\frac{df(z)}{dz} = 0$ **D.** $\frac{df(z)}{dz} \neq 0$

Q5.23.1 MCQ 2023 2M

Let $f(z) = j\frac{1-z}{1+z}$, where z denotes a complex number and j denotes $\sqrt{-1}$. The inverse function $f^{-1}(z)$ maps the real axis to the __________.

A. unit circle with centre at the origin

B. unit circle with centre not at the origin

C. imaginary axis

D. real axis

Q5.22.1 MSQ 2022 2M

For the complex number $z = \frac{a+jb}{a-jb}$, where $a > 0$ and $b > 0$. Which of the following statement(s) is/are true?

- ☐ **A.** The phase is $2\tan^{-1}\frac{b}{a}$
- ☐ **B.** The phase is $\tan^{-1}\frac{2b}{a}$
- ☐ **C.** The magnitude is 1
- ☐ **D.** The magnitude is $\sqrt{\frac{a^2+b^2}{a^2-b^2}}$

Q5.22.2 NAT 2022 2M

Consider the function $f(z) = \frac{1}{(z+1)(z+2)(z+3)}$. The residue of $f(z)$ at $z = -1$, is ________.

Q5.21.1 NAT 2021 1M

Let $f(z) = \frac{1}{z^2+6z+9}$ defined in the complex plane. The integral $\oint_c f(z)\,dz$ over the contour of a circle c with center at the origin and unit radius is ________.

Q5.21.2 MCQ 2021 2M

$f(z) = (z-1)^{-1} - 1 + (z-1) - (z-1)^2 + \cdots$ is the series expansion of

A. $\dfrac{-1}{z(z-1)}$ for $|z-1| < 1$

B. $\dfrac{1}{z(z-1)}$ for $|z-1| < 1$

C. $\dfrac{1}{(z-1)^2}$ for $|z-1| < 1$

D. $\dfrac{-1}{(z-1)}$ for $|z-1| < 1$

Q5.20.1 MCQ 2020 1M

Let $f(z) = \frac{1}{z+a}, a > 0$. The value of the integral $\oint f(z)dz$ over a circle C with center $(-a, 0)$ and radius $R > 0$ evaluated in the anti-clockwise direction is ________

A. 0 **B.** $2\pi i$ **C.** $-2\pi i$ **D.** $4\pi i$

Q5.19.1 MCQ 2019 2M

A complex function $f(z) = u(x, y) + i\,v(x, y)$ and its complex conjugate $f^*(z) = u(x, y) - i\,v(x, y)$ are both analytic in the entire complex plane, where $z = x + i\,y$ and $i = \sqrt{-1}$. The function f is then given by

A. $f(z) = x + i\,y$ **B.** $f(z) = x^2 - y^2 + i\,2xy$

C. $f(z) =$ constant **D.** $f(z) = x^2 + y^2$

Q5.18.1 MCQ 2018 1M

Let $f_1(z) = z^2$ and $f_2(z) = \bar{z}$ be two complex variable functions. Here $\bar{z}$ is the complex conjugate of z. Choose the correct answer.

A. Both $f_1(z)$ and $f_2(z)$ are analytic

B. Only $f_1(z)$ is analytic

C. Only $f_2(z)$ is analytic

D. Both $f_1(z)$ and $f_2(z)$ are not analytic

Q5.17.1 MCQ 2017 1M

Let $z = x + jy$ where $j = \sqrt{-1}$. Then $\overline{\cos z} =$

A. $\cos z$ **B.** $\cos \bar{z}$ **C.** $\sin z$ **D.** $\sin \bar{z}$

Q5.16.1 NAT 2016 2M

The value of the integral $\dfrac{1}{2\pi j}\oint_C \dfrac{z^2+1}{z^2-1}\,dz$ where z is a complex number and C is a unit circle with center at $1 + 0j$ in the complex plane is ________.

Q5.15.1 MCQ 2015 1M

The value of $\oint \dfrac{1}{z^2}\,dz$, where the contour is the unit circle traversed clockwise, is

A. $-2\pi i$ **B.** 0 **C.** $2\pi i$ **D.** $4\pi i$

Q5.13.1 MCQ 2013 1M

The complex function $\tanh(s)$ is analytic over a region of the imaginary axis of the complex s-plane if the following is **TRUE** everywhere in the region for all integers n

A. $Re(s) = 0$ **B.** $Im(s) \neq n\pi$

C. $Im(s) \neq \dfrac{n\pi}{3}$ **D.** $Im(s) \neq \dfrac{(2n+1)\pi}{2}$

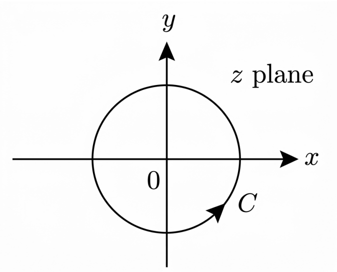
Q5.11.1 MCQ 2011 1M

The contour integral $\oint_C e^{1/z}\,dz$ with C as the counter-clockwise unit circle in the z-plane is equal to

A. 0 **B.** 2π **C.** $2\pi\sqrt{-1}$ **D.** ∞

Q5.10.1 MCQ 2010 1M

The contour C in the adjoining figure is described by $x^2 + y^2 = 16$. The value of $\oint_C \dfrac{z^2+8}{0.5z - 1.5j}\,dz$ is ________.(Note: $j = \sqrt{-1}$)



A. $-2\pi j$ **B.** $2\pi j$ **C.** $4\pi j$ **D.** $-4\pi j$

Q5.09.1 MCQ 2009 1M

If $z = x + jy$, where x and y are real, the value of $\left|e^{jz}\right|$ is

A. 1 **B.** $e^{\sqrt{x^2+y^2}}$ **C.** e^y **D.** e^{-y}

Q5.09.2 MCQ 2009 1M

The value of $\oint \frac{\sin z}{z}\, dz$, where the contour of integration is a simple closed curve around the origin, is

A. 0 **B.** $2\pi j$ **C.** ∞ **D.** $1/(2\pi j)$

Q5.09.3 MCQ 2009 2M

One of the roots of the equation $x^3 = j$, where j is the positive square root of -1, is

A. j

B. $\frac{\sqrt{3}}{2} + j\frac{1}{2}$

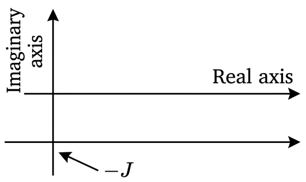
C. $\frac{\sqrt{3}}{2} - j\frac{1}{2}$

D. $-\frac{\sqrt{3}}{2} - j\frac{1}{2}$

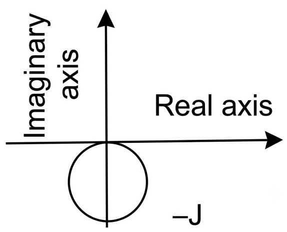
Q5.08.1 MCQ 2008 2M

A complex variable $z = x + j\,0.1$ has its real part x varying in the range $-\infty$ to $+\infty$. Which one of the following is the locus (shown in thick lines) of $\frac{1}{z}$ in the complex plane?

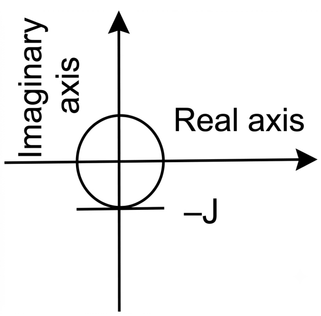
A.



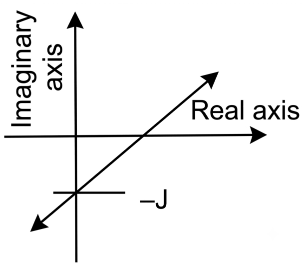
B.



C.



D.



Q5.07.1 MCQ 2007 2M

For the function $\frac{\sin z}{z^3}$ of a complex variable z, the point $z = 0$ is

A. a pole of order 3

B. a pole of order 2

C. a pole of order 1

D. not a singularity

Q5.05.1 MCQ 2005 2M

Consider the circle $|Z - 5 - 5i| = 2$ in the complex plane (x, y) with $Z = x + iy$. The minimum distance from the origin to the circle is

A. $5\sqrt{2} - 2$ **B.** $\sqrt{54}$ **C.** $\sqrt{34}$ **D.** $5\sqrt{2}$

Q5.05.2 MCQ 2005 2M

Let $Z^3 = \overline{Z}$, where Z is a complex number not equal to zero. Then Z is a solution of

A. $Z^2 = 1$

B. $Z^3 = 3$

C. $Z^4 = 1$

D. $Z^9 = 1$

Q5.02.1 MCQ 2002 1M

The bilinear transformation $w = \frac{z-1}{z+1}$

A. maps the inside of the unit circle in the z-plane to the left half of the w-plane

B. maps the outside the unit circle in the z-plane to the left half of the w-plane

C. maps the inside of the unit circle in the z-plane to right half of the w-plane

D. maps the outside of the unit circle in the z-plane to the right half of the w-plane

Q5.97.1 MCQ 1997 1M

The complex number $z = x + jy$ which satisfy the equation $|z + 1| = 1$ lie on

A. a circle with $(1, 0)$ as the centre and radius 1

B. a circle with $(-1, 0)$ as the centre and radius 1

C. y-axis

D. x-axis

Q5.94.1 MCQ 1994 1M

The real part of the complex number $z = x + iy$ is given by

A. $\text{Re}(z) = z - z^*$ **B.** $\text{Re}(z) = \dfrac{z - z^*}{2}$

C. $\text{Re}(z) = \dfrac{z + z^*}{2}$ **D.** $\text{Re}(z) = z + z^*$

Q5.94.2 MCQ 1994 1M

$\cos\phi$ can be represented as

A. $\dfrac{e^{i\phi} - e^{-i\phi}}{2}$ **B.** $\dfrac{e^{i\phi} - e^{-i\phi}}{2i}$

C. $\dfrac{e^{i\phi} + e^{-i\phi}}{i}$ **D.** $\dfrac{e^{i\phi} + e^{-i\phi}}{2}$

Mechanical Engineering

Q5.26.1 MCQ 2026 2M

If $w = \log_e z = \log_e(x + iy)$, where $i = \sqrt{-1}$, then which one of the following statements is correct?

A. w is analytic everywhere except at $z = 0$

B. w is non-analytic everywhere

C. The conjugate functions of w are $\log_e(x^2 + y^2)$ and $\log_e(x^2 - y^2)$

D. The conjugate functions of w are $\tan^{-1}(x/y)$ and $\tan^{-1}(y/x)$

Q5.25.1 NAT 2025 2M

If C is the unit circle in the complex plane with its center at the origin, then the value of n in the equation given below is __________ (rounded off to 1 decimal place).

$$\oint_C \frac{z^3}{(z^2 + 4)(z^2 - 4)} dz = 2\pi i n$$

Q5.24.1 MCQ 2024 1M

Let $f(z)$ be an analytic function, where $z = x + iy$. If the real part of $f(z)$ is $\cosh x \cos y$, and the imaginary part of $f(z)$ is zero for $y = 0$, then $f(z)$ is

A. $\cosh x \exp(-iy)$ **B.** $\cosh z \exp z$

C. $\cosh z \cos y$ **D.** $\cosh z$

Q5.23.1 NAT 2023 1M

The value of k that makes the complex-valued function

$$f(z) = e^{-kx}(\cos 2y - i \sin 2y)$$

analytic, where $z = x + iy$, is __________. (Answer in integer)

Q5.22.1 MCQ 2022 2M

The value of the integral

$$\oint \left(\frac{6z}{2z^4 - 3z^3 + 7z^2 - 3z + 5} \right) dz$$

evaluated over a counter-clockwise circular contour in the complex plane enclosing only the pole $z = i$, where i is the imaginary unit, is

A. $(-1 + i)\pi$ **B.** $(1 + i)\pi$

C. $2(1 - i)\pi$ **D.** $(2 + i)\pi$

Q5.22.2 NAT 2022 2M

Given $z = x + iy$, $i = \sqrt{-1}$. C is a circle of radius 2 with the centre at the origin. If the contour C is traversed anticlockwise, then the value of the integral $\frac{1}{2\pi}\int_C \frac{1}{(z-i)(z+4i)} dz$ is __________ (round off to one decimal place).

Q5.21.1 MCQ 2021 2M

Let C represent the unit circle centered at origin in the complex plane, and complex variable, $z = x + iy$. The value of the contour integral $\oint_C \frac{\cosh 3z}{2z} dz$ (where integration is taken counter clockwise) is

A. 0 **B.** 2 **C.** πi **D.** $2\pi i$

Q5.21.2 MCQ 2021 1M

Value of $(1 + i)^8$, where $i = \sqrt{-1}$, is equal to

A. 4 **B.** 16 **C.** $4i$ **D.** $16i$

Q5.20.1 MCQ 2020 1M

Which of the following function $f(z)$, of the complex variable z, is **NOT** analytic at all the points of the complex plane?

A. $f(z) = z^2$ **B.** $f(z) = e^z$

C. $f(z) = \sin z$ **D.** $f(z) = \log z$

Q5.20.2 NAT 2020 2M

An analytic function of a complex variable $z = x + iy$ ($i = \sqrt{-1}$) is defined as

$$f(z) = x^2 - y^2 + i\psi(x, y),$$

where $\psi(x, y)$ is a real function. The value of the imaginary part of $f(z)$ at $z = (1 + i)$ is __________ (round off to 2 decimal places).

Q5.20.3 MCQ 2020 2M

The function $f(z)$ of complex variable $z = x + iy$, where $i = \sqrt{-1}$, is given as $f(z) = (x^3 - 3xy^2) + iv(x, y)$. For this function to be analytic, $v(x, y)$ should be

A. $(3xy^2 - y^3)$ + constant

B. $(3x^2y^2 - y^3)$ + constant

C. $(x^3 - 3x^2y)$ + constant

D. $(3x^2y - y^3)$ + constant

Q5.19.1 MCQ 2019 1M

An analytic function $f(z)$ of complex variable $z = x + iy$ may be written as $f(z) = u(x, y) + iv(x, y)$. Then, $u(x, y)$ and $v(x, y)$ must satisfy

A. $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$ and $\dfrac{\partial u}{\partial y} = \dfrac{\partial v}{\partial x}$

B. $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$ and $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

C. $\dfrac{\partial u}{\partial x} = -\dfrac{\partial v}{\partial y}$ and $\dfrac{\partial u}{\partial y} = \dfrac{\partial v}{\partial x}$

D. $\dfrac{\partial u}{\partial x} = -\dfrac{\partial v}{\partial y}$ and $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

Q5.18.1 MCQ 2018 2M

Let z be a complex variable. For a counter-clockwise integration around a unit circle C, centred at origin,

$$\oint_C \frac{1}{5z - 4} dz = A\pi i,$$

the value of A is

A. $2/5$ **B.** $1/2$ **C.** 2 **D.** $4/5$

Q5.17.1 MCQ 2017 2M

If $f(z) = (x^2 + ay^2) + ibxy$ is a complex analytic function of $z = x + iy$, where $i = \sqrt{-1}$, then

A. $a = -1, b = -1$ **B.** $a = -1, b = 2$

C. $a = 1, b = 2$ **D.** $a = 2, b = 2$

Q5.16.1 MCQ 2016 1M

$f(z) = u(x, y) + iv(x, y)$ is an analytic function of complex variable $z = x + iy$ where $i = \sqrt{-1}$. If $u(x, y) = 2xy$, then $v(x, y)$ may be expressed as

A. $-x^2 + y^2$ + constant **B.** $x^2 - y^2$ + constant

C. $x^2 + y^2$ + constant **D.** $-(x^2 + y^2)$ + constant

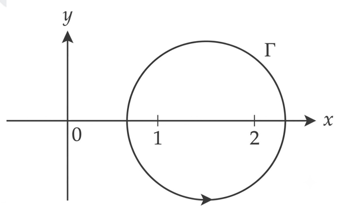
Q5.16.2 NAT 2016 1M

A function f of the complex variable $z = x + iy$, is given as $f(x, y) = u(x, y) + iv(x, y)$, where $u(x, y) = 2kxy$ and $v(x, y) = x^2 - y^2$. The value of k, for which the function is analytic, is __________.

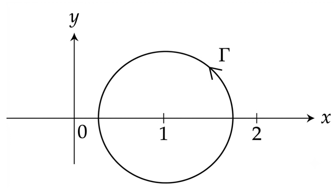
Q5.16.3 MCQ 2016 2M

The value of $\oint_\Gamma \frac{3z-5}{(z-1)(z-2)} dz$ along a closed path Γ is equal to $(4\pi i)$, where $z = x + iy$ and $i = \sqrt{-1}$. The correct path Γ is

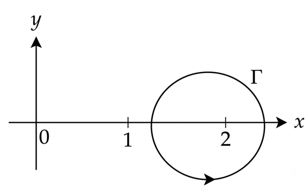
A.



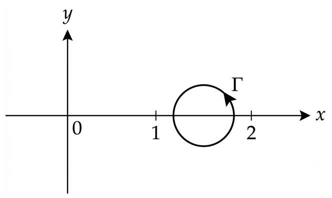
B.



C.



D.



Q5.15.1 MCQ 2015 1M

Given two complex numbers $z_1 = 5 + (5\sqrt{3})i$ and $z_2 = \frac{2}{\sqrt{3}} + 2i$, the argument of $\frac{z_1}{z_2}$ in degrees is

A. 0 **B.** 30 **C.** 60 **D.** 90

Q5.14.1 MCQ 2014 1M

The argument of the complex number $\frac{1+i}{1-i}$, where $i = \sqrt{-1}$, is

A. $-\pi$ **B.** $-\frac{\pi}{2}$ **C.** $\frac{\pi}{2}$ **D.** π

Q5.14.2 MCQ 2014 2M

An analytic function of a complex variable $z = x + iy$ is expressed as $f(z) = u(x, y) + iv(x, y)$, where $i = \sqrt{-1}$. If $u(x, y) = x^2 - y^2$, then expression for $v(x, y)$ in terms of x, y and a general constant c would be

A. $xy + c$

B. $\dfrac{x^2 + y^2}{2} + c$

C. $2xy + c$

D. $\dfrac{(x - y)^2}{2} + c$

Q5.11.1 MCQ 2011 1M

The product of two complex numbers $1 + i$ and $2 - 5i$ is

A. $7 - 3i$

B. $3 - 4i$

C. $-3 - 4i$

D. $7 + 3i$

Q5.10.1 MCQ 2010 1M

The modulus of the complex number $\left(\frac{3+4i}{1-2i}\right)$ is

A. 5 **B.** $\sqrt{5}$ **C.** $1/\sqrt{5}$ **D.** $1/5$

Q5.09.1 MCQ 2009 2M

An analytic function of a complex variable $z = x + iy$ is expressed as $f(z) = u(x, y) + iv(x, y)$ where $i = \sqrt{-1}$. If $u = xy$, the expression for v should be

A. $\dfrac{(x + y)^2}{2} + k$

B. $\dfrac{x^2 - y^2}{2} + k$

C. $\dfrac{y^2 - x^2}{2} + k$

D. $\dfrac{(x - y)^2}{2} + k$

Q5.08.1 MCQ 2008 2M

The integral $\oint f(z)\, dz$ evaluated around the unit circle on the complex plane for $f(z) = \dfrac{\cos z}{z}$ is

A. $2\pi i$ **B.** $4\pi i$ **C.** $-2\pi i$ **D.** 0

Q5.07.1 MCQ 2007 1M

If $\phi(x, y)$ and $\psi(x, y)$ are functions with continuous second derivatives, then $\phi(x, y) + i\psi(x, y)$ can be expressed as an analytic function of $x + iy$ $(i = \sqrt{-1})$, when

A. $\dfrac{\partial \phi}{\partial x} = -\dfrac{\partial \psi}{\partial x}; \dfrac{\partial \phi}{\partial y} = \dfrac{\partial \psi}{\partial y}$

B. $\dfrac{\partial \phi}{\partial y} = -\dfrac{\partial \psi}{\partial x}; \dfrac{\partial \phi}{\partial x} = \dfrac{\partial \psi}{\partial y}$

C. $\dfrac{\partial^2 \phi}{\partial x^2} + \dfrac{\partial^2 \phi}{\partial y^2} = \dfrac{\partial^2 \psi}{\partial x^2} + \dfrac{\partial^2 \psi}{\partial y^2} = 1$

D. $\dfrac{\partial \phi}{\partial x} + \dfrac{\partial \phi}{\partial y} = \dfrac{\partial \psi}{\partial x} + \dfrac{\partial \psi}{\partial y} = 0$

Q5.06.1 MCQ 2006 2M

Assuming $i = \sqrt{-1}$ and t is a real number $\int_0^{\pi/3} e^{it}\, dt$ is

A. $\dfrac{\sqrt{3}}{2} + i\dfrac{1}{2}$

B. $\dfrac{\sqrt{3}}{2} - i\dfrac{1}{2}$

C. $\dfrac{1}{2} + i\left(\dfrac{\sqrt{3}}{2}\right)$

D. $\dfrac{1}{2} + i\left(1 + \dfrac{\sqrt{3}}{2}\right)$

Civil Engineering

Q5.11.1 MCQ 2011 1M

For an analytic function, $f(x+iy) = u(x, y) + iv(x, y)$, u is given by $u = 3x^2 - 3y^2$. The expression for v, considering K to be constant is

A. $3y^2 - 3x^2 + K$

B. $6x - 6y + K$

C. $6y - 6x + K$

D. $6xy + K$

Q5.10.1 MCQ 2010 1M

The value of the integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ is

A. $-\pi$ **B.** $-\pi/2$ **C.** $\pi/2$ **D.** π

Q5.09.1 MCQ 2009 1M

The analytic function has singularities at, where $\mathrm{f}(z) = \frac{z-1}{z^2+1}$

A. 1 & -1 **B.** 1 and i

C. 1 and $-$i **D.** i and $-$i

Q5.09.2 MCQ 2009 2M

The value of the integral $\oint_C \frac{\cos(2\pi z)}{(2z-1)(z-3)} dz$ where c is a closed curve given by $|z| = 1$ is

A. $-\pi\mathrm{i}$ **B.** $\frac{\pi\mathrm{i}}{5}$ **C.** $\frac{2\pi\mathrm{i}}{5}$ **D.** $\pi\mathrm{i}$

Q5.07.1 MCQ 2007 2M

Potential function ϕ is given as $\phi = x^2 - y^2$, what will be the stream function ψ with the condition $\psi = 0$ at $x = y = 0$?

A. $2xy$ **B.** $x^2 + y^2$

C. $x^2 - y^2$ **D.** $2x^2y^2$

Q5.06.1 MCQ 2006 1M

Using Cauchy's integral theorem, the value of the integral (integration being taken in counter clockwise direction) $\oint_C \frac{z^3-6}{3z-1} dz$ is

A. $\frac{2\pi}{81} - 4\pi\mathrm{i}$ **B.** $\frac{\pi}{8} - 6\pi\mathrm{i}$ **C.** $\frac{4\pi}{81} - 6\pi\mathrm{i}$ **D.** 1

Q5.05.1 MCQ 2005 1M

Which one of the following is Not true for complex numbers Z_1 and Z_2 ?

A. $\frac{Z_1}{Z_2} = \frac{Z_1\overline{Z_2}}{|Z_2|^2}$

B. $|Z_1 + Z_2| \leq |Z_1| + |Z_2|$

C. $|Z_1 - Z_2| \leq |Z_1| - |Z_2|$

D. $|Z_1 + Z_2|^2 + |Z_1 - Z_2|^2 = 2|Z_1|^2 + 2|Z_2|^2$

Q5.05.2 MCQ 2005 2M

Consider likely applicability of cauchy's Integral theorem to evaluate the following integral counter clockwise around the unit circle C.

$$I = \oint_C \sec z \, dz$$

z being a complex variable. The value of I will be

A. $I = 0$: singularities set $= \phi$

B. $I = 0$: singularities set $= \left\{\pm\frac{2n+1}{2}\pi / n = 0, 1, 2, \ldots\right\}$

C. $I = \pi/2$: singularities set $= \{\pm n\pi : n = 0, 1, 2, \ldots\}$

D. None of above

Chemical Engineering

Q5.26.1 MCQ 2026 2M

Consider the following complex numbers

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$$

$$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

where r_1, r_2 are real numbers, $0 \leq \theta_1 \leq \frac{\pi}{2}$, $0 \leq \theta_2 \leq \frac{\pi}{2}$, and $i = \sqrt{-1}$

If $|z_1 + z_2| = |z_1| + |z_2|$, which one of the following conditions is necessarily CORRECT?

A. $\theta_1 = 0, \theta_2 = \frac{\pi}{2}$ **B.** $\theta_1 = \frac{\pi}{2}, \theta_2 = 0$

C. $r_1 = r_2$ **D.** $\theta_1 = \theta_2$

Q5.25.1 MCQ 2025 1M

Consider two complex numbers $z_1 = 1 - i$ and $z_2 = i$. The argument of $z_1 z_2$ is

A. 0 **B.** $\frac{\pi}{4}$ **C.** $\frac{\pi}{2}$ **D.** π

Q5.24.1 MCQ 2024 1M

If $z_1 = -1 + i$ and $z_2 = 2i$, where $i = \sqrt{-1}$, then $\mathrm{Arg}(z_1/z_2)$ is

A. $\frac{3\pi}{4}$ **B.** $\frac{\pi}{4}$ **C.** $\frac{\pi}{2}$ **D.** $\frac{\pi}{3}$

Q5.23.1 MCQ | 2023 | 1M

Given that

$$F = \frac{|z_1 + z_2|}{|z_1| + |z_2|},$$

where $z_1 = 2 + 3i$ and $z_2 = -2 + 3i$ with $i = \sqrt{-1}$, which one of the following options is CORRECT?

A. $F < 0$ **B.** $F < 1$ **C.** $F > 1$ **D.** $F = 1$

Q5.20.1 MCQ | 2020 | 1M

Consider the hyperbolic functions in **Group – 1** and their definitions in **Group – 2**.

	Group – 1		**Group – 2**
P	$\tanh x$	**I**	$\frac{e^x + e^{-x}}{e^x - e^{-x}}$
Q	$\coth x$	**II**	$\frac{2}{e^x + e^{-x}}$
R	$\operatorname{sech} x$	**III**	$\frac{2}{e^x - e^{-x}}$
S	$\operatorname{cosech} x$	**IV**	$\frac{e^x - e^{-x}}{e^x + e^{-x}}$

The correct combination is

A. P – IV, Q – I, R – III, S – II

B. P – II, Q – III, R – I, S – IV

C. P – IV, Q – I, R – II, S – III

D. P – I, Q – II, R – IV, S – III

Q5.19.1 MCQ | 2019 | 2M

The value of the complex number $i^{-1/2}$ (where $i = \sqrt{-1}$) is

A. $\frac{1}{\sqrt{2}}(1 - i)$ **B.** $-\frac{1}{\sqrt{2}}i$ **C.** $\frac{1}{\sqrt{2}}i$ **D.** $\frac{1}{\sqrt{2}}(1 + i)$

Q5.17.1 NAT | 2017 | 1M

The real part of $6e^{i\pi/3}$ is ________.

Q5.16.1 MCQ | 2016 | 1M

What are the modulus (r) and argument (θ) of the complex number $3 + 4i$?

A. $r = \sqrt{7}, \quad \theta = \tan^{-1}\left(\frac{4}{3}\right)$ **B.** $r = \sqrt{7}, \quad \theta = \tan^{-1}\left(\frac{3}{4}\right)$

C. $r = 5, \quad \theta = \tan^{-1}\left(\frac{3}{4}\right)$ **D.** $r = 5, \quad \theta = \tan^{-1}\left(\frac{4}{3}\right)$

Q5.15.1 MCQ | 2015 | 1M

A complex-valued function, $f(z)$, given below is analytic in domain D:

$$f(z) = u(x, y) + iv(x, y) \qquad z = x + iy$$

Which of the following is NOT correct?

A. $\frac{df}{dz} = \frac{\partial v}{\partial y} + i\frac{\partial u}{\partial y}$ **B.** $\frac{df}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$

C. $\frac{df}{dz} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$ **D.** $\frac{df}{dz} = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$

Q5.15.2 NAT | 2015 | 2M

For complex variable z, the value of the contour integral $\frac{1}{2\pi i}\oint_C \frac{e^{-2z}}{z(z-3)}dz$ along the clockwise contour $C : |z| = 2$ (up to two decimal places) is ________.

Q5.14.1 MCQ | 2014 | 1M

If $f^*(x)$ is the complex conjugate of $f(x) = \cos(x) + i\sin(x)$, then for real a and b,

$$\int_a^b f^*(x)f(x)\,dx \text{ is } \textbf{ALWAYS}$$

A. positive **B.** negative **C.** real **D.** imaginary

Q5.12.1 MCQ | 2012 | 2M

If $i = \sqrt{-1}$, the value of the integral

$$\oint_c \frac{7z + i}{z\,(z^2 + 1)}\,dz \qquad |z| < 2,$$

using the Cauchy residue theorem is

A. $2\pi i$ **B.** 0 **C.** -6π **D.** 6π

Q5.08.1 MCQ 2008 2M

An analytic function $w(z)$ is defined as $w = u + iv$, where $i = \sqrt{-1}$ and $z = x + iy$. If the real part is given by $u = \dfrac{y}{x^2 + y^2}$, $w(z)$ is

A. $\dfrac{1}{z}$ **B.** $\dfrac{1}{z^2}$ **C.** $\dfrac{i}{z}$ **D.** $\dfrac{1}{iz}$

Q5.07.1 MCQ 2007 2M

If $z = x + iy$ is a complex number, where $i = \sqrt{-1}$ then the derivative of $z\bar{z}$ at $2 + i$ is

A. 0 **B.** 2 **C.** 4 **D.** does not exist

Production & Industrial Engineering

Q5.26.1 MCQ 2026 1M

Which ONE of the following is the value of $\dfrac{1}{i^n}$? where $i = \sqrt{-1}$, and n is an even positive integer.

A. $+1$ or -1 **B.** $+i$ or $-i$

C. only $+1$ **D.** only $-i$

Q5.25.1 MCQ 2025 1M

The eigenvalues of the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are

A. $-\sqrt{-1}$ and $\sqrt{-1}$ **B.** -1 and 1

C. $1 + \sqrt{-1}$ and $1 - \sqrt{-1}$ **D.** $-1 + \sqrt{-1}$ and $-1 - \sqrt{-1}$

Q5.25.2 MCQ 2025 2M

Which one of the following functions is analytic, given $i = \sqrt{-1}$?

A. $e^x(\cos y + i \sin y)$ **B.** $e^x(\cos y - i \sin y)$

C. $e^x(-\cos y + i \sin y)$ **D.** $e^{-x}(\cos y + i \sin y)$

Q5.24.1 MSQ 2024 1M

In a complex function

$$f(x, y) = u(x, y) + i\, v(x, y),$$

i is the imaginary unit, and $x, y, u(x, y)$ and $v(x, y)$ are real.If $f(x, y)$ is analytic then which of the following equations is/are TRUE?

☐ **A.** $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$

☐ **B.** $\dfrac{\partial^2 v}{\partial x^2} + \dfrac{\partial^2 v}{\partial y^2} = 0$

☐ **C.** $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 v}{\partial y^2} = 0$

☐ **D.** $\left(\dfrac{\partial u}{\partial x}\right)\left(\dfrac{\partial v}{\partial x}\right) + \left(\dfrac{\partial u}{\partial y}\right)\left(\dfrac{\partial v}{\partial y}\right) = 0$

Q5.21.1 NAT 2021 1M

If $(3i + 1)x + (4i + 4)y + 5 = 0$ with x, y being real and $i = \sqrt{-1}$, then $x =$ ________. (correct up to one decimal place)

Q5.20.1 MCQ 2020 1M

For the complex numbers $z_1 = 2 + 3i$ and $z_2 = 4 - 5i$, the value of $(z_1 + z_2)^2$ is

A. $32 - 24i$ **B.** $-32 - 24i$

C. $32 + 24i$ **D.** $-32 + 24i$

Q5.18.1 MCQ 2018 2M

Consider the analytic function $f(z) = x^2 - y^2 + i2xy$ of the complex variable $z = x + iy$, where $i = \sqrt{-1}$. The derivative $f'(z)$ is

A. $2x + i2y$ **B.** $x^2 + iy^2$

C. $x + iy$ **D.** $2x - i2y$

Q5.17.1 MCQ 2017 1M

The product of a complex number $z = x+iy$ and its complex conjugate $\bar{z}$ is

A. x^2 **B.** y^2 **C.** $x^2 - y^2$ **D.** $x^2 + y^2$

Q5.16.1 MCQ 2016 1M

The function

$$f(z) = \frac{z^2 + 1}{z^2 + 4}$$

is singular at

A. $z = \pm 2$ **B.** $z = \pm 1$ **C.** $z = \pm i$ **D.** $z = \pm 2i$

Q5.13.1 MCQ 2013 1M

The eigenvalues of a symmetric matrix are all

A. complex with non-zero positive imaginary part

B. complex with non-zero negative imaginary part

C. real

D. pure imaginary

Q5.11.1 MCQ 2011 2M

The value of $\oint_C \frac{z^2}{z^4-1}dz$ using Cauchy's integral around the circle $|z+1|=1$, where $z=x+iy$, is

A. $2\pi i$ **B.** $-\frac{\pi i}{2}$ **C.** $-\frac{3\pi i}{2}$ **D.** $\pi^2 i$

Q5.10.1 MCQ 2010 1M

If a complex number w satisfies the equation $w^3=1$ then the value of $1+w+\frac{1}{w}$ is

A. 0 **B.** 1 **C.** 2 **D.** 4

Q5.08.1 MCQ 2008 1M

The value of the expression $\frac{-5+10\mathrm{i}}{3+4\mathrm{i}}$ is

A. $1-2\mathrm{i}$ **B.** $1+2\mathrm{i}$

C. $2-\mathrm{i}$ **D.** $2+\mathrm{i}$

Q5.07.1 MCQ 2007 1M

If a complex variable $z=\frac{\sqrt{3}}{2}+\mathrm{i}\frac{1}{2}$ then z^4 is

A. $2\sqrt{3}+2\mathrm{i}$ **B.** $\frac{-1}{2}+\mathrm{i}\frac{\sqrt{3}}{2}$

C. $\frac{\sqrt{3}}{2}-\mathrm{i}\frac{1}{2}$ **D.** $\frac{\sqrt{3}}{8}+\mathrm{i}\frac{1}{8}$

Q5.05.5 MCQ 2005 2M

The function

$$w=u+\mathrm{i}v=\frac{1}{2}\log(x^2+y^2)+\mathrm{i}\tan^{-1}(y/x)$$

is not analytic at the point

A. $(0,0)$ **B.** $(0,1)$ **C.** $(1,0)$ **D.** $(2,0)$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q5.25.1 A,B	Q5.24.1 D	Q5.23.1 A	Q5.23.2 B	Q5.22.1 0.5	Q5.21.1 *
Q5.19.1 D	Q5.19.2 0	Q5.18.1 2	Q5.17.1 B	Q5.17.2 D	Q5.16.1 -136 to -132
Q5.16.2 0	Q5.16.3 B	Q5.15.1 D	Q5.15.2 B	Q5.15.3 9.9 to 10.1	Q5.15.4 0.5
Q5.15.5 B	Q5.14.1 B	Q5.14.2 B	Q5.14.3 -0.6 to -0.4	Q5.12.1 C	Q5.11.1 A
Q5.10.1 C	Q5.09.1 D	Q5.08.1 D	Q5.08.2 A	Q5.07.1 A	Q5.06.1 B
Q5.06.2 D					
ELECTRICAL ENGINEERING					
Q5.26.1 25 to 25.6	Q5.25.1 0	Q5.24.1 D	Q5.24.2 0	Q5.21.1 D	Q5.21.2 C
Q5.20.1 C	Q5.19.1 B	Q5.19.2 A	Q5.18.1 A	Q5.18.2 B	Q5.17.1 D
Q5.17.2 B	Q5.16.1 B	Q5.16.2 B	Q5.15.1 B	Q5.14.1 C	Q5.14.2 B
Q5.14.3 C	Q5.13.1 B	Q5.13.2 A	Q5.12.1 A	Q5.11.1 D	Q5.11.2 D
Q5.08.1 D	Q5.07.1 B	Q5.97.1 B			
INSTRUMENTATION ENGINEERING					
Q5.26.1 A	Q5.25.1 2	Q5.25.2 A,B	Q5.24.1 B	Q5.24.2 A,C	Q5.23.1 A
Q5.22.1 A,C	Q5.22.2 0.49 to 0.51	Q5.21.1 0	Q5.21.2 B	Q5.20.1 B	Q5.19.1 C
Q5.18.1 B	Q5.17.1 B	Q5.16.1 1	Q5.15.1 B	Q5.13.1 D	Q5.11.1 C
Q5.10.1 D	Q5.09.1 D	Q5.09.2 A	Q5.09.3 B	Q5.08.1 B	Q5.07.1 B
Q5.05.1 A	Q5.05.2 C	Q5.02.1 A	Q5.97.1 B	Q5.94.1 C	Q5.94.2 D
MECHANICAL ENGINEERING					
Q5.26.1 A	Q5.25.1 -0.1 to 0.1	Q5.24.1 D	Q5.23.1 1.999 to 2.001	Q5.22.1 A	Q5.22.2 0.2

Q5.21.1 C	Q5.21.2 B	Q5.20.1 D	Q5.20.2 1.99 to 2.01	Q5.20.3 D	Q5.19.1 B
Q5.18.1 A	Q5.17.1 B	Q5.16.1 A	Q5.16.2 -1.1 to -0.9	Q5.16.3 B	Q5.15.1 A
Q5.14.1 C	Q5.14.2 C	Q5.11.1 A	Q5.10.1 B	Q5.09.1 C	Q5.08.1 A
Q5.07.1 B	Q5.06.1 A				
CIVIL ENGINEERING					
Q5.11.1 D	Q5.10.1 D	Q5.09.1 D	Q5.09.2 C	Q5.07.1 A	Q5.06.1 A
Q5.05.1 C	Q5.05.2 A				
CHEMICAL ENGINEERING					
Q5.26.1 D	Q5.25.1 B	Q5.24.1 B	Q5.23.1 B	Q5.20.1 C	Q5.19.1 A
Q5.17.1 2.99 to 3.01	Q5.16.1 D	Q5.15.1 A	Q5.15.2 0.31 to 0.35	Q5.14.1 C	Q5.12.1 B
Q5.08.1 C	Q5.07.1 D				
PRODUCTION & INDUSTRIAL ENGINEERING					
Q5.26.1 A	Q5.25.1 A	Q5.25.2 A	Q5.24.1 A,B,D	Q5.21.1 2.4 to 2.6	Q5.20.1 A
Q5.18.1 A	Q5.17.1 D	Q5.16.1 D	Q5.13.1 C	Q5.11.1 B	Q5.10.1 A
Q5.08.1 B	Q5.07.1 B	Q5.05.5 A			

UNIT

VI

Differential Equations

Topics

1. First Order Differential Equations
2. Higher Order Differential Equations
3. Initial Value Problems
4. Boundary Value Problems
5. Laplace Transform Method

Note to Reader

Differential Equations are present in branches with paper codes AE, AG, BT, BM, CE, CH, EC, EE, ES, IN, MA, ME, MN, NM, PE, PI, XE according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 38 **MCQ:** 32 **MSQ:** 1 **NAT:** 5
1M: 22 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2020)

EE — Electrical Engineering

Questions: 17 **MCQ:** 11 **MSQ:** 1 **NAT:** 5
1M: 8 **2M:** 9 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–2M(2026)

IN — Instrumentation Engineering

Questions: 17 **MCQ:** 15 **MSQ:** 1 **NAT:** 1
1M: 5 **2M:** 12 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–3M(2026)

ME — Mechanical Engineering

Questions: 56 **MCQ:** 48 **MSQ:** 0 **NAT:** 8
1M: 23 **2M:** 31 **Desc:** 2 **Avg Weightage** ~2M **Range:** 1M(2016)–3M(2026)

CE — Civil Engineering

Questions: 64 **MCQ:** 51 **MSQ:** 3 **NAT:** 10
1M: 27 **2M:** 36 **Desc:** 1 **Avg Weightage** ~3M **Range:** 1M(2022)–5M(2010)

CH — Chemical Engineering

Questions: 36 **MCQ:** 29 **MSQ:** 1 **NAT:** 6
1M: 11 **2M:** 25 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2023)–4M(2014)

PI — Production & Industrial Engineering

Questions: 24 **MCQ:** 20 **MSQ:** 0 **NAT:** 4
1M: 12 **2M:** 12 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2023)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q6.26.1 MCQ 2026 1M

Consider the differential equation $\dot{\vec{w}} = A\vec{w}$, with $\vec{w}(t = 0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

If $\vec{w}(t) = e^t \vec{u}_x + e^{-2t}\vec{u}_y$ be the solution to the equation where $\vec{u}_x$ and $\vec{u}_y$ are unit vectors along the positive x and y axes respectively, then which of the following options is the correct matrix representing A?

A. $\begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$ **B.** $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$

C. $\begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix}$

Q6.25.1 NAT 2025 1M

The function $y(t)$ satisfies

$$t^2 y''(t) - 2ty'(t) + 2y(t) = 0,$$

where $y'(t)$ and $y''(t)$ denote the first and second derivatives of $y(t)$, respectively.

Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0, 1]$ is _____ *(rounded off to two decimal places)*.

Q6.24.1 MCQ 2024 1M

The general form of the complementary function of a differential equation is given by $y(t) = (At+B)e^{-2t}$, where A and B are real constants determined by the initial condition. The corresponding differential equation is _____.

A. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = f(t)$

B. $\frac{d^2y}{dt^2} + 4y = f(t)$

C. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = f(t)$

D. $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = f(t)$

Q6.22.1 MSQ 2022 1M

Consider the following partial differential equation (PDE)

$$a\frac{\partial^2 f(x,y)}{\partial x^2} + b\frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y),$$

where a and b are distinct positive real numbers. Select the combination(s) of values of the real parameters ξ and η such that $f(x,y) = e^{(\xi x+\eta y)}$ is a solution of the given PDE.

☐ **A.** $\xi = \frac{1}{\sqrt{2a}}, \eta = \frac{1}{\sqrt{2b}}$

☐ **B.** $\xi = \frac{1}{\sqrt{a}}, \eta = 0$

☐ **C.** $\xi = 0, \eta = 0$

☐ **D.** $\xi = \frac{1}{\sqrt{a}}, \eta = \frac{1}{\sqrt{b}}$

Q6.21.1 MCQ 2021 1M

Consider the differential equation given below.

$$\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$$

The integrating factor of the differential equation is

A. $(1-x^2)^{-3/4}$ **B.** $(1-x^2)^{-1/4}$

C. $(1-x^2)^{-3/2}$ **D.** $(1-x^2)^{-1/2}$

Q6.20.1 MCQ 2020 1M

The partial derivative of the function

$$f(x,y,z) = e^{1-x\cos y} + xze^{-1/\left(1+y^2\right)}$$

with respect to x at the point $(1, 0, e)$ is

A. -1 **B.** 0 **C.** 1 **D.** $\frac{1}{e}$

Q6.20.2 MCQ 2020 1M

The general solution of $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$ is

A. $y = C_1e^{3x} + C_2e^{-3x}$ **B.** $y = (C_1 + C_2x)e^{-3x}$

C. $y = (C_1 + C_2x)e^{3x}$ **D.** $y = C_1e^{3x}$

Q6.20.3 MCQ 2020 2M

Which one of the following options contain two solutions of the differential equation $\frac{dy}{dx} = (y-1)x$?

A. $\ln|y-1| = 0.5x^2 + C$ and $y = 1$

B. $\ln|y-1| = 2x^2 + C$ and $y = 1$

C. $\ln|y-1| = 0.5x^2 + C$ and $y = -1$

D. $\ln|y-1| = 2x^2 + C$ and $y = -1$

Q6.19.1 MCQ 2019 1M

The families of curves represented by the solution of the equation

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^n$$

for $n = -1$ and $n = +1$, respectively, are

A. Parabolas and Circles

B. Circles and Hyperbolas

C. Hyperbolas and Circles

D. Hyperbolas and Parabolas

Q6.19.2 NAT 2019 2M

Consider the homogeneous ordinary differential equation

$$x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 3y = 0, \qquad x > 0$$

with $y(x)$ as a general solution. Given that $y(1) = 1$ and $y(2) = 14$. The value of $y(1.5)$, rounded off to two decimal places, is ______

Q6.18.1 MCQ 2018 2M

A curve passes through the point $(x = 1, y = 0)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{x^2+y^2}{2y} + \frac{y}{x}$. The equation that describes the curve is

A. $\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$

B. $\frac{1}{2}\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$

C. $\ln\left(1 + \frac{y}{x}\right) = x - 1$

D. $\frac{1}{2}\ln\left(1 + \frac{y}{x}\right) = x - 1$

Q6.18.2 NAT 2018 2M

The position of a particle $y(t)$ is described by the differential equation:

$$\frac{d^2y}{dt^2} = -\frac{dy}{dt} - \frac{5y}{4}.$$

The initial conditions are $y(0) = 1$ and $\left.\frac{dy}{dt}\right|_{t=0} = 0$. The position (accurate to two decimal places) of the particle at $t = \pi$ is ______

Q6.17.1 MCQ 2017 2M

Which one of the following is the general solution of the first order differential equation

$$\frac{dy}{dx} = (x+y-1)^2,$$

where x, y are real?

A. $y = 1 + x + \tan^{-1}(x+c)$, where c is a constant

B. $y = 1 + x + \tan(x+c)$, where c is a constant

C. $y = 1 - x + \tan^{-1}(x+c)$, where c is a constant

D. $y = 1 - x + \tan(x+c)$, where c is a constant

Q6.17.2 MCQ 2017 1M

The general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 5y = 0$$

in terms of arbitrary constants K_1 and K_2 is

A. $K_1e^{(-1+\sqrt{6})x} + K_2e^{(-1-\sqrt{6})x}$

B. $K_1e^{(-1+\sqrt{8})x} + K_2e^{(-1-\sqrt{8})x}$

C. $K_1e^{(-2+\sqrt{6})x} + K_2e^{(-2-\sqrt{6})x}$

D. $K_1e^{(-2+\sqrt{8})x} + K_2e^{(-2-\sqrt{8})x}$

Q6.16.1 MCQ 2016 2M

The particular solution of the Initial value problem given below is

$$\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0$$

with $y(0) = 3$ and $\left.\frac{dy}{dx}\right|_{x=0} = -36$

A. $(3 - 18x)\,e^{-6x}$ **B.** $(3 + 25x)\,e^{-6x}$

C. $(3 + 20x)\,e^{-6x}$ **D.** $(3 - 12x)\,e^{-6x}$

Q6.15.1 MCQ 2015 2M

The solution of the differential equation $\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=0$ with $y(0)=y'(0)=1$ is

A. $(2-t)\,e^{t}$ **B.** $(1+2t)\,e^{-t}$

C. $(2+t)\,e^{-t}$ **D.** $(1-2t)\,e^{t}$

Q6.15.2 MCQ 2015 1M

The general solution of the differential equation $\frac{dy}{dx}=\frac{1+\cos 2y}{1-\cos 2x}$ is

A. $\tan y-\cot x=c$ (c is a constant)

B. $\tan x-\cot y=c$ (c is a constant)

C. $\tan y+\cot x=c$ (c is a constant)

D. $\tan x+\cot y=c$ (c is a constant)

Q6.15.3 MCQ 2015 2M

Consider the differential equation $\frac{dx}{dt}=10-0.2x$ with initial condition $x(0)=1$. The response $x(t)$ for $t>0$ is

A. $2-e^{-0.2t}$ **B.** $2-e^{0.2t}$

C. $50-49e^{-0.2t}$ **D.** $50-49e^{0.2t}$

Q6.15.4 NAT 2015 2M

Consider the differential equation

$$\frac{d^2x(t)}{dt^2}+3\frac{dx(t)}{dt}+2x(t)=0.$$

with $y(x)$ Given $x(0)=20$ and $x(1)=10/e$, where $e=2.718$, the value of $x(2)$ is _____

Q6.14.1 MCQ 2014 1M

If the characteristic equation of the differential equation

$$\frac{d^2y}{dx^2}+2\alpha\frac{dy}{dx}+y=0$$

has two equal roots, then the values of α are

A. ± 1 **B.** $0, 0$ **C.** $\pm j$ **D.** $\pm 1/2$

Q6.14.2 MCQ 2014 1M

Which ONE of the following is a linear non-homogeneous differential equation, where x and y are the independent and dependent variables respectively?

A. $\frac{dy}{dx}+xy=e^{-x}$ **B.** $\frac{dy}{dx}+xy=0$

C. $\frac{dy}{dx}+xy=e^{-y}$ **D.** $\frac{dy}{dx}+e^{-y}=0$

Q6.14.3 MCQ 2014 1M

If $z=xy\ln(xy)$, then

A. $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=0$ **B.** $y\frac{\partial z}{\partial x}=x\frac{\partial z}{\partial y}$

C. $x\frac{\partial z}{\partial x}=y\frac{\partial z}{\partial y}$ **D.** $y\frac{\partial z}{\partial x}+x\frac{\partial z}{\partial y}=0$

Q6.14.4 MCQ 2014 1M

If a and b are constants, the most general solution of the differential equation

$$\frac{d^2x}{dt^2}+2\frac{dx}{dt}+x=0 \text{ is}$$

A. ae^{-t} **B.** $ae^{-t}+bte^{-t}$

C. $ae^{t}+bte^{-t}$ **D.** ae^{-2t}

Q6.14.5 NAT 2014 2M

With initial values $y(0)=y'(0)=1$, the solution of the differential equation

$$\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=0$$

at $x=1$ is _____.

Q6.13.1 MCQ 2013 2M

A system is described by a linear, constant coefficient, ordinary, first order differential equation has an exact solution given by $y(t)$ for $t>0$, when the forcing function is $x(t)$ and the initial condition is $y(0)$. If one wishes to modify the system so that the solution becomes $-2y(t)$ for $t>0$, we need to

A. change the initial condition to $-y(0)$ and the forcing function to $2x(t)$

B. change the initial condition to $2y(0)$ and the forcing function to $-x(t)$

C. change the initial condition to $j\sqrt{2}\,y(0)$ and the forcing function to $j\sqrt{2}\,x(t)$

D. change the initial condition to $-2y(0)$ and the forcing function to $-2x(t)$

Q6.12.1 MCQ 2012 1M

With initial condition $x(1) = 0.5$, the solution of the differential equation, $t\frac{dx}{dt} + x = t$ is

A. $x = t - \frac{1}{2}$ **B.** $x = t^2 - \frac{1}{2}$

C. $x = \frac{t^2}{2}$ **D.** $x = \frac{t}{2}$

Q6.12.2 MCQ 2012 2M

Consider the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = \delta(t) \text{ with } \left. y(t)\right|_{t=0^-} = -2$$

and $\left.\frac{dy}{dt}\right|_{t=0^-} = 0$. The numerical value of $\left.\frac{dy}{dt}\right|_{t=0^+}$ is

A. -2 **B.** -1 **C.** 0 **D.** 1

Q6.11.1 MCQ 2011 1M

The solution of the differential equation $\frac{dy}{dx} = ky,\ y(0) = c$ is

A. $x = ce^{-ky}$ **B.** $x = ke^{cy}$

C. $y = ce^{kx}$ **D.** $y = ce^{-kx}$

Q6.10.1 MCQ 2010 1M

A function $n(x)$ satisfies the differential equation

$$\frac{d^2n(x)}{dx^2} - \frac{n(x)}{L^2} = 0$$

where L is a constant. The boundary conditions are: $n(0) = K$ and $n(\infty) = 0$. The solution of this equation is

A. $n(x) = Ke^{x/L}$ **B.** $n(x) = Ke^{-x/\sqrt{L}}$

C. $n(x) = K^2e^{-x/L}$ **D.** $n(x) = Ke^{-x/L}$

Q6.09.1 MCQ 2009 1M

The order of the differential equation

$$\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^3 + y^4 = e^{-t} \text{ is}$$

A. 1 **B.** 2 **C.** 3 **D.** 4

Q6.09.2 MCQ 2009 2M

Match each differential equation in Group I to its family of solution curves from Group II.

	Group I		Group II
P.	$\frac{dy}{dx} = \frac{y}{x}$	1.	Circles
Q.	$\frac{dy}{dx} = -\frac{y}{x}$	2.	Straight lines
R.	$\frac{dy}{dx} = \frac{x}{y}$	3.	Hyperbolas
S.	$\frac{dy}{dx} = -\frac{x}{y}$		

A. P-2, Q-3, R-3, S-1 **B.** P-1, Q-3, R-2, S-1

C. P-2, Q-1, R-3, S-3 **D.** P-3, Q-2, R-1, S-2

Q6.08.1 MCQ 2008 1M

Which of the following is a solution of the differential equation $\frac{dx(t)}{dt} + 3x(t) = 0$?

A. $x(t) = 3e^{-t}$ **B.** $x(t) = 2e^{-3t}$

C. $-\frac{3}{2}t^2$ **D.** $3t^2$

Q6.07.1 MCQ 2007 2M

The solution of the differential equation $k^2\frac{d^2y}{dx^2} = y - y_2$ under the boundary conditions (i) $y = y_1$ at $x = 0$ and (ii) $y = y_2$ at $x = \infty$, where k, y_1 and y_2 are constants, is

A. $y = (y_1 - y_2)exp(-x/k^2) + y_2$

B. $y = (y_2 - y_1)exp(-x/k) + y_1$

C. $y = (y_1 - y_2)sinh(x/k) + y_1$

D. $y = (y_1 - y_2)exp(-x/k) + y_2$

Q6.06.1 MCQ 2006 1M

A solution for the differential equation $\dot{x}(t) + 2x(t) = \delta(t)$ with initial condition $x(0^-) = 0$ is

A. $e^{-2t}\, u(t)$

B. $e^{2t}\, u(t)$

C. $e^{-t}\, u(t)$

D. $e^{t}\, u(t)$

Q6.06.2 MCQ 2006 2M

For the differential equation $\frac{d^2y}{dx^2} + k^2 y = 0$ the boundary conditions are (i) $y = 0$ for $x = 0$ and (ii) $y = 0$ for $x = a$.

The form of non-zero solution of y (where m varies over all integers) are

A. $y = \sum_m A_m \sin \frac{m\pi x}{a}$

B. $y = \sum_m A_m \cos \frac{m\pi x}{a}$

C. $y = \sum_m A_m x^{\frac{m\pi}{a}}$

D. $y = \sum_m A_m e^{\frac{-m\pi x}{a}}$

Q6.05.1 MCQ 2005 1M

The following differential equation has

$$3\left(\frac{d^2y}{dx^2}\right) + 4\left(\frac{dy}{dx}\right)^3 + y^2 + 2 = x$$

A. degree=2, order=1

B. degree=1, order=2

C. degree=4, order=3

D. degree=2, order=3

Q6.05.2 MCQ 2005 1M

A solution of the following differential equation is given by $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

A. $y = e^{2x} + e^{-3x}$

B. $y = e^{2x} + e^{3x}$

C. $y = e^{-2x} + e^{3x}$

D. $y = e^{-2x} + e^{-3x}$

Q6.94.1 MCQ 1994 2M

Match each of the items A, B and C with an appropriate item from 1, 2, 3, 4 and 5.

(A) $a_1\frac{d^2y}{dx^2} + a_2\, y\frac{dy}{dx} + a_3\, y = a_4$

(B) $a_1\frac{d^3y}{dx^3} + a_2\, y = a_3$

(C) $a_1\frac{d^2y}{dx^2} + a_2\, x\frac{dy}{dx} + a_3\, x^2 y = 0$

1. Non-linear differential equation
2. Linear differential equation with constant coefficient
3. Linear homogeneous differential equation
4. Non-linear homogeneous differential equation
5. Non-linear first order differential equation

A. A–1, B–2, C–4

B. A–3, B–4, C–2

C. A–2, B–5, C–3

D. A–3, B–1, C–2

Electrical Engineering

Q6.26.1 MCQ 2026 2M

Consider the second-order differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

with initial conditions

$$y(0) = 1, \qquad \left.\frac{dy}{dx}\right|_{x=0} = 1$$

The solution is given by

A. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

B. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

C. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

D. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) - \sqrt{3}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

Q6.25.1 NAT 2025 2M

Consider ordinary differential equations given by

$$\dot{x_1}(t) = 2x_2(t)$$
$$\dot{x_2}(t) = r(t)$$

with initial conditions $x_1(0) = 1$ and $x_2(0) = 0$.

$$\text{If } r(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}, \text{ then at } t = 1,\ x_1(t) = \_\_\_\_\_\_$$

(round off to the nearest integer).

Q6.24.1 MSQ 2024 2M

Which of the following differential equations is/are nonlinear?

- [ ] **A.** $t\,x(t) + \frac{dx(t)}{dt} = t^2 e^t, \qquad x(0) = 0$
- [ ] **B.** $\frac{1}{2}e^t + x(t)\frac{dx(t)}{dt} = 0, \qquad x(0) = 0$
- [ ] **C.** $x(t)\cos t - \frac{dx(t)}{dt}\sin t = 1, \; x(0) = 0$
- [ ] **D.** $x(t) + e^{\left(\frac{dx(t)}{dt}\right)} = 1, \qquad x(0) = 0$

Q6.20.1 NAT 2020 1M

Consider the initial value problem below. The value of y at $x = \ln 2$, (rounded off to 3 decimal places) is ______.

$$\frac{dy}{dx} = 2x - y, \qquad y(0) = 1$$

Q6.19.1 MCQ 2019 1M

The partial differential equation

$$\frac{\partial^2 u}{\partial t^2} - c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0; \text{ where } c \neq 0$$

is known as

A. Heat Equation **B.** Wave Equation

C. Poisson's Equation **D.** Laplace Equation

Q6.17.1 MCQ 2017 2M

Consider the differential equation $(t^2 - 81)\frac{dy}{dt} + 5t\,y = \sin(t)$ with $y(1) = 2\pi$. There exists a unique solution for this differential equation when t belongs to the interval

A. $(-2, 2)$ **B.** $(-10, 10)$ **C.** $(-10, 2)$ **D.** $(0, 10)$

Q6.16.1 MCQ 2016 1M

A function $y(t)$, such that $y(0) = 1$ and $y(1) = 3e^{-t}$ is a solution of the differential equation $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$. Then $y(2)$ is

A. $5e^{-t}$ **B.** $5e^{-2}$ **C.** $7e^{-t}$ **D.** $7e^{-2}$

Q6.16.2 MCQ 2016 1M

The solution of the differential equation, for $t > 0$, $y''(t) + 2y'(t) + y(t) = 0$ with initial conditions $y(0) = 0$ and $y'(0) = 1$, is ($u(t)$ denotes the unit step function),

A. $te^{-t}u(t)$ **B.** $e^{-t} - te^{-t}u(t)$

C. $(-e^{-t} + te^{-t})u(t)$ **D.** $e^{-t}u(t)$

Q6.16.3 NAT 2016 2M

Let $y(x)$ be the solution of the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$ with initial conditions $y(0) = 0$ and $\left.\frac{dy}{dx}\right|_{x=0} = 1$. Then the value of $y(1)$ is ______.

Q6.15.1 NAT 2015 2M

A solution of the ordinary differential equation $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0$ is such that $y(0) = 2$ and $y(1) = -\frac{1-3e}{e^3}$. The value of $\frac{dy}{dt}(0)$ is ______.

Q6.15.2 NAT 2015 2M

A differential equation $\frac{di}{dt} - 0.2i = 0$ is applicable over $-10 < t < 10$. If $i(4) = 10$, then $i(-5)$ is ______.

Q6.14.1 MCQ 2014 1M

The solution for the differential equation

$$\frac{d^2x}{dt^2} = -9x,$$

with initial conditions $x(0) = 1$ and $\left.\frac{dx}{dt}\right|_{t=0} = 1$, is

A. $t^2 + t + 1$ **B.** $\sin 3t + \frac{1}{3}\cos 3t + \frac{2}{3}$

C. $\frac{1}{3}\sin 3t + \cos 3t$ **D.** $\cos 3t + t$

Q6.14.2 MCQ 2014 1M

Consider the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.

Which of the following is a solution to this differential equation for $x > 0$?

A. e^x **B.** x^2 **C.** $1/x$ **D.** $\ln x$

Q6.11.1 MCQ 2011 1M

With K as a constant, the possible solution for the first order differential equation $\frac{dy}{dx} = e^{-3x}$ is

A. $-\frac{1}{3}e^{-3x} + K$ **B.** $-\frac{1}{3}e^{3x} + K$

C. $-3e^{-3x} + K$ **D.** $-3e^{-x} + K$

Q6.10.1 MCQ 2010 2M

For the differential equation $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 0$ with initial conditions $x(0) = 1$ and $\left.\frac{dx}{dt}\right|_{t=0} = 0$, the solution is

A. $x(t) = 2e^{-6t} - e^{-2t}$ **B.** $x(t) = 2e^{-2t} - e^{-4t}$

C. $x(t) = -e^{-6t} + 2e^{-4t}$ **D.** $x(t) = e^{-2t} + 2e^{-4t}$

Q6.05.1 MCQ 2005 1M

The solution of the first order differential equation $\dot{x}(t) = -3x(t),\ x(0) = x_0$ is:

A. $x(t) = x_0\, e^{-3t}$ **B.** $x(t) = x_0\, e^{-3}$

C. $x(t) = x_0\, e^{-t/3}$ **D.** $x(t) = x_0\, e^{-t}$

Q6.05.2 MCQ 2005 2M

For the equation $\ddot{x}(t) + 3\dot{x}(t) + 2x(t) = 5$, the solution $x(t)$ approaches the following values at $t \to \infty$

A. 0 **B.** $\frac{5}{2}$ **C.** 5 **D.** 10

Instrumentation Engineering

Q6.26.1 MCQ 2026 1M

Among the following, the differential equation(s) not representative of a linear dynamical system is/are ______.

(i) $\frac{d^2x}{dt^2} + 2t\frac{dx}{dt} + x(t) = u(t)$

(ii) $\left(\frac{dx}{dt}\right)^2 + 2\frac{dx}{dt} + x(t) = u(t)$

(iii) $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x(t) = u(t)$

A. (i) and (ii) only **B.** (i) only

C. (ii) only **D.** (ii) and (iii) only

Q6.26.2 MSQ 2026 2M

Given a differential equation

$$\frac{d^2x}{dt^2} + x = 0 \text{ with } x(0) \neq 0.$$

Which of the following statements is/are true?

- [ ] **A.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = 0$ for all $t \geq 0$
- [ ] **B.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = c$ for all $t \geq 0$, for some real constant $c > 0$
- [ ] **C.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = e^{jt}$ for all $t \geq 0$
- [ ] **D.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = \sin t$ for all $t \geq 0$

Q6.25.1 MCQ 2025 1M

The solution of the differential equation $\frac{dy}{dx} = 9\frac{x}{y}$ represents

A. a hyperbola **B.** a parabola

C. an ellipse **D.** a circle

Q6.24.1 NAT 2024 2M

The solution of an ordinary differential equation $y''' + 3y'' + 3y' + y = 30\,e^{-t}$ is

$$y(t) = (c_0 + c_1 t - c_2 t^2 + c_3 t^3)e^{-t}$$

Given $y(0) = 3,\ y'(0) = -3$ and $y''(0) = -47$, the value of $(c_0 + c_1 + c_2 + c_3)$ is ______ **(rounded off to nearest integer).**

Note: $y''' = d^3y/dt^3,\ y'' = d^2y/dt^2,\ y' = dy/dt$ and c_0, c_1, c_2, c_3 are constants.

Q6.23.1 MCQ 2023 1M

The solution $x(t),\ t \geq 0$, to the differential equation $\ddot{x} = -k\dot{x},\ k > 0$ with initial conditions $x(0) = 1$ and $\dot{x}(0) = 0$ is

A. $x(t) = 2e^{-kt} + 2kt - 1$

B. $x(t) = 2e^{-kt} - 1$

C. $x(t) = 1$

D. $x(t) = 2e^{-kt} - kt - 1$

Q6.22.1 MCQ 2022 2M

Consider the differential equation

$$\frac{dy}{dx} + y\ln(y) = 0$$

If $y(0) = e$, then $y(1)$ is ______

A. e^{e} **B.** e^{-e} **C.** $e^{(1/e)}$ **D.** $e^{(-1/e)}$

Q6.20.1 MCQ 2020 2M

Consider the differential equation $\frac{dx}{dt} = \sin(x)$, with the initial condition $x(0) = 0$. The solution to this ordinary differential equation is ______

A. $x(t) = 0$

B. $x(t) = \sin(t)$

C. $x(t) = \cos(t)$

D. $x(t) = \sin(t) - cos(t)$

Q6.18.1 MCQ 2018 2M

Consider the following equation

$$\frac{\partial V(x,y)}{\partial x} = px^2 + y^2 + 2xy$$

$$\frac{\partial V(x,y)}{\partial y} = x^2 + qy^2 + 2xy$$

where p and q are constants. $V(x,y)$ that satisfies the above equations is

A. $p\frac{x^3}{3} + q\frac{y^3}{3} + 2xy + 6$

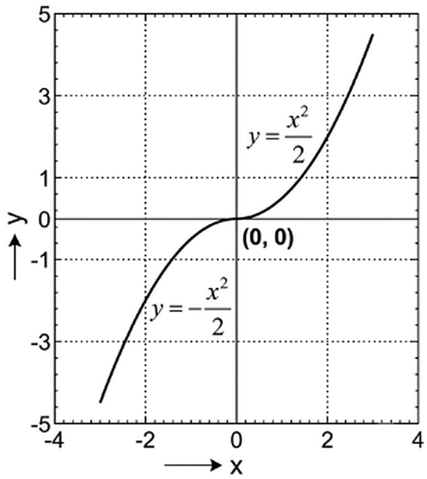
B. $p\frac{x^3}{3} + q\frac{y^3}{3} + 5$

C. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2 + xy$

D. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2$

Q6.14.1 MCQ 2014 1M

The figure shows the plot of y as a function of x



The function shown is the solution of the differential equation (assuming all initial conditions to be zero) is:

A. $\frac{d^2y}{dx^2} = 1$

B. $\frac{dy}{dx} = x$

C. $\frac{dy}{dx} = -x$

D. $\frac{dy}{dx} = |x|$

Q6.13.1 MCQ 2013 1M

The type of the partial differential equation $\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$ is

A. Parabolic

B. Elliptic

C. Hyperbolic

D. Nonlinear

Q6.13.2 MCQ 2013 2M

The maximum value of the solution $y(t)$ of the differential equation $y(t) + \ddot{y}(t) = 0$ with initial conditions $\dot{y}(0) = 1$ and $y(0) = 1$, for $t \geq 0$ is

A. 1 **B.** 2 **C.** π **D.** $\sqrt{2}$

Q6.11.1 MCQ 2011 2M

Consider the differential equation $\ddot{y} + 2\dot{y} + y = 0$ with boundary conditions $y(0) = 1$, $y(1) = 0$. The value of $y(2)$ is

A. -1 **B.** $-e^{-1}$ **C.** $-e^{-2}$ **D.** $-e^{2}$

Q6.10.1 MCQ 2010 2M

Consider the differential equation $\frac{dy}{dx} + y = e^{x}$ with $y(0) = 1$. The value of $y(1)$ is

A. $e + e^{-1}$

B. $\frac{1}{2}\left(e - e^{-1}\right)$

C. $\frac{1}{2}\left(e + e^{-1}\right)$

D. $2\left(e - e^{-1}\right)$

Q6.08.1 MCQ 2008 2M

Consider the differential equation $\frac{dy}{dx} = 1 + y^2$. Which of the following can be a particular solution of this differential equation?

A. $y = \tan(x + 3)$ **B.** $y = \tan x + 3$

C. $x = \tan(y + 3)$ **D.** $x = \tan y + 3$

Q6.07.1 MCQ 2007 2M

The boundary-value problem $y'' + \lambda y = 0,\ y(0) = y(\pi) = 0$ will have non-zero solutions if and only if the values of λ are

A. $0, \pm1, \pm2, \ldots$ **B.** $1, 2, 3, \ldots$

C. $1, 4, 9, \ldots$ **D.** $1, 9, 25, \ldots$

Q6.06.1 MCQ 2006 2M

For an initial value problem $\ddot{y} + 2\dot{y} + 101y = 10.4\,e^x$, $y(0) = 1.1$ and $\dot{y} = -0.9$. Various solutions are written in the following groups. Match the type of solution with the correct expression.

	Group 1		Group 2
P.	General solution of homogeneous equations	1.	$0.1e^x$
Q.	Particular integral	2.	$e^{-x}(A\cos 10x + B\sin 10x)$
R.	Total solution satisfying boundary conditions	3.	$e^{-x}\cos 10x + 0.1e^x$

A. P–2, Q–1, R–3 **B.** P–1, Q–3, R–2

C. P–1, Q–2, R–3 **D.** P–3, Q–2, R–1

Q6.05.1 MCQ 2005 2M

The general solution of the differential equation $(D^2-4D+4)y = 0$, is of the form (given $D = d/dx$), and C_1 and C_2 are constants

A. C_1e^{2x} **B.** $C_1e^{2x} + C_2e^{-2x}$

C. $C_1e^{2x} + C_2xe^{2x}$ **D.** $C_1e^{2x} + C_2xe^{-2x}$

Mechanical Engineering

Q6.26.1 MCQ 2026 1M

The order and degree of the following differential equation are m and n, respectively

$$\frac{\partial^3\varphi}{\partial x^3} + \frac{\partial^2\varphi}{\partial y^2}\frac{\partial\varphi}{\partial x} + \left(\frac{\partial^2\varphi}{\partial x^2}\right)^2 + \frac{\partial\varphi}{\partial y} = 0$$

The value of $(m - n)$ is

A. 2 **B.** 3 **C.** 1 **D.** 0

Q6.26.2 MCQ 2026 2M

Consider the following differential equation

$$\frac{dy}{dx} = 3\frac{dy}{dt} + y$$

If $y(x, 0) = 10\,e^{-2x}$, then the solution of the differential equation is

A. $y(x,t) = 10\,e^{-2x-t}$ **B.** $y(x,t) = 10\,e^{-2x+t}$

C. $y(x,t) = 10\,e^{-2x-2t}$ **D.** $y(x,t) = 10\,e^{-2x+2t}$

Q6.25.1 MCQ 2025 1M

For the differential equation given below, which one of the following options is correct?

$$\frac{\partial^2u}{\partial x^2} + \frac{\partial^2u}{\partial y^2} = 0$$

$0 \le x \le 1,\ 0 \le y \le 1$

A. $u = e^{x+y}$ is a solution for all x and y

B. $u = e^x \sin y$ is a solution for all x and y

C. $u = \sin x \sin y$ is a solution for all x and y

D. $u = \cos x \cos y$ is a solution for all x and y

Q6.25.2 NAT 2025 2M

Let y be the solution of the differential equation with the initial conditions given below. If $y(x = 2) = A\ln 2$, then the value of A is _____ (rounded off to 2 decimal places).

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0$$

$$y(x = 1) = 0 \qquad \frac{dy}{dx}(x = 1) = 1$$

Q6.24.1 NAT 2024 2M

If $x(t)$ satisfies the differential equation

$$t\frac{dx}{dt} + (t - x) = 0$$

subject to the condition $x(1) = 0$, then the value of $x(2)$ is _____ (rounded off to 2 decimal places)

Q6.23.1 NAT 2023 2M

Consider the second order linear ordinary differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$, $x \geq 1$ with initial condition $y(x = 1) = 6$, $\left.\frac{dy}{dx}\right|_{x=1} = 2$. The value of y at $x = 2$ equals _____ (Answer in integer).

Q6.22.1 MCQ 2022 1M

Solution of $\nabla^2 T = 0$ in a square domain $(0 < x < 1$ and $0 < y < 1)$ with boundary conditions:

$$T(x, 0) = x;\ T(0, y) = y;$$

$$T(x, 1) = 1 + x;\ T(1, y) = 1 + y$$

is

A. $T(x, y) = x - xy + y$

B. $T(x, y) = x + y$

C. $T(x, y) = -x + y$

D. $T(x, y) = x + xy + y$

Q6.22.2 MCQ 2022 2M

For the exact differential equation,

$$\frac{du}{dx} = \frac{-xu^2}{2 + x^2u}$$

Which one of the following is the solution?

A. $u^2 + 2x^2 =$ Constant

B. $xu^2 + u =$ Constant

C. $\frac{1}{2}x^2u^2 + 2u =$ Constant

D. $\frac{1}{2}ux^2 + 2x =$ Constant

Q6.21.1 MCQ 2021 1M

If $y(x)$ satisfies the differential equation

$$(\sin x)\frac{dy}{dx} + y\cos x = 1,$$

subject to the condition $y(\pi/2) = \pi/2$, then $y(\pi/6)$ is

A. 0 **B.** $\frac{\pi}{6}$ **C.** $\frac{\pi}{3}$ **D.** $\frac{\pi}{2}$

Q6.21.2 MCQ 2021 2M

Consider the following differential equation

$$(1 + y)\frac{dy}{dx} = y.$$

The solution of the equation that satisfies the condition $y(1) = 1$ is

A. $2ye^y = e^x + e$

B. $y^2e^y = e^x$

C. $ye^y = e^x$

D. $(1 + y)e^y = 2e^x$

Q6.19.1 MCQ 2019 1M

For the equation $\frac{dy}{dx} + 7x^2y = 0$, if $y(0) = 3/7$, then the value of $y(1)$ is

A. $\frac{7}{3}e^{-7/3}$ **B.** $\frac{7}{3}e^{-3/7}$ **C.** $\frac{3}{7}e^{-7/3}$ **D.** $\frac{3}{7}e^{-3/7}$

Q6.19.2 MCQ 2019 1M

The differential equation $\frac{dy}{dx} + 4y = 5$ is valid in the domain $0 \leq x \leq 1$ with $y(0) = 2.25$. The solution of the differential equation is

A. $y = e^{-4x} + 5$

B. $y = e^{-4x} + 1.25$

C. $y = e^{4x} + 5$

D. $y = e^{4x} + 1.25$

Q6.19.3 MCQ 2019 2M

A differential equation is given as

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 4.$$

The solution of the differential equation in terms of arbitrary constants C_1 and C_2 is

A. $y = C_1x^2 + C_2x + 2$

B. $y = C_1x^2 + C_2x + 4$

C. $y = \frac{C_1}{x^2} + C_2x + 2$

D. $y = \frac{C_1}{x^2} + C_2x + 4$

Q6.18.1 MCQ 2018 1M

If y is the solution of the differential equation $y^3 \frac{dy}{dx} + x^3 = 0,\ y(0) = 1$, the value of $y(-1)$ is

A. -2 **B.** -1 **C.** 0 **D.** 1

Q6.18.2 NAT 2018 2M

Given the ordinary differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

with $y(0) = 0$ and $\frac{dy}{dx}(0) = 1$, the value of $y(1)$ is _____ (corrected to two decimal places).

Q6.17.1 MCQ 2017 1M

Consider the partial differential equation for $u(x, y)$ with the constant $c > 1$:

$$\frac{\partial u}{\partial y} + c\frac{\partial u}{\partial x} = 0$$

Solution of this equation is

A. $u(x, y) = f(x + cy)$ **B.** $u(x, y) = f(x - cy)$

C. $u(x, y) = f(cx + y)$ **D.** $u(x, y) = f(cx - y)$

Q6.17.2 MCQ 2017 1M

The differential equation $\frac{d^2y}{dx^2} + 16y = 0$ for $y(x)$ with the two boundary conditions $\left.\frac{dy}{dx}\right|_{x=0} = 1$ and $\left.\frac{dy}{dx}\right|_{x=\pi/2} = -1$ has

A. no solution **B.** exactly two solutions

C. exactly one solution **D.** infinitely many solutions

Q6.17.3 NAT 2017 2M

Consider the differential equation $3y''(x) + 27y(x) = 0$ with initial conditions $y(0) = 0$ and $y'(0) = 2000$. The value of y at $x = 1$ is _____.

Q6.16.1 NAT 2016 2M

If $y = f(x)$ satisfies the boundary value problem $y'' + 9y = 0$. $y(0) = 0,\ y(\pi/2) = \sqrt{2}$, then $y(\pi/4)$ is _____.

Q6.15.1 MCQ 2015 2M

Find the solution of $\frac{d^2y}{dx^2} = y$ which passes through the origin and the point $\left(\ln 2, \frac{3}{4}\right)$.

A. $y = (1/2)e^x - e^{-x}$ **B.** $y = (1/2)(e^x + e^{-x})$

C. $y = (1/2)(e^x - e^{-x})$ **D.** $y = (1/2)e^x + e^{-x}$

Q6.15.2 MCQ 2015 2M

Consider the following differential equation:

$$\frac{dy}{dt} = -5y;$$

initial condition: $y = 2$ at $t = 0$. The value of y at $t = 3$ is

A. $-5e^{-10}$ **B.** $2e^{-10}$ **C.** $2e^{-15}$ **D.** $-15e^2$

Q6.14.1 MCQ 2014 1M

The matrix form of the linear system $\frac{dx}{dt} = 3x - 5y$ and $\frac{dy}{dt} = 4x + 8y$

A. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

B. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & 8 \\ 4 & -5 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

C. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 4 & -5 \\ 3 & 8 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

D. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 4 & 8 \\ 3 & -5 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

Q6.14.2 NAT 2014 2M

If $y = f(x)$ is the solution of $\frac{d^2y}{dx^2} = 0$ with the boundary conditions $y = 5$ at $x = 0$, and $\frac{dy}{dx} = 2$ at $x = 10$, $f(15) =$ _____

Q6.14.3 MCQ 2014 2M

The general solution of the differential equation $\frac{dy}{dx} = \cos(x+y)$, with c as a constant, is

A. $y + \sin(x+y) = x + c$

B. $\tan\left(\frac{x+y}{2}\right) = y + c$

C. $\cos\left(\frac{x+y}{2}\right) = x + c$

D. $\tan\left(\frac{x+y}{2}\right) = x + c$

Q6.14.4 MCQ 2014 2M

Consider two solutions $x(t) = x_1(t)$ and $x(t) = x_2(t)$ of the differential equation

$$\frac{d^2x(t)}{dt^2} + x(t) = 0,\ t > 0,$$

such that $x_1(0) = 1$, $\left.\frac{dx_1(t)}{dt}\right|_{t=0} = 0$, $x_2(0) = 0$, $\left.\frac{dx_2(t)}{dt}\right|_{t=0} = 1$. The Wronskian

$$W(t) = \begin{vmatrix} x_1(t) & x_2(t) \\ \frac{dx_1(t)}{dt} & \frac{dx_2(t)}{dt} \end{vmatrix}$$ at $t = \pi/2$ is

A. 1 **B.** -1 **C.** 0 **D.** $\pi/2$

Q6.14.5 MCQ 2014 1M

The solution of the initial value problem $\frac{dy}{dx} = -2xy$; $y(0) = 2$ is

A. $1 + e^{-x^2}$ **B.** $2e^{-x^2}$ **C.** $1 + e^{x^2}$ **D.** $2e^{x^2}$

Q6.13.1 MCQ 2013 1M

The partial differential equation $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2}$ is a

A. Linear equation of order 2

B. Non-linear equation of order 1

C. Linear equation of order 1

D. Non-linear equation of order 2

Q6.13.2 MCQ 2013 2M

The solution to the differential equation $\frac{d^2u}{dx^2} - k\frac{du}{dx} = 0$ where k is a constant, subjected to the boundary conditions $u(0) = 0$ and $u(L) = U$, is

A. $u = U\frac{x}{L}$ **B.** $u = U\left(\frac{1-e^{kx}}{1-e^{kL}}\right)$

C. $u = U\left(\frac{1-e^{-kx}}{1-e^{-kL}}\right)$ **D.** $u = U\left(\frac{1+e^{-kx}}{1+e^{-kL}}\right)$

Q6.12.1 MCQ 2012 2M

Consider the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 4y = 0$ with the boundary conditions of $y(0) = 0$ and $y(1) = 1$. The complete solution of the differential equation is

A. x^2 **B.** $\sin\left(\frac{\pi x}{2}\right)$

C. $e^x \sin\left(\frac{\pi x}{2}\right)$ **D.** $e^{-x}\sin\left(\frac{\pi x}{2}\right)$

Q6.11.1 MCQ 2011 2M

Consider the differential equation $\frac{dy}{dx} = (1+y^2)x$. The general solution with constant c is

A. $y = \tan\frac{x^2}{2} + \tan c$ **B.** $y = \tan^2\left(\frac{x}{2} + c\right)$

C. $y = \tan^2\left(\frac{x^2}{2}\right) + c$ **D.** $y = \tan\left(\frac{x^2}{2} + c\right)$

Q6.10.1 MCQ 2010 1M

The Blasius equation $\frac{d^3f}{d\eta^3} + \frac{f}{2}\frac{d^2f}{d\eta^2} = 0$, is a

A. second order nonlinear ordinary differential equation

B. third order nonlinear ordinary differential equation

C. third order linear ordinary differential equation

D. mixed order nonlinear ordinary differential equation

Q6.09.1 MCQ 2009 2M

The solution of $x\frac{dy}{dx} + y = x^4$ with the condition $y(1) = \frac{6}{5}$ is

A. $y = \frac{x^4}{5} + \frac{1}{x}$ **B.** $y = \frac{4x^4}{5} + \frac{4}{5x}$

C. $y = \frac{x^4}{5} + 1$ **D.** $y = \frac{x^5}{5} + 1$

Q6.08.1 MCQ 2008 1M

Given that $\ddot{x} + 3x = 0$, and $x(0) = 1,\ \dot{x} = 0$, what is $x(1)$?

A. -0.99 **B.** -0.16 **C.** 0.16 **D.** 0.99

Q6.08.2 MCQ 2008 2M

Let $f = y^x$. What is $\dfrac{\partial^2 f}{\partial x \partial y}$ at $x = 2,\ y = 1$?

A. 0 **B.** $\ln 2$ **C.** 1 **D.** $\dfrac{1}{\ln 2}$

Q6.08.3 MCQ 2008 2M

It is given that $y'' + 2y' + y = 0,\ y(0) = 0,\ y(1) = 0$. What is $y(0.5)$?

A. 0 **B.** 0.37 **C.** 0.62 **D.** 1.13

Q6.07.1 MCQ 2007 1M

The partial differential equation

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial y} = 0$$

has

A. degree 1 order 2 **B.** degree 1 order 1

C. degree 2 order 1 **D.** degree 2 order 2

Q6.07.2 MCQ 2007 2M

The solution of $\dfrac{dy}{dx} = y^2$ with initial value $y(0) = 1$ bounded in the interval

A. $-\infty \le x \le \infty$ **B.** $-\infty \le x \le 1$

C. $x < 1,\ x > 1$ **D.** $-2 \le x \le 2$

Q6.06.1 MCQ 2006 1M

The solution of the differential equation $\dfrac{dy}{dx} + 2xy = e^{-x^2}$ with $y(0) = 1$ is

A. $(1 + x)e^{+x^2}$ **B.** $(1 + x)e^{-x^2}$

C. $(1 - x)e^{+x^2}$ **D.** $(1 - x)e^{-x^2}$

Q6.06.2 MCQ 2006 2M

For $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 3y = 3e^{2x}$, the particular integral is:

A. $\dfrac{1}{15}e^{2x}$ **B.** $\dfrac{1}{5}e^{2x}$

C. $3e^{2x}$ **D.** $C_1e^{-x} + C_2e^{-3x}$

Q6.05.1 MCQ 2005 2M

If $x^2\dfrac{dy}{dx} + 2xy = \dfrac{2\ln x}{x}$, and $y(1) = 0$, then what is $y(e)$?

A. e **B.** 1 **C.** $\dfrac{1}{e}$ **D.** $\dfrac{1}{e^2}$

Common Data Questions

Answer the next two questions using the following data. The complete solution for the ordinary differential equation $\dfrac{d^2y}{dx^2} + p\dfrac{dy}{dx} + qy = 0$ is $y = c_1e^{-x} + c_2e^{-3x}$

Q6.05.2 MCQ 2005 2M

Then, p and q are

A. $p = 3,\ q = 3$ **B.** $p = 3,\ q = 4$

C. $p = 4,\ q = 3$ **D.** $p = 4,\ q = 4$

Q6.05.3 MCQ 2005 2M

Which of the following is a solution of the differential equation

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + (q + 1)y = 0?$$

A. e^{-3x} **B.** xe^{-x} **C.** xe^{-2x} **D.** x^2e^{-2x}

Q6.03.1 MCQ 2003 2M

The solution of the differential equation $\dfrac{dy}{dx} + y^2 = 0$ is

A. $y = \dfrac{1}{x + c}$

B. $y = \dfrac{-x^3}{3} + c$

C. ce^x

D. unsolvable as equation is non-linear

Q6.01.1 MCQ 2001 5M

Solve the differential equation

$$\frac{d^2y}{dx^2} + y = x$$

with the following conditions:

(i) at $x = 0$, $y = 1$

(ii) at $x = \frac{\pi}{2}$, $y = \frac{\pi}{2}$

A. $\cos x + x^2$ **B.** $\cos x - x$

C. $\cos x + x$ **D.** $\sin x + x^2$

Q6.00.1 MCQ 2000 1M

The solution of the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$ is

A. $Ae^x + Be^{-x}$

B. $e^x(Ax + B)$

C. $e^{-x}\left\{A\cos\left(\frac{\sqrt{3}}{2}\right)x + B\cos\left(\frac{\sqrt{3}}{2}\right)x\right\}$

D. $e^{-x/2}\left\{A\cos\left(\frac{\sqrt{3}}{2}\right)x + B\sin\left(\frac{\sqrt{3}}{2}\right)x\right\}$

Q6.99.1 MCQ 1999 2M

$$\frac{d^2y}{dx^2} + (x^2 + 4x)\frac{dy}{dx} + y = x^8 - 8$$

The above equation is a

A. partial differential equation

B. nonlinear differential equation

C. non-homogeneous differential equation

D. ordinary differential equation

Q6.98.1 MCQ 1998 2M

The general solution of the differential equation $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ is:

A. $Ax + Bx^2$ (A, B are constants)

B. $Ax + B\log x$ (A, B are constants)

C. $Ax + Bx^2\log x$ (A, B are constants)

D. $Ax + Bx\log x$ (A, B are constants)

Q6.98.2 NAT 1998 5M

The radial displacement in a rotating disc is governed by the differential equation

$$\frac{d^2u}{dx^2} + \frac{1}{x}\frac{du}{dx} - \frac{u}{x^2} = 8x$$

where u is the displacement and x is the radius.

If $u = 0$ at $x = 0$, and $u = 2$ at $x = 1$, calculate the displacement at $x = \frac{1}{2}$

Q6.96.1 MCQ 1996 2M

The particular solution for the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 5\cos x \text{ is:}$$

A. $0.5\cos x + 1.5\sin x$ **B.** $1.5\cos x + 0.5\sin x$

C. $1.5\sin x$ **D.** $0.5\cos x$

Q6.95.1 MCQ 1995 1M

A differential equation of form $\frac{dy}{dx} = f(x, y)$ is homogeneous if the function $f(x, y)$ depends only on the ratio $\frac{y}{x}$ or $\frac{x}{y}$

A. True

B. False

Q6.95.2 MCQ 1995 1M

The solution to the differential equation $f''(x) + 4f'(x) + 4f(x) = 0$ is:

A. $f_1(x) = e^{-2x}$

B. $f_1(x) = e^{2x}$, $f_2(x) = e^{-2x}$

C. $f_1(x) = e^{-2x}$, $f_2(x) = xe^{-2x}$

D. $f_1(x) = e^{-2x}$, $f_2(x) = e^{-x}$

Q6.94.1 MCQ 1994 1M

For the differential equation $\frac{dy}{dx}+5y=0$ with $y(0)=1$, the general solution is

A. e^{5t} **B.** e^{-5t} **C.** $5e^{-5t}$ **D.** $e^{\sqrt{-5t}}$

Q6.94.2 MCQ 1994 1M

If $H(x, y)$ is a homogeneous function of degree n, then $x\frac{\partial H}{\partial x}+y\frac{\partial H}{\partial y}=nH$.

A. True

B. False

Q6.94.3 MCQ 1994 2M

Solve for y, if $\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=0$ with $y(0)=1$ and $y'(0)=2$.

A. $2e^{-t}+te^{-t}$ **B.** $e^{-t}+2te^{-2t}$

C. $e^{-t}+3te^{-2t}$ **D.** $e^{-t}+3te^{-t}$

Q6.93.1 MCQ 1993 1M

The differential equation $y''+y=0$ is subjected to the conditions $y(0)=0$, $y(\lambda)=0$. In order that the equation has non-trivial solutions, the general value of λ is

A. $\frac{m\pi}{2}$ **B.** $m\pi$ **C.** $\frac{m\pi}{4}$ **D.** $\frac{m\pi}{6}$

Q6.93.2 MCQ 1993 1M

The differential equation $\frac{d^2y}{dx^2}+\frac{dy}{dx}+\sin y=0$ is

A. Linear **B.** Non-linear

C. Homogeneous **D.** of degree 2

Civil Engineering

Q6.26.1 MCQ 2026 2M

An ordinary differential equation is given below.

$$x^2\frac{d^2y}{dx^2}=6y$$

Considering a and b as arbitrary constants, the general solution of the equation is

A. $y(x)=ax^3+\frac{b}{x^2}$ **B.** $y(x)=ax^2+\frac{b}{x^3}$

C. $y(x)=ax^2+b\ln x$ **D.** $y(x)=ax^3+b\ln x$

Q6.26.2 MSQ 2026 1M

A partial differential equation is given below.

$$\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}=0$$

Possible solutions(s) is/are:

- **A.** $(x+y)^5$
- **B.** $(x-2y)^3$
- **C.** $\cos(x+y)$
- **D.** $\sin(x-2y)$

Q6.26.3 NAT 2026 2M

Consider differential equation $\frac{dy}{dx}+xy=x$ with the condition as $y=0$ at $x=0$. The value of y at $x=1.0$ is _____ *(rounded off to two decimal places).*

Q6.25.1 MSQ 2025 1M

Which of the following equations belong/belongs to the class of second-order, linear, homogeneous partial differential equations:

- **A.** $\frac{\partial^2 u}{\partial t^2}=c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)+xy$
- **B.** $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=0$
- **C.** $\frac{\partial u}{\partial t}=c\frac{\partial u}{\partial x}$
- **D.** $\left(\frac{\partial^2 u}{\partial t^2}\right)^2=c^2\frac{\partial^2 u}{\partial x^2}$

Q6.25.2 NAT 2025 2M

Let y be the solution of the initial value problem $y''+0.8y'+0.16y=0$, where $y(0)=3$ and $y'(0)=4.5$. Then, $y(1)$ is equal to _____ *(rounded off to 1 decimal place).*

Q6.25.3 NAT 2025 1M

The "order" of the following ordinary differential equation is ______.

$$\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^6 + \left(\frac{dy}{dx}\right)^4 + y = 0$$

Q6.25.4 MCQ 2025 2M

Pick the **CORRECT** solution for the following differential equation

$$\frac{dy}{dx} = e^{x-y}$$

A. $y = \ln(e^x + \text{Constant})$

B. $\ln(y) = x + \text{Constant}$

C. $\ln(y) = \ln(e^x) + \text{Constant}$

D. $y = x + \text{Constant}$

Q6.24.1 MCQ 2024 1M

The second-order differential equation in an unknown function $u : u(x, y)$ is

$$\frac{\partial^2 u}{\partial x^2} = 2$$

Assuming $g : g(x)$, $f : f(y)$, and $h : h(y)$, the general solution of the above differential equation is

A. $u = x^2 + f(y) + g(x)$

B. $u = x^2 + x f(y) + h(y)$

C. $u = x^2 + x f(y) + g(x)$

D. $u = x^2 + f(y) + y\, g(x)$

Q6.24.2 MSQ 2024 1M

For the following partial differential equation,

$$x\frac{\partial^2 f}{\partial x^2} + y\frac{\partial^2 f}{\partial y^2} = \frac{x^2 + y^2}{2}$$

which of the following option(s) is/are CORRECT?

- [ ] **A.** elliptic for $x > 0$ and $y > 0$
- [ ] **B.** parabolic for $x > 0$ and $y > 0$
- [ ] **C.** elliptic for $x = 0$ and $y > 0$
- [ ] **D.** hyperbolic for $x < 0$ and $y > 0$

Q6.24.3 MCQ 2024 1M

Consider two Ordinary Differential Equations (ODEs):

$$\text{P: } \frac{dy}{dx} = \frac{x^4 + 3x^2y^2 + 2y^4}{x^3 y}$$

$$\text{Q: } \frac{dy}{dx} = \frac{-y^2}{x^2}$$

Which one of the following options is CORRECT?

A. P is a homogeneous ODE and Q is an exact ODE

B. P is a homogeneous ODE and Q is not an exact ODE

C. P is a nonhomogeneous ODE and Q is an exact ODE

D. P is a nonhomogeneous ODE and Q is not an exact ODE

Q6.23.1 NAT 2023 1M

In the differential equation $\frac{dy}{dx} + \alpha xy = 0$, α is a positive constant. If $y = 1.0$ at $x = 0.0$, and $y = 0.8$ at $x = 1.0$, the value of α is ______ (rounded off to three decimal places).

Q6.23.2 MCQ 2023 2M

The solution of the differential equation

$$\frac{d^3y}{dx^3} - 5.5\frac{d^2y}{dx^2} + 9.5\frac{dy}{dx} - 5y = 0$$

is expressed as $y = C_1e^{2.5x} + C_2e^{\alpha x} + C_3e^{\beta x}$, where C_1, C_2, C_3, α, and β are constants, with α and β being distinct and not equal to 2.5. Which of the following options is correct for the values of α and β ?

A. 1 and 2 **B.** -1 and -2

C. 2 and 3 **D.** -2 and -3

Q6.22.1 MCQ 2022 1M

Consider the following expression:

$$z = \sin(y + it) + \cos(y - it)$$

where z, y, and t are variables, and $i = \sqrt{-1}$ is a complex number. The partial differential equation derived from the above expression is

A. $\frac{\partial^2 z}{\partial t^2} + \frac{\partial^2 z}{\partial y^2} = 0$ **B.** $\frac{\partial^2 z}{\partial t^2} - \frac{\partial^2 z}{\partial y^2} = 0$

C. $\frac{\partial z}{\partial t} - i\frac{\partial z}{\partial y} = 0$ **D.** $\frac{\partial z}{\partial t} + i\frac{\partial z}{\partial y} = 0$

Q6.22.2 MCQ 2022 1M

For the equation

$$\frac{d^3y}{dx^3} + x\left(\frac{dy}{dx}\right)^{3/2} + x^2y = 0$$

the correct description is

A. an ordinary differential equation of order 3 and degree 2

B. an ordinary differential equation of order 3 and degree 3

C. an ordinary differential equation of order 2 and degree 3

D. an ordinary differential equation of order 3 and degree 3/2

Q6.21.1 MCQ 2021 2M

The solution of the second-order differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ with boundary conditions $y(0) = 1$ and $y(1) = 3$ is

A. $e^{-x} + (3e - 1)xe^{-x}$

B. $e^{-x} - (3e - 1)xe^{-x}$

C. $e^{-x} + \left[3e\sin\left(\frac{\pi x}{2}\right) - 1\right]xe^{-x}$

D. $e^{-x} - \left[3e\sin\left(\frac{\pi x}{2}\right) - 1\right]xe^{-x}$

Q6.21.2 MCQ 2021 2M

If k is a constant, the general solution of $\frac{dy}{dx} - \frac{y}{x} = 1$ will be in the form of

A. $y = x\ln(kx)$

B. $y = k\ln(kx)$

C. $y = x\ln(x)$

D. $y = xk\ln(k)$

Q6.20.1 MCQ 2020 1M

In the following partial differential equation, θ is a function of 't' and 'z' and 'D' and 'K' are functions of θ

$$D(\theta)\frac{\partial^2\theta}{\partial z^2} + \frac{\partial K(\theta)}{\partial z} - \frac{\partial\theta}{\partial t} = 0$$

The above equation is

A. a second order linear equation

B. a second order non-linear equation

C. a second-degree non-linear equation

D. a second-degree linear equation

Q6.20.2 MCQ 2020 2M

For the ordinary differential equation $\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$, with initial condition $x(0) = 0$ and $\frac{dx}{dt}(0) = 10$, the solution is

A. $-10\,e^{2t} + 10\,e^{3t}$

B. $5\,e^{2t} + 6\,e^{3t}$

C. $10\,e^{2t} + 10\,e^{3t}$

D. $-5\,e^{2t} + 6\,e^{3t}$

Q6.20.3 MCQ 2020 1M

The ordinary differential equation $\frac{d^2u}{dx^2} - 2x^2u + \sin x = 0$ is

A. linear and homogeneous

B. non-linear and homogeneous

C. non-linear and nonhomogeneous

D. linear and nonhomogeneous

Q6.20.4 MCQ 2020 1M

The following partial differential equation is defined for $u : u(x, y)$

$$\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial x^2};\ y \geq 0;\ x_1 \leq x \leq x_2$$

The set of auxiliary conditions necessary to solve the equation uniquely, is

A. one initial condition and two boundary conditions

B. three initial conditions

C. two initial conditions and one boundary condition

D. three boundary conditions

Q6.20.5 MCQ 2020 2M

An ordinary differential equation is given below:

$$6\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$$

The general solution of the above equation (with constants C_1 and C_2), is

A. $y(x) = C_1 e^{x/3} + C_2 e^{-x/2}$

B. $y(x) = C_1 x e^{-x/3} + C_2 e^{x/2}$

C. $y(x) = C_1 e^{-x/3} + C_2 e^{x/2}$

D. $y(x) = C_1 e^{-x/3} + C_2 x e^{x/2}$

Q6.19.1 MCQ 2019 1M

Consider a two-dimensional flow through isotropic soil along x direction and z direction. If h is the hydraulic head, the Laplace's equation of continuity is expressed as

A. $\frac{\partial h}{\partial x} + \frac{\partial h}{\partial z} = 0$ **B.** $\frac{\partial h}{\partial x} + \frac{\partial h}{\partial x}\frac{\partial h}{\partial z} + \frac{\partial h}{\partial z} = 0$

C. $\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial z^2} = 0$ **D.** $\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial x \partial z} + \frac{\partial^2 h}{\partial z^2} = 0$

Q6.19.2 NAT 2019 2M

Consider the ordinary differential equation $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$. Given the values of $y(1) = 0$ and $y(2) = 2$, the value of $y(3)$ (round off to 1 decimal place), is ______

Q6.19.3 MCQ 2019 2M

An ordinary differential equation is given below.

$$\left(\frac{dy}{dx}\right)(x \ln x) = y$$

The solution for the above equation is (*Note: K denotes a constant in the options*)

A. $y = Kx \ln x$ **B.** $y = Kxe^x$

C. $y = Kxe^{-x}$ **D.** $y = K \ln x$

Q6.18.1 MCQ 2018 2M

The solution at $x = 1, t = 1$ of the partial differential equation $\frac{\partial^2 u}{\partial x^2} = 25\frac{\partial^2 u}{\partial t^2}$ subject to initial conditions of $u(0) = 3x$ and $\frac{\partial u}{\partial t}(0) = 3$ is ______

A. 1 **B.** 2 **C.** 4 **D.** 6

Q6.18.2 NAT 2018 2M

The solution (up to three decimal places) at $x = 1$ of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ subject to boundary conditions $y(0) = 1$ and $\frac{dy}{dx}(0) = -1$ is ______

Q6.18.3 MCQ 2018 1M

The solution of the equation $x\frac{dy}{dx} + y = 0$ passing through the point $(1, 1)$ is

A. x **B.** x^2 **C.** x^{-1} **D.** x^{-2}

Q6.17.1 NAT 2017 1M

Consider the following partial differential equation

$$3\frac{\partial^2 \phi}{\partial x^2} + B\frac{\partial^2 \phi}{\partial x \partial y} + 3\frac{\partial^2 \phi}{\partial y^2} + 4\phi = 0$$

For this equation to be classified as parabolic, the value of B^2 must be ______

Q6.17.2 MCQ 2017 2M

The solution of the equation $\frac{dQ}{dt} + Q = 1$ with $Q = 0$ at $t = 0$ is

A. $Q(t) = e^{-t} - 1$ **B.** $Q(t) = 1 + e^{-t}$

C. $Q(t) = 1 - e^{t}$ **D.** $Q(t) = 1 - e^{-t}$

Q6.17.3 MCQ 2017 2M

Consider the following second-order differential equation:

$$y'' - 4y' + 3y = 2t - 3t^2$$

The particular solution of the differential equation is

A. $-2 - 2t - t^2$ **B.** $-2t - t^2$

C. $2t - 3t^2$ **D.** $-2 - 2t - 3t^2$

Q6.16.1 MCQ 2016 1M

The type of partial differential equation $\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + 3\frac{\partial^2 P}{\partial x \partial y} + 2\frac{\partial P}{\partial x} - \frac{\partial P}{\partial y} = 0$ is

A. elliptic **B.** parabolic

C. hyperbolic **D.** none of these

Q6.16.2 MCQ 2016 1M

The solution of the partial differential equation $\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}$ is of the form

A. $C\cos(kt)\left[C_1 e^{(\sqrt{k/\alpha})x} + C_2 e^{-(\sqrt{k/\alpha})x}\right]$

B. $C\,e^{kt}\left[C_1\,e^{(\sqrt{k/\alpha})x} + C_2\,e^{-(\sqrt{k/\alpha})x}\right]$

C. $C\,e^{kt}\left[C_1\,\cos(\sqrt{k/\alpha})x + C_2\,\sin(-\sqrt{k/\alpha})x\right]$

D. $C\,\sin(kt)\left[C_1\,\cos(\sqrt{k/\alpha})x + C_2\,\sin(-\sqrt{k/\alpha})x\right]$

Q6.16.3 MCQ 2016 2M

The respective expressions for complimentary function and particular integral part of the solution of the differential equation $\frac{d^4y}{dx^4} + 3\frac{d^2y}{dx^2} = 108x^2$ are

A. $[c_1 + c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[3x^4 - 12x^2 + c]$

B. $[c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[5x^4 - 12x^2 + c]$

C. $[c_1 + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[3x^4 - 12x^2 + c]$

D. $[c_1 + c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[5x^4 - 12x^2 + c]$

Q6.15.1 MCQ 2015 2M

Consider the following differential equation:

$$x(y\,dx + x\,dy)\cos\frac{y}{x} = y(x\,dy - y\,dx)\sin\frac{y}{x}$$

Which of the following is the solution of the above equation (c is an arbitrary constant)?

A. $\frac{x}{y}\cos\frac{y}{x} = c$ **B.** $\frac{x}{y}\sin\frac{y}{x} = c$

C. $xy\cos\frac{y}{x} = c$ **D.** $xy\sin\frac{y}{x} = c$

Q6.15.2 NAT 2015 2M

Consider the following second order linear differential equation

$$\frac{d^2y}{dx^2} = -12x^2 + 24x - 20$$

The boundary conditions are: at $x = 0$, $y = 5$ and at $x = 2$, $y = 21$. The value of y at $x = 1$ is ______.

Q6.14.1 MCQ 2014 1M

The integrating factor for the differential equation $\frac{dP}{dt} + k_2P = k_1L_0e^{-k_1t}$ is

A. e^{-k_1t} **B.** e^{-k_2t} **C.** e^{k_1t} **D.** e^{k_2t}

Q6.14.2 NAT 2014 2M

Water is flowing at a steady rate through a homogeneous and saturated horizontal soil strip of 10m length. The strip is being subjected to a constant water head (H) of 5m at the beginning and 1m at the end. If the governing equation of flow in the soil strip is $\frac{d^2H}{dx^2} = 0$ (where x is the distance along the soil strip), the value of H (in m) at the middle of the strip is ______.

Q6.13.1 NAT 2013 2M

Laplace equation for water flow in soil is given by

$$\frac{\partial^2H}{\partial x^2} + \frac{\partial^2H}{\partial y^2} + \frac{\partial^2H}{\partial z^2} = 0$$

Head H does not vary in y and z-direction. Boundary condition $x = 0$, $H = 5m$, $\frac{dH}{dx} = -1$. What is the value of H at $x = 1.2$?

Q6.12.1 MCQ 2012 2M

The solution of the ordinary differential equation $\frac{dy}{dx} + 2y = 0$ for the boundary condition, $y = 5$ at $x = 1$ is

A. $y = e^{-2x}$ **B.** $y = 2e^{-2x}$

C. $y = 10.95e^{-2x}$ **D.** $y = 36.95e^{-2x}$

Q6.11.1 MCQ 2011 2M

The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = x$, with the condition that $y = 1$ at $x = 1$, is

A. $y = \frac{2}{3x^2} + \frac{x}{3}$ **B.** $y = \frac{x}{2} + \frac{1}{2x}$

C. $y = \frac{2}{3} + \frac{x}{3}$ **D.** $y = \frac{2}{3x} + \frac{x^2}{3}$

Q6.10.1 MCQ 2010 1M

The order and degree of the differential equation

$$\frac{d^3y}{dx^3} + 4\sqrt{\left(\frac{dy}{dx}\right)^3} + y^2 = 0$$

are respectively

A. 3 and 2 **B.** 2 and 3

C. 3 and 3 **D.** 3 and 1

Q6.10.2 MCQ 2010 2M

The solution to the ordinary differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$ is

A. $y = c_1 e^{3x} + c_2 e^{-2x}$
B. $y = c_1 e^{3x} + c_2 e^{2x}$
C. $y = c_1 e^{-3x} + c_2 e^{2x}$
D. $y = c_1 e^{-3x} + c_2 e^{-2x}$

Q6.10.3 MCQ 2010 2M

The partial differential equation that can be formed from $z = ax + by + ab$ has the form $\left(\text{with } p = \frac{\partial z}{\partial x} \text{ and } q = \frac{\partial z}{\partial y}\right)$

A. $z = px + qy$
B. $z = px + pq$
C. $z = px + qy + pq$
D. $z = qy + pq$

Q6.09.1 MCQ 2009 2M

Solution of the differential equation $3y\frac{dy}{dx} + 2x = 0$ represent a family of

A. ellipses
B. circles
C. parabolas
D. hyperbolas

Q6.08.1 MCQ 2008 1M

The general solution of $\frac{d^2y}{dx^2} + y = 0$ is

A. $y = P\cos x + Q\sin x$
B. $y = P\cos x$
C. $y = P\sin x$
D. $y = P\sin^2 x$

Q6.08.2 MCQ 2008 2M

The equation $k_x\frac{\partial^2 h}{\partial x^2} + k_z\frac{\partial^2 h}{\partial z^2} = 0$ can be transformed to $\frac{\partial^2 h}{\partial x_1^2} + \frac{\partial^2 h}{\partial z^2} = 0$ by substituting

A. $x_1 = x\frac{k_z}{k_x}$
B. $x_1 = x\frac{k_x}{k_z}$
C. $x_1 = x\sqrt{\frac{k_x}{k_z}}$
D. $x_1 = x\sqrt{\frac{k_z}{k_x}}$

Q6.08.3 MCQ 2008 2M

Solution of $\frac{dy}{dx} = -\frac{x}{y}$ at $x = 1$ and $y = \sqrt{3}$ is

A. $x - y^2 = 2$
B. $x + y^2 = 4$
C. $x^2 - y^2 = -2$
D. $x^2 + y^2 = 4$

Q6.07.1 MCQ 2007 1M

The degree of the differential equation $\frac{d^2x}{dt^2} + 2x^3 = 0$ is

A. 0
B. 1
C. 2
D. 3

Q6.07.2 MCQ 2007 1M

The solution for the differential equation $\frac{dy}{dx} = x^2y$ with condition that $y = 1$ at $x = 0$ is

A. $y = e^{1/2x}$
B. $\ln(y) = \frac{x^3}{3} + 4$
C. $\ln(y) = \frac{x^2}{2}$
D. $y = e^{x^3/3}$

Q6.06.1 MCQ 2006 2M

The solution of the differential equation, $x^2\frac{dy}{dx} + 2xy - x + 1 = 0$, given that at $x = 1,\ y = 0$ is

A. $\frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}$
B. $\frac{1}{2} - \frac{1}{x} - \frac{1}{2x^2}$
C. $\frac{1}{2} + \frac{1}{x} + \frac{1}{2x^2}$
D. $-\frac{1}{2} + \frac{1}{x} + \frac{1}{2x^2}$

Q6.05.1 MCQ 2005 2M

Transformation to linear form by substituting $v = y^{1-n}$ of the equation $\frac{dy}{dt} + p(t)y = q(t)\,y^n,\ n > 0$ will be

A. $\frac{dv}{dt} + (1-n)pv = (1-n)q$
B. $\frac{dv}{dt} + (1-n)pv = (1+n)q$
C. $\frac{dv}{dt} + (1+n)pv = (1-n)q$
D. $\frac{dv}{dt} + (1+n)pv = (1+n)q$

Q6.05.2 MCQ 2005 2M

The solution of $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 17y = 0;\ y(0) = 1,$ $\frac{dy}{dx}\left(\frac{\pi}{4}\right) = 0$ in the range $0 < x < \pi/4$ is given by

A. $e^{-x}\left(\cos 4x + \frac{1}{4}\sin 4x\right)$

B. $e^{x}\left(\cos 4x - \frac{1}{4}\sin 4x\right)$

C. $e^{-4x}\left(\cos x - \frac{1}{4}\sin x\right)$

D. $e^{-4x}\left(\cos 4x - \frac{1}{4}\sin 4x\right)$

Q6.04.1 MCQ 2004 2M

Biotransformation of an organic compound having concentration (x) can be modeled using an ordinary differential equation $\frac{dx}{dt} + kx^2 = 0$, where k is the reaction rate constant. If $x = a$ at $t = 0$, the solution of the equation is

A. $x = ae^{-kt}$

B. $\frac{1}{x} = \frac{1}{a} + kt$

C. $x = a(1 - e^{-kt})$

D. $x = a + kt$

Q6.01.1 MCQ 2001 1M

The number of boundary conditions required to solve the differential equation $\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$ is

A. 2 **B.** 0 **C.** 4 **D.** 1

Q6.01.2 MCQ 2001 2M

The solution for the following differential equation with boundary conditions $y(0) = 2$ and $y'(1) = -3$ is

$$\frac{d^2y}{dx^2} = 3x - 2$$

A. $y = \frac{x^3}{3} - \frac{x^2}{2} + 3x - 6$

B. $y = 3x^3 - \frac{x^2}{2} - 5x + 2$

C. $y = \frac{x^3}{2} - x^2 - \frac{5x}{2} + 2$

D. $y = x^3 - \frac{x^2}{2} + 5x + \frac{3}{2}$

Q6.00.1 MCQ 2000 2M

If $f(x,y,z) = (x^2 + y^2 + z^2)^{-1/2}$, $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ is equal to

A. Zero

B. 1

C. 2

D. $-3(x^2 + y^2 + z^2)^{5/2}$

Q6.99.1 MCQ 1999 2M

If c is a constant, solution of the equation $\frac{dy}{dx} = 1 + y^2$ is

A. $y = \sin(x + c)$

B. $y = \cos(x + c)$

C. $y = \tan(x + c)$

D. $y = e^x + c$

Q6.98.1 MCQ 1998 5M

Solve $\frac{d^4y}{dx^4} - y = 15\cos 2x$

A. $C_1e^{x} + C_2e^{-x} + C_3\cos x + C_4\sin x + \cos 2x$

B. $C_1e^{2x} + C_2e^{-x} + C_3\cos x + C_4\sin x + \cos x$

C. $C_1e^{2x} + C_2e^{-2x} + C_3\cos x + C_4\sin x + \cos 2x$

D. $C_1e^{-2x} + C_2e^{-x} + C_3\cos x + C_4\sin x + \cos x$

Q6.97.1 MCQ 1997 1M

For the differential equation, $f(x,y)\frac{dy}{dx} + g(x,y) = 0$ to be exact,

A. $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$

B. $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}$

C. $f = g$

D. $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 g}{\partial y^2}$

Q6.97.2 MCQ 1997 1M

The differential equation $\frac{dy}{dx} + Py = Q$, is a linear equation of first order only if

A. P is a constant but Q is a function of y

B. P and Q are functions of y or constants

C. P is a function of y but Q is a constant

D. P and Q are functions of x or constants

Q6.95.1 MCQ 1995 2M

The solution of a differential equation $y'' + 3y' + 2y = 0$ is of the form

A. $c_1e^x + c_2e^{2x}$ **B.** $c_1e^{-x} + c_2e^{3x}$

C. $c_1e^{-x} + c_2e^{-2x}$ **D.** $c_1e^{x-2} + c_2 2^{-x}$

Q6.95.2 MCQ 1995 1M

The differential equation $y'' + (x^3 \sin x)^5 y' + y = \cos x^3$ is

A. homogeneous

B. non-linear

C. second order linear

D. non-homogeneous with constant coefficients

Q6.94.1 MCQ 1994 2M

The differential equation $\frac{d^4y}{dx^4} + P\frac{d^2y}{dx^2} + ky = 0$ is

A. Linear of fourth order

B. Non-Linear of fourth order

C. Non-Homogeneous

D. Linear and fourth degree

Q6.94.2 MCQ 1994 1M

The necessary and sufficient condition for the differential equation of the form $M(x,y)\,dx + N(x,y)\,dy = 0$ to be exact is

A. $M = N$ **B.** $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

C. $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ **D.** $\frac{\partial^2 M}{\partial x^2} = \frac{\partial^2 N}{\partial y^2}$

Chemical Engineering

Q6.26.1 NAT 2026 2M

Consider a differential equation

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} - 3y = 0$$

$$\left.\begin{aligned} y &= 3 \\ \frac{dy}{dx} &= -5 \end{aligned}\right\} \text{at } x = 1$$

The value of y at $x = 2$ is _____ *(rounded off to two decimal places)*.

Q6.25.1 MSQ 2025 2M

Consider the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = 0$$

Choose the CORRECT option(s) for the solution of y.

☐ **A.** $y = x + c$; c is a constant

☐ **B.** $y = \frac{c}{x}$; c is a constant

☐ **C.** $y = -x + c$; c is a constant

☐ **D.** $y = 0$

Q6.24.1 NAT 2024 2M

Consider the ordinary differential equation

$$x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0,$$

with the boundary conditions $y(x = 1) = 2$ and $y(x = 2) = 17/2$. The solution $y(x)$ at $x = 3/2$, rounded off to 2 decimal places, is _____

Q6.23.1 NAT 2023 1M

The position $x(t)$ of a particle, at constant ω, is described by the equation

$$\frac{d^2x}{dt^2} = -\omega^2 x.$$

The initial conditions are $x(t = 0) = 1$ and $\left.\frac{dx}{dt}\right|_{t=0} = 0$.

The position of the particle at $t = (3\pi/\omega)$ is _____ (in integer).

Q6.22.1 MCQ 2022 2M

The partial differential equation

$$\frac{\partial u}{\partial t} = \frac{1}{\pi^2}\frac{\partial^2 u}{\partial x^2}$$

where, $t \geq 0$ and $x \in [0, 1]$, is subjected to the following initial and boundary conditions:

$$u(x, 0) = \sin(\pi x)$$

$$u(0, t) = 0$$

$$u(1, t) = 0$$

The value of t at which $\frac{u(0.5, t)}{u(0.5, 0)} = \frac{1}{e}$ is

A. 1 **B.** e **C.** π **D.** $1/e$

Q6.21.1 NAT 2021 2M

For the ordinary differential equation

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = 1$$

with the initial conditions $y(0) = y'(0) = y''(0) = y'''(0) = 0$, the value of $\lim_{t\to\infty} y(t) =$ _____ (round off to 3 decimal places).

Q6.20.1 NAT 2020 2M

Given $\frac{dy}{dx} = y - 20$, and $y|_{x=0} = 40$, the value of y at $x = 2$ is _____ (round off to nearest integer).

Q6.19.1 MCQ 2019 2M

The solution of the ordinary differential equation $\frac{dy}{dx} + 3y = 1$, subject to the initial condition $y = 1$ at $x = 0$, is

A. $\frac{1}{3}(1 + 2e^{-x/3})$ **B.** $\frac{1}{3}(5 - 2e^{-x/3})$

C. $\frac{1}{3}(5 - 2e^{-3x})$ **D.** $\frac{1}{3}(1 + 2e^{-3x})$

Q6.18.1 MCQ 2018 1M

Consider the following two equations:

$$\frac{dx}{dt} + x + y = 0$$

$$\frac{dy}{dt} - x = 0$$

The above set of equations is represented by

A. $\frac{d^2y}{dt^2} - \frac{dy}{dt} - y = 0$ **B.** $\frac{d^2x}{dt^2} - \frac{dx}{dt} - y = 0$

C. $\frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0$ **D.** $\frac{d^2x}{dt^2} + \frac{dx}{dt} + y = 0$

Q6.17.1 NAT 2017 2M

For the initial value problem

$$\frac{dx}{dt} = \sin(t),\ x(0) = 0$$

the value of x at $t = \pi/3$, is _____.

Q6.16.1 MCQ 2016 2M

What is the solution for the second order differential equation $\frac{d^2y}{dx^2} + y = 0$, with the initial condition $y|_{x=0} = 5$ and $\left.\frac{dy}{dx}\right|_{x=0} = 10$?

A. $y = 5 + 10\sin x$ **B.** $y = 5\cos x - 5\sin x$

C. $y = 5\cos x + 10x$ **D.** $y = 5\cos x + 10\sin x$

Q6.15.1 MCQ 2015 1M

Consider a linear ordinary differential equation: $\frac{dy}{dx} + p(x)y = r(x)$. Functions $p(x)$ and $r(x)$ are defined and have a continuous first derivative. The integrating factor of this equation is non-zero. Multiplying this equation by its integrating factor converts this into a:

A. Homogeneous differential equation

B. Non-linear differential equation

C. Second order differential equation

D. Exact differential equation

Q6.14.1 MCQ 2014 2M

The integrating factor for the differential equation $\frac{dy}{dx} - \frac{y}{1+x} = (1 + x)$ is

A. $\frac{1}{1+x}$ **B.** $(1 + x)$ **C.** $x(1 + x)$ **D.** $\frac{x}{1+x}$

Q6.14.2 MCQ 2014 2M

The differential equation $\frac{d^2y}{dx^2} + x^2\frac{dy}{dx} + x^3y = e^x$ is a

A. non-linear differential equation of first degree

B. linear differential equation of first degree

C. linear differential equation of second degree

D. non-linear differential equation of second degree

Q6.13.1 MCQ 2013 2M

The solution of the differential equation $\frac{dy}{dx} - y^2 = 0$, given $y = 1$ at $x = 0$ is

A. $\frac{1}{1+x}$ **B.** $\frac{1}{1-x}$ **C.** $\frac{1}{(1-x)^2}$ **D.** $\frac{x^3}{3} + 1$

Q6.13.2 MCQ 2013 2M

The solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} + 0.25y = 0$, given $y = 0$ at $x = 0$ and $\frac{dy}{dx} = 1$ at $x = 0$ is

A. $x\,e^{0.5x} - x\,e^{-0.5x}$
B. $0.5x\,e^{x} - 0.5x\,e^{-x}$
C. $x\,e^{0.5x}$
D. $-x\,e^{0.5x}$

Q6.12.1 MCQ 2012 1M

If a and b are arbitrary constants, then the solution to the ordinary differential equation $\frac{d^2y}{dx^2} - 4y = 0$ is

A. $y = ax + b$
B. $y = ae^{-x}$
C. $y = a\sin 2x + b\cos 2x$
D. $y = a\cosh 2x + b\sinh 2x$

Q6.11.1 MCQ 2011 2M

Which ONE of the following choices is a solution of the differential equation given below?

$$\frac{dy}{dx} = \frac{y^2}{x} + \frac{y}{x} - \frac{2}{x}$$

Note: c is a real constant

A. $y = \frac{c - x^2}{c + 2x^2}$
B. $y = \frac{c + 2x^2}{c - x^2}$
C. $y = \frac{c - x^3}{c + 2x^3}$
D. $y = \frac{c + 2x^3}{c - x^3}$

Q6.10.1 MCQ 2010 2M

The solution of the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = 0$$

with the initial conditions $y(0) = 0$, $\left.\frac{dy}{dx}\right|_{t=0} = -1$, is

A. $-t\sin t$
B. $-e^{-t}(1 - \cos t)$
C. $-(t + \sin t)/2$
D. $-e^{-t}\sin t$

Q6.09.1 MCQ 2009 2M

The general solution of the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0,$$

with C_1 and C_2 are constants of integration, is

A. $C_1\,e^{-3x} + C_2\,e^{-2x}$
B. $C_1\,e^{3x} + C_2\,e^{-2x}$
C. $C_1\,e^{3x} + C_2\,e^{2x}$
D. $C_1\,e^{-3x} + C_2\,e^{2x}$

Q6.08.1 MCQ 2008 1M

Which **ONE** of the following is **NOT** an integrating factor for the differential equation $x\,dy - y\,dx = 0$?

A. $\frac{1}{x^2}$
B. $\frac{1}{y^2}$
C. $\frac{1}{xy}$
D. $\frac{1}{(x+y)}$

Q6.08.2 MCQ 2008 1M

Which **ONE** of the following is **NOT** a solution of the differential equation $\frac{d^2y}{dx^2} + y = 1$?

A. $y = 1$
B. $y = 1 + \cos x$
C. $y = 1 + \sin x$
D. $y = 2 + \sin x + \cos x$

Q6.08.3 MCQ 2008 2M

Which **ONE** of the following transformations $\{u = f(y)\}$ reduces $\frac{dy}{dx} + Ay^3 + By = 0$ to a linear differential equation? (A and B are positive constants)

A. $u = y^{-3}$
B. $u = y^{-2}$
C. $u = y^{-1}$
D. $u = y^2$

Q6.07.1 MCQ 2007 1M

The initial condition for which the following equation

$$(x^2 + 2x)\frac{dy}{dx} = 2(x+1)y;\;\; y(x_0) = y_0$$

has infinitely many solutions, is

A. $y(x = 0) = 5$
B. $y(x = 0) = 1$
C. $y(x = 2) = 1$
D. $y(x = -2) = 0$

Q6.07.2 MCQ 2007 2M

The solution of the following differential equation

$$x\frac{dy}{dx} + y(x^2 - 1) = 2x^3$$

is

A. 0 **B.** $2 + ce^{-x^2/2}$

C. $c_1x + c_2x^2$ **D.** $2x + cxe^{-x^2/2}$

Q6.06.1 MCQ 2006 2M

The solution to the following equation is

$$x^2\frac{d^3y}{dx^3} + 2x\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$$

is given by

A. $y = C_1x + C_2x^{-2} + C_3$

B. $y = C_1x^2 + C_2x^{-2} + C_3$

C. $y = C_1x^2 + C_2x^{-1} + C_3$

D. $y = C_1x + C_2x^{-1} + C_3$

Q6.05.1 MCQ 2005 1M

Match the following, where x is the spatial coordinate and t is time

	Group - I		Group - II
P.	Wave equation	1.	$\frac{\partial c}{\partial t} = \alpha\frac{\partial c}{\partial x}$
Q.	Heat equation	2.	$\frac{\partial c}{\partial t} = \alpha^2\frac{\partial^2 c}{\partial x^2}$
		3.	$\frac{\partial^2 c}{\partial t^2} = \alpha^2\frac{\partial c}{\partial x}$
		4.	$\frac{\partial^2 c}{\partial t^2} = \alpha^2\frac{\partial^2 c}{\partial x^2}$

A. P-4, Q-2 **B.** P-2, Q-4

C. P-3, Q-1 **D.** P-1, Q-3

Q6.05.2 MCQ 2005 2M

What condition is to be satisfied so that the solution of the differential equation

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$$

is of the form $y = (C_1 + C_2x)e^{mx}$, where C_1 and C_2 are constants of integration?

A. $a^2 = b$ **B.** $b^2 = a$

C. $a^2 = 4b$ **D.** $b^2 = 4a$

Q6.05.3 MCQ 2005 2M

If $f(x)$ is the solution of the equation

$$\frac{dy}{dx} + 2xy + 2x = 0$$

and $g(x)$ is the solution of the equation

$$\frac{dy}{dx} + 2xy - 2x = 0,$$

and the constant of integration in $f(x)$ is equal to that in $g(x)$, then which of the following is true?

A. $g(x) = f(x) + 2$ **B.** $g(x) = f(x) + 1$

C. $g(x) = f(x) - 1$ **D.** $g(x) = f(x) - 2$

Q6.04.1 MCQ 2004 1M

The differential equation $\frac{d^2y}{dx^2} + \sin x\frac{dy}{dx} + ye^x = \sinh x$

A. first order and linear

B. first order and non-linear

C. second order and linear

D. second order and non-linear

Q6.04.2 MCQ 2004 2M

The differential equation for the variation of the amount of salt X in a tank with time t is given by $\frac{dX}{dt} + \frac{X}{20} = 10$. X is in kg and t is in minutes. Assuming that there is no salt in the tank initially, the time (in min) at which the amount of salt increases to 100 kg is

A. $10\ln 2$ **B.** $20\ln 2$

C. $50\ln 2$ **D.** $100\ln 2$

Q6.04.3 MCQ 2004 2M

The differential equation $\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2} = 0$ can be reduced to (where α is a constant)

A. $\left(\frac{dy}{dx}\right)^3 = \alpha - \frac{3y^2}{2}$ **B.** $\left(\frac{dy}{dx}\right)^2 = \alpha - 2y$

C. $\left(\frac{dy}{dx}\right) = \frac{\alpha}{y^2}$ **D.** $\left(\frac{dy}{dx}\right) = \frac{\alpha}{y}$

Q6.03.1 MCQ 2003 2M

The value of y as $t \to \infty$ for the following differential equation for an initial value of $y(1) = 0$ is:

$$(4t^2 + 1)\frac{dy}{dx} + 8yt - t = 0$$

A. 1 **B.** 1/2 **C.** 1/4 **D.** 1/8

Q6.03.2 MCQ 2003 1M

The differential equation, $\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$ will have a solution of the form (where C_1 and C_2 are constants).

A. $(C_1 + C_2 t)\, e^{-5t}$ **B.** $C_1\, e^{-2t}$

C. $C_1\, e^{-5t} + C_2\, e^{5t}$ **D.** $C_1\, e^{5t} + C_2\, e^{2t}$

Q6.00.1 MCQ 2000 1M

The integrating factor of differential equation

$$\cos^2 x \frac{dy}{dx} + y = \tan x \text{ is}$$

A. $e^{\tan x}$ **B.** $\cos 2x$

C. $e^{-\tan x}$ **D.** $\sin 2x$

Q6.00.2 MCQ 2000 2M

The general solution of

$$\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0 \text{ is}$$

A. $(c_1x + c_2)e^x + (c_3 + c_4x)e^{-x}$

B. $c_1 \cos x + c_2 \sin x$

C. $c_1\, e^{2x} + c_2\, e^{-2x}$

D. $(c_1 + c_2x)\cos x + (c_3 + c_4x)\sin x$

Production & Industrial Engineering

Q6.26.1 NAT 2026 2M

If $\frac{dy}{dx} = 2y$, and the value of y at $t = 0$ is 2, then the value of y at $t = 1$ is _____ (*rounded off to two decimal places*).

Q6.25.1 MCQ 2025 1M

Which one of the following equations is a linear differential equation?

A. $\frac{dy}{dx} + 2x = y^2$ **B.** $x^3\frac{dy}{dx} + xy = x^2$

C. $x\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} = 0$ **D.** $\left(\frac{dy}{dx}\right)^2 + 2x = y$

Q6.25.2 MCQ 2025 2M

The solution of the linear differential equation

$$\frac{dy}{dx} + y = e^x,$$

when $y(0) = 0$, is

A. $y = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$ **B.** $y = \frac{1}{2}e^x + \frac{1}{2}e^{-x}$

C. $y = e^x - e^{-x}$ **D.** $y = e^x + e^{-x}$

Q6.24.1 NAT 2024 2M

The following differential equation governs the evolution of variable $x(t)$ with time t, $t \geq 0$.

$$\frac{d^2x}{dt^2} + 4x = e^{-t}$$

Given the initial conditions $x = 0$ and $\frac{dx}{dt} = 0$ at $t = 0$, the value of x at $t = \pi/8$ is _____. (*Rounded off to* 3 *decimal places*)

Q6.23.1 MCQ 2023 1M

The non-linear differential equation from the following options is

A. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 10y = 0$

B. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 10y = 0$

C. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 10x = 0$

D. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 10xy = 0$

Q6.22.1 NAT 2022 1M ○○○

The order of the following differential equation is ______. [*in integer*]

$$\left(\frac{dy}{dx}\right)^2 + 5\frac{dy}{dx} + 4y = 5x^3$$

Q6.22.2 MCQ 2022 2M ○○○

Consider the following ordinary differential equation:

$$4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 0.$$

Given that c_1 and c_2 are constants, the general solution of the differential equation is

A. $y = (c_1 + c_2x)e^x$ **B.** $y = c_1e^{x/2} + c_2e^x$

C. $y = c_1e^x + c_2e^{2x}$ **D.** $y = (c_1 + c_2x)e^{x/2}$

Q6.20.1 MCQ 2020 1M ○○○

An integrating factor for the differential equation $\frac{dy}{dx} + my = e^{-mx}$ is

A. e^m **B.** e^{-m} **C.** e^{-mx} **D.** e^{mx}

Q6.20.2 MCQ 2020 2M ○○○

General solution of $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$ is

A. $y = \frac{C_1}{x} + \frac{C_2}{x^3}$ **B.** $y = C_1x^2 + \frac{C_2}{x^2}$

C. $y = C_1x + \frac{C_2}{x}$ **D.** $y = C_1x + \frac{C_2}{x^3}$

Q6.19.1 MCQ 2019 1M ○○○

If roots of the auxiliary equation of $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$ are real and equal, the general solution of the differential equation is

A. $y = c_1e^{-ax/2} + c_2e^{ax/2}$

B. $y = (c_1 + c_2x)e^{-ax/2}$

C. $y = (c_1 + c_2\ln x)e^{-ax/2}$

D. $y = (c_1\cos x + c_2\sin x)e^{-ax/2}$

Q6.19.2 MCQ 2019 2M ○○○

General solution of the Cauchy-Euler equation is $x^2\frac{d^2y}{dx^2} - 7x\frac{dy}{dx} + 16y = 0$ is

A. $y = c_1x^2 + c_2x^4$ **B.** $y = c_1x^2 + c_2x^{-4}$

C. $y = (c_1 + c_2\ln x)x^4$ **D.** $y = c_1x^4 + c_2x^{-4}\ln x$

Q6.18.1 NAT 2018 2M ○○○

Consider the differential equation

$$2\frac{d^2y}{dt^2} + 8y = 0$$

with initial conditions: at $t = 0$, $y = 0$ and $\frac{dy}{dt} = 10$. The value of y (up to two decimal places) at $t = 1$ is ______.

Q6.16.1 MCQ 2016 1M ○○○

For the two functions

$$f(x, y) = x^3 - 3xy^2 \text{ and } g(x, y) = 3x^2y - y^3$$

which one of the following options is correct?

A. $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial x}$ **B.** $\frac{\partial f}{\partial x} = -\frac{\partial g}{\partial y}$

C. $\frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}$ **D.** $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$

Q6.15.1 MCQ 2015 2M ○○○

The solution of $6yy' - 25x = 0$ represents a

A. family of circles **B.** family of ellipses

C. family of parabolas **D.** family of hyperbolas

Q6.13.1 MCQ 2013 1M ○○○

The partial differential equation $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2}$ is a

A. linear equation of order 2

B. non-linear equation of order 1

C. linear equation of order 1

D. non-linear equation of order 2

Q6.11.1 MCQ 2011 1M

The solution of the differential equation $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 9x + 6$ with C_1 and C_2 as constants is

A. $y = (C_1x + C_2)e^{-3x}$

B. $y = C_1e^{3x} + C_2e^{-3x} + x$

C. $y = (C_1x + C_2)e^{-3x} + x$

D. $y = (C_1x + C_2)e^{3x} + x$

Q6.10.1 MCQ 2010 1M

Which one of the following differential equations has a solution given by the function $y = 5\sin\left(3x + \frac{\pi}{3}\right)$?

A. $\frac{dy}{dx} - \frac{5}{3}\cos(3x) = 0$ **B.** $\frac{dy}{dx} + \frac{5}{3}\cos(3x) = 0$

C. $\frac{d^2y}{dx^2} + 9y = 0$ **D.** $\frac{d^2y}{dx^2} - 9y = 0$

Q6.10.2 MCQ 2010 2M

The solution of the differential equation

$$\frac{dy}{dx} - y^2 = 1$$

satisfying the condition $y(0) = 1$ is

A. $y = e^{x^2}$ **B.** $y = \sqrt{x}$

C. $y = \cot\left(x + \frac{\pi}{4}\right)$ **D.** $y = \tan\left(x + \frac{\pi}{4}\right)$

Q6.09.1 MCQ 2009 1M

The homogeneous part of the differential equation

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = r$$

(p, q and r are constants) has real and distinct roots if

A. $p^2 - 4q > 0$ **B.** $p^2 - 4q < 0$

C. $p^2 - 4q = 0$ **D.** $p^2 - 4q = r$

Q6.09.2 MCQ 2009 2M

The solution of the differential equation $\frac{d^2y}{dx^2} = 0$ with boundary conditions:

(i) $\frac{dy}{dx} = 1$ at $x = 0$ and

(ii) $\frac{dy}{dx} = 1$ at $x = 1$, is

A. $y = 1$

B. $y = x$

C. $y = x + C$, where C is an arbitrary constant.

D. $y = C_1x + C_2$, where C_1 and C_2 are arbitrary constants.

Q6.08.1 MCQ 2008 2M

For the partial differential equation $\frac{\partial^2 u}{\partial x^2} = \pi^2\frac{\partial u}{\partial t}$ in the domain $0 \le x \le 1$ with boundary conditions $u(0,t) = 0$ and $u(1,t) = 0$ and initial condition $u(x,0) = \sin(\pi x)$, the solution of the differential equation is

A. $e^{-t}\sin(\pi x)$ **B.** $e^{t}\sin(\pi x)$

C. $e^{-\pi t}\sin(\pi x)$ **D.** $e^{\pi t}\sin(\pi x)$

Q6.08.2 MCQ 2008 2M

The solution of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$ are

A. $e^{-(1+i)x},\ e^{-(1-i)x}$ **B.** $e^{(1+i)x},\ e^{(1-i)x}$

C. $e^{-(1+i)x},\ e^{(1+i)x}$ **D.** $e^{(1+i)x},\ e^{-(1+i)x}$

Q6.05.1 MCQ 2005 1M

The differential equation

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = C^2\left[\frac{d^2y}{dx^2}\right]^2$$

is of

A. 2nd order and 3rd degree

B. 3rd order and 2nd degree

C. 2nd order and 2nd degree

D. 3rd order and 3rd degree

Q6.94.1 MCQ 1994 1M

Solve for y if $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ with $y(0) = 1$ and $y'(0) = -2$

A. $(1 - t)e^{-t}$ **B.** $(1 + t)e^{t}$

C. $(1 + t)e^{-t}$ **D.** $(1 - t)e^{t}$

Answer Key

ELECTRONICS & COMMUNICATION ENGINEERING					
Q6.26.1 A	Q6.25.1 0.25	Q6.24.1 A	Q6.22.1 A,B	Q6.21.1 B	Q6.20.1 B
Q6.20.2 C	Q6.20.3 A	Q6.19.1 C	Q6.19.2 5.24 to 5.26	Q6.18.1 A	Q6.18.2 -0.23 to -0.19
Q6.17.1 D	Q6.17.2 A	Q6.16.1 A	Q6.15.1 B	Q6.15.2 C	Q6.15.3 C
Q6.15.4 0.83 to 0.88	Q6.14.1 A	Q6.14.2 A	Q6.14.3 C	Q6.14.4 B	Q6.14.5 0.53 to 0.55
Q6.13.1 D	Q6.12.1 D	Q6.12.2 D	Q6.11.1 C	Q6.10.1 D	Q6.09.1 B
Q6.09.2 A	Q6.08.1 B	Q6.07.1 D	Q6.06.1 A	Q6.06.2 A	Q6.05.1 B
Q6.05.2 B	Q6.94.1 A				
ELECTRICAL ENGINEERING					
Q6.26.1 A	Q6.25.1 2	Q6.24.1 B,D	Q6.20.1 0.8774 to 0.8952	Q6.19.1 B	Q6.17.1 A
Q6.16.1 B	Q6.16.2 A	Q6.16.3 7.389	Q6.15.1 -3	Q6.15.2 1.652	Q6.14.1 C
Q6.14.2 C	Q6.11.1 A	Q6.10.1 B	Q6.05.1 A	Q6.05.2 B	
INSTRUMENTATION ENGINEERING					
Q6.26.1 C	Q6.26.2 B	Q6.25.1 A	Q6.24.1 33	Q6.23.1 C	Q6.22.1 C
Q6.20.1 A	Q6.18.1 D	Q6.14.1 D	Q6.13.1 A	Q6.13.2 D	Q6.11.1 C
Q6.10.1 C	Q6.08.1 A	Q6.07.1 C	Q6.06.1 A	Q6.05.1 C	
MECHANICAL ENGINEERING					
Q6.26.1 A	Q6.26.2 A	Q6.25.1 B	Q6.25.2 0.45 to 0.55	Q6.24.1 -1.40 to -1.38	Q6.23.1 8.999 to 9.001
Q6.22.1 B	Q6.22.2 C	Q6.21.1 C	Q6.21.2 C	Q6.19.1 C	Q6.19.2 B
Q6.19.3 A	Q6.18.1 C	Q6.18.2 1.47	Q6.17.1 B	Q6.17.2 A	Q6.17.3 93 to 95
Q6.16.1 -1.05 to -0.95	Q6.15.1 C	Q6.15.2 C	Q6.14.1 A	Q6.14.2 34 to 36	Q6.14.3 D

Q6.14.4 A	Q6.14.5 B	Q6.13.1 D	Q6.13.2 B	Q6.12.1 A	Q6.11.1 D
Q6.10.1 B	Q6.09.1 A	Q6.08.1 B	Q6.08.2 C	Q6.08.3 A	Q6.07.1 A
Q6.07.2 C	Q6.06.1 B	Q6.06.2 B	Q6.05.1 D	Q6.05.2 C	Q6.05.3 C
Q6.03.1 A	Q6.01.1 C	Q6.00.1 D	Q6.99.1 C	Q6.98.1 D	Q6.98.2 0.625
Q6.96.1 A	Q6.95.1 A	Q6.95.2 C	Q6.94.1 B	Q6.94.2 A	Q6.94.3 D
Q6.93.1 B	Q6.93.2 B				
CIVIL ENGINEERING					
Q6.26.1 A	Q6.26.2 A,C	Q6.26.3 0.37 to 0.42	Q6.25.1 B	Q6.25.2 5.7 to 5.9	Q6.25.3 3
Q6.25.4 A	Q6.24.1 B	Q6.24.2 A,D	Q6.24.3 B	Q6.23.1 0.446	Q6.23.2 A
Q6.22.1 A	Q6.22.2 A	Q6.21.1 A	Q6.21.2 A	Q6.20.1 B	Q6.20.2 A
Q6.20.3 D	Q6.20.4 A	Q6.20.5 A	Q6.19.1 C	Q6.19.2 5.9 to 6.1	Q6.19.3 D
Q6.18.1 D	Q6.18.2 0.36 to 0.37	Q6.18.3 C	Q6.17.1 35.9 to 36.1	Q6.17.2 D	Q6.17.3 A
Q6.16.1 C	Q6.16.2 B	Q6.16.3 A	Q6.15.1 C	Q6.15.2 18	Q6.14.1 D
Q6.14.2 3.0	Q6.13.1 3.8	Q6.12.1 D	Q6.11.1 D	Q6.10.1 A	Q6.10.2 C
Q6.10.3 C	Q6.09.1 A	Q6.08.1 A	Q6.08.2 D	Q6.08.3 D	Q6.07.1 B
Q6.07.2 D	Q6.06.1 A	Q6.05.1 A	Q6.05.2 A	Q6.04.1 B	Q6.01.1 C
Q6.01.2 C	Q6.00.1 A	Q6.99.1 C	Q6.98.1 A	Q6.97.1 B	Q6.97.2 D
Q6.95.1 C	Q6.95.2 C	Q6.94.1 A	Q6.94.2 C		
CHEMICAL ENGINEERING					
Q6.26.1 2.24 to 2.26	Q6.25.1 B,D	Q6.24.1 4.00 to 4.08	Q6.23.1 -1	Q6.22.1 A	Q6.21.1 0.161 to 0.169
Q6.20.1 162 to 170	Q6.19.1 D	Q6.18.1 C	Q6.17.1 0.49 to 0.51	Q6.16.1 D	Q6.15.1 D
Q6.14.1 A	Q6.14.2 B	Q6.13.1 B	Q6.13.2 C	Q6.12.1 D	Q6.11.1 D

Q6.10.1 D	Q6.09.1 B	Q6.08.1 D	Q6.08.2 D	Q6.08.3 B	Q6.07.1 D
Q6.07.2 D	Q6.06.1 C	Q6.05.1 A	Q6.05.2 C	Q6.05.3 A	Q6.04.1 C
Q6.04.2 B	Q6.04.3 D	Q6.03.1 D	Q6.03.2 A	Q6.00.1 A	Q6.00.2 D
PRODUCTION & INDUSTRIAL ENGINEERING					
Q6.26.1 14.5 to 14.9	Q6.25.1 B	Q6.25.2 A	Q6.24.1 0.064 to 0.065	Q6.23.1 B	Q6.22.1 1
Q6.22.2 D	Q6.20.1 D	Q6.20.2 C	Q6.19.1 B	Q6.19.2 C	Q6.18.1 4.52 to 4.56
Q6.16.1 C	Q6.15.1 D	Q6.13.1 D	Q6.11.1 C	Q6.10.1 C	Q6.10.2 D
Q6.09.1 A	Q6.09.2 C	Q6.08.1 A	Q6.08.2 A	Q6.05.1 C	Q6.94.1 A

UNIT

VII

Transform Theory

Topics

1. Laplace Transform
2. Inverse Laplace Transform
3. Fourier Series
4. Fourier Transform
5. Z-Transform
6. Applications

Note to Reader

Transform theory are present directly or indirectly in branches with paper codes EC, EE, IN in subject Signal and System; MA, PH, ME, CE, CH, AE, BT, AG, MT, MN, PI, TF, BM, PE, CY in form of Laplace Transform according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Tranform theory for EC, EE, IN are solved extensively in Signal and System

Branches: ME,CE,CH,PI

Branch Snapshots*

ME — Mechanical Engineering

Questions: 16 **MCQ:** 16 **MSQ:** 0 **NAT:** 0
1M: 14 **2M:** 2 **Desc:** 0 **Avg Weightage** ~2M **Range:** 0M(2014)–2M(2026)

CE — Civil Engineering

Questions: 18 **MCQ:** 16 **MSQ:** 1 **NAT:** 1
1M: 14 **2M:** 4 **Desc:** 0 **Avg Weightage** ~1M **Range:** 1M(2025)–3M(2011)

CH — Chemical Engineering

Questions: 8 **MCQ:** 8 **MSQ:** 0 **NAT:** 0
1M: 5 **2M:** 3 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–2M(2017)

PI — Production & Industrial Engineering

Questions: 3 **MCQ:** 3 **MSQ:** 0 **NAT:** 0
1M: 3 **2M:** 0 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2012)–1M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Mechanical Engineering

Q8.26.1 MCQ 2026 2M

Let $f(t)$ be a function of t defined for all positive values of t. The Laplace transform of $f(t)$ denoted by $L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$, provided that the integral exists where s is a parameter which may be a real or complex number. The Laplace transform of $f(t) = \sin 2t \sin 4t$ is

A. $\dfrac{16s}{(s^2+4)(s^2+36)}$ **B.** $\dfrac{32s}{(s^2+4)(s^2+36)}$

C. $\dfrac{16}{(s^2+4)(s^2+36)}$ **D.** $\dfrac{32}{(s^2+4)(s^2+36)}$

Q8.23.1 MCQ 2023 2M

Which one of the options given is the inverse Laplace transform of $\dfrac{1}{s^3 - s}$?

$u(t)$ denotes the unit-step function.

A. $(-1 + \frac{1}{2}e^{-t} + \frac{1}{2}e^{t})u(t)$

B. $(\frac{1}{2}e^{-t} - e^{t})u(t)$

C. $(-1 + \frac{1}{2}e^{-(t-1)} + \frac{1}{2}e^{(t-1)})u(t-1)$

D. $(-1 - \frac{1}{2}e^{-(t-1)} - \frac{1}{2}e^{(t-1)})u(t-1)$

Q8.22.1 MCQ 2022 1M

$F(t)$ is a periodic square wave function as shown. It takes only two values, 4 and 0, and stays at each of these values for 1 second before changing. What is the constant term in the Fourier series expansion of $F(t)$?

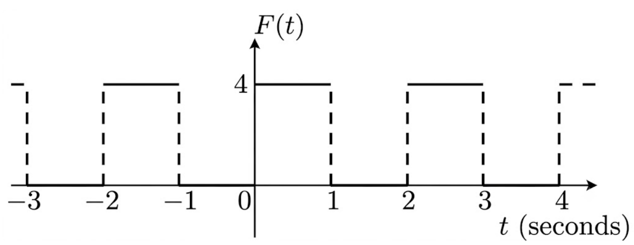


Fig. Q8.22.1

A. 1 **B.** 2 **C.** 3 **D.** 4

Q8.22.2 MCQ 2022 1M

The Fourier series expansion of x^3 in the interval $-1 \le x < 1$ with periodic continuation has

A. only sine terms

B. only cosine terms

C. both sine and cosine terms

D. only sine terms and a non-zero constant

Q8.15.1 MCQ 2015 1M

Laplace transform of $\cos(\omega t)$ is

A. $\dfrac{s}{s^2+\omega^2}$ **B.** $\dfrac{\omega}{s^2+\omega^2}$ **C.** $\dfrac{s}{s^2-\omega^2}$ **D.** $\dfrac{\omega}{s^2-\omega^2}$

Q8.15.2 MCQ 2015 1M

The Laplace transform of e^{i5t} where $i = \sqrt{-1}$, is

A. $\dfrac{s-5i}{s^2-25}$ **B.** $\dfrac{s+5i}{s^2+25}$ **C.** $\dfrac{s+5i}{s^2-25}$ **D.** $\dfrac{s-5i}{s^2+25}$

Q8.15.3 MCQ 2015 1M

Laplace transform of the function $f(t)$ is given by $f(s) = L\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt$. Laplace transform of the function shown below is given by

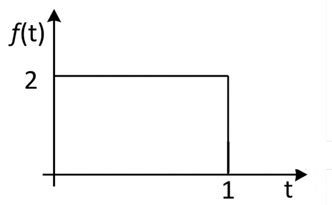


Fig. Q8.15.3

A. $\dfrac{1-e^{-2s}}{s}$ **B.** $\dfrac{1-e^{-s}}{s}$ **C.** $\dfrac{2-2e^{-s}}{s}$ **D.** $\dfrac{1-2e^{-s}}{s}$

Q8.15.4 MCQ 2015 1M

The Laplace Transform of $f(t) = e^{2t}\sin(5t)\,u(t)$ is

A. $\dfrac{5}{s^2-4s+29}$ **B.** $\dfrac{5}{s^2+5}$ **C.** $\dfrac{s-2}{s^2-4s+29}$ **D.** $\dfrac{5}{s+5}$

Q8.14.1 MCQ 2014 1M

Laplace transform of $\cos(\omega t)$ is $\frac{s}{s^2+\omega^2}$. The Laplace transform of $e^{-2t}\cos(4t)$ is

A. $\frac{s-2}{(s-2)^2+16}$ **B.** $\frac{s+2}{(s-2)^2+16}$

C. $\frac{s-2}{(s+2)^2+16}$ **D.** $\frac{s+2}{(s+2)^2+16}$

Q8.13.1 MCQ 2013 1M

The function $f(t)$ satisfies the differential equation $\frac{d^2f}{dt^2}+f=0$ and the auxiliary conditions, $f(0)=0$, $\frac{df}{dt}(0)=4$. The Laplace transform of $f(t)$ is given by

A. $\frac{2}{s+1}$ **B.** $\frac{4}{s+1}$ **C.** $\frac{4}{s^2+1}$ **D.** $\frac{2}{s^4+1}$

Q8.12.1 MCQ 2012 1M

The inverse Laplace transform of the function $F(s)=\frac{1}{s(s+1)}$ is given by

A. $f(t)=\sin t$ **B.** $f(t)=e^{-t}\sin t$

C. $f(t)=e^{-t}$ **D.** $f(t)=1-e^{-t}$

Q8.10.1 MCQ 2010 1M

The Laplace transform of $f(t)$ is $\frac{1}{s^2(s+1)}$. The function is

A. $t-1+e^{-t}$ **B.** $t+1+e^{-t}$

C. $-1+e^{-t}$ **D.** $2t+e^{t}$

Q8.07.1 MCQ 2007 1M

If $F(s)$ is the Laplace transform of the function $f(t)$ then Laplace transform of $\int_0^t f(x)\,dx$ is

A. $\frac{1}{s}F(s)$ **B.** $\frac{1}{s}F(s)-f(0)$

C. $sF(s)-f(0)$ **D.** $\int F(s)\,ds$

Q8.99.1 MCQ 1999 1M

Laplace transform of $(a+bt)^2$ where 'a' and 'b' are constants is given by:

A. $(a+bs)^2$

B. $1/(a+bs)^2$

C. $(a^2/s)+(2ab/s^2)+(2b^2/s^3)$

D. $(a^2/s)+(2ab/s^2)+(b^2/s^3)$

Q8.94.1 MCQ 1994 1M

If $f(t)$ is a finite and continuous Function for $t \geq 0$, the laplace transformation is given by $F=\int_0^\infty e^{-st}f(t)\,dt$. Then, for $f(t)=\cosh mt$, the laplace transformation is

A. $\frac{s}{s^2-m^2}$ **B.** $\frac{m}{s^2-m^2}$ **C.** $\frac{s}{s^2+m^2}$ **D.** $\frac{m}{s^2+m^2}$

Q8.93.1 MCQ 1993 1M

The laplace transform of the periodic function $f(t)$ described by the curve below i.e.

$$f(t)=\begin{cases}\sin t, & \text{if } (2n-1)\pi < t < 2n\pi \ (n=1,2,3,\ldots)\\ 0 & \text{otherwise}\end{cases}$$

is

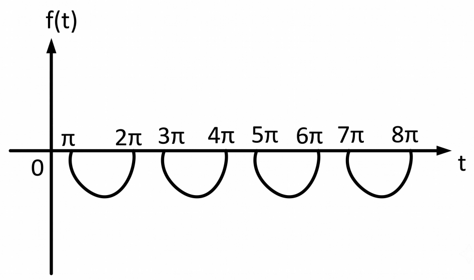


Fig. Q8.93.1

A. $\frac{e^{-\pi s}}{(1-s^2)(e^{-\pi s}-1)}$ **B.** $\frac{e^{-\pi s}}{(1+s^2)(e^{-\pi s}-1)}$

C. $\frac{e^{-\pi s}}{(1+s^2)(e^{-\pi s}+1)}$ **D.** $\frac{e^{\pi s}}{(1+s^2)(e^{-\pi s}-1)}$

Civil Engineering

Q8.25.1 MCQ 2025 1M

Which one of the following options is the correct Fourier series of the periodic function $f(x)$ described below:

$$f(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ 2k & \text{if } -1 < x < 1; \text{ period} = 4 \\ 0 & \text{if } -1 < x < 2 \end{cases}$$

A. $f(x) = \frac{k}{2} + \frac{2k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - + \ldots\right)$

B. $f(x) = \frac{k}{2} + \frac{2k}{\pi}\left(\sin\frac{\pi}{2}x - \frac{1}{3}\sin\frac{3\pi}{2}x + \frac{1}{5}\sin\frac{5\pi}{2}x - + \ldots\right)$

C. $f(x) = k + \frac{4k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - + \ldots\right)$

D. $f(x) = k + \frac{4k}{\pi}\left(\sin\frac{\pi}{2}x - \frac{1}{3}\sin\frac{3\pi}{2}x + \frac{1}{5}\sin\frac{5\pi}{2}x - + \ldots\right)$

Q8.23.1 MSQ 2023 1M

The following function is defined over the interval $[-L, L]$:

$$f(x) = px^4 + qx^5$$

If it is expressed as a Fourier series,

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left\{a_n \sin\left(\frac{n\pi x}{L}\right) + b_n \cos\left(\frac{n\pi x}{L}\right)\right\},$$

which options amongst the following are true?

A. $a_n,\ n = 1, 2, \ldots, \infty$ depend on p

B. $a_n,\ n = 1, 2, \ldots, \infty$ depend on q

C. $b_n,\ n = 1, 2, \ldots, \infty$ depend on p

D. $b_n,\ n = 1, 2, \ldots, \infty$ depend on q

Q8.22.1 NAT 2022 1M

The Fourier cosine series of a function is given by:

$$f(x) = \sum_{n=0}^{\infty} f_n \cos nx$$

For $f(x) = \cos^4 x$, the numerical value of $(f_4 + f_5)$ is ________ (round off to three decimal places).

Q8.11.1 MCQ 2011 1M

Given two continuous time signals $x(t) = e^{-t}$ and $y(t) = e^{-2t}$ which exists for $t > 0$ then the convolution $z(t) = x(t) * y(t)$ is

A. $e^{-t} - e^{-2t}$ **B.** e^{-2t} **C.** e^{-t} **D.** $e^{-t} + e^{-3t}$

Q8.11.2 MCQ 2011 2M

If $F(s) = L\{f(t)\} = \dfrac{2(s+1)}{s^2 + 4s + 7}$ then the initial and final values of $f(t)$ are respectively

A. $0,\ 2$ **B.** $2,\ 0$

C. $0,\ \frac{2}{7}$ **D.** $\frac{2}{7},\ 0$

Q8.09.1 MCQ 2009 1M

Laplace transform of $f(x) = \cosh(ax)$ is

A. $\dfrac{a}{s^2 - a^2}$ **B.** $\dfrac{s}{s^2 - a^2}$ **C.** $\dfrac{a}{s^2 + a^2}$ **D.** $\dfrac{s}{s^2 + a^2}$

Q8.05.1 MCQ 2005 2M

Laplace transform of $f(t) = \cos(pt + q)$ is

A. $\dfrac{s\cos q - p\sin q}{s^2 + p^2}$ **B.** $\dfrac{s\cos q + p\sin q}{s^2 + p^2}$

C. $\dfrac{s\sin q - p\cos q}{s^2 + p^2}$ **D.** $\dfrac{s\sin q + p\cos q}{s^2 + p^2}$

Q8.05.2 MCQ 2005 1M

The Laplace transform of a function $f(t)$ is

$$F(s) = \frac{5s^2 + 23s + 6}{s(s^2 + 2s + 2)}$$

As $t \to \infty$, $f(t)$ approaches

A. 3 **B.** 5 **C.** $17/2$ **D.** ∞

Q8.04.1 MCQ 2004 1M

A delayed unit step function is defined as

$$u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

Its Laplace transform is

A. $a\,e^{-as}$ **B.** e^{-as}/s **C.** e^{as}/s **D.** e^{-as}/a

Q8.03.1 MCQ 2003 1M

If L denotes the laplace transform of a function, $L\{\sin at\}$ will be equal to

A. $\frac{a}{s^2 - a^2}$ **B.** $\frac{a}{s^2 + a^2}$ **C.** $\frac{s}{s^2 + a^2}$ **D.** $\frac{s}{s^2 - a^2}$

Q8.02.1 MCQ 2002 2M

The Laplace transform of the following function is

$$f(t) = \begin{cases} \sin t & \text{for } 0 \le t \le \pi \\ 0 & \text{for } t > \pi \end{cases}$$

A. $1/(1+s^2)$ for all $s > 0$

B. $1/(1+s^2)$ for all $s < \pi$

C. $(1+e^{-\pi s})/(1+s^2)$ for all $s > 0$

D. $e^{-\pi s}/(1+s^2)$ for all $s > 0$

Q8.01.1 MCQ 2001 1M

The inverse Laplace transform of $1/(s^2 + 2s)$ is

A. $(1 - e^{-2t})$ **B.** $(1 + e^{2t})/2$

C. $(1 - e^{2t})/2$ **D.** $(1 - e^{-2t})/2$

Q8.00.1 MCQ 2000 2M

Let $F(s) = L[f(t)]$ denote the Laplace transform of the function $f(t)$. Which of the following statements is correct?

A. $L[df/dt] = 1/s\, F(s)$; $L\left\{\int_0^t f(\tau)\, d\tau\right\} = sF(s) - f(0)$

B. $L[df/dt] = sF(s) - F(0)$; $L\left\{\int_0^t f(\tau)\, d\tau\right\} = -dF/ds$

C. $L[df/dt] = sF(s) - F(0)$; $L\left\{\int_0^t f(\tau)\, d\tau\right\} = F(s-a)$

D. $L[df/dt] = sF(s) - f(0)$; $L\left\{\int_0^t f(\tau)\, d\tau\right\} = \frac{1}{s}F(s)$

Q8.99.1 MCQ 1999 1M

The Laplace transform of the function

$$\begin{aligned} f(t) &= k, \quad 0 < t < c \\ &= 0, \quad c < t < \infty, \end{aligned}$$

is

A. $(k/s)e^{-sc}$ **B.** $(k/s)e^{sc}$

C. $k\, e^{-sc}$ **D.** $(k/s)(1 - e^{-sc})$

Q8.98.1 MCQ 1998 1M

The Laplace Transform of a unit step function $u_a(t)$, defined as

$$\begin{aligned} u_a(t) &= 0 \quad \text{for } t < a \\ &= 1 \quad \text{for } t > a, \end{aligned}$$

is

A. e^{-as}/s **B.** se^{-as} **C.** $s - u(0)$ **D.** $se^{-as} - 1$

Q8.98.2 MCQ 1998 1M

A discontinuous real function can be expressed as

A. Taylor's series and Fourier's series

B. Taylor's series and not by Fourier's series

C. Neither Taylor's series nor Fourier's series

D. not by Taylor's series, but by Fourier's series

Q8.96.1 MCQ 1996 1M

$(s+1)^{-2}$ is laplace transform of

A. t^2 **B.** t^3 **C.** e^{-2t} **D.** te^{-t}

Q8.95.1 MCQ 1995 1M

The inverse Laplace transform of $\frac{(s+9)}{(s^2+6s+13)}$ is

A. $\cos 2t + 9 \sin 2t$

B. $e^{-3t}\cos 2t - 3e^{-3t}\sin 2t$

C. $e^{-3t}\sin 2t + 3e^{-3t}\cos 2t$

D. $e^{-3t}\cos 2t + 3e^{-3t}\sin 2t$

Chemical Engineering

Q8.20.1 MCQ 2020 1M

Consider the following unit step function.

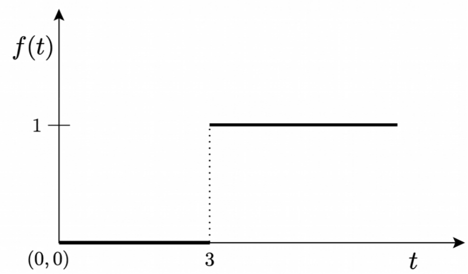


Fig. Q8.20.1

The Laplace transform of this function is

A. $\frac{e^{-3s}}{s}$ **B.** $\frac{e^{-3s}}{s^2}$ **C.** $\frac{e^{-3s}}{3s}$ **D.** $\frac{e^{-6s}}{s}$

Q8.17.1 MCQ 2017 2M

The Laplace transform of function is $\frac{s+1}{s(s+2)}$. The initial and final values, respectively, of the function are

A. 0 and 1 **B.** 1 and $\frac{1}{2}$ **C.** $\frac{1}{2}$ and 1 **D.** $\frac{1}{2}$ and 0

Q8.16.1 MCQ 2016 1M

The Laplace transform of $e^{at}\sin(bt)$ is

A. $\frac{b}{(s-a)^2+b^2}$ **B.** $\frac{(s-a)}{(s-a)^2+b^2}$

C. $\frac{(s-a)}{(s-a)^2-b^2}$ **D.** $\frac{b}{(s-a)^2-b^2}$

Q8.14.1 MCQ 2014 1M

For the time domain function $f(t)=t^2$, which ONE of the following is the Laplace transform of $\int_0^t f(t)\,dt$?

A. $\frac{3}{s^4}$ **B.** $\frac{1}{4s^2}$ **C.** $\frac{2}{s^3}$ **D.** $\frac{2}{s^4}$

Q8.12.1 MCQ 2012 1M

The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2+s+1}$. The unilateral Laplace transform of $t\,f(t)$ is

A. $\frac{-s}{(s^2+s+1)^2}$ **B.** $\frac{-(2s+1)}{(s^2+s+1)^2}$

C. $\frac{s}{(s^2+s+1)^2}$ **D.** $\frac{2s+1}{(s^2+s+1)^2}$

Q8.08.1 MCQ 2008 2M

The Laplace transform of the function $f(t)=t\sin t$ is

A. $\frac{2s}{(s^2+1)^2}$ **B.** $\frac{1}{s^2(s^2+1)}$

C. $\frac{1}{s^2}+\frac{1}{(s^2+1)}$ **D.** $\frac{1}{(s-1)^2+1}$

Q8.07.1 MCQ 2007 2M

The Laplace transform of $f(t)=\frac{1}{\sqrt{t}}$ is

A. $\sqrt{\frac{\pi}{s}}$ **B.** $\frac{1}{\sqrt{s}}$ **C.** $\frac{1}{s^{3/2}}$ **D.** does not exist

Q8.98.1 MCQ 1998 1M

The Laplace transform of the function e^{-at} has the form:

A. $\frac{1}{s+1}$ **B.** $\frac{1}{s(s+a)}$ **C.** $\frac{a}{s}$ **D.** $\frac{1}{s+a}$

Production & Industrial Engineering

Q8.12.1 MCQ 2012 1M

The inverse Laplace transform of the function $F(s)=\frac{1}{s(s+1)}$ is given by

A. $f(t)=\sin t$ **B.** $f(t)=e^{-t}\sin t$

C. $f(t)=e^{-t}$ **D.** $f(t)=1-e^{-t}$

Q8.08.1 MCQ 2008 1M

Laplace transform of $\sin ht$ is

A. $\frac{1}{s^2-1}$ **B.** $\frac{1}{1-s^2}$ **C.** $\frac{s}{s^2-1}$ **D.** $\frac{s}{1-s^2}$

Q8.08.2 MCQ 2008 1M

Laplace transform of $8\,t^3$ is

A. $\frac{8}{s^4}$ **B.** $\frac{16}{s^4}$ **C.** $\frac{24}{s^4}$ **D.** $\frac{48}{s^4}$

Answer Key

MECHANICAL ENGINEERING					
Q8.26.1 A	Q8.23.1 A	Q8.22.1 B	Q8.22.2 A	Q8.15.1 A	Q8.15.2 B
Q8.15.3 C	Q8.15.4 A	Q8.14.1 D	Q8.13.1 C	Q8.12.1 D	Q8.10.1 A
Q8.07.1 A	Q8.99.1 C	Q8.94.1 A	Q8.93.1 B		
CIVIL ENGINEERING					
Q8.25.1 C	Q8.23.1 B, C	Q8.22.1 0.120 to 0.130	Q8.11.1 A	Q8.11.2 B	Q8.09.1 B
Q8.05.1 A	Q8.05.2 A	Q8.04.1 B	Q8.03.1 B	Q8.02.1 C	Q8.01.1 D
Q8.00.1 D	Q8.99.1 D	Q8.98.1 A	Q8.98.2 D	Q8.96.1 D	Q8.95.1 D
CHEMICAL ENGINEERING					
Q8.20.1 A	Q8.17.1 B	Q8.16.1 A	Q8.14.1 D	Q8.12.1 D	Q8.08.1 A
Q8.07.1 A	Q8.98.1 D				
PRODUCTION & INDUSTRIAL ENGINEERING					
Q8.12.1 D	Q8.08.1 A	Q8.08.2 D			

UNIT

VIII

Numerical Methods

Topics

1. Numerical Solutions of Linear Algebraic Equations
2. Numerical Solutions of Nonlinear Algebraic Equations
3. Numerical Integration using Trapezoidal Rule and Simpson's Rule
4. Single-Step and Multi-Step Methods for Ordinary Differential Equations

Note to Reader

Numerical methods are present in branches with paper codes AG, BM, BT, CE, CH, IN, TF, PI, PE, MT, MN, ME according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 8 **MCQ:** 4 **MSQ:** 0 **NAT:** 4
1M: 4 **2M:** 4 **Desc:** 0 **Avg Weightage** ~1M **Range:** 0M(2015)–2M(2010)

EE — Electrical Engineering

Questions: 9 **MCQ:** 7 **MSQ:** 0 **NAT:** 2
1M: 6 **2M:** 3 **Desc:** 0 **Avg Weightage** ~1M **Range:** 0M(2014)–2M(2023)

IN — Instrumentation Engineering

Questions: 4 **MCQ:** 4 **MSQ:** 0 **NAT:** 0
1M: 4 **2M:** 0 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2014)–1M(2014)

ME — Mechanical Engineering

Questions: 21 **MCQ:** 9 **MSQ:** 0 **NAT:** 12
1M: 10 **2M:** 11 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2023)

CE — Civil Engineering

Questions: 26 **MCQ:** 15 **MSQ:** 0 **NAT:** 11
1M: 10 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2015)

CH — Chemical Engineering

Questions: 31 **MCQ:** 20 **MSQ:** 0 **NAT:** 11
1M: 12 **2M:** 19 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–5M(2016)

PI — Production & Industrial Engineering

Questions: 10 **MCQ:** 8 **MSQ:** 0 **NAT:** 2
1M: 6 **2M:** 4 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2016)–2M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q7.17.1 NAT 2017 2M

Starting with $x = 1$, the solution of the equation $x^3 + x = 1$, after two iterations of Newton-Raphson's method (up to two decimal places) is ________

Q7.16.1 NAT 2016 2M

Consider the first order initial value problem $y' = y + 2x - x^2$, $y(0) = 1$, $(0 \le x < \infty)$ with exact solution $y(x) = x^2 + e^x$. For $x = 0.1$, the percentage difference between the exact solution and the solution obtained using a single iteration of the second-order Runge-Kutta method with step-size $h = 0.1$ is ________

Q7.16.2 NAT 2016 2M

The ordinary differential equation $\frac{dx}{dt} = -3x + 2$, with $x(0) = 1$ is to be solved using the forward Euler method. The largest time step that can be used to solve the equation without making the numerical solution unstable is ________

Q7.15.1 NAT 2015 1M

The Newton-Raphson method is used to solve the equation $f(x) = x^3 - 5x^2 + 6x - 8 = 0$. Taking the initial guess as $x = 5$, the solution obtained at the end of the first iteration is ________

Q7.14.1 MCQ 2014 1M

Match the application to appropriate numerical method:

Applications

$P1$ Numerical integration

$P2$ Solution to a transcendental equation

$P3$ Solution to a system of linear equations

$P4$ Solution to a differential equation

Numerical Method

$M1$ Newton-Raphson Method

$M2$ Runge-Kutta Method

$M3$ Simpson's 1/3-rule

$M4$ Gauss Elimination Method

A. $P1 - M3,\ P2 - M2,\ P3 - M4,\ P4 - M1$

B. $P1 - M3,\ P2 - M1,\ P3 - M4,\ P4 - M2$

C. $P1 - M4,\ P2 - M1,\ P3 - M3,\ P4 - M2$

D. $P1 - M2,\ P2 - M1,\ P3 - M3,\ P4 - M4$

Q7.11.1 MCQ 2011 1M

A numerical solution of the equation $f(x) = x + \sqrt{x} - 3 = 0$ can be obtained using Newton-Raphson method. If the starting value is $x = 2$ for the iteration, then the value of x that is to be used in the next step is

A. 0.306 **B.** 0.739 **C.** 1.694 **D.** 2.306

Q7.10.1 MCQ 2010 2M

Consider a differential equation $\frac{dy(x)}{dx} - y(x) = x$ with initial condition $y(0) = 0$. Using Euler's first order method with a step size of 0.1, then the value of $y(0.3)$ is

A. 0.01 **B.** 0.031 **C.** 0.0631 **D.** 0.1

Q7.08.1 MCQ 2008 1M

The recursion relation to solve $x = e^{-x}$ using Newton-Raphson method is

A. $x_{n+1} = e^{-x_n}$

B. $x_{n+1} = x_n - e^{-x_n}$

C. $x_{n+1} = \frac{(1 + x_n)e^{-x_n}}{(1 + e^{-x_n})}$

D. $x_{n+1} = \frac{x_n^2 - e^{-x_n}(1 + x_n) - 1}{x_n - e^{-x_n}}$

Electrical Engineering

Q7.23.1 MCQ 2023 1M

One million random numbers are generated from a statistically stationary process with a Gaussian distribution with mean zero and standard deviation σ_0. The σ_0 is estimated by randomly drawing out 10,000 numbers of samples (x_n). The estimates $\widehat{\sigma}_1, \widehat{\sigma}_2$ are computed in the following two ways:

$$\widehat{\sigma}_1^2 = \frac{1}{100000} \sum_{n=1}^{10000} x_n^2 \qquad \widehat{\sigma}_2^2 = \frac{1}{9999} \sum_{n=1}^{10000} x_n^2$$

Which of the following statements is true?

A. $E(\widehat{\sigma}_2^2) = \sigma_0^2$

B. $E(\widehat{\sigma}_2) = \sigma_0$

C. $E(\widehat{\sigma}_1^2) = \sigma_0^2$

D. $E(\widehat{\sigma}_1) = E(\widehat{\sigma}_2)$

Q7.23.2 NAT 2023 1M

Three points in the x-y plane are $(-1, 0.8)$, $(0, 2.2)$ and $(1, 2.8)$. The value of the slope of the best fit straight line in the least square sense is ________ (Round off to 2 decimal places).

Q7.15.1 MCQ 2015 1M

If a continuous function $f(x)$ does not have a root in the interval $[a, b]$, then which one of the following statements is TRUE?

A. $f(a) \cdot f(b) = 0$

B. $f(a) \cdot f(b) < 0$

C. $f(a) \cdot f(b) > 0$

D. $f(a)/f(b) \le 0$

Q7.11.1 MCQ 2011 2M

Solution, the variable x_1 and x_2 for the following equations is to be obtained by employing the Newton-Raphson iteration method:

$$10x_2 \sin x_1 - 0.8 = 0$$
$$10x_2^2 - 10x_2 \cos x_1 - 0.6 = 0$$

Assuming the initial values $x_1 = 0.0$ and $x_2 = 1.0$, the Jacobian matrix is

A. $\begin{bmatrix} 10 & -0.8 \\ 0 & -0.6 \end{bmatrix}$

B. $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$

C. $\begin{bmatrix} 0 & -0.8 \\ 10 & -0.6 \end{bmatrix}$

D. $\begin{bmatrix} 10 & 0 \\ 10 & -10 \end{bmatrix}$

Q7.14.1 NAT 2014 1M

The function $f(x) = e^x - 1$ is to be solved using Newton-Raphson method. If the initial value of x_0 is taken 1.0, then the absolute error observed at 2nd iteration is ________

Q7.09.1 MCQ 2009 1M

Let $x^2 - 117 = 0$. The iterative steps for the solution using Newton-Raphson's method is given by

A. $x_{k+1} = \frac{1}{2}\left(x_k + \frac{117}{x_k}\right)$

B. $x_{k+1} = x_k - \frac{117}{x_k}$

C. $x_{k+1} = x_k - \frac{x_k}{117}$

D. $x_{k+1} = x_k - \frac{1}{2}\left(x_k + \frac{117}{x_k}\right)$

Q7.08.1 MCQ 2008 1M

Equation $e^x - 1 = 0$ is required to be solved using Newton's method with an initial guess $x_0 = -1$. Then after one step of Newton's method estimate x_1 of the solution will be given by

A. 0.71828 **B.** 0.36784 **C.** 0.20587 **D.** 0.0000

Q7.08.2 MCQ 2008 2M

A differential equation $\frac{dx}{dt} = e^{-2t}u(t)$ has to be solved using trapezoidal rule of integration with a step size $h = 0.01$ sec. Function $u(t)$ indicates a unit step function. If $x(0) = 0$ then the value of x at $t = 0.01$ sec will be given by

A. 0.00099 **B.** 0.00495 **C.** 0.0099 **D.** 0.0198

Q7.98.2 MCQ 1998 2M

The value of $\int_1^2 \frac{1}{x}\,dx$ computed using Simpson's rule with a step size of $h = 0.25$ is

A. 0.69430 **B.** 0.69385 **C.** 0.69325 **D.** 0.69415

Instrumentation Engineering

Q7.14.1 MCQ 2014 1M

The iteration step in order to solve for the cube roots of a given number 'N' using the Newton-Raphson's method is

A. $x_{k+1} = x_k + \frac{1}{3}\left(N - x_k^3\right)$

B. $x_{k+1} = \frac{1}{3}\left(2x_k + \frac{N}{x_k^2}\right)$

C. $x_{k+1} = x_k - \frac{1}{3}\left(N - x_k^3\right)$

D. $x_{k+1} = \frac{1}{3}\left(2x_k - \frac{N}{x_k^2}\right)$

Q7.13.1 MCQ 2013 1M

While numerically solving the differential equation $\frac{dy}{dx} + 2xy^2 = 0, y(0) = 1$ using Euler's predictor corrector (improved Euler-Cauchy) method with a step size of 0.2, the value of y after the first step is

A. 1.00 **B.** 1.03 **C.** 0.97 **D.** 0.96

Q7.08.1 MCQ 2008 1M

It is known that two roots of the non-linear equation $x^3 - 6x^2 + 11x - 6 = 0$ are 1 and 3. The third root will be

A. j **B.** -j **C.** 2 **D.** 4

Q7.07.1 MCQ 2007 1M

Identify the Newton-Raphson iteration scheme for finding the square root of 2

A. $x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$

B. $x_{n+1} = \frac{1}{2}\left(x_n - \frac{2}{x_n}\right)$

C. $x_{n+1} = \frac{2}{x_n}$

D. $x_{n+1} = \sqrt{2 + x_n}$

Mechanical Engineering

Q7.26.1 MCQ 2026 1M

Newton-Raphson method for solving algebraic equations is based on

A. Taylor series

B. Fourier series

C. Laurent series

D. Power series

Q7.26.2 MCQ 2026 1M

The exact solution of $\int_0^4 \frac{dx}{1+x}$ is represented as n. If m represents numerically evaluated value of the above integral using Trapezoidal rule by considering four equal subintervals in the range of x, then $(m - n)$ is

A. 0.074 **B.** −0.074 **C.** −0.003 **D.** 0.003

Q7.25.1 NAT 2025 1M

The values of a function f obtained for different values of x are shown in the table below.

x	0	0.25	0.5	0.75	1.0
$f(x)$	0.9	2.0	1.5	1.8	0.4

Using Simpson's one-third rule,

$$\int_0^1 f(x)\,dx \approx \_\_\_\_\_\_$$

[Rounded off to 2 decimal places]

Q7.24.1 MCQ 2024 1M

In order to numerically solve the ordinary differential equation $dy/dt = -y$ for $t > 0$, with an initial condition $y(0) = 1$, the following scheme is employed:

$$\frac{y_{n+1} - y_n}{\Delta t} = -\frac{1}{2}(y_{n+1} + y_n).$$

Here, Δt is the time step and $y_n = y(n\Delta t)$ for $n = 0, 1, 2, \ldots$. This numerical scheme will yield a solution with non-physical oscillations for $\Delta t > h$. The value of h is

A. $\frac{1}{2}$ **B.** 1 **C.** $\frac{3}{2}$ **D.** 2

Q7.23.1 NAT 2023 2M

The smallest perimeter that a rectangle with area of 4 square units can have is ______ units.

[Answer in integer]

Q7.23.2 NAT 2023 2M

The initial value problem

$$\frac{dy}{dt} + 2y = 0, y(0) = 1$$

is solved numerically using the forward Euler's method with a constant and positive time step of Δt.

Let y_n represent the numerical solution obtained after n steps. The condition $|y_{n+1}| \le |y_n|$ is satisfied if and only if Δt does not exceed ______.

[Answer in integer]

Q7.17.1 MCQ | 2017 | 2M

$P(0,3)$, $Q(0.5,4)$, and $R(1,5)$ are three points on the curve defined by $f(x)$. Numerical integration is carried out using both Trapezoidal rule and Simpson's rule within limits $x = 0$ and $x = 1$ for the curve. The difference between the two results will be

A. 0 **B.** 0.25 **C.** 0.5 **D.** 1

Q7.16.1 MCQ | 2016 | 1M

The root of the function $f(x) = x^3 + x - 1$ obtained after first iteration on application of Newton-Raphson scheme using an initial guess of $x_0 = 1$ is

A. 0.682 **B.** 0.686 **C.** 0.750 **D.** 1.000

Q7.16.2 MCQ | 2016 | 1M

Numerical integration using trapezoidal rule gives the best result for a single variable function, which is

A. linear **B.** parabolic

C. logarithmic **D.** hyperbolic

Q7.16.3 NAT | 2016 | 2M

Gauss-Seidel method is used to solve the following equations (as per the given order).

$$x_1 + 2x_2 + 3x_3 = 5$$

$$2x_1 + 3x_2 + x_3 = 1$$

$$3x_1 + 2x_2 + x_3 = 3$$

Assuming initial guess as $x_1 = x_2 = x_3 = 0$, the value of x_3 after the first iteration is ________.

Q7.16.4 NAT | 2016 | 2M

The error in numerically computing the integral $\int_0^{\pi}(\sin x + \cos x)\,dx$ using the trapezoidal rule with three intervals of equal length between 0 and π is ________.

Q7.16.5 NAT | 2016 | 2M

Solve the equation $x = 10\cos(x)$ using the Newton-Raphson method. The initial guess is $x = \frac{\pi}{4}$. The value of the predicted root after the first iteration, up to second decimal, is ________.

Q7.15.1 NAT | 2015 | 2M

The values of function $f(x)$ at 5 discrete points are given below:

x	0	0.1	0.2	0.3	0.4
f(x)	0	10	40	90	160

Using Trapezoidal rule with step size of 0.1, the value of $\int_0^{0.4} f(x)\,dx$ is ________.

Q7.15.2 NAT | 2015 | 2M

Simpson's $\frac{1}{3}$ rule is used to integrate the function $f(x) = \frac{3}{5}x^2 + \frac{9}{5}$ between $x = 0$ and $x = 1$ using the least number of equal sub-intervals. The value of the integral is ________.

Q7.15.3 NAT | 2015 | 2M

Using a unit step size, the value of integral $\int_1^2 x \ln x\,dx$ by trapezoidal rule is ________.

Q7.14.1 NAT | 2014 | 1M

Using the trapezoidal rule, and dividing the interval of integration into three equal sub-intervals, the definite integral $\int_{-1}^{+1} |x|\,dx$ is ________

Q7.14.2 NAT | 2014 | 1M

The real root of the equation $5x - 2\cos x = 0$ (up to two decimal accuracy) is ________

Q7.14.3 MCQ | 2014 | 2M

Consider an ordinary differential equation $\frac{dx}{dt} = 4t + 4$. If $x = x_0$ at $t = 0$, the increment in x calculated using Runge-Kutta fourth order multi-step method with a step size of $\Delta t = 0.2$ is

A. 0.22 **B.** 0.44 **C.** 0.66 **D.** 0.88

Q7.14.4 NAT | 2014 | 2M

The value of $\int_{2.5}^{4} \ln(x)\,dx$ calculated using the Trapezoidal rule with five sub-intervals is ________.

Q7.13.1 MCQ | 2013 | 1M

Match the CORRECT pairs.

Numerical Integration scheme	Fitting Polynomial order
P. Simpson's 3/8 Rule	1. First
Q. Trapezoidal Rule	2. Second
R. Simpson's 1/3 Rule	3. Third

A. $P-2,\ Q-1,\ R-3$

B. $P-3,\ Q-2,\ R-1$

C. $P-1,\ Q-2,\ R-3$

D. $P-3,\ Q-1,\ R-2$

Q7.11.1 MCQ 2011 1M

The integral $\int_1^3 \frac{1}{x}\, dx$ when evaluated by using Simpson's $1/3^{\text{rd}}$ rule on two equal sub intervals each of length 1, equals to

A. 1.000 **B.** 1.008 **C.** 1.1111 **D.** 1.120

Civil Engineering

Q7.26.1 NAT 2026 2M

Starting with the first approximation as $x = 0.5$, the second approximation for the root of the following function by the Newton-Raphson method is ________ (rounded off to two decimal places).

$$f(x) = e^{-x} - x$$

Q7.26.2 NAT 2026 2M

Values of y for different values of x are tabulated below.

x	-2	1	2
y	28	4	16

If a second-degree interpolating polynomial $P_2(x)$ is used to represent y, the value of $P_2(0)$ is ________ (rounded off to the nearest integer).

Q7.26.3 MCQ 2026 1M

A fifth-degree polynomial in x is defined for $x > 0$. All coefficients of the polynomial are positive. The first derivative of the polynomial is obtained numerically at a point by using the first-order forward as well as the first-order backward difference methods. Identical step lengths are used for both the methods.
Following statements are made.

(I) Forward difference method underestimates the true derivative.

(II) Backward difference method overestimates the true derivative.

Which one of the following options is CORRECT?

A. Both statements (I) and (II) are FALSE.

B. Both statements (I) and (II) are TRUE.

C. Statement (I) is TRUE and statement (II) is FALSE.

D. Statement (I) is FALSE and statement (II) is TRUE.

Q7.25.1 NAT 2025 2M

Consider the differential equation given below. Using the Euler method with the step size (h) of 0.5, the value of y at $x = 1.0$ is equal to ________ (rounded off to 1 decimal place).

$$\frac{dy}{dx} = y + 2x - x^2;\ y(0) = 1 \quad (0 \le x < \infty)$$

Q7.24.1 MCQ 2024 1M

The second derivative of a function f is computed using the fourth-order Central Divided Difference method with a step length h. The CORRECT expression for the second derivative is

A. $\frac{1}{12h^2}[-f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} - f_{i-2}]$

B. $\frac{1}{12h^2}[f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} - f_{i-2}]$

C. $\frac{1}{12h^2}[-f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} + f_{i-2}]$

D. $\frac{1}{12h^2}[-f_{i+2} - 16f_{i+1} + 30f_i - 16f_{i-1} - f_{i-2}]$

Q7.24.2 NAT 2024 1M

Consider the data of $f(x)$ given in the table.

i	0	1	2
x_i	1	2	3
$f(x_i)$	0	0.3010	0.4771

The value of $f(1.5)$ estimated using second-order Newton's interpolation formula is ________ (rounded off to 2 decimal places).

Q7.23.1 MCQ | 2023 | 2M

A function $f(x)$, that is smooth and convex-shaped between interval (x_l, x_u) is shown in the figure. This function is observed at odd number of regularly spaced points. If the area under the function is computed numerically, then.

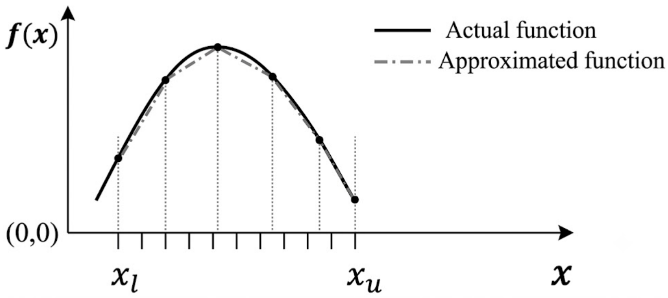


Fig. Q7.23.1

A. the numerical value of the area obtained using the trapezoidal rule will be less than the actual

B. the numerical value of the area obtained using the trapezoidal rule will be more than the actual

C. the numerical value of the area obtained using the trapezoidal rule will be exactly equal to the actual

D. with the given details, the numerical value of area cannot be obtained using trapezoidal rule

Q7.22.1 NAT | 2022 | 1M

Consider the following recursive iteration scheme for different values of variable P with the initial guess $x_1 = 1$:

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{P}{x_n}\right), n = 1, 2, 3, 4, 5$$

For $P = 2$, x_5 is obtained to be 1.414, rounded-off to three decimal places. For $P = 3$, x_5 is obtained to be 1.732, rounded-off to three decimal places. If $P = 10$, the numerical value of x_5 is ________ (round off to three decimal places).

Q7.17.1 NAT | 2017 | 2M

Consider the equation $\frac{du}{dt} = 3t^2 + 1$ with $u = 0$ at $t = 0$. This is numerically solved by using the forward Euler method with a step size $\Delta t = 2$. The absolute error in the solution at the end of the first time step is ________.

Q7.16.1 NAT | 2016 | 2M

Newton-Raphson method is to be used to find root of equation $3x - e^x + \sin x = 0$. If the initial trial value for the root is taken as 0.333, the next approximation for the root would be ________ (note: answer up to three decimal).

Q7.16.2 MCQ | 2016 | 1M

The quadratic approximation of $f(x) = x^3 - 3x^2 - 5$ at the point $x = 0$ is

A. $3x^2 - 6x - 5$

B. $-3x^2 - 5$

C. $-3x^2 + 6x - 5$

D. $3x^2 - 5$

Q7.15.1 NAT | 2015 | 2M

For step-size, $\Delta x = 0.4$, the value of following integral using Simpson's 1/3 rule is ________.

$$\int_0^{0.8} \left(0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5\right) dx$$

Q7.15.2 NAT | 2015 | 2M

In Newton-Raphson iterative method, the initial guess value (x_{ini}) is considered as zero while finding the roots of the equation:

$$f(x) = -2 + 6x - 4x^2 + 0.5x^3$$

The correction, Δx, to be added to x_{ini} in the first iteration is ________.

Q7.15.3 MCQ | 2015 | 2M

The integral $\int_{x_1}^{x_2} x^2\, dx$ with $x_2 > x_1 > 0$ is evaluated analytically as well as numerically using a single application of the trapezoidal rule. If I is the exact value of the integral obtained analytically and J is the approximate value obtained using the trapezoidal rule, which of the following statements is correct about their relationship?

A. J > I

B. J < I

C. J = I

D. Insufficient data to determine the relationship

Q7.15.4 NAT 2015 2M

The quadratic equation $x^2 - 4x + 4 = 0$ is to be solved numerically, starting with the initial guess $x_0 = 3$. The Newton-Raphson method is applied once to get a new estimate and then the Secant method is applied once using in the initial guess and this new estimate. The estimated value of the root after the application of the Secant method is ________.

Q7.13.1 NAT 2013 2M

There is no value of x that can simultaneously satisfy both the given equations. Therefore, find the 'least squares error' solution to the two equations, i.e., find the value of x that minimizes the sum of squares of the errors in the two equations:

$$2x = 3$$

$$4x = 1$$

Q7.12.1 MCQ 2012 1M

The estimate of $\int_{0.5}^{1.5} \frac{dx}{x}$ obtained using Simpson's rule with three-point function evaluation exceeds the exact value by

A. 0.235 **B.** 0.068 **C.** 0.024 **D.** 0.012

Q7.11.1 MCQ 2011 2M

The square root of a number N is to be obtained by applying the Newton – Raphson iteration to the equation $x^2 - N = 0$. If i denotes the iteration index, the correct iterative scheme will be

A. $x_{i+1} = \frac{1}{2}\left[x_i + \frac{N}{x_i}\right]$ **B.** $x_{i+1} = \frac{1}{2}\left[x_i^2 - \frac{N}{x_i^2}\right]$

C. $x_{i+1} = \frac{1}{2}\left[x_i + \frac{N^2}{x_i}\right]$ **D.** $x_{i+1} = \frac{1}{2}\left[x_i - \frac{N}{x_i}\right]$

Q7.10.1 MCQ 2010 2M

The table below gives values of a function $f(x)$ obtained for values of x at intervals of 0.25

X	0	0.25	0.50	0.75	1
F(x)	1	0.9412	0.88	0.64	0.5

The value of the integral of the function between the limits 0 to 1, using Simpson's rule is

A. 0.7854 **B.** 2.3562 **C.** 3.1416 **D.** 7.5000

Q7.09.1 MCQ 2009 2M

The area under the curve shown between $x = 1$ and $x = 3$ is to be evaluated using the trapezoidal rule. The following points on the curve are given

Point	X – coordinate (m)	Y – coordinate (m)
1	1	1
2	2	4
3	3	9

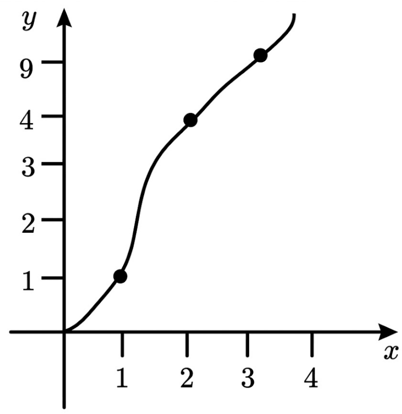


Fig. Q7.09.1

The evaluated area (in m^2) will be

A. 7 **B.** 8.67 **C.** 9 **D.** 18

Q7.08.1 MCQ 2008 1M

The Newton-Raphson iteration $x_{n+1} = \frac{1}{2}\left(x_n + \frac{R}{x_n}\right)$ can be used to compute

A. square of R **B.** reciprocal of R

C. square root of R **D.** logarithm of R

Q7.08.2 MCQ 2008 2M

If the interval of integration is divided into two equal intervals of width 1.0, the value of the definite integral $\int_1^3 \log_e x \, dx$ using simpson's one – third rule will be

A. 0.5 **B.** 0.80 **C.** 1.00 **D.** 0.29

Q7.07.1 MCQ 2007 1M

The following equation needs to be numerically solved using the Newton – Raphson method $x^3 + 4x - 9 = 0$. The iterative equation for this purpose is (k indicates the iteration level)

A. $X_{k+1} = \dfrac{2X_k^3 + 9}{3X_k^2 + 4}$ **B.** $X_{k+1} = \dfrac{3X_k^3 + 9}{2X_k^2 + 9}$

C. $X_{k+1} = X_k - 3_k^2 + 4$ **D.** $X_{k+1} = \frac{4X_k^2+3}{9X_k^2+2}$

Q7.07.2 MCQ 2007 1M

Given that one root of the equation $x^3 - 10x^2 + 31x - 30 = 0$ is 5 then other roots are

A. 2 and 3 **B.** 2 and 4

C. 3 and 4 **D.** -2 and -3

Q7.05.1 MCQ 2005 2M

Given $a > 0$, we wish to calculate its reciprocal value $\frac{1}{a}$ by using Newton - Raphson method for $f(x) = 0$. For $a = 7$ and starting with $x_0 = 0.2$ the first two iterations will be

A. 0.11, 0.1299 **B.** 0.12, 0.1392

C. 0.12, 0.1416 **D.** 0.13, 0.1428

Q7.05.2 MCQ 2005 1M

Given $a > 0$, we wish to calculate its reciprocal value $\frac{1}{a}$ by using Newton - Raphson method for $f(x) = 0$. The Newton - Raphson algorithm for the function will be

A. $X_{k+1} = \dfrac{1}{2}\left(X_k + \dfrac{a}{X_k}\right)$ **B.** $X_{k+1} = X_k + \dfrac{a}{2}X_k^2$

C. $X_{k+1} = 2X_k - aX_k^2$ **D.** $X_{k+1} = 2X_k - \dfrac{a}{2}X_k^2$

Chemical Engineering

Q7.26.1 NAT 2026 1M

Using single step trapezoidal rule, the value of

$$\int_0^{0.5} (1 + 16x - 20x^2)\, dx$$

is ________ (rounded off to two decimal places).

Q7.25.1 NAT 2025 2M

The Newton-Raphson method is used to find the root of

$$f(x) = x^2 - x - 1 = 0$$

Starting with an initial guess $x(0) = 1$, the second iterate $x(2)$ is

________ (rounded off to 2 decimal places).

Q7.24.1 NAT 2024 2M

The Newton-Raphson method is used to solve $f(x) = 0$, where $f(x) = e^x - 5x$. If the initial guess $x(0) = 1.0$, the value of the next iterate, $x(1)$, rounded off to 2 decimal places, is

____

Q7.24.2 MCQ 2024 1M

The first non-zero term in the Taylor series expansion of $(1 - x) - e^{-x}$ about $x = 0$ is

A. 1 **B.** -1 **C.** $\dfrac{x^2}{2}$ **D.** $-\dfrac{x^2}{2}$

Q7.23.1 MCQ 2023 2M

Simpson's one-third rule is used to estimate the definite integral

$$I = \int_{-1}^{1} \sqrt{(1 - x^2)}\, dx$$

with an interval length of 0.5. Which one of the following is the CORRECT estimate of I obtained using this rule?

A. $\dfrac{1}{3} - \dfrac{1}{\sqrt{3}}$ **B.** $\dfrac{1}{3} + \dfrac{2}{\sqrt{3}}$ **C.** $\dfrac{1}{3} + \dfrac{1}{\sqrt{3}}$ **D.** $\dfrac{1}{3} - \dfrac{2}{\sqrt{3}}$

Q7.22.1 NAT 2022 2M

The equation $\dfrac{dy}{dx} = xy^2 + 2y + x - 4.5$ with the initial condition $y(x = 0) = 1$ is to be solved using a predictor-corrector approach. Use a predictor based on the implicit Euler's method and a corrector based on the trapezoidal rule of integration, each with a full-step size of 0.5. Considering only positive values of y, the value of y at $x = 0.5$ is ______(rounded off to three decimal places).

Q7.21.1 MCQ 2021 1M

An ordinary differential equation (ODE), $\dfrac{dy}{dx} = 2y$, with an initial condition $y(0) = 1$, has the analytical solution $y = e^{2x}$. Using Runge-Kutta second order method, numerically integrate the ODE to calculate y at $x = 0.5$ using a step size of $h = 0.5$. If the relative percentage error is defined as,

$$\varepsilon = \left|\frac{y_{\text{analytical}} - y_{\text{numerical}}}{y_{\text{analytical}}}\right| \times 100$$

then the value of ε at $x = 0.5$ is

A. 0.06 **B.** 0.8 **C.** 4.0 **D.** 8.0

Q7.21.2 MCQ 2021 1M

The function $\cos(x)$ is approximated using Taylor series around $x = 0$ as $\cos(x) \approx 1 + ax + bx^2 + cx^3 + dx^4$. The values of a, b, c and d are

A. $a = 1, \quad b = -0.5, \quad c = -1, \quad d = -0.25$

B. $a = 0, \quad b = -0.5, \quad c = 0, \quad d = 0.042$

C. $a = 0, \quad b = 0.5, \quad c = 0, \quad d = 0.042$

D. $a = -0.5, \quad b = 0, \quad c = 0.042, \quad d = 0$

Q7.21.3 NAT 2021 2M

To solve an algebraic equation $f(x) = 0$, an iterative scheme of the type $x_{n+1} = g(x_n)$ is proposed, where $g(x) = x - \dfrac{f(x)}{f'(x)}$. At the solution $x = s$, $g'(s) = 0$ and $g''(s) \neq 0$. The order of convergence for this iterative scheme near the solution is ______

Q7.20.1 MCQ 2020 1M

Which one of the following methods requires specifying an initial interval containing the root (i.e., bracketing) to obtain the solution of $f(x) = 0$, where $f(x)$ is a continuous non-linear algebraic function?

A. Newton-Raphson method

B. regula falsi method

C. secant method

D. fixed point iteration method

Q7.20.2 NAT 2020 2M

Consider the following dataset.

x	1	3	5	15	25
$f(x)$	6	8	10	12	5

The value of the integral $\int_1^{25} f(x)\,dx$ using Simpson's $1/3^{\text{rd}}$ rule is ______(round off to 1 decimal place).

Q7.19.1 NAT 2019 2M

The Newton-Raphson method is used to determine the root of the equation $f(x) = e^{-x} - x$. If the initial guess for the root is 0, the estimate of the root after two iterations is ______ (rounded off to three decimal places).

Q7.18.1 MCQ 2018 1M

The fourth order Runge-Kutta (RK4) method to solve an ordinary differential equation $\dfrac{dy}{dx} = f(x, y)$ is given as

$$y(x+h) = y(x) + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = hf(x, y)$$

$$k_2 = hf\left(x + \frac{h}{2}, y + \frac{k_1}{d}\right)$$

$$k_3 = hf\left(x + \frac{h}{2}, y + \frac{k_2}{d}\right)$$

$$k_4 = hf(x + h, y + k_3)$$

For a special case when the function f depends solely on x, the above RK4 method reduces to

A. Euler's explicit method

B. Trapezoidal rule

C. Euler's implicit method

D. Simpson's 1/3 rule

Q7.17.1 NAT 2017 1M

The number of positive roots of the function $f(x)$ shown below in the range $0 < x < 6$ is

______

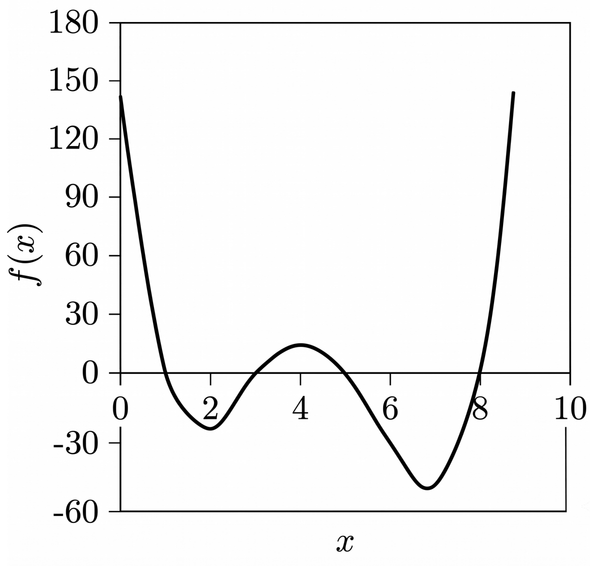


Fig. Q7.17.1

Q7.17.2 MCQ 2017 2M

Match the problem type in Group-1 with the numerical method in Group-2.

Group-1	Group-2
P) System of linear algebraic equations	I) Newton-Raphson
Q) Non-linear algebraic equations	II) Gauss-Seidel
R) Ordinary differential equations	III) Simpson's rule
S) Numerical integration	IV) Runge-Kutta

A. P-II, Q-I, R-III, S-IV **B.** P-I, Q-II, R-IV, S-III

C. P-IV, Q-III, R-II, S-I **D.** P-II, Q-I, R-IV, S-III

Q7.16.1 MCQ 2016 1M

Which one of the following is an iterative technique for solving a system of simultaneous linear algebraic equations?

A. Gauss elimination **B.** Gauss-Jordan

C. Gauss-Seidel **D.** LU decomposition

Q7.16.2 NAT 2016 2M

The model $y = mx^2$ is to be fit to the data given below.

x	1	$\sqrt{2}$	$\sqrt{3}$
y	2	5	8

Using linear regression, the value (rounded off to the second decimal place) of m is

____

Q7.16.3 NAT 2016 2M

Values of $f(x)$ in the interval $[0, 4]$ are given below.

x	0	1	2	3	4
$f(x)$	3	10	21	36	55

Using Simpson's 1/3 rule with a step size of 1, the numerical approximation (rounded off to the second decimal place) of $\int_0^4 f(x)\,dx$ is

____

Q7.15.1 MCQ 2015 2M

The solution of the non-linear equation $x^3 - x = 0$ is to be obtained using Newton-Raphson method. If the initial guess is $x = 0.5$, the method converges to which one of the following values?

A. -1 **B.** 0 **C.** 1 **D.** 2

Q7.13.1 NAT 2013 2M

The value of the integral $\int_{0.1}^{0.5} e^{-x^3}\,dx$ evaluated by Simpson's rule using 4 subintervals (up to 3 digits after the decimal point) is

____

Q7.12.1 MCQ 2012 2M

The Newton-Raphson method is used to find the roots of the equation

$$f(x) = x - \cos \pi x \quad 0 \le x \le 1$$

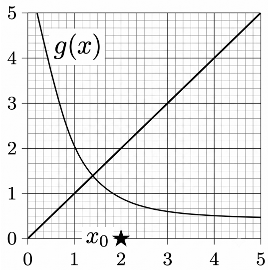
If the initial guess for the root is 0.5, then the value of x after the first iteration is

A. 1.02 **B.** 0.62 **C.** 0.55 **D.** 0.38

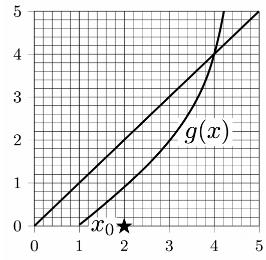
Q7.11.1 MCQ 2011 2M

In the fixed point iteration method for solving equations of the form $x = g(x)$, the $(n+1)^{th}$ iteration value is $x_{n+1} = g(x_n)$, where x_n represents the n^{th} iteration value. $g(x)$ and corresponding initial guess value x_0 in the domain of interest are shown in the following choices. Which ONE of these choices leads to a converged solution for x?

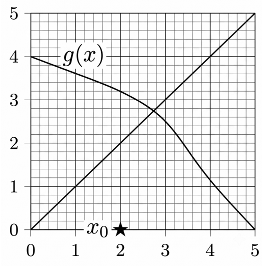
A.



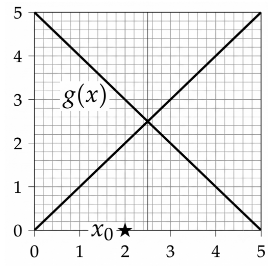
B.



C.



D.



Q7.10.1 | MCQ | 2010 | 1M

A root of the equation $x^4 - 3x + 1 = 0$ needs to be found using the Newton-Raphson method. If the initial guess, x_0, is taken as 0, then the new estimate x_1, after the first iteration is

A. $\frac{1}{3}$ **B.** $-\frac{1}{3}$ **C.** 3 **D.** -3

Q7.09.1 | MCQ | 2009 | 2M

Using the trapezoidal rule and 4 equal intervals ($n = 4$), the calculated value of the integral (rounded to the first place of decimal) $\int_0^{\pi} \sin\theta \, d\theta$ is

A. 1.7 **B.** 1.9 **C.** 2.0 **D.** 2.1

Q7.08.1 | MCQ | 2008 | 1M

A nonlinear function $f(x)$ is defined in the interval $-1.2 < x < 4$ as illustrated in the figure below. The equation $f(x) = 0$ is solved for x within this interval by using the Newton-Raphson iterative scheme. Among the initial guesses (I_1, I_2, I_3 and I_4), the guess that is likely to lead to the root most rapidly is

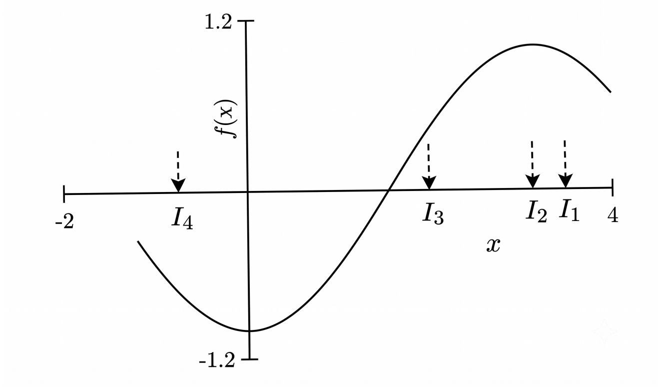


Fig. Q7.08.1

A. I_1 **B.** I_2 **C.** I_3 **D.** I_4

Q7.08.2 | MCQ | 2008 | 2M

Using Simpson's 1/3 rule and FOUR equally spaced intervals ($n = 4$), estimate the value of the integral $\int_0^{\pi/4} \frac{\sin x}{\cos^3 x} dx$

A. 0.3887 **B.** 0.4384 **C.** 0.5016 **D.** 0.5527

Q7.08.3 | MCQ | 2008 | 2M

The following differential equation is to be solved numerically by the Euler's explicit method.

$$\frac{dy}{dx} = x^2 y - 1.2y \quad \text{with } y(0) = 1$$

A step size of 0.1 is used. The solution for y at $x = 0.1$ is

A. 0.880 **B.** 0.905 **C.** 1.000 **D.** 1.100

Q7.06.1 | MCQ | 2006 | 1M

The ordinary differential equation $dY/dt = f(Y)$ is solved using the approximation $Y(t + \Delta t) = Y(t) + f[Y(t)]/\Delta t$. The numerical error introduced by the approximation at each step is

A. proportional to Δt

B. proportional to $(\Delta t)^2$

C. independent of Δt

D. proportional to $(1/\Delta t)$

Q7.06.2 MCQ 2006 1M

The trepezoidal rule of integration when applied to $\int_a^b f(x)\,dx$ will give the exact value of the integral

A. if $f(x)$ is a linear function of x

B. if $f(x)$ is a quadratic function of x

C. for any $f(x)$

D. for no $f(x)$

Q7.06.3 MCQ 2006 2M

The Newton-Raphson method is used to solve the equation, $(x-1)^2 + x - 3 = 0$. The method will fail in the very first iteration if the initial guess is

A. 0 **B.** 0.5 **C.** 1 **D.** 3

Q7.05.1 MCQ 2005 2M

The function $f(x)$ satisfies the equation $f(x) = 0$ at $x = x_e$. The Newton-Raphson iterative method converges to the solution in one step, regardless of the initial guess, if

A. $f(x)$ is a linear function of x

B. $f(x)$ is a quadratic function of x

C. $f(x)$ is a cubic function of x

D. $f(x)$ is an exponential function of x

Production & Industrial Engineering

Q7.16.1 MCQ 2016 1M

To solve the equation $2\sin x = x$ by Newton-Raphson method, the initial guess was chosen to be $x = 2.0$. Consider x in radian only. The value of x (in radian) obtained after one iteration will be closest to

A. -8.101 **B.** 1.901 **C.** 2.099 **D.** 12.101

Q7.15.1 MCQ 2015 2M

In numerical integration using Simpson's rule, the approximating function in the interval is a

A. constant **B.** straight line

C. cubic B-Spline **D.** Parabola

Q7.14.1 NAT 2014 1M

Using the Simpson's $1/3^{\text{rd}}$ rule, the value of $\int y\,dx$ computed, for the data given below, is ________.

x	1	3	5
y	2	6	4

Q7.14.2 NAT 2014 1M

If the equation $\sin(x) = x^2$ is solved by Newton Raphson's method with the initial guess of $x = 1$, then the value of x after 2 iterations would be ________.

Q7.10.1 MCQ 2010 2M

The following algorithm computes the integral $J = \int_a^b f(x)\,dx$ from the given values $f_j = f(x_j)$ at equidistant points $x_0 = a,\ x_1 = x_0 + h,\ x_2 = x_0 + 2h, \ldots\ x_{2m} = x_0 + 2mh = b$ compute

$$S_0 = f_0 + f_{2m}$$
$$S_1 = f_1 + f_3 + \cdots + f_{2m-1}$$
$$S_2 = f_2 + f_4 + \cdots + f_{2m-2}$$
$$J = \frac{h}{3}\left[S_0 + 4(S_1) + 2(S_2)\right]$$

The rule of numerical integration, which uses the above algorithm is

A. Rectangle rule **B.** Trapezoidal

C. Four – point rule **D.** Simpson's rule

Q7.09.1 MCQ 2009 1M

During the numerical solution of a first order differential equation using the Euler (also known as Euler Cauchy) method with step size h, the local truncation error is of the order of

A. h^2 **B.** h^3 **C.** h^4 **D.** h^5

Q7.07.1 MCQ 2007 2M

Matching exercise choose the correct one out of the alternatives A, B, C, D

Group – I

P. 2^{nd} order differential equations

Q. Non-linear algebraic equations

R. Linear algebraic equations

S. Numerical integration

Group – II

(1) Runge – Kutta method

(2) Newton – Raphson method

(3) Gauss Elimination

(4) Simpson's Rule

A. $P-3,\ Q-2,\ R-4,\ S-1$

B. $P-2,\ Q-4,\ R-3,\ S-1$

C. $P-1,\ Q-2,\ R-3,\ S-4$

D. $P-1,\ Q-3,\ R-2,\ S-4$

Q7.05.1 MCQ 2005 2M

The real root of the equation $xe^x = 2$ is evaluated using Newton – Raphson's method. If the first approximation of the value of x is 0.8679, the 2^{nd} approximation of the value of x correct to three decimal places is

A. 0.865 **B.** 0.853 **C.** 0.849 **D.** 0.838

Q7.05.2 MCQ 2005 1M

For solving algebraic and transcendental equation which one of the following is used?

A. Coulomb's theorem

B. Newton - Raphson method

C. Euler's method

D. Stoke's theorem

Q7.05.3 MCQ 2005 1M

Newton – Raphson formula to find the roots of an equation $f(x) = 0$ is given by

A. $X_{n+1} = X_n - \dfrac{f(X_n)}{f^1(X_n)}$ **B.** $X_{n+1} = X_n + \dfrac{f(X_n)}{f^1(X_n)}$

C. $X_{n+1} = \dfrac{f(X_n)}{X_n f^1(X_n)}$ **D.** $X_{n+1} = \dfrac{X_n f(X_n)}{f^1(X_n)}$

Answer Key

ELECTRONICS & COMMUNICATION ENGINEERING					
Q7.17.1 0.68 to 0.70	Q7.16.1 0.06	Q7.16.2 0.66	Q7.15.1 4.30	Q7.14.1 B	Q7.11.1 C
Q7.10.1 B	Q7.08.1 C				
ELECTRICAL ENGINEERING					
Q7.23.1 C	Q7.23.2 0.99 to 1.01	Q7.15.1 C	Q7.11.1 B	Q7.14.1 0.06	Q7.09.1 A
Q7.08.1 A	Q7.08.2 C	Q7.98.2 C			
INSTRUMENTATION ENGINEERING					
Q7.14.1 B	Q7.13.1 D	Q7.08.1 C	Q7.07.1 A		
MECHANICAL ENGINEERING					
Q7.26.1 A	Q7.26.2 A	Q7.25.1 1.61 to 1.65	Q7.24.1 D	Q7.23.1 7.999 to 8.001	Q7.23.2 0.999 to 1.001
Q7.17.1 A	Q7.16.1 C	Q7.16.2 A	Q7.16.3 -6	Q7.16.4 0.186	Q7.16.5 1.56
Q7.15.1 22	Q7.15.2 2	Q7.15.3 0.69	Q7.14.1 1.1	Q7.14.2 0.3726	Q7.14.3 D
Q7.14.4 1.7532	Q7.13.1 D	Q7.11.1 C			
CIVIL ENGINEERING					
Q7.26.1 0.55 to 0.58	Q7.26.2 2	Q7.26.3 A	Q7.25.1 2.4 to 2.8	Q7.24.1 A	Q7.24.2 0.16 to 0.18
Q7.23.1 A	Q7.22.1 3.1 to 3.2	Q7.17.1 8	Q7.16.1 0.3601	Q7.16.2 B	Q7.15.1 1.367
Q7.15.2 0.33	Q7.15.3 A	Q7.15.4 2.33	Q7.13.1 0.5	Q7.12.1 D	Q7.11.1 A
Q7.10.1 A	Q7.09.1 C	Q7.08.1 C	Q7.08.2 D	Q7.07.1 A	Q7.07.2 A
Q7.05.1 B	Q7.05.2 C				
CHEMICAL ENGINEERING					
Q7.26.1 1.25	Q7.25.1 1.63	Q7.24.1 -0.01 to 0.01	Q7.24.2 D	Q7.23.1 B	Q7.22.1 0.875
Q7.21.1 D	Q7.21.2 B	Q7.21.3 2	Q7.20.1 B	Q7.20.2 241.4 to 242.4	Q7.19.1 0.565 to 0.567

Q7.18.1 D	Q7.17.1 3	Q7.17.2 D	Q7.16.1 C	Q7.16.2 2.57	Q7.16.3 94.67
Q7.15.1 A	Q7.13.1 0.385	Q7.12.1 D	Q7.11.1 A	Q7.10.1 A	Q7.09.1 B
Q7.08.1 C	Q7.08.2 C	Q7.08.3 A	Q7.06.1 B	Q7.06.2 A	Q7.06.3 B
Q7.05.1 A					
PRODUCTION & INDUSTRIAL ENGINEERING					
Q7.16.1 B	Q7.15.1 D	Q7.14.1 20	Q7.14.2 0.86 to 0.89	Q7.10.1 D	Q7.09.1 A
Q7.07.1 C	Q7.05.1 B	Q7.05.2 B	Q7.05.3 A		

Also in this Series

Every PrepFusion GATE title is built the same way: the real previous-year papers, nothing paraphrased or reconstructed, arranged so the pattern the exam repeats is visible on the page. Each questions volume has a companion solutions volume that reuses the same question identifiers, so you can move between the two without a lookup table.

NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.